

[illegible]

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```
<ipython-input-22-427afd947222>:54: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
```

```
augmented_signal_df['label'] = labels[0] # Assuming a single label for each signal, adjust if there are multiple
```

```
In [ ]: import os

# Folder path
folder_path = '/content/PPG_Data/AugmentedDatawithlabels' # Replace with your folder path

# List all items (files and directories) in the folder
items = os.listdir(folder_path)

# Count the number of items
item_count = len(items)

print(f'Total number of items in the folder: {item_count}')
```

Total number of items in the folder: 268

```
In [ ]: df=pd.read_csv('/content/PPG_Data/AugmentedDatawithlabels/augmented_signal_01_0001_filtered_downs
df
```

```
Out[ ]:
```

	signal	label
0	0.001321	1
1	-0.003048	1
2	-0.000869	1
3	0.108074	1
4	7.170749	1
...	...	...
301	4.456760	1
302	6.393533	1
303	5.484700	1
304	1.625583	1
305	-2.911665	1

306 rows × 2 columns

```
In [ ]: import numpy as np
import pandas as pd
import os

# Folder containing the original PPG signals
signals_folder = '/content/PPG_Data/DownsampledData'

# Folder to save augmented data
augmented_folder = '/content/PPG_Data/AugmentedDatawithlabels'

# Folder containing the labels (if they are separate, else integrate this into the signal files)
```



```

labels_folder = '/content/PPG_Data/Labels'

# Create the output folder if it doesn't exist
if not os.path.exists(augmented_folder):
    os.makedirs(augmented_folder)

# List all CSV files containing PPG signals
signal_files = [f for f in os.listdir(signals_folder) if f.endswith('.csv')]
signal_files.sort()

# List all label files (assuming one label file per signal file)
label_files = [f for f in os.listdir(labels_folder) if f.endswith('.csv')]
label_files.sort()

# Define the augmentation function (e.g., adding Gaussian noise)
def add_gaussian_noise(signal, std_dev=0.01):
    """Add Gaussian noise to the signal."""
    noise = np.random.normal(0, std_dev, len(signal))
    return signal + noise

# Function to augment data and map labels
def augment_data_with_labels(signal_files, label_files, std_devs=[0.01, 0.05, 0.1, 0.2], num_aug=1):
    for idx, signal_file in enumerate(signal_files):
        # Read the signal data
        signal_data = pd.read_csv(os.path.join(signals_folder, signal_file), header=None).values

        # Get the corresponding label file (same index as the signal file)
        label_file = label_files[idx]

        # Read the labels from the corresponding label file
        labels_df = pd.read_csv(os.path.join(labels_folder, label_file))

        # Assuming the label file has the following columns: 'ID', 'Gender', 'Age', 'Height', 'Weight', 'Glucose'
        # We can get the first row of the labels, assuming it corresponds to the signal
        label_row = labels_df.iloc[0] # Get the first row of the label file for corresponding signal

        # Extract each label
        ID = label_row['ID']
        Gender = label_row['Gender']
        Age = label_row['Age']
        Height = label_row['Height']
        Weight = label_row['Weight']
        Glucose = label_row['Glucose']

        # Augment the signal by adding Gaussian noise with different standard deviations
        for std_dev in std_devs:
            augmented_signal = add_gaussian_noise(signal_data, std_dev=std_dev) # Apply Gaussian noise

            # Convert to DataFrame and add the label(s) + additional information
            augmented_signal_df = pd.DataFrame(augmented_signal, columns=['signal'])

            # Add extra columns for the labels (ID, Gender, Age, Height, Weight, Glucose)
            augmented_signal_df['ID'] = ID
            augmented_signal_df['Gender'] = Gender
            augmented_signal_df['Age'] = Age
            augmented_signal_df['Height'] = Height
            augmented_signal_df['Weight'] = Weight
            augmented_signal_df['Glucose'] = Glucose

```



```
# Save the augmented data to a new CSV file
augmented_file_name = f"augmented_{signal_file.split('.')[0]}_std_{std_dev}_aug.csv"
augmented_signal_df.to_csv(os.path.join(augmented_folder, augmented_file_name), index=False)

print(f"Augmented {signal_file} with different std_devs.")

# Run augmentation and save to a new folder
augment_data_with_labels(signal_files, label_files, std_devs=[0.01, 0.05, 0.1, 0.2])
```

[illegible]

Augmented signal_20_0003_filtered_downsampled.csv with different std_devs.
Augmented signal_21_0001_filtered_downsampled.csv with different std_devs.
Augmented signal_21_0002_filtered_downsampled.csv with different std_devs.
Augmented signal_22_0001_filtered_downsampled.csv with different std_devs.
Augmented signal_22_0002_filtered_downsampled.csv with different std_devs.
Augmented signal_22_0003_filtered_downsampled.csv with different std_devs.
Augmented signal_23_0001_filtered_downsampled.csv with different std_devs.
Augmented signal_23_0002_filtered_downsampled.csv with different std_devs.

```
In [ ]: import numpy as np
import pandas as pd
import os
from sklearn.preprocessing import MinMaxScaler

# Folder containing the augmented PPG signals with labels
augmented_folder = '/content/PPG_Data/AugmentedDatawithlabels'

# Folder to save normalized data
normalized_folder = '/content/PPG_Data/NormalizedDataafteraugmentation'

# Create the output folder if it doesn't exist
if not os.path.exists(normalized_folder):
    os.makedirs(normalized_folder)

# List all CSV files containing the augmented data
augmented_files = [f for f in os.listdir(augmented_folder) if f.endswith('.csv')]
augmented_files.sort()

# Function to normalize the signal data
def normalize_signal_data(signal_df):
    # Initialize Min-Max scaler to normalize between 0 and 1
    scaler = MinMaxScaler()

    # Normalize the 'signal' column (assuming it's the first column)
    signal_df['signal'] = scaler.fit_transform(signal_df[['signal']])

    return signal_df

# Function to apply normalization to all augmented files
def normalize_all_files(augmented_files):
    for augmented_file in augmented_files:
        # Read the augmented signal data
        augmented_signal_df = pd.read_csv(os.path.join(augmented_folder, augmented_file))

        # Normalize the signal data
        normalized_signal_df = normalize_signal_data(augmented_signal_df)

        # Save the normalized data to a new CSV file
        normalized_file_name = f"normalized_{augmented_file}"
        normalized_signal_df.to_csv(os.path.join(normalized_folder, normalized_file_name), index=False)

        print(f"Normalized {augmented_file} and saved to {normalized_folder}")

# Run normalization and save to a new folder
normalize_all_files(augmented_files)
```

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PG_Data/NormalizedDataafteraugmentation

Normalized augmented_signal_23_0002_filtered_downsampled_std_0.1_aug.csv and saved to /content/PP

G_Data/NormalizedDataafteraugmentation

Normalized augmented_signal_23_0002_filtered_downsampled_std_0.2_aug.csv and saved to /content/PP

G_Data/NormalizedDataafteraugmentation

```
In [ ]: import os
import pandas as pd

# Folder containing the augmented data with labels
augmented_folder = '/content/PPG_Data/AugmentedDatawithlabels' # Change this to your path

# Function to apply the manual encoding (Male to 0, Female to 1)
def encode_gender_column(df):
    # Check if 'Gender' column exists
    if 'Gender' in df.columns:
        # Convert 'Male' to 0 and 'Female' to 1
        df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})
    return df

# List all CSV files containing the augmented data with labels
augmented_files = [f for f in os.listdir(augmented_folder) if f.endswith('.csv')]
augmented_files.sort()

# Iterate over each file
for file in augmented_files:
    file_path = os.path.join(augmented_folder, file)

    # Read the data
    df = pd.read_csv(file_path)

    # Apply the gender encoding
    df = encode_gender_column(df)

    # Save the modified file back to the same folder (or to a new folder if required)
    df.to_csv(file_path, index=False)
    print(f"Processed and saved: {file}")
```

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Processed and saved: augmented_signal_20_0003_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_20_0003_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_20_0003_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_20_0003_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_21_0001_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_21_0001_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_21_0001_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_21_0001_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_21_0002_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_21_0002_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_21_0002_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_21_0002_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_22_0001_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_22_0001_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_22_0001_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_22_0001_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_22_0002_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_22_0002_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_22_0002_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_22_0002_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_22_0003_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_22_0003_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_22_0003_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_22_0003_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_23_0001_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_23_0001_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_23_0001_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_23_0001_filtered_downsampled_std_0.2_aug.csv
Processed and saved: augmented_signal_23_0002_filtered_downsampled_std_0.01_aug.csv
Processed and saved: augmented_signal_23_0002_filtered_downsampled_std_0.05_aug.csv
Processed and saved: augmented_signal_23_0002_filtered_downsampled_std_0.1_aug.csv
Processed and saved: augmented_signal_23_0002_filtered_downsampled_std_0.2_aug.csv

```
In [ ]: df=pd.read_csv('/content/PPG_Data/AugmentedDatawithlabels/augmented_signal_01_0001_filtered_downs
df
```



```
Out[ ]:
```

	signal	ID	Gender	Age	Height	Weight	Glucose
0	-0.001689	1	0	38	180	53	99
1	0.006433	1	0	38	180	53	99
2	0.005465	1	0	38	180	53	99
3	0.113350	1	0	38	180	53	99
4	7.175986	1	0	38	180	53	99
...	...	...	...	...	...	...	...
301	4.445192	1	0	38	180	53	99
302	6.387539	1	0	38	180	53	99
303	5.492667	1	0	38	180	53	99
304	1.655626	1	0	38	180	53	99
305	-2.909335	1	0	38	180	53	99

306 rows × 7 columns

```
In [ ]: df['Gender'].value_counts()
```

```
Out[ ]:
```

Gender	count
0	306

dtype: int64

```
In [ ]: import matplotlib.pyplot as plt
import pandas as pd
import os

# Folder containing the normalized PPG signals with Labels
normalized_folder = '/content/PPG_Data/NormalizedDataafteraugmentation'

# List all CSV files containing the normalized data
normalized_files = [f for f in os.listdir(normalized_folder) if f.endswith('.csv')]
normalized_files.sort()

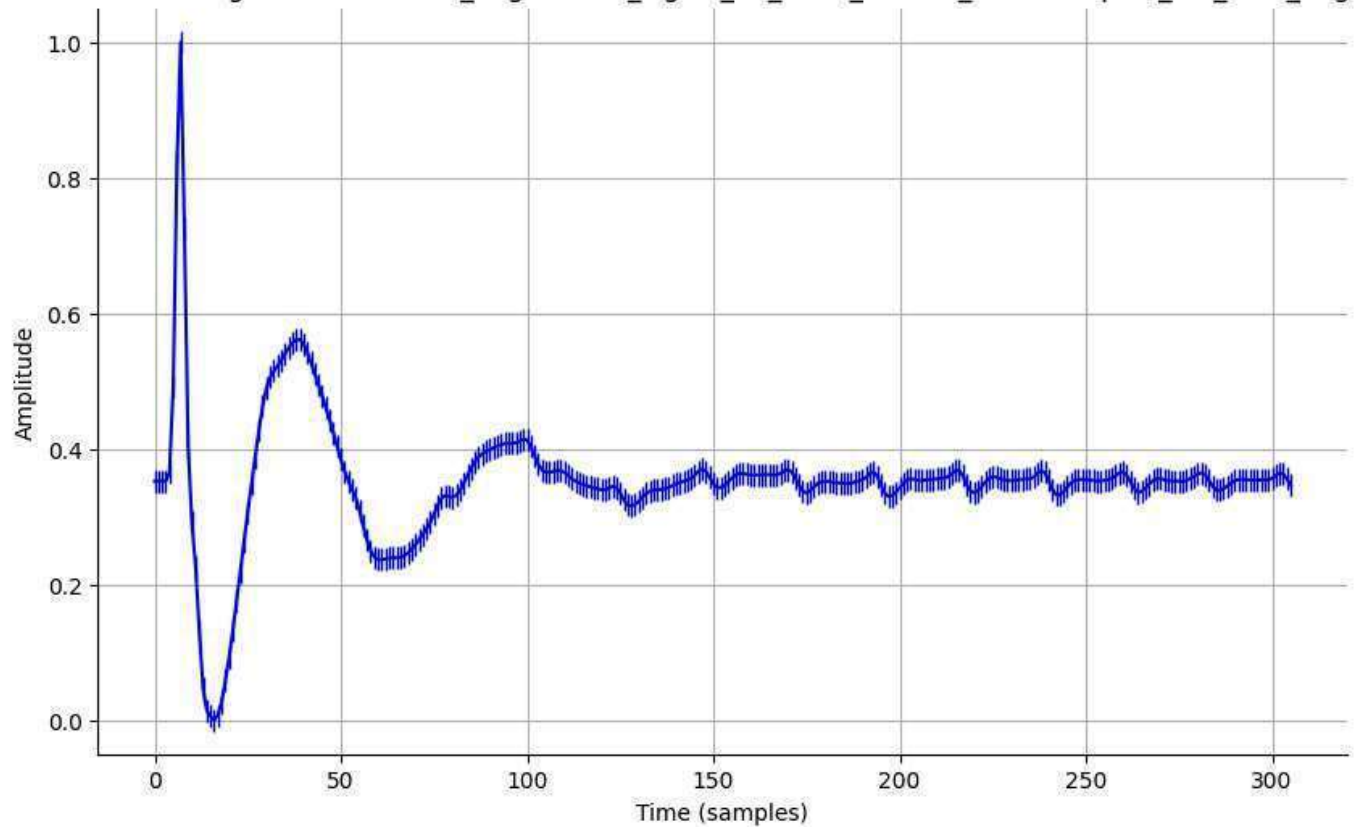
# Function to plot the spike chart for each normalized signal
def plot_spike_chart(normalized_files):
    for normalized_file in normalized_files:
        # Read the normalized signal data
        normalized_signal_df = pd.read_csv(os.path.join(normalized_folder, normalized_file))

        # Plot the 'signal' column as a spike plot
        plt.figure(figsize=(10, 6))
        plt.plot(normalized_signal_df['signal'], marker='|', markersize=10, color='b') # Spikes
        plt.title(f'Normalized Signal - {normalized_file}')
        plt.xlabel('Time (samples)')
        plt.ylabel('Amplitude')
```

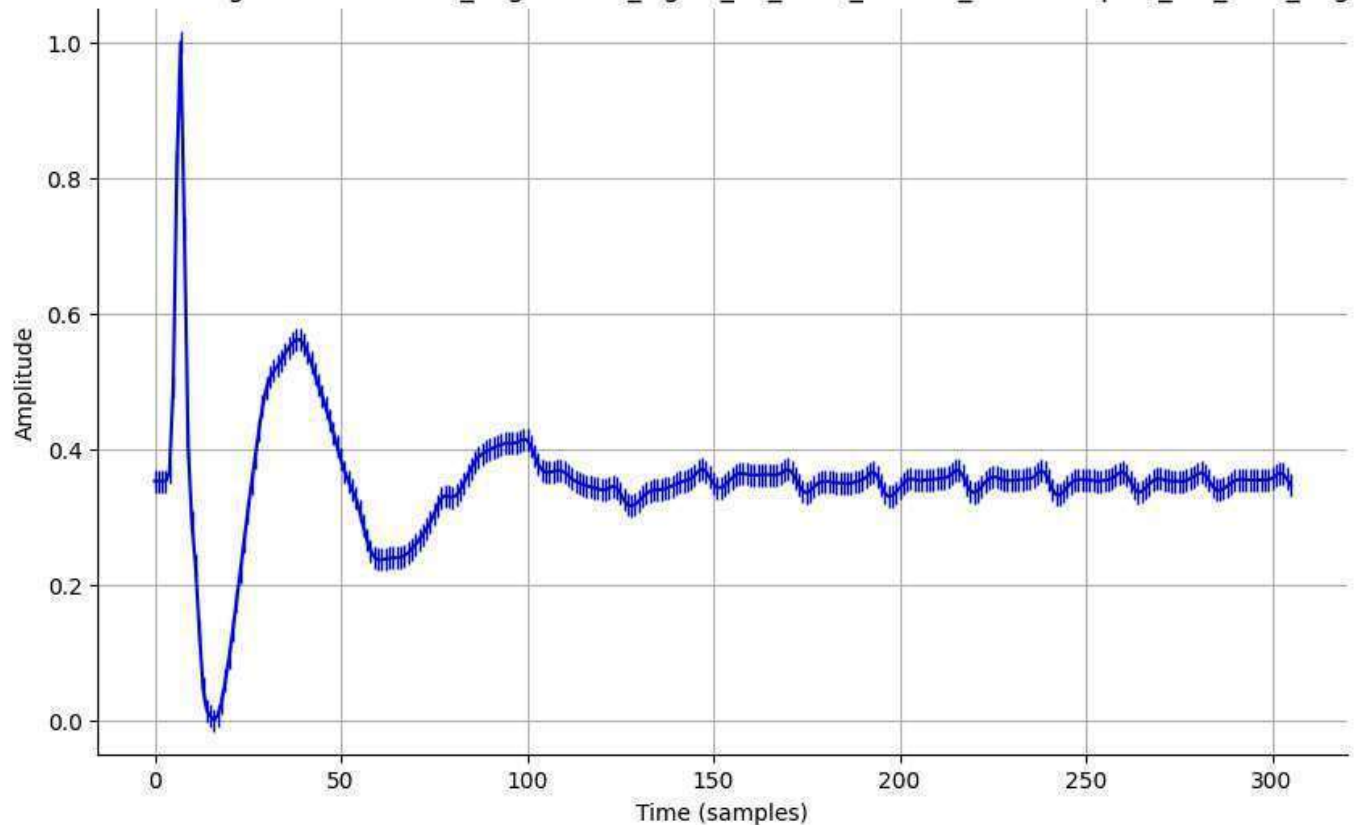
```
plt.grid(True)
plt.gca().spines[['top', 'right']].set_visible(False) # Remove top and right spines
plt.show()
```

Run the plotting function for all normalized files
plot_spike_chart(normalized_files)

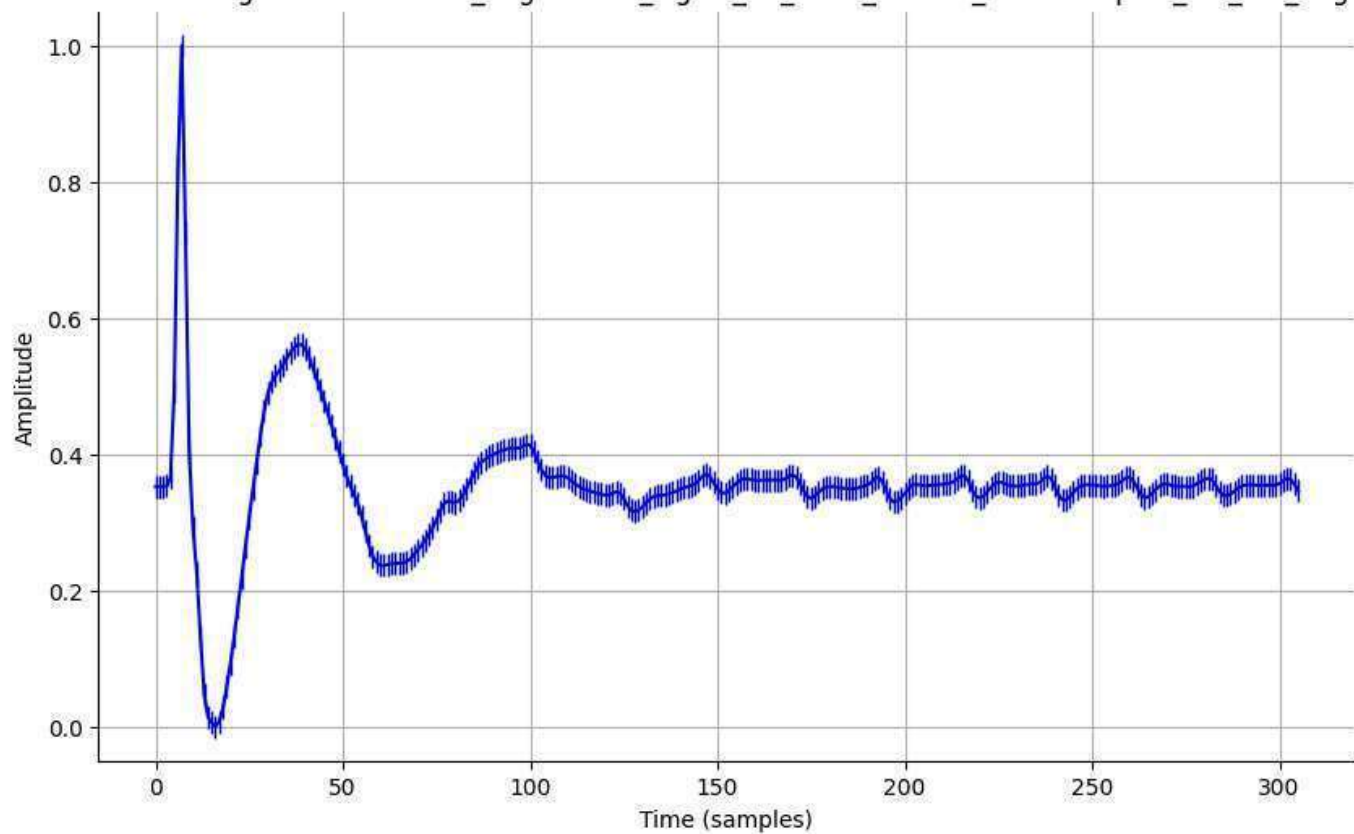
Normalized Signal - normalized_augmented_signal_01_0001_filtered_downsampled_std_0.01_aug.csv



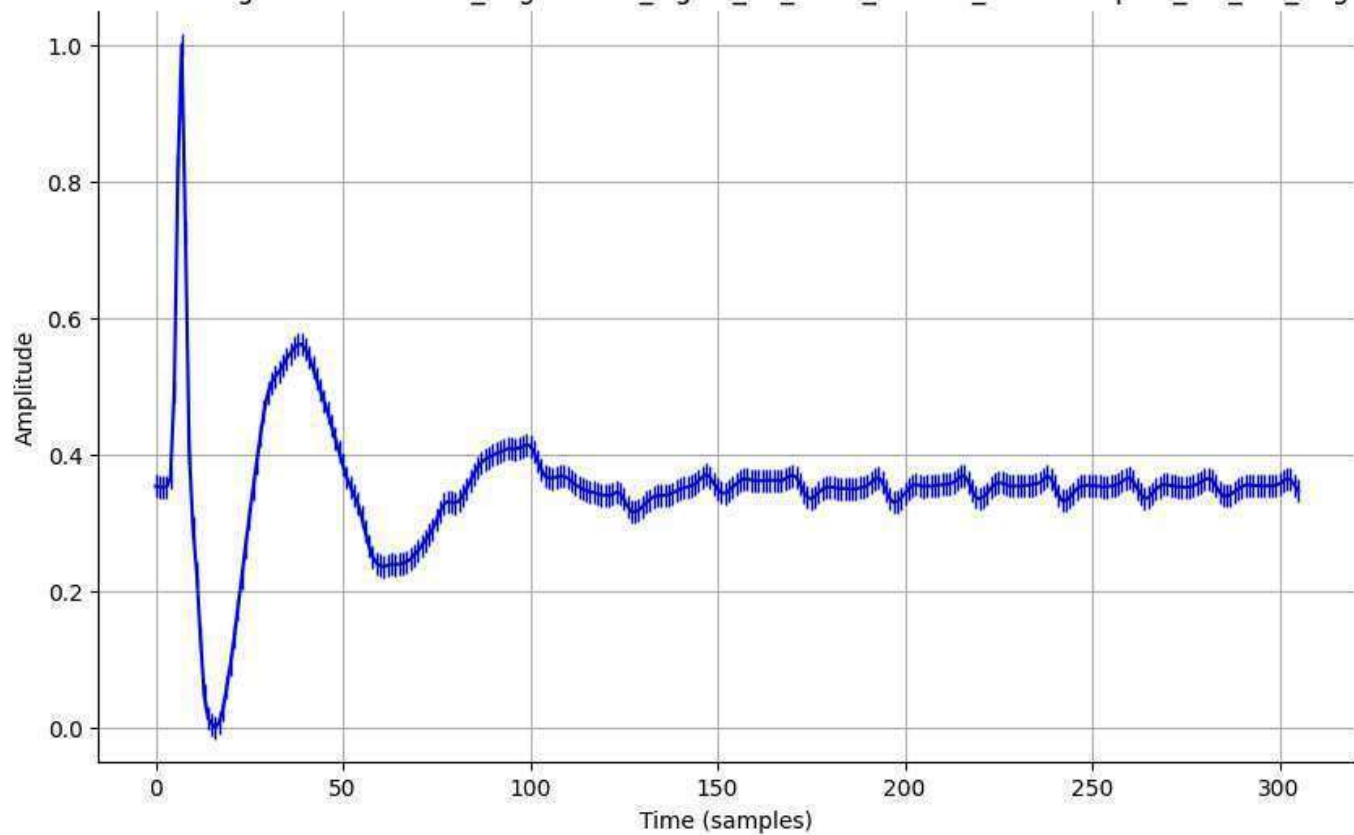
Normalized Signal - normalized_augmented_signal_01_0001_filtered_downsampled_std_0.05_aug.csv



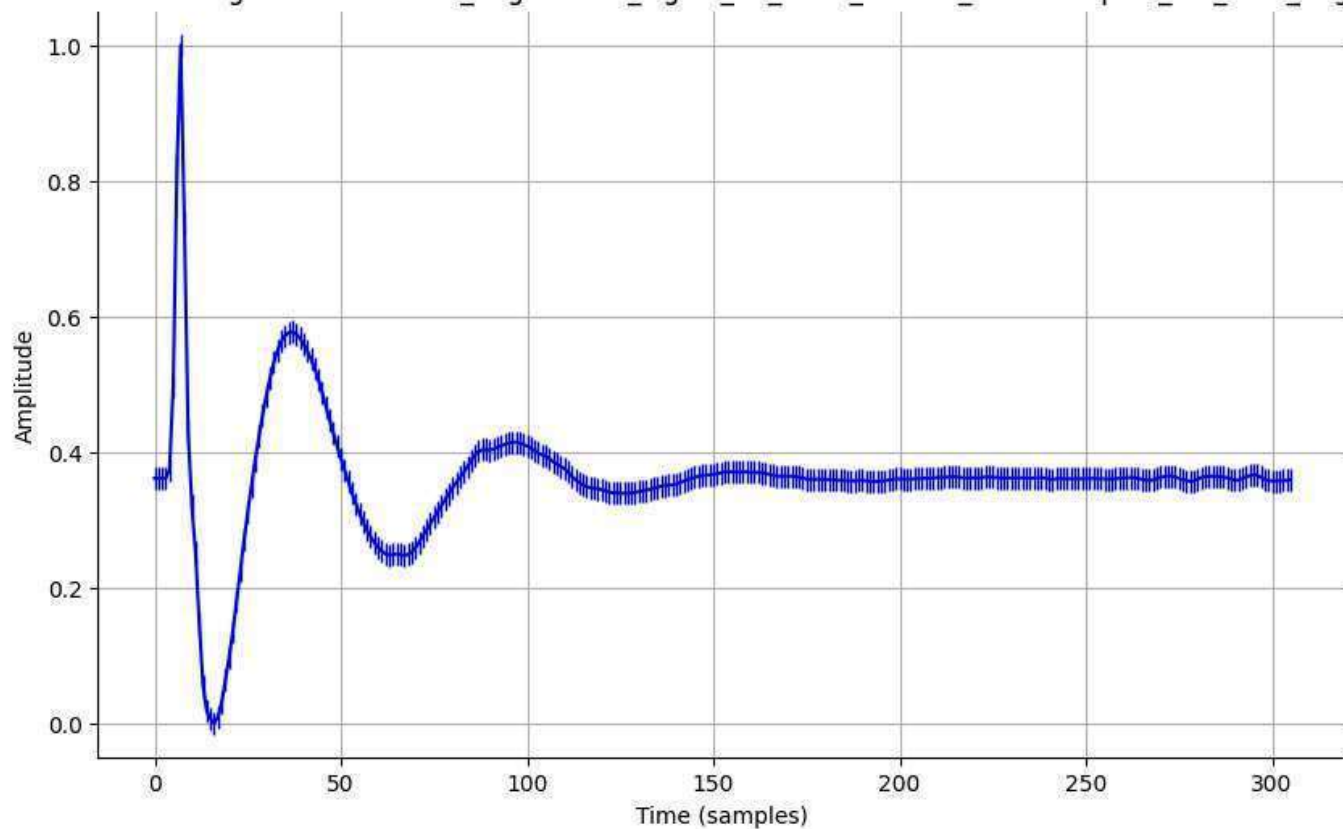
Normalized Signal - normalized_augmented_signal_01_0001_filtered_downsampled_std_0.1_aug.csv



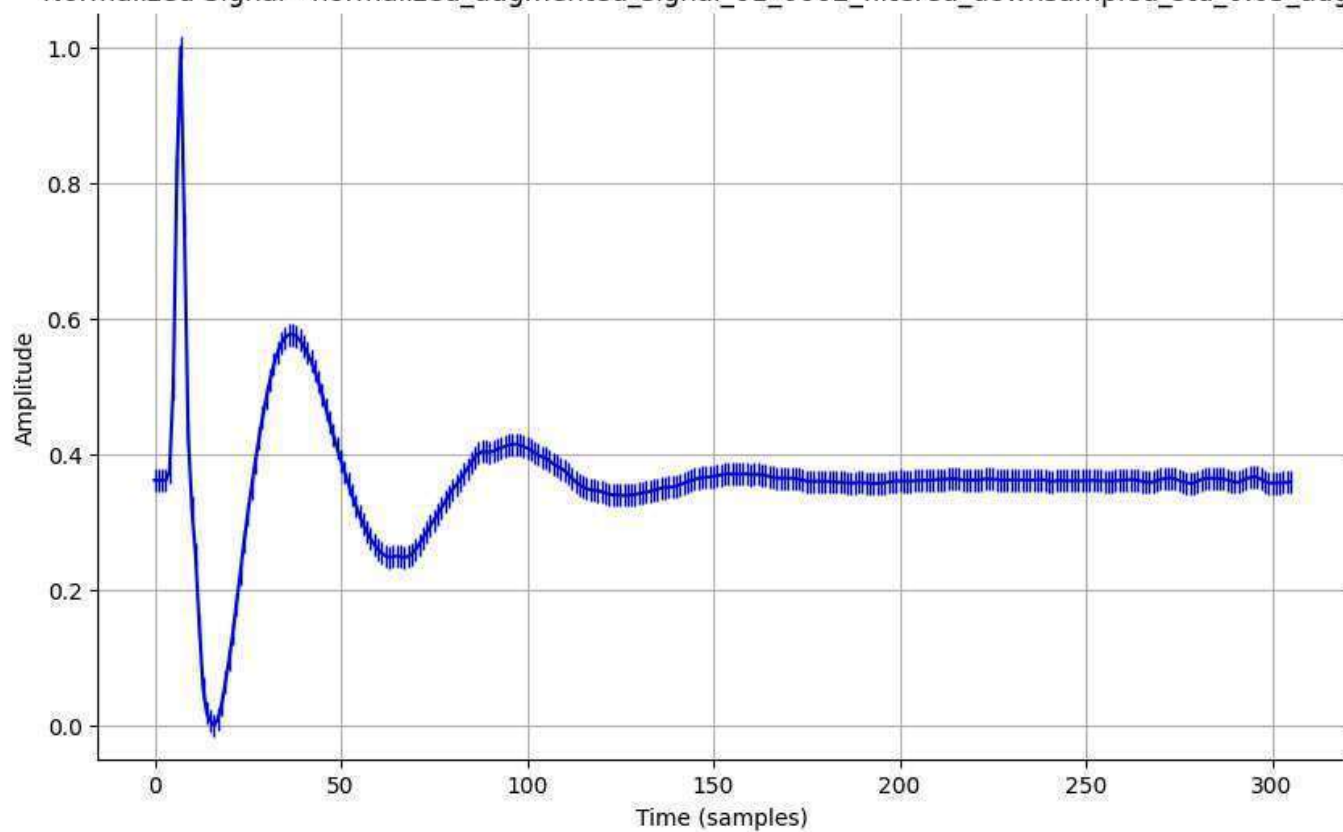
Normalized Signal - normalized_augmented_signal_01_0001_filtered_downsampled_std_0.2_aug.csv



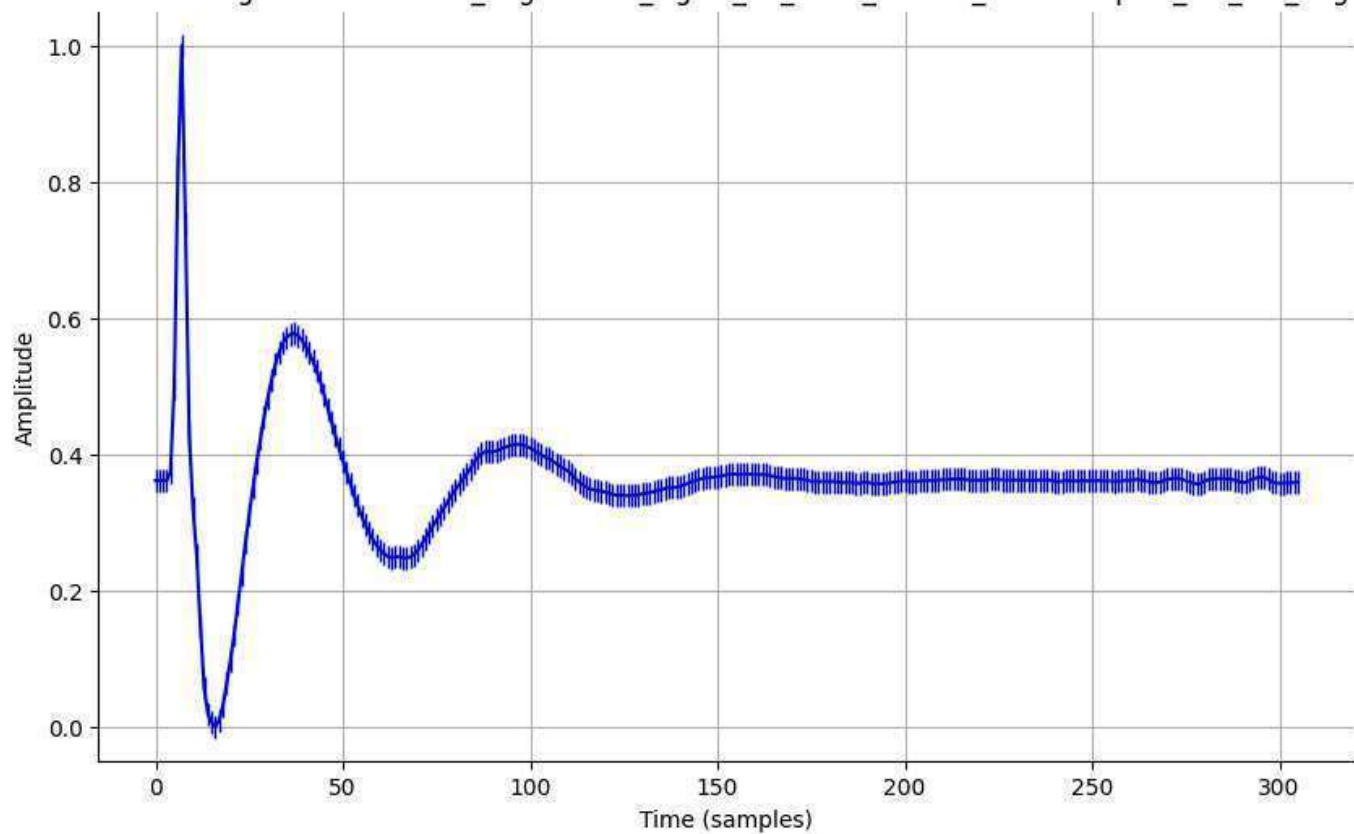
Normalized Signal - normalized_augmented_signal_01_0002_filtered_downsampled_std_0.01_aug.csv



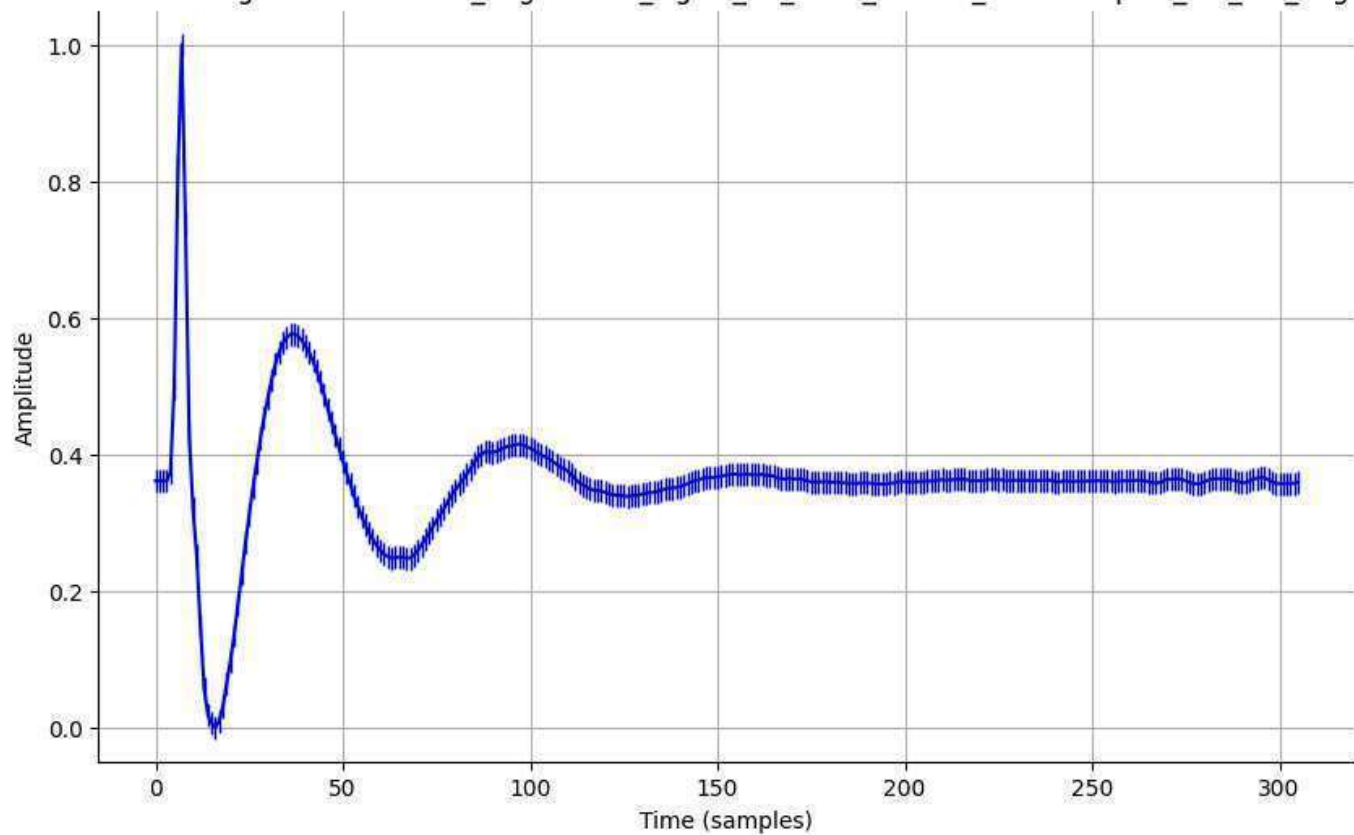
Normalized Signal - normalized_augmented_signal_01_0002_filtered_downsampled_std_0.05_aug.csv



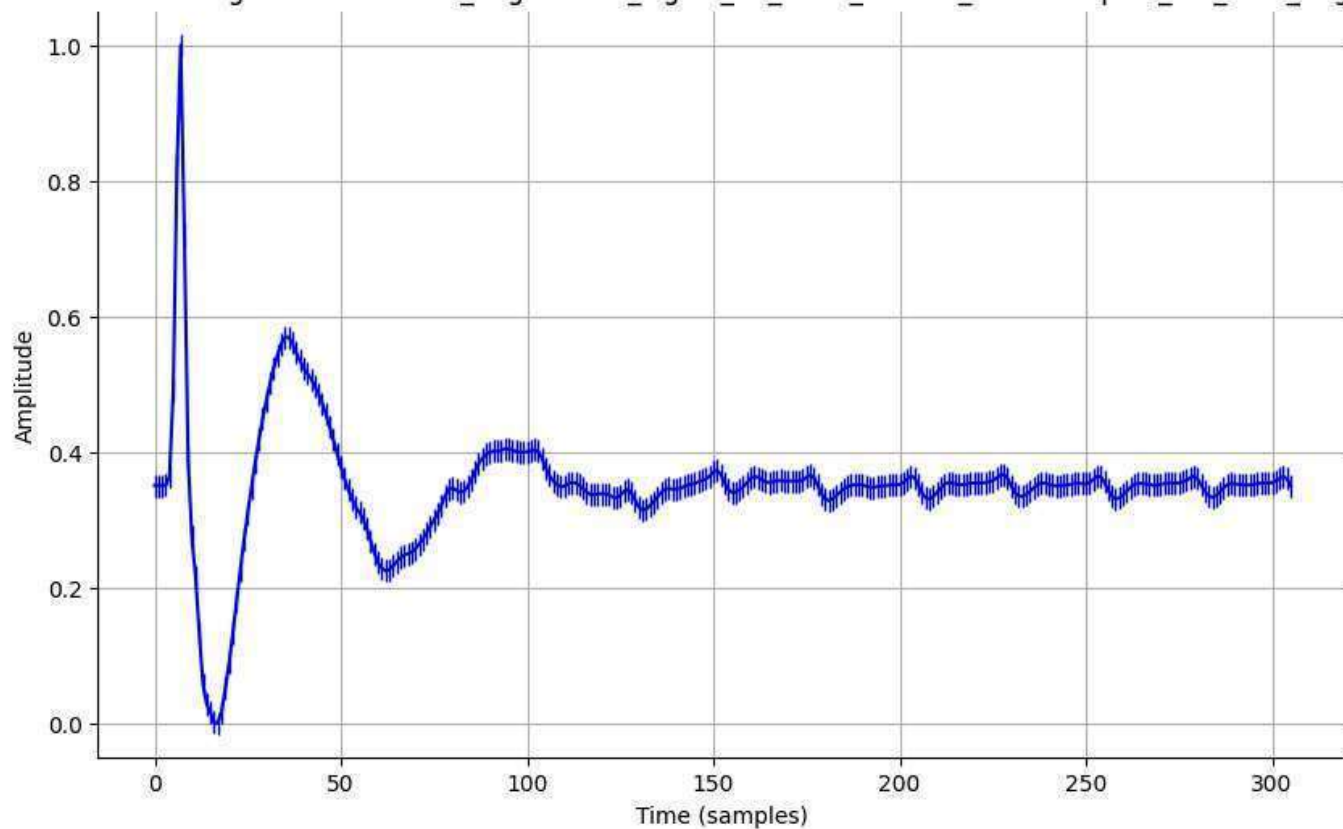
Normalized Signal - normalized_augmented_signal_01_0002_filtered_downsampled_std_0.1_aug.csv



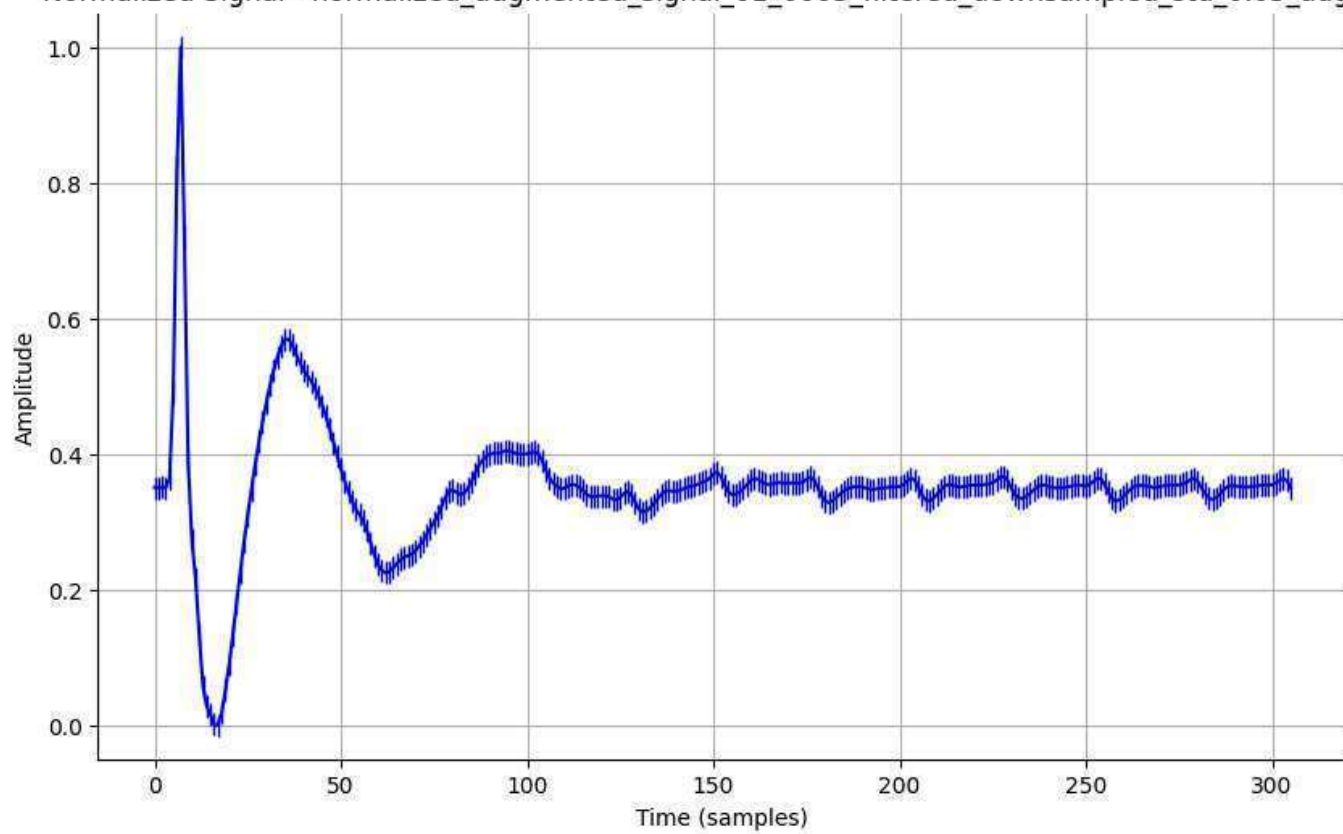
Normalized Signal - normalized_augmented_signal_01_0002_filtered_downsampled_std_0.2_aug.csv



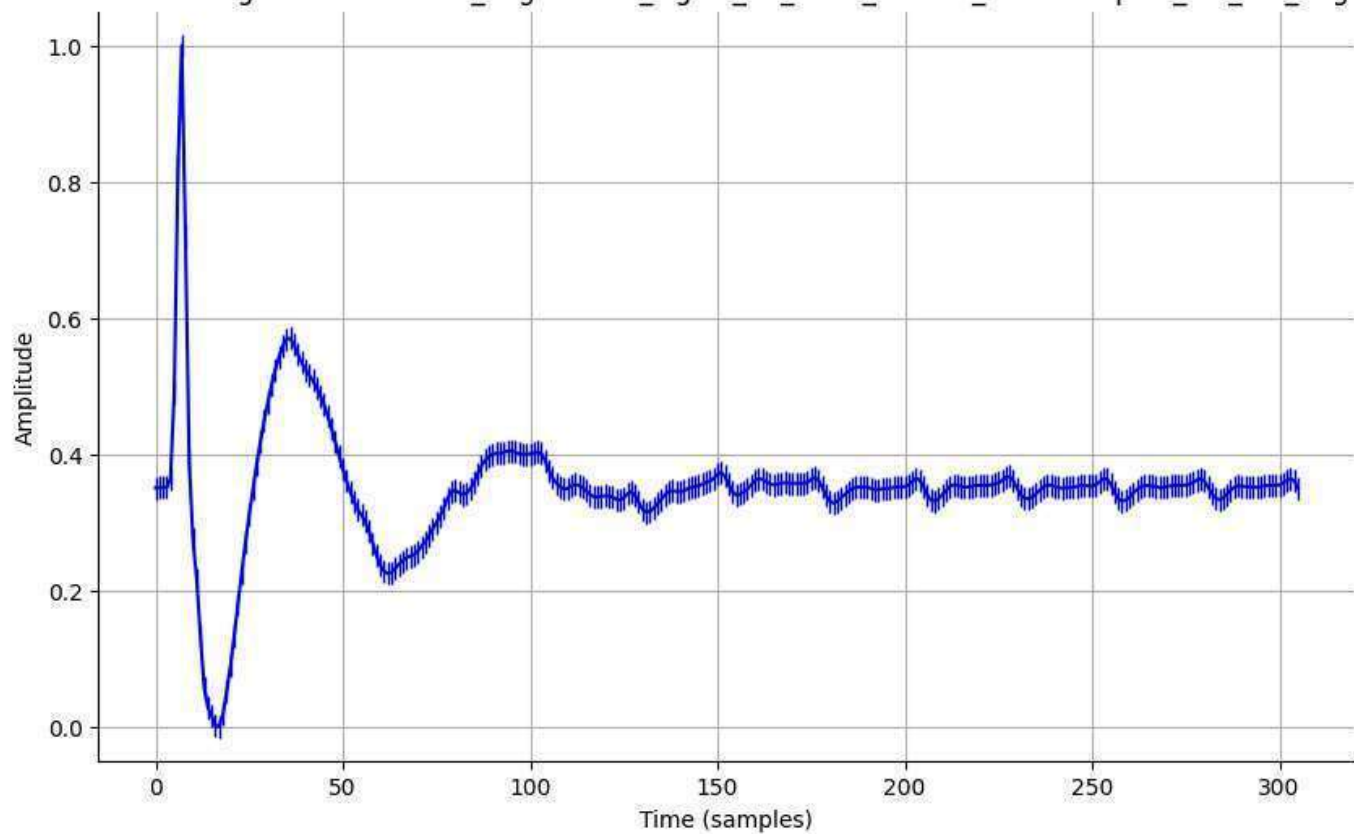
Normalized Signal - normalized_augmented_signal_01_0003_filtered_downsampled_std_0.01_aug.csv



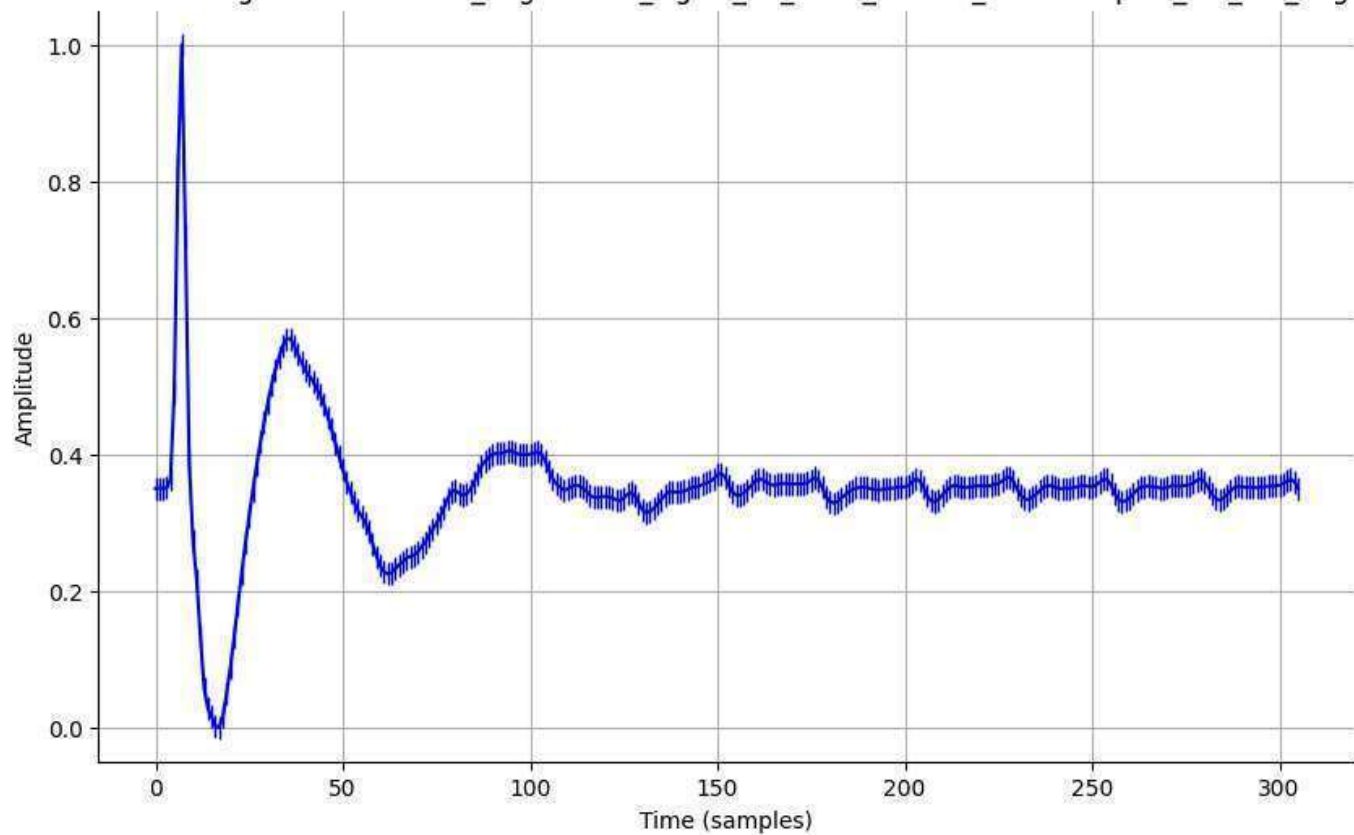
Normalized Signal - normalized_augmented_signal_01_0003_filtered_downsampled_std_0.05_aug.csv



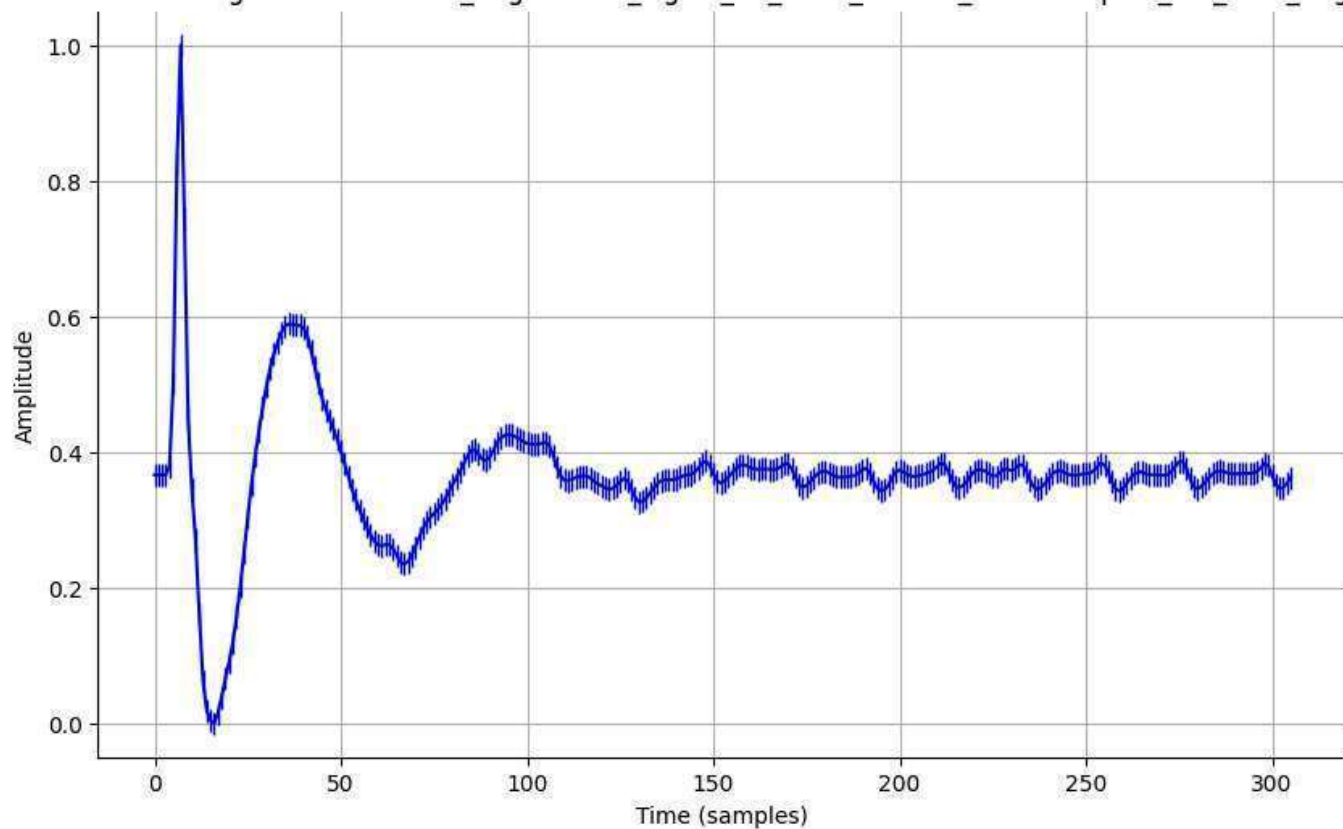
Normalized Signal - normalized_augmented_signal_01_0003_filtered_downsampled_std_0.1_aug.csv



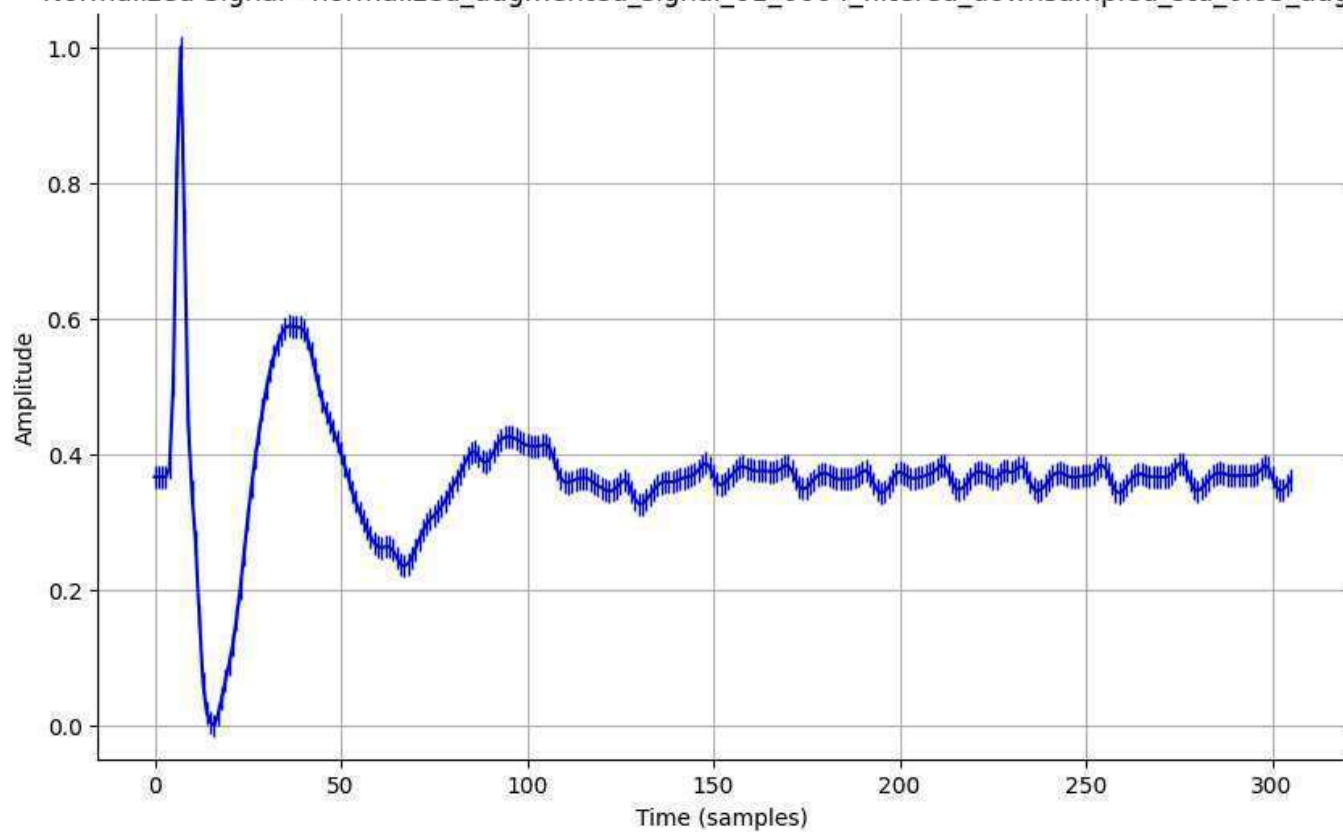
Normalized Signal - normalized_augmented_signal_01_0003_filtered_downsampled_std_0.2_aug.csv



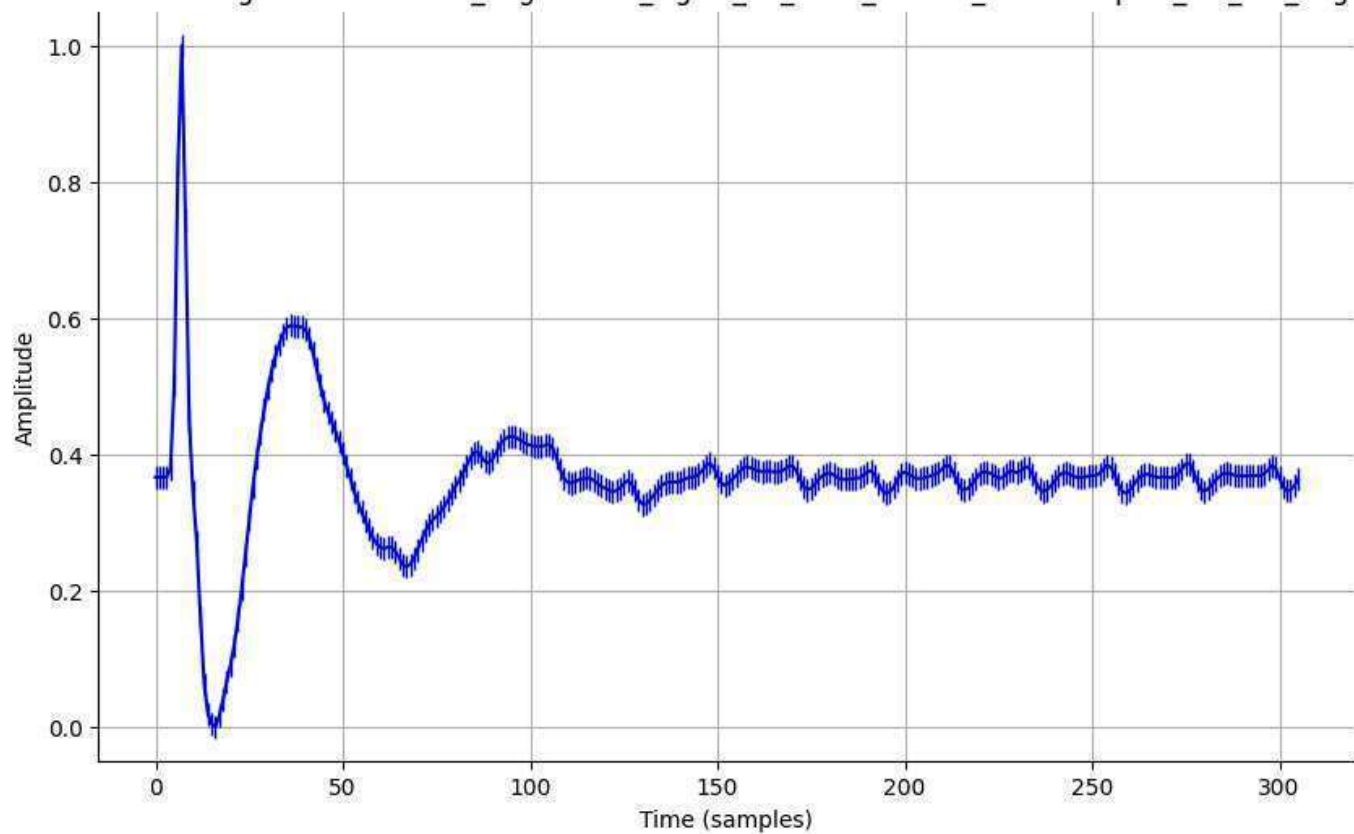
Normalized Signal - normalized_augmented_signal_01_0004_filtered_downsampled_std_0.01_aug.csv



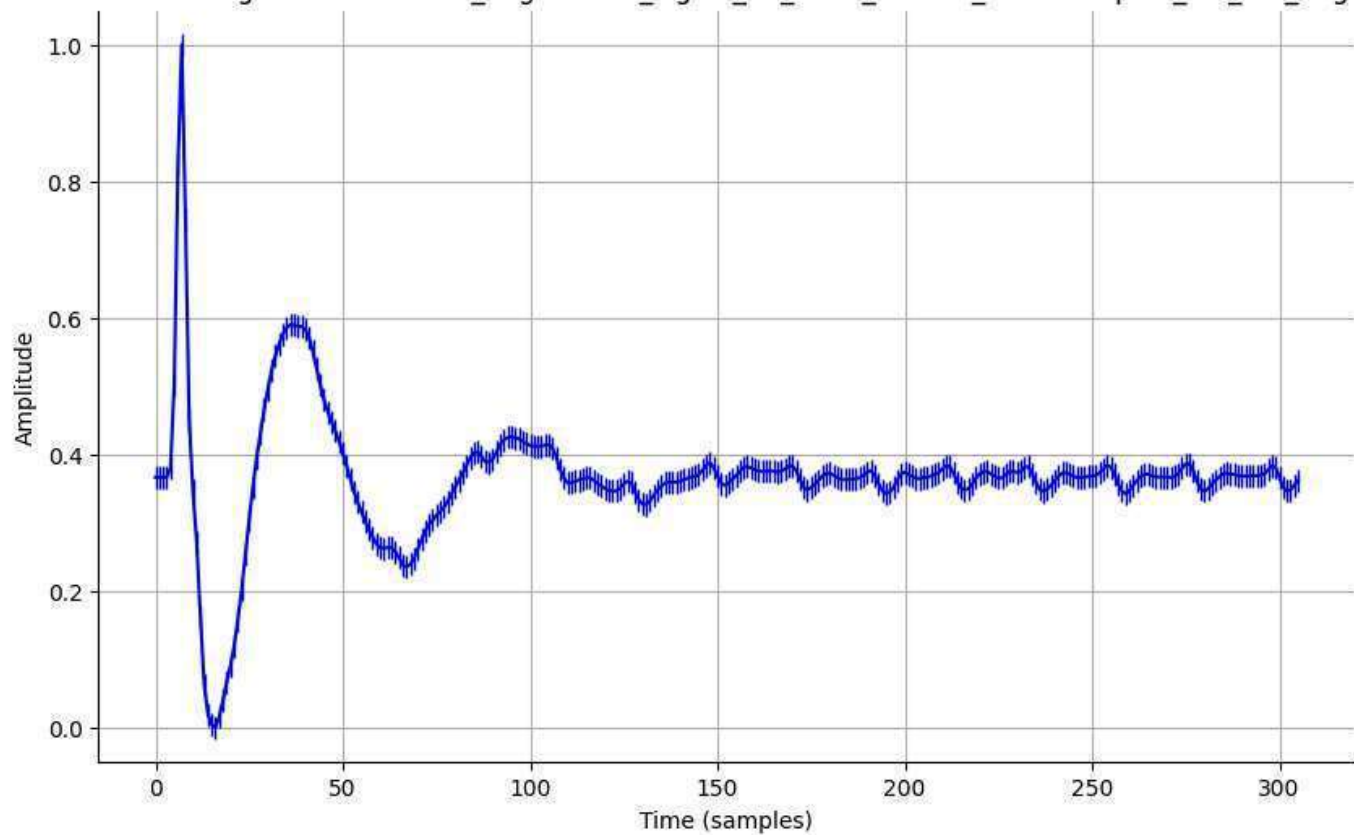
Normalized Signal - normalized_augmented_signal_01_0004_filtered_downsampled_std_0.05_aug.csv



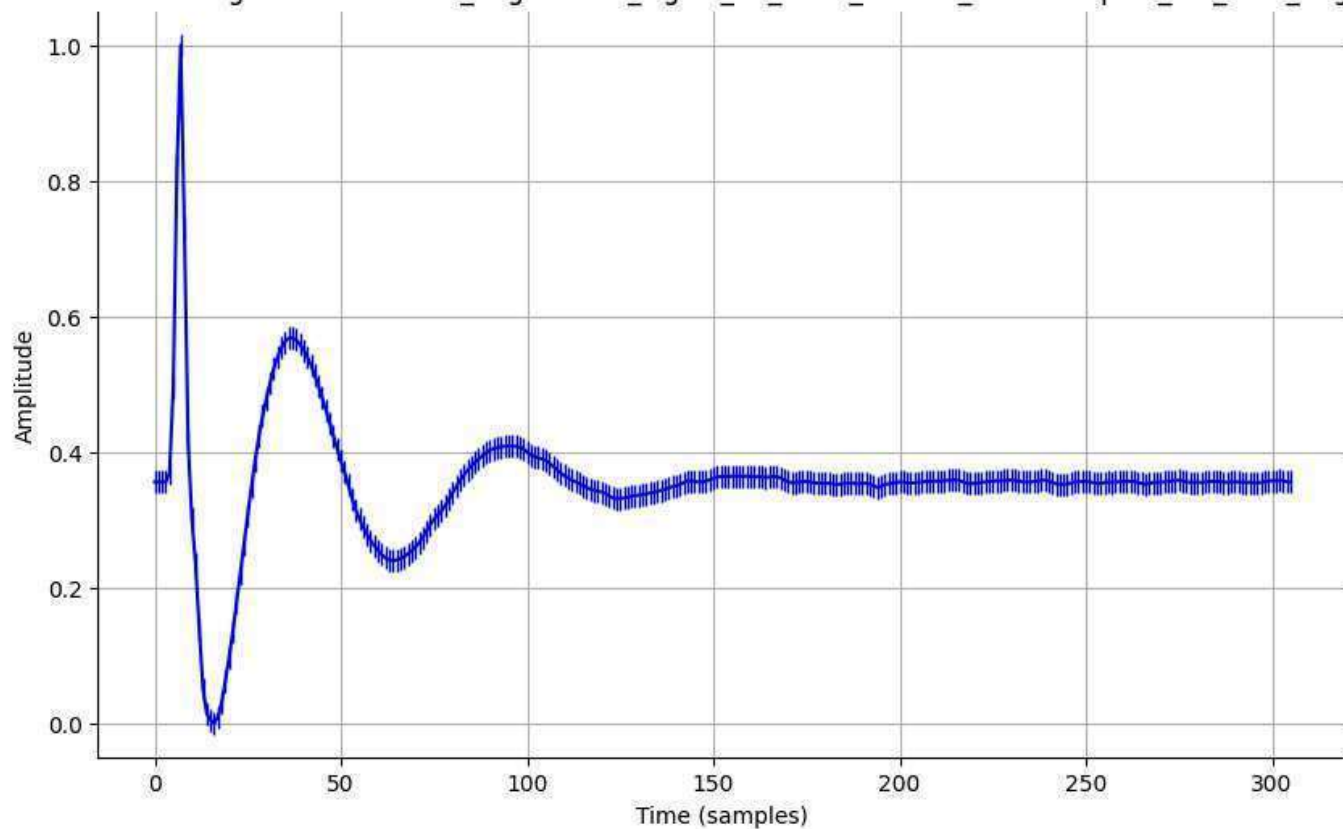
Normalized Signal - normalized_augmented_signal_01_0004_filtered_downsampled_std_0.1_aug.csv



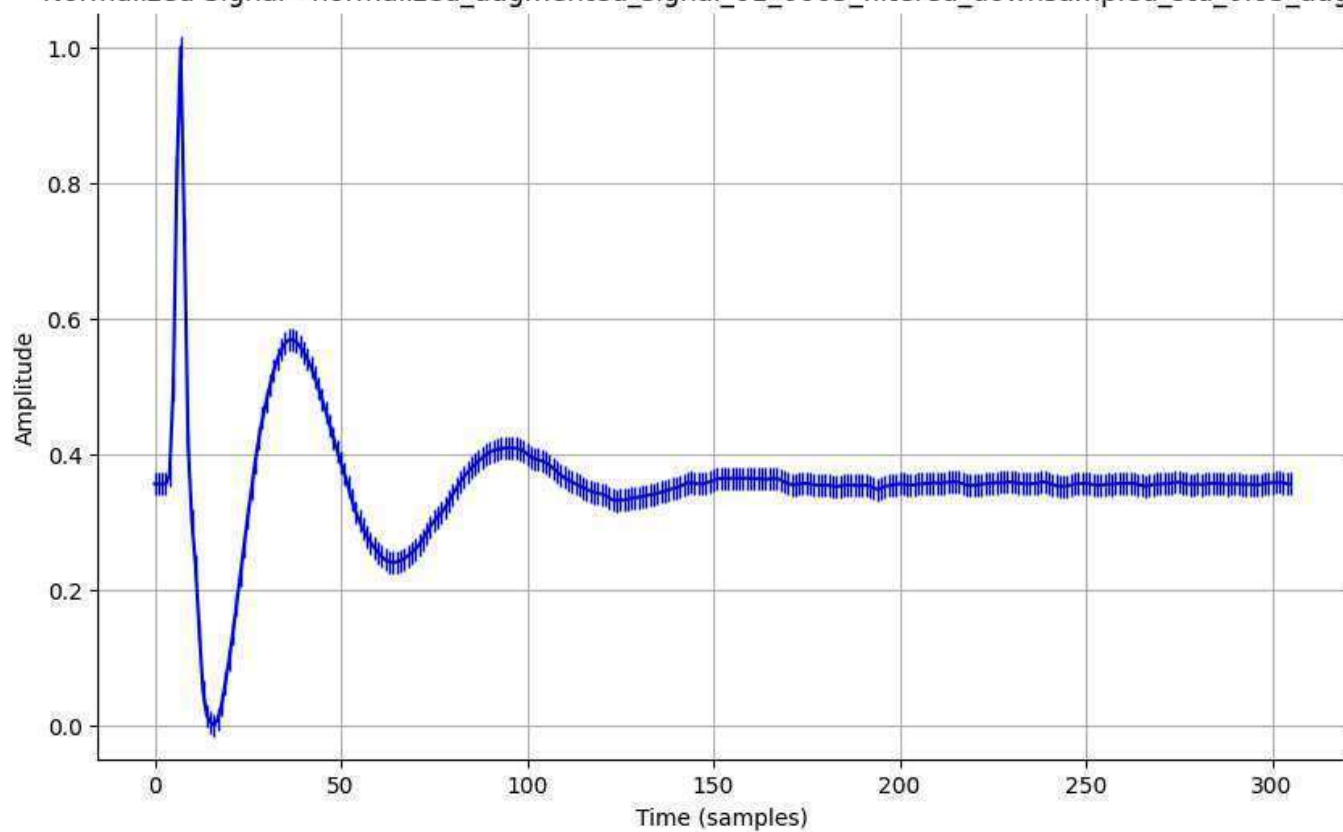
Normalized Signal - normalized_augmented_signal_01_0004_filtered_downsampled_std_0.2_aug.csv



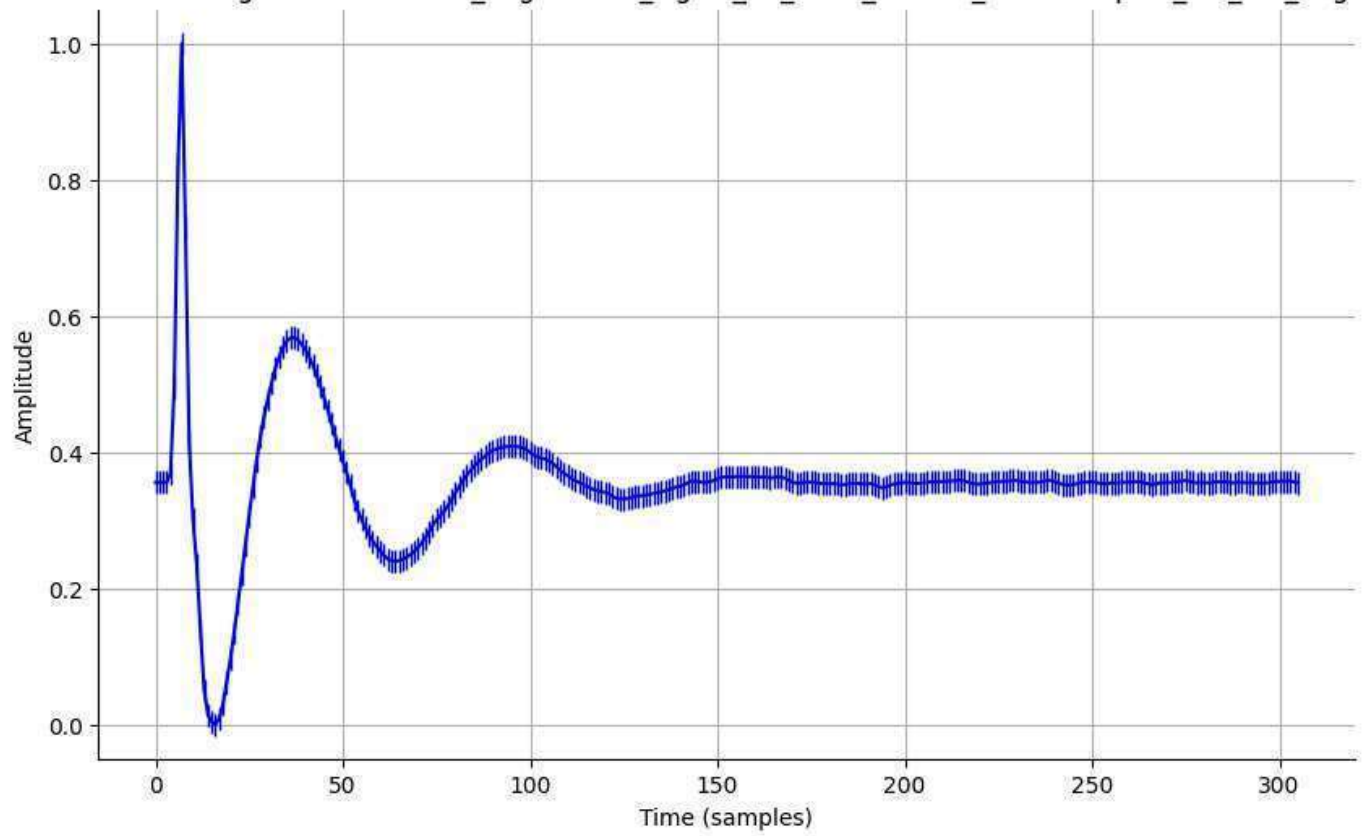
Normalized Signal - normalized_augmented_signal_01_0005_filtered_downsampled_std_0.01_aug.csv



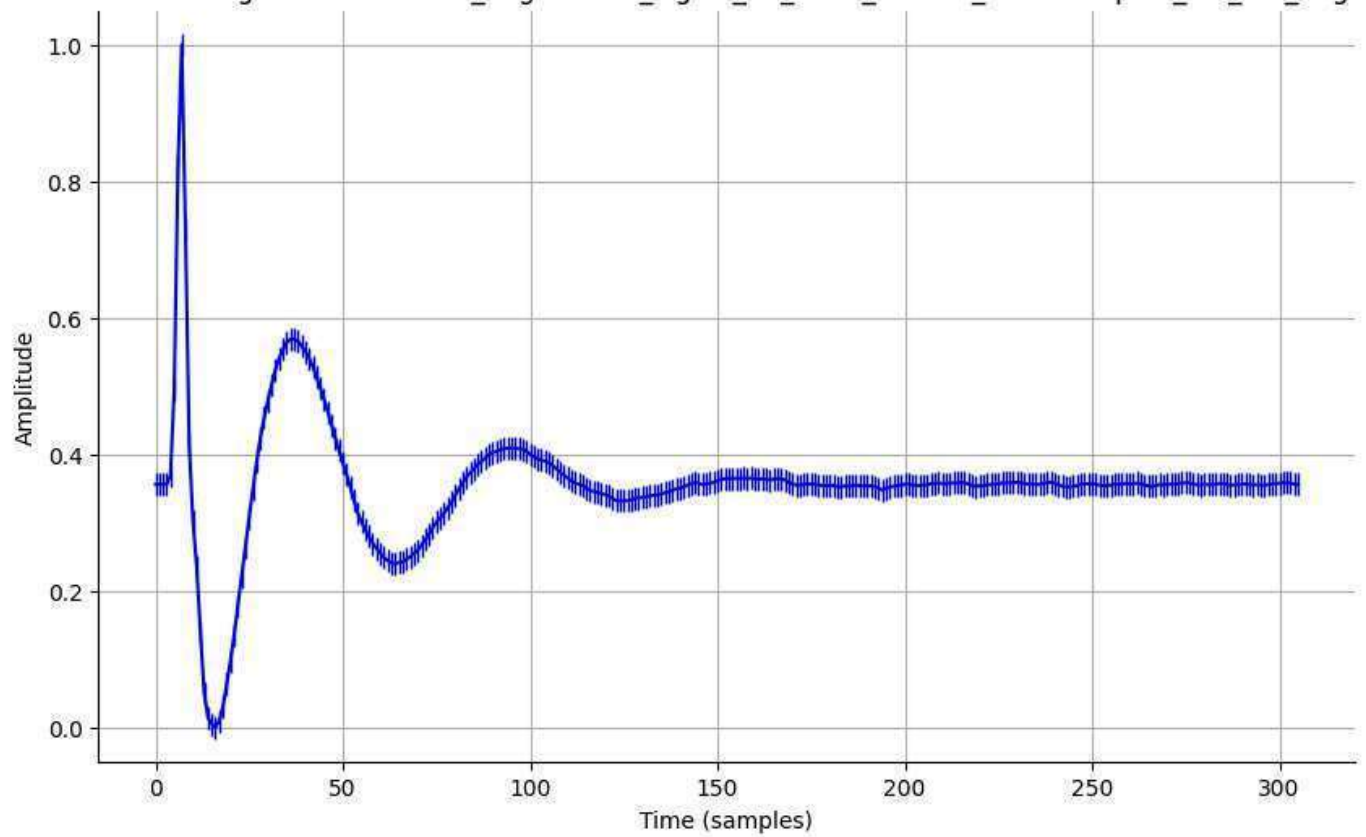
Normalized Signal - normalized_augmented_signal_01_0005_filtered_downsampled_std_0.05_aug.csv



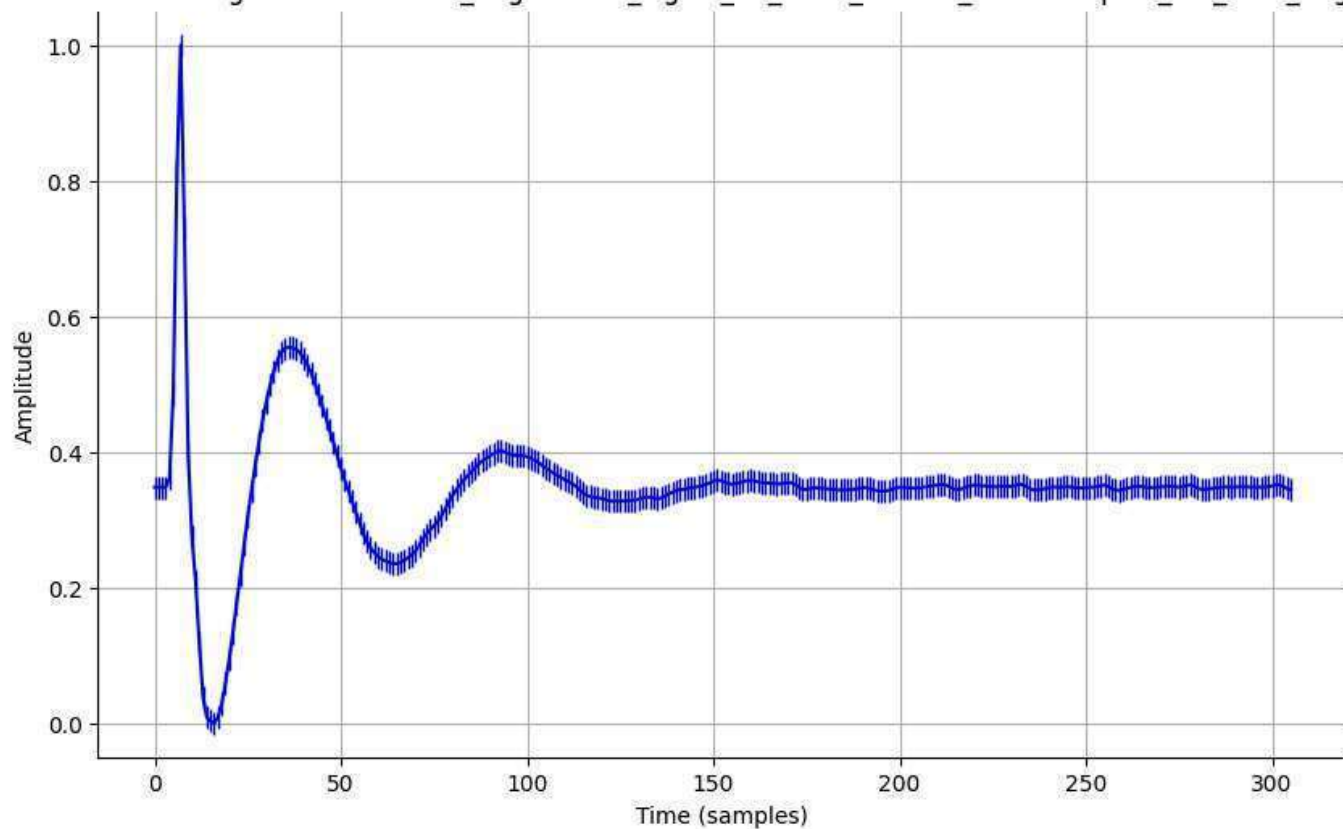
Normalized Signal - normalized_augmented_signal_01_0005_filtered_downsampled_std_0.1_aug.csv



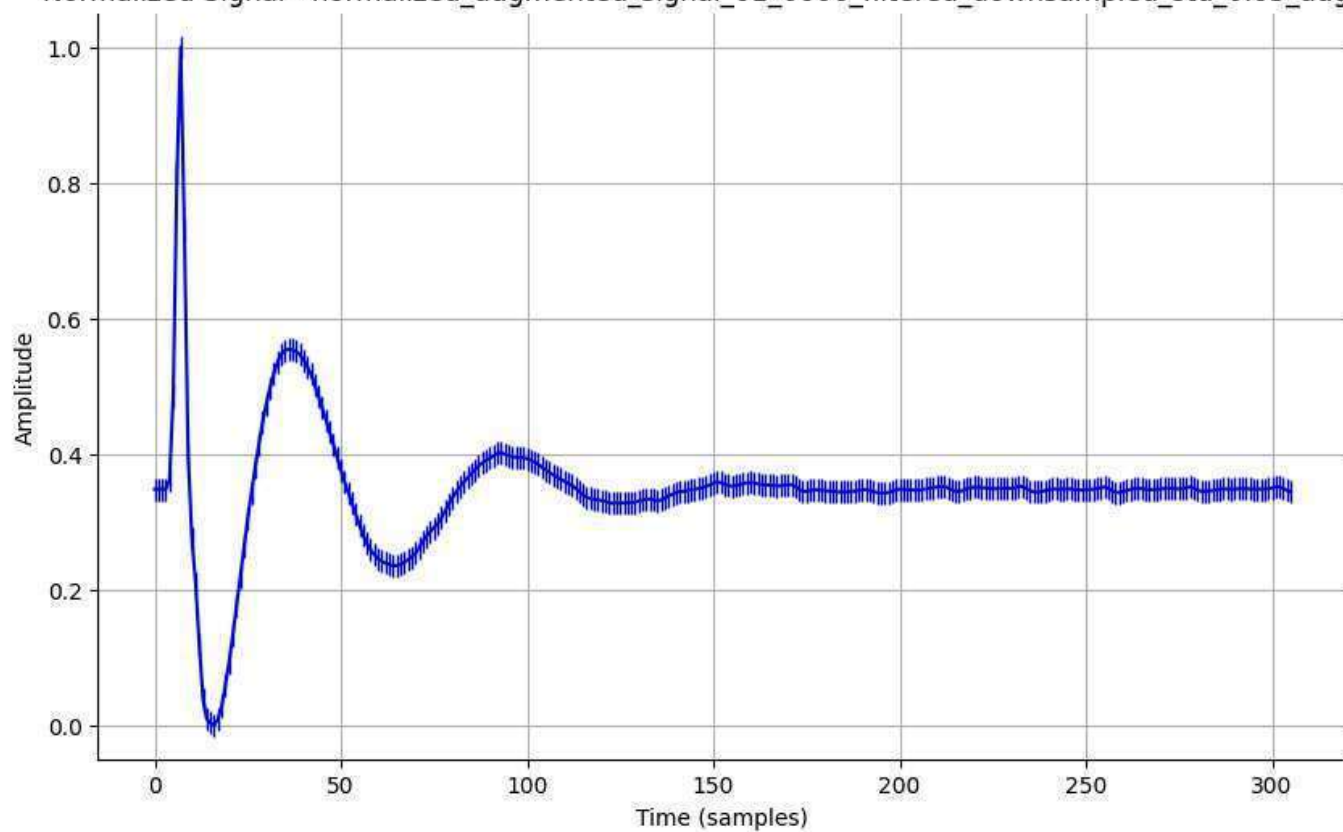
Normalized Signal - normalized_augmented_signal_01_0005_filtered_downsampled_std_0.2_aug.csv



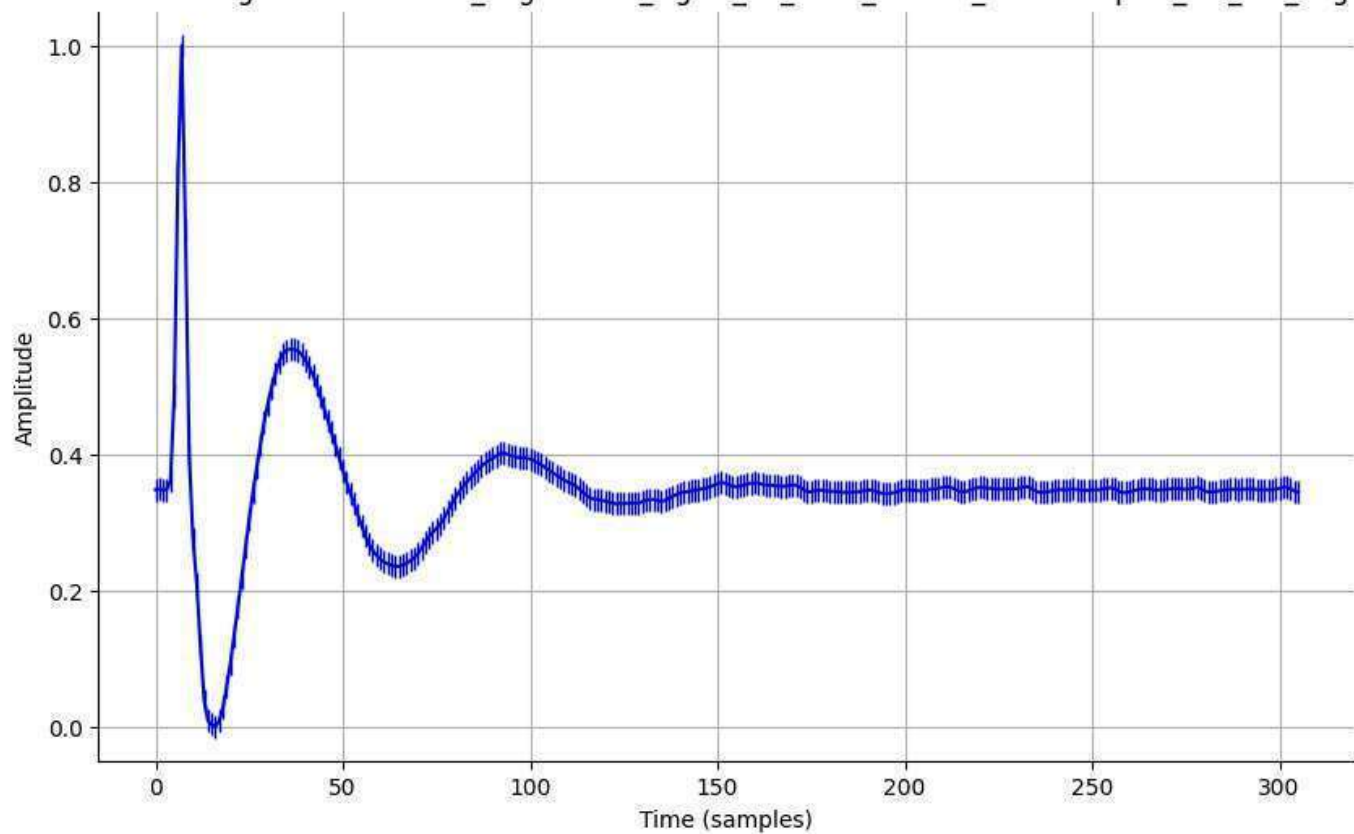
Normalized Signal - normalized_augmented_signal_01_0006_filtered_downsampled_std_0.01_aug.csv



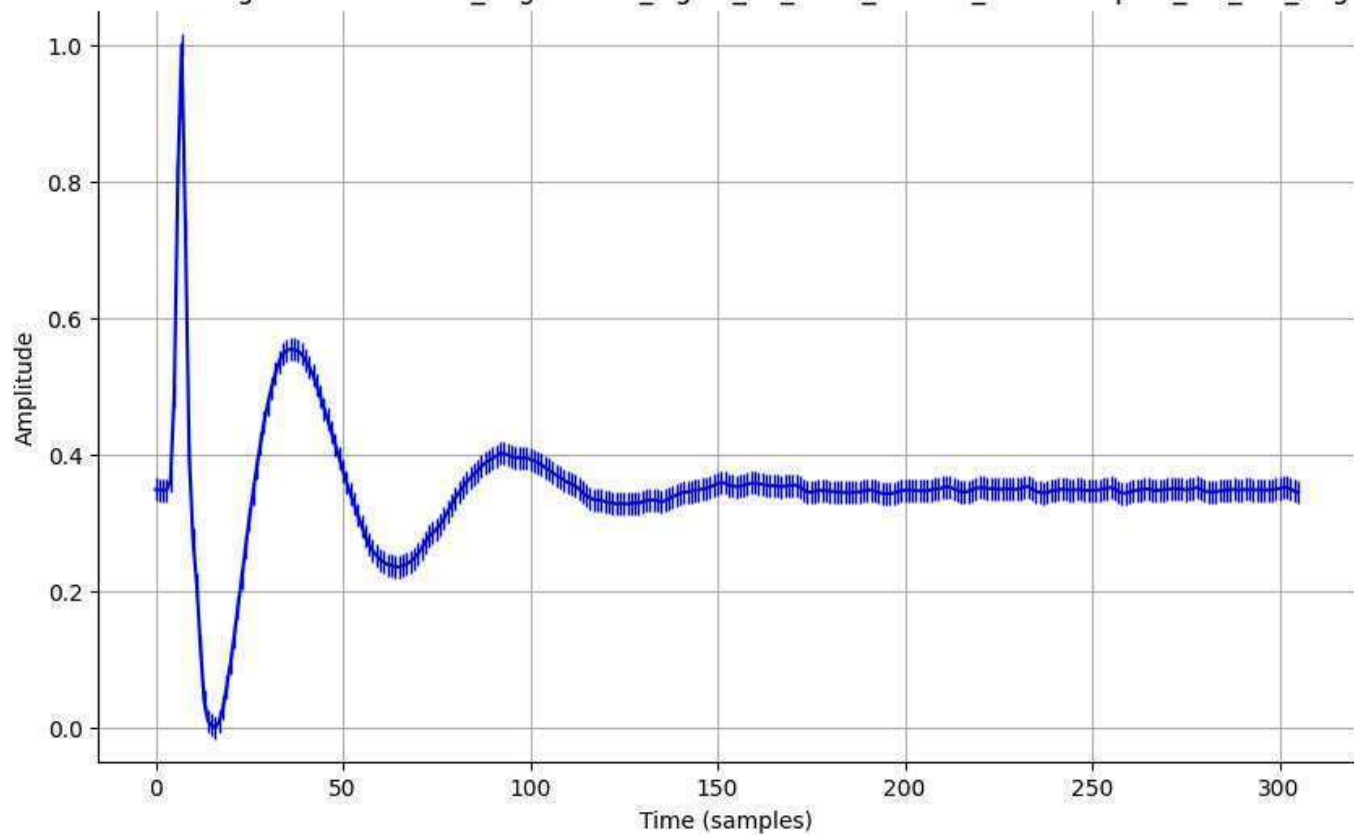
Normalized Signal - normalized_augmented_signal_01_0006_filtered_downsampled_std_0.05_aug.csv



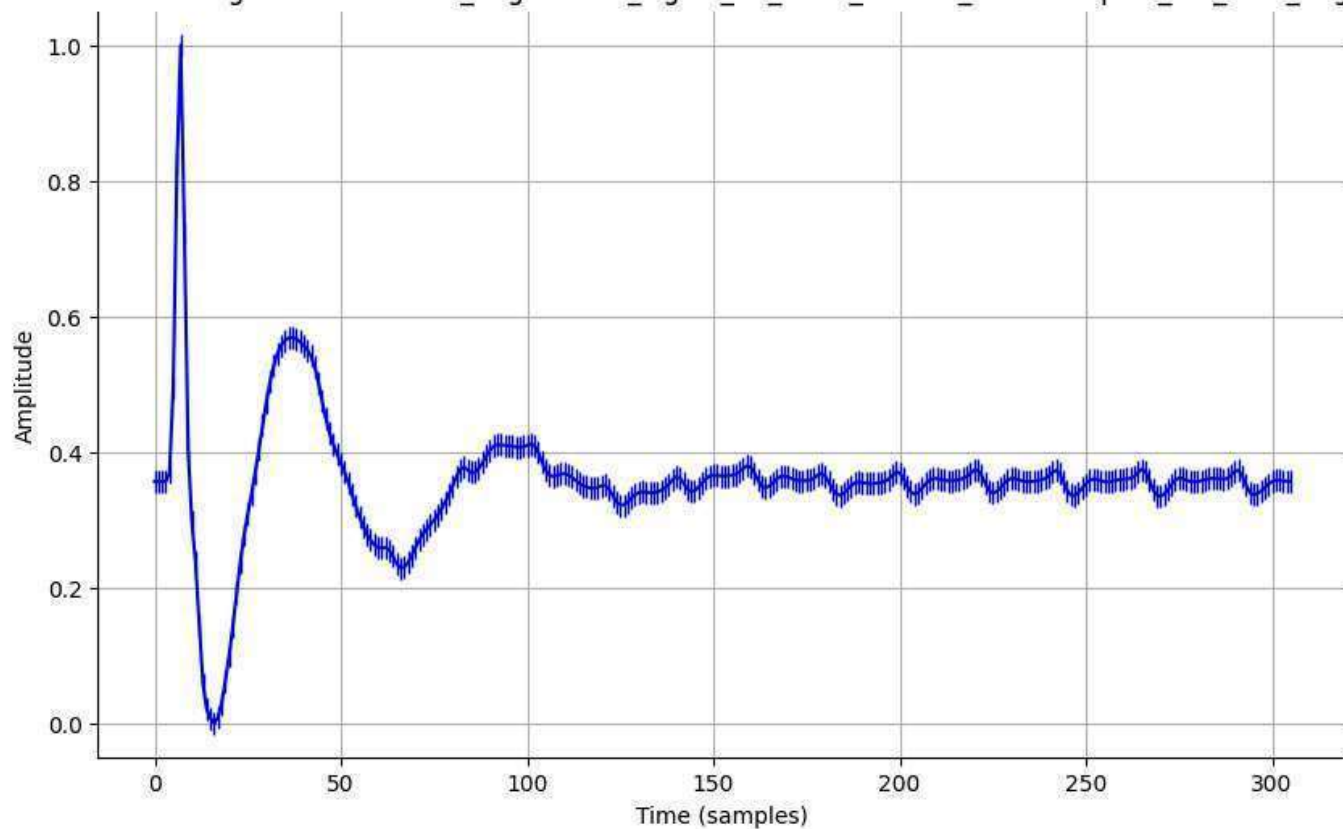
Normalized Signal - normalized_augmented_signal_01_0006_filtered_downsampled_std_0.1_aug.csv



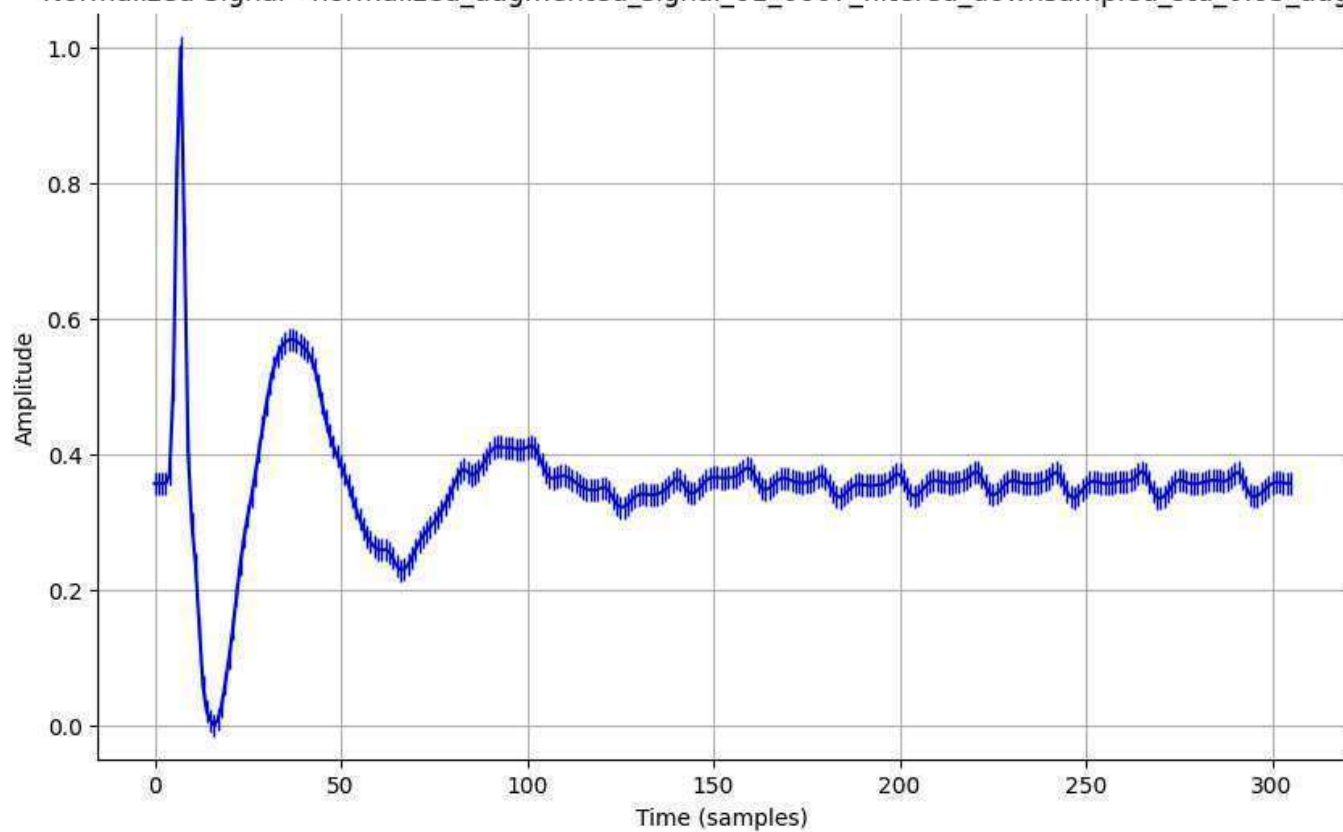
Normalized Signal - normalized_augmented_signal_01_0006_filtered_downsampled_std_0.2_aug.csv



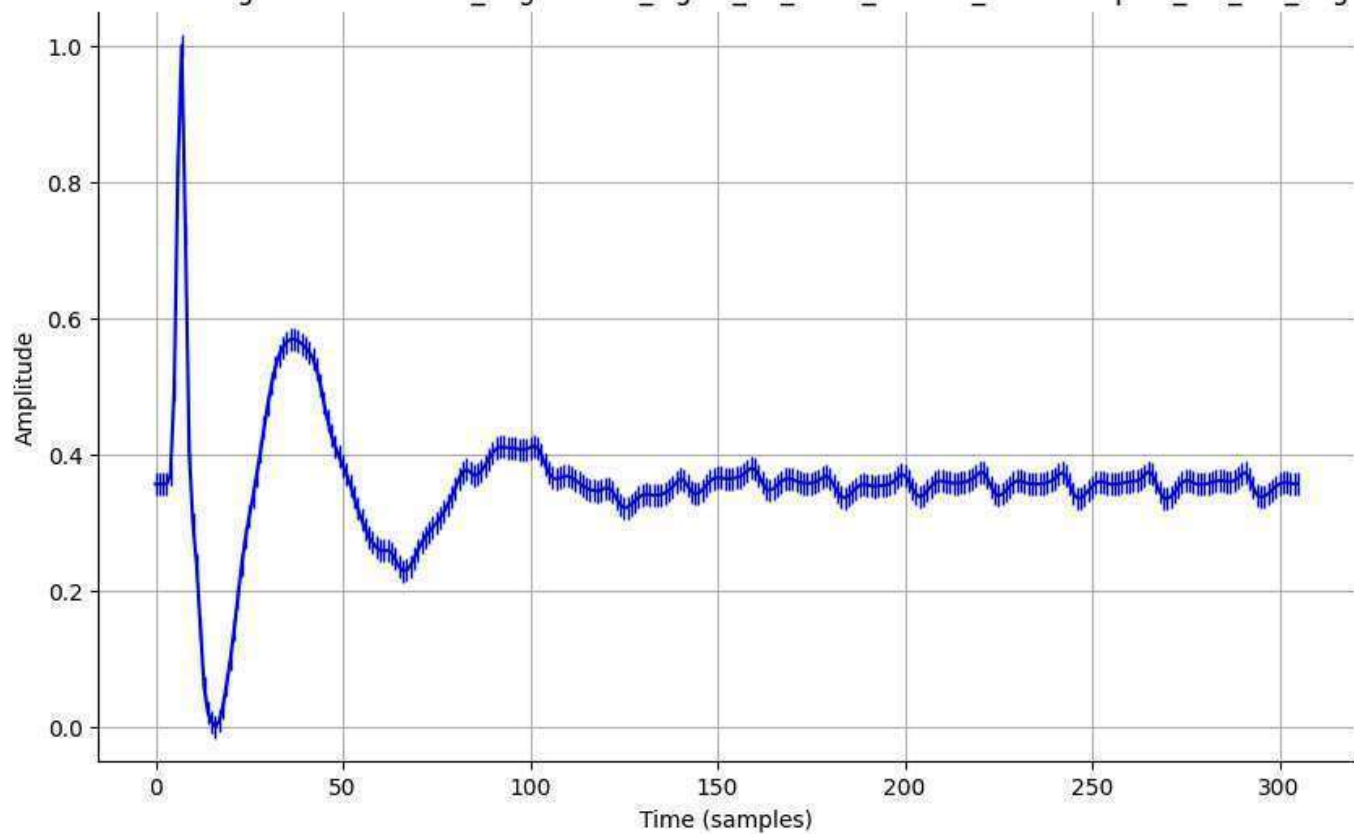
Normalized Signal - normalized_augmented_signal_01_0007_filtered_downsampled_std_0.01_aug.csv



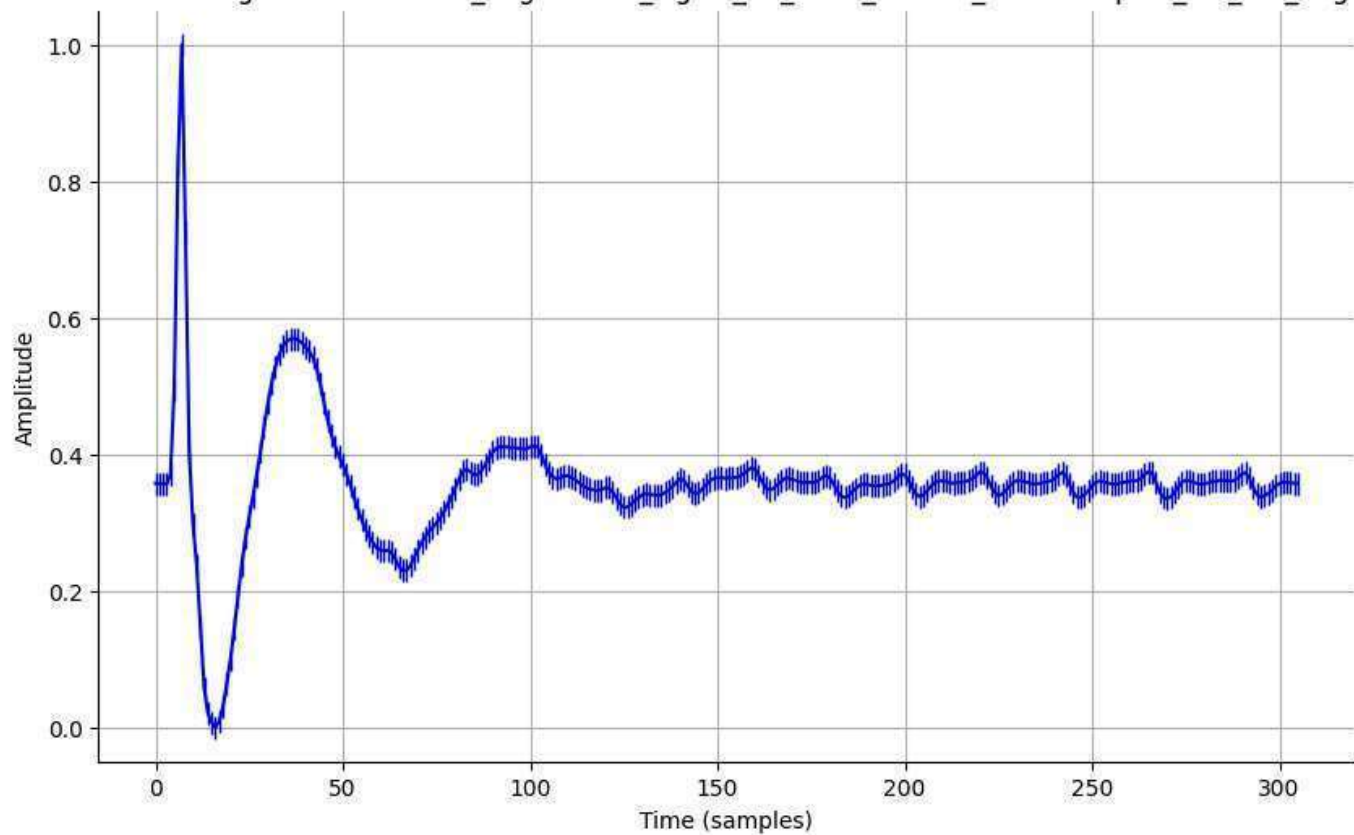
Normalized Signal - normalized_augmented_signal_01_0007_filtered_downsampled_std_0.05_aug.csv



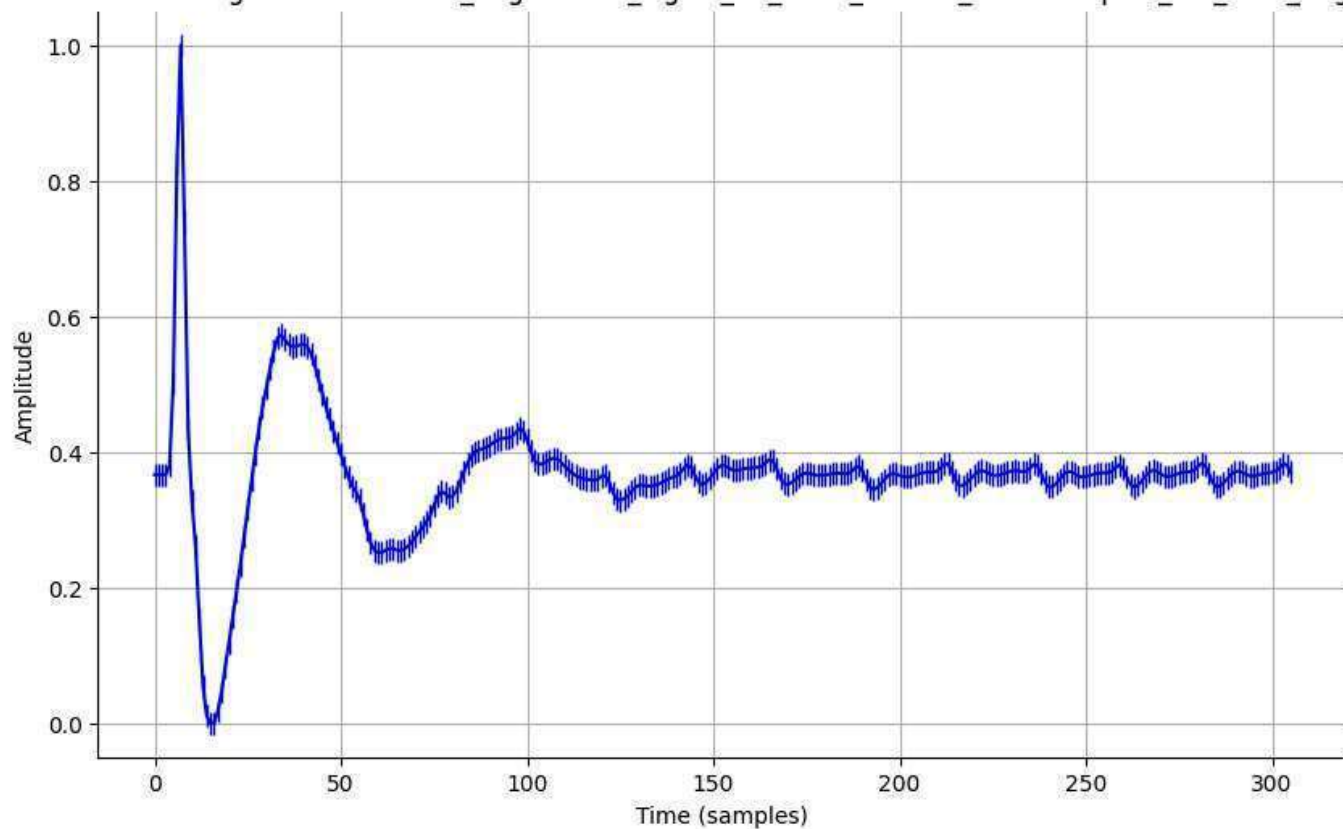
Normalized Signal - normalized_augmented_signal_01_0007_filtered_downsampled_std_0.1_aug.csv



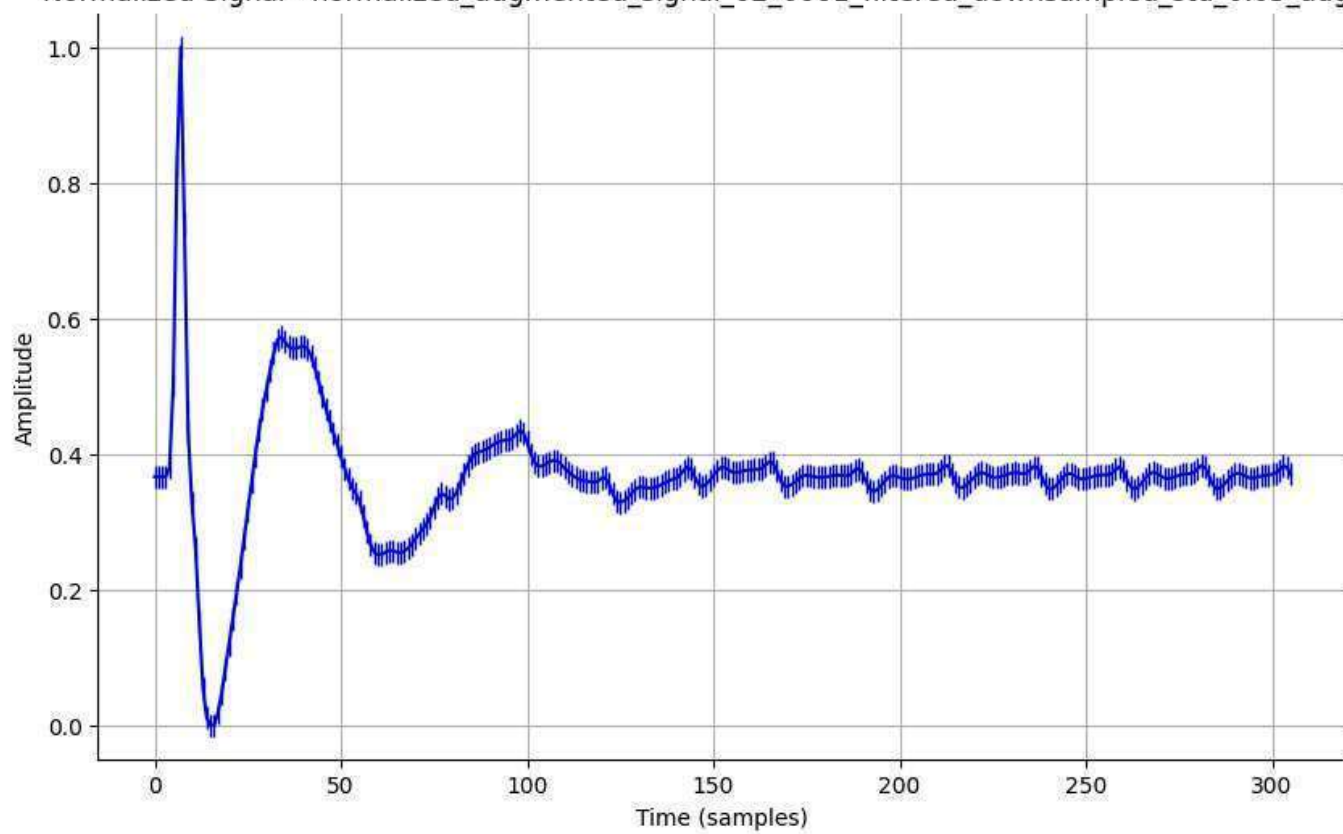
Normalized Signal - normalized_augmented_signal_01_0007_filtered_downsampled_std_0.2_aug.csv



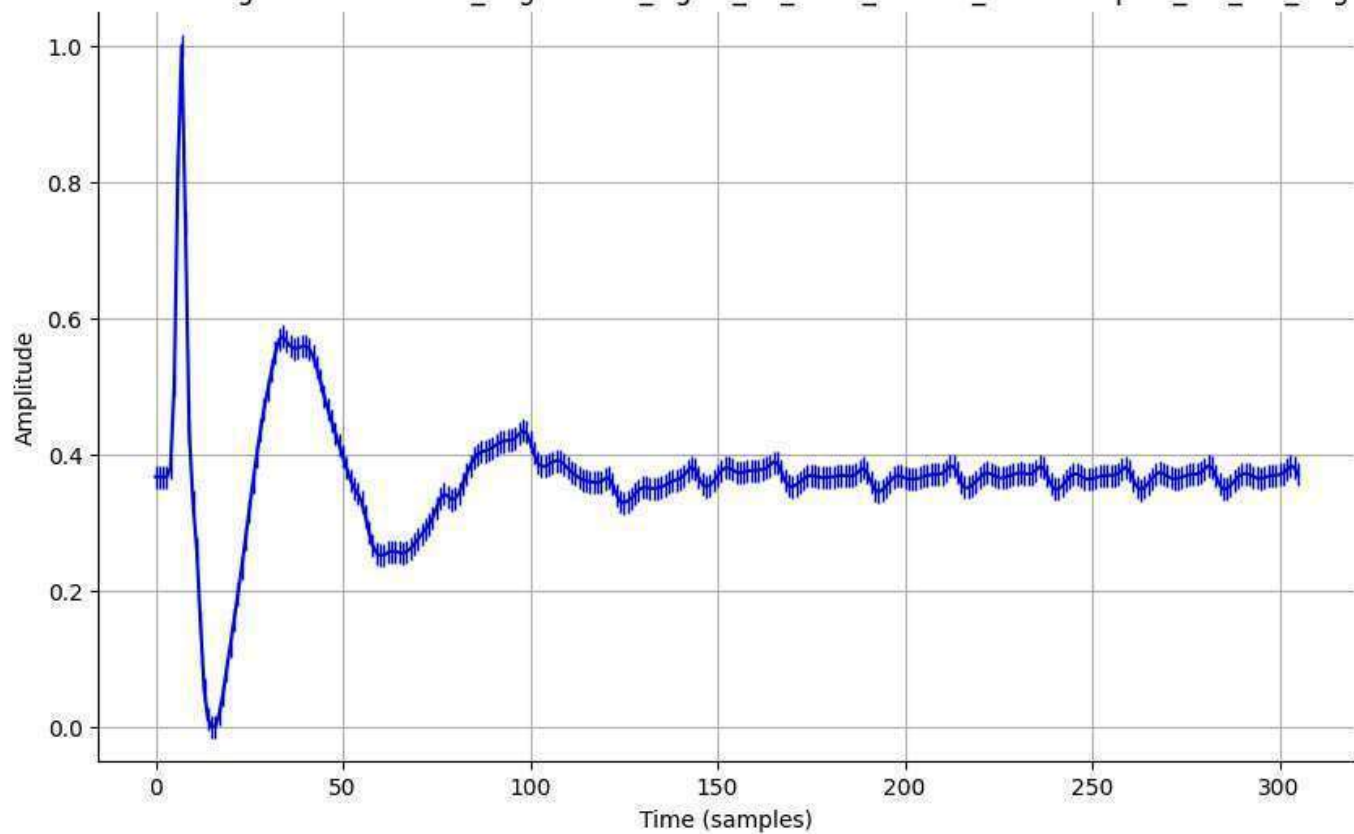
Normalized Signal - normalized_augmented_signal_02_0001_filtered_downsampled_std_0.01_aug.csv



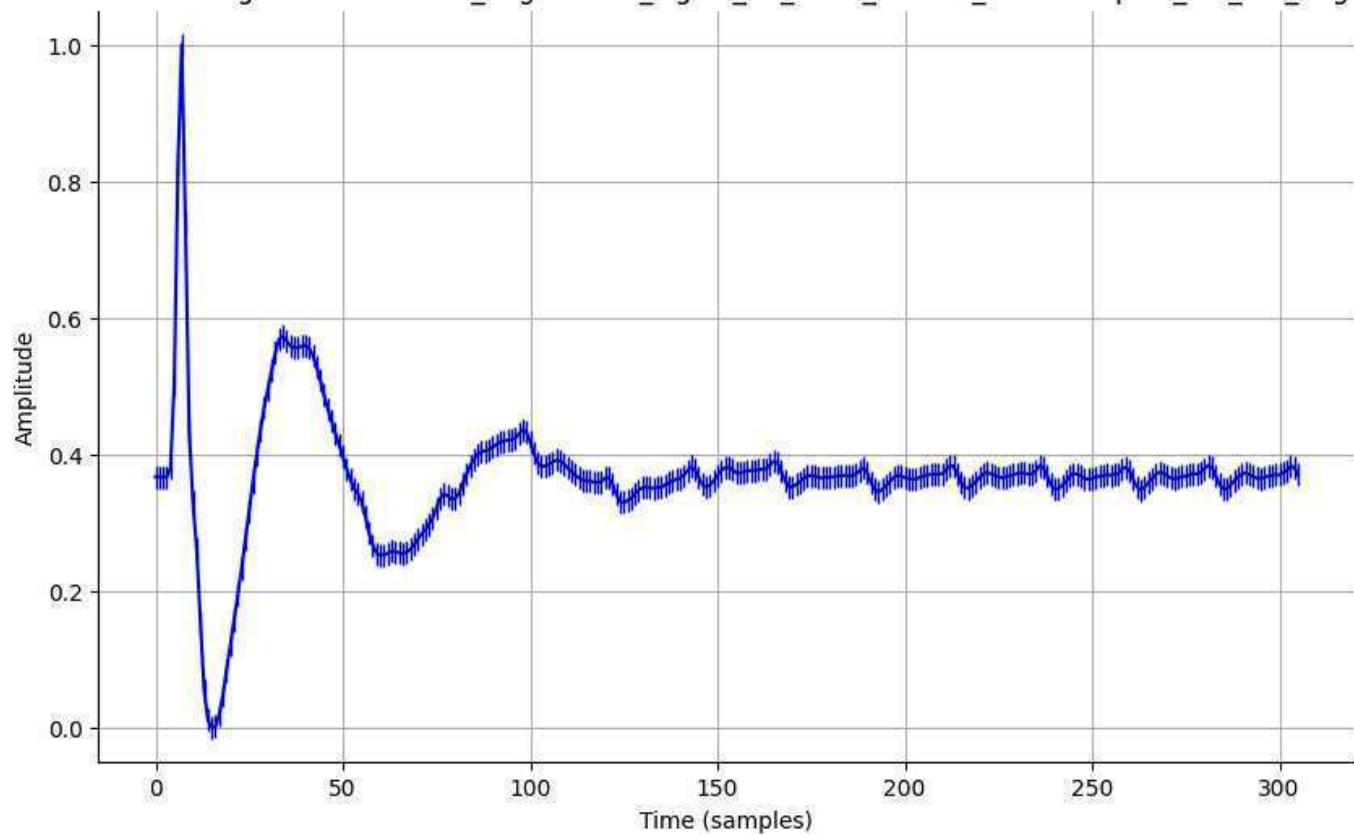
Normalized Signal - normalized_augmented_signal_02_0001_filtered_downsampled_std_0.05_aug.csv



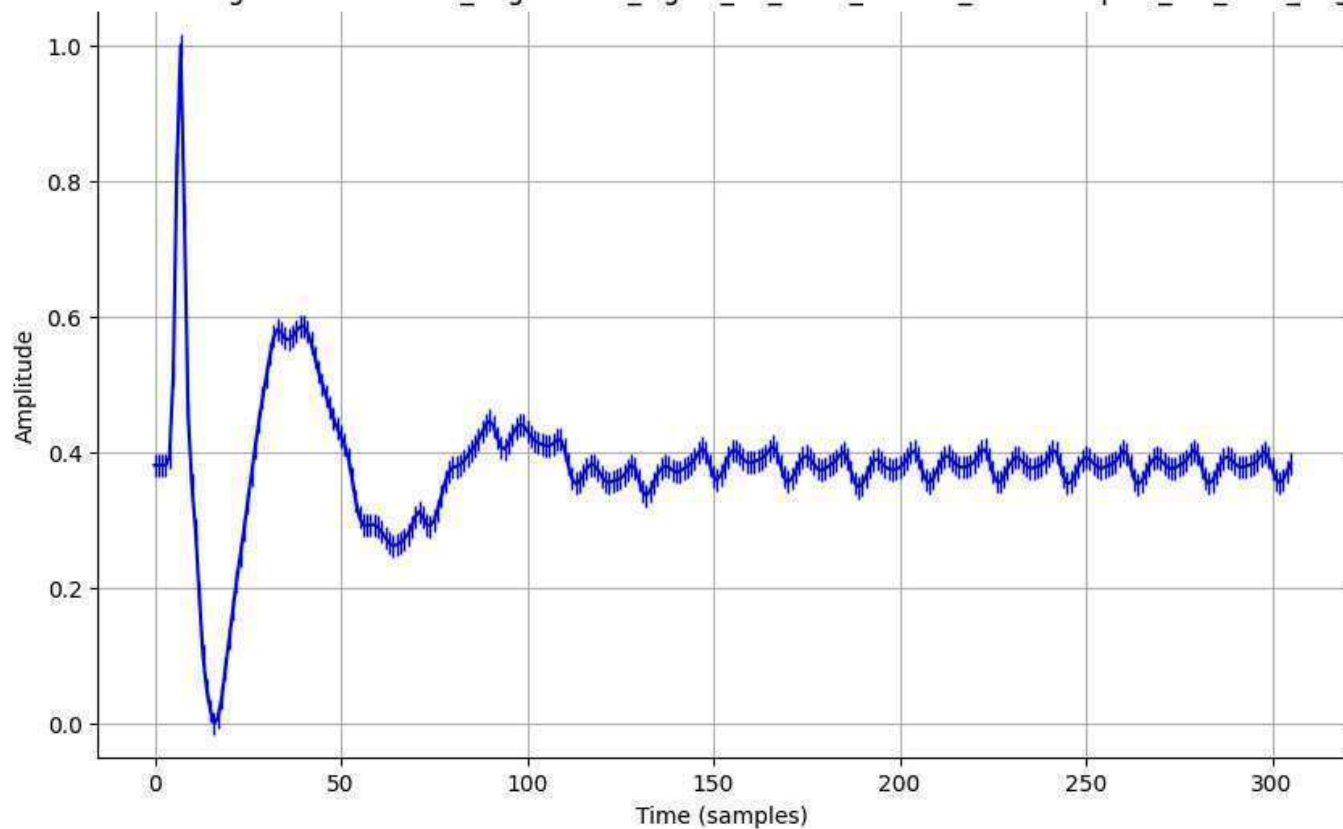
Normalized Signal - normalized_augmented_signal_02_0001_filtered_downsampled_std_0.1_aug.csv



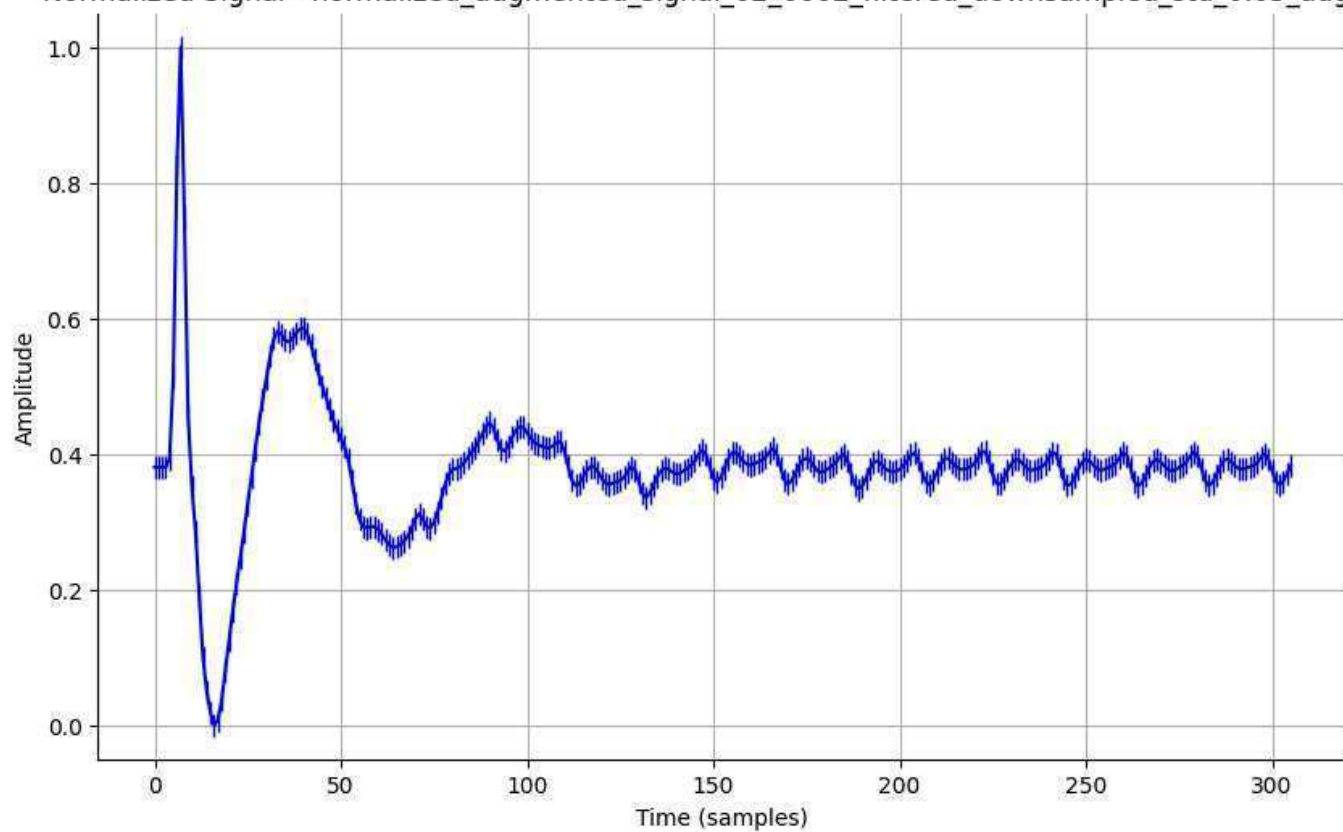
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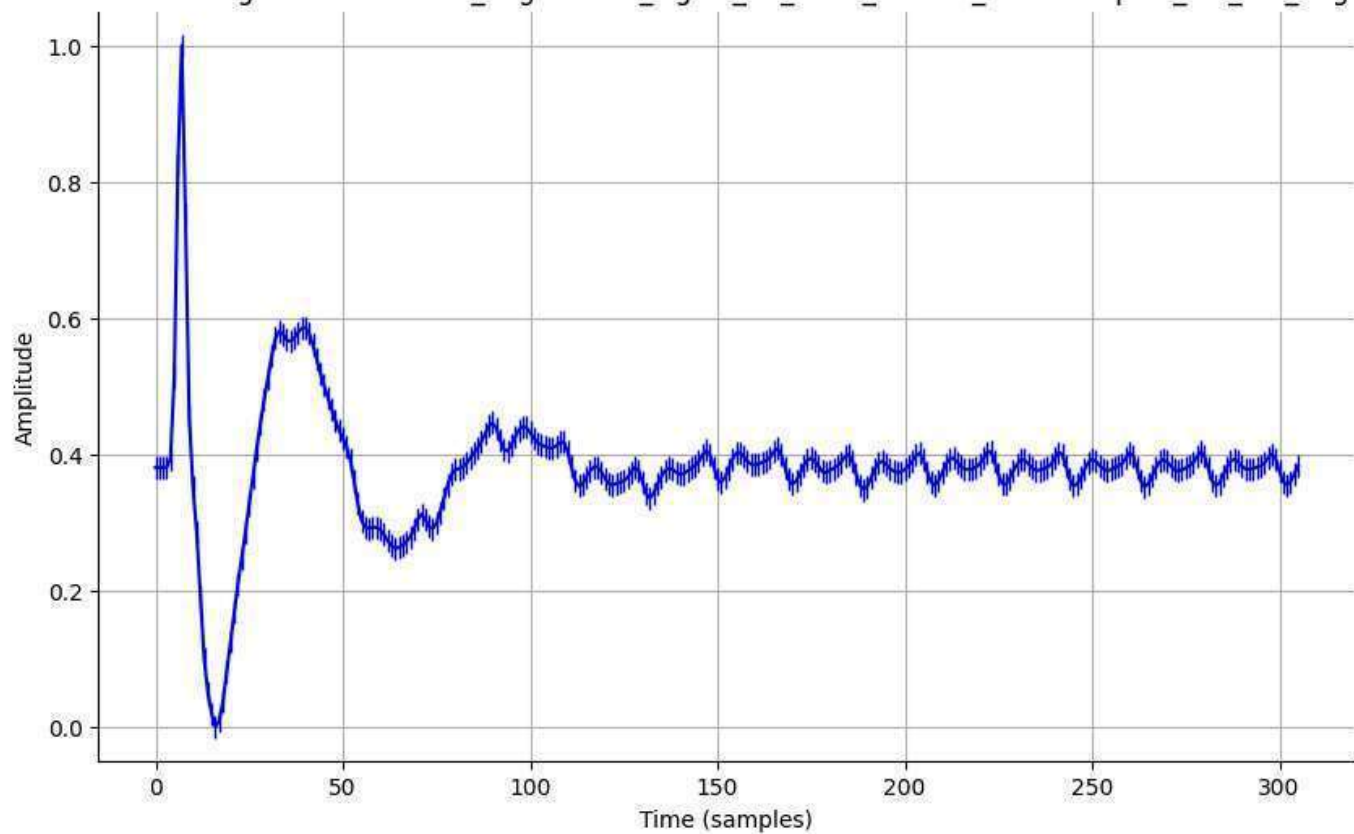
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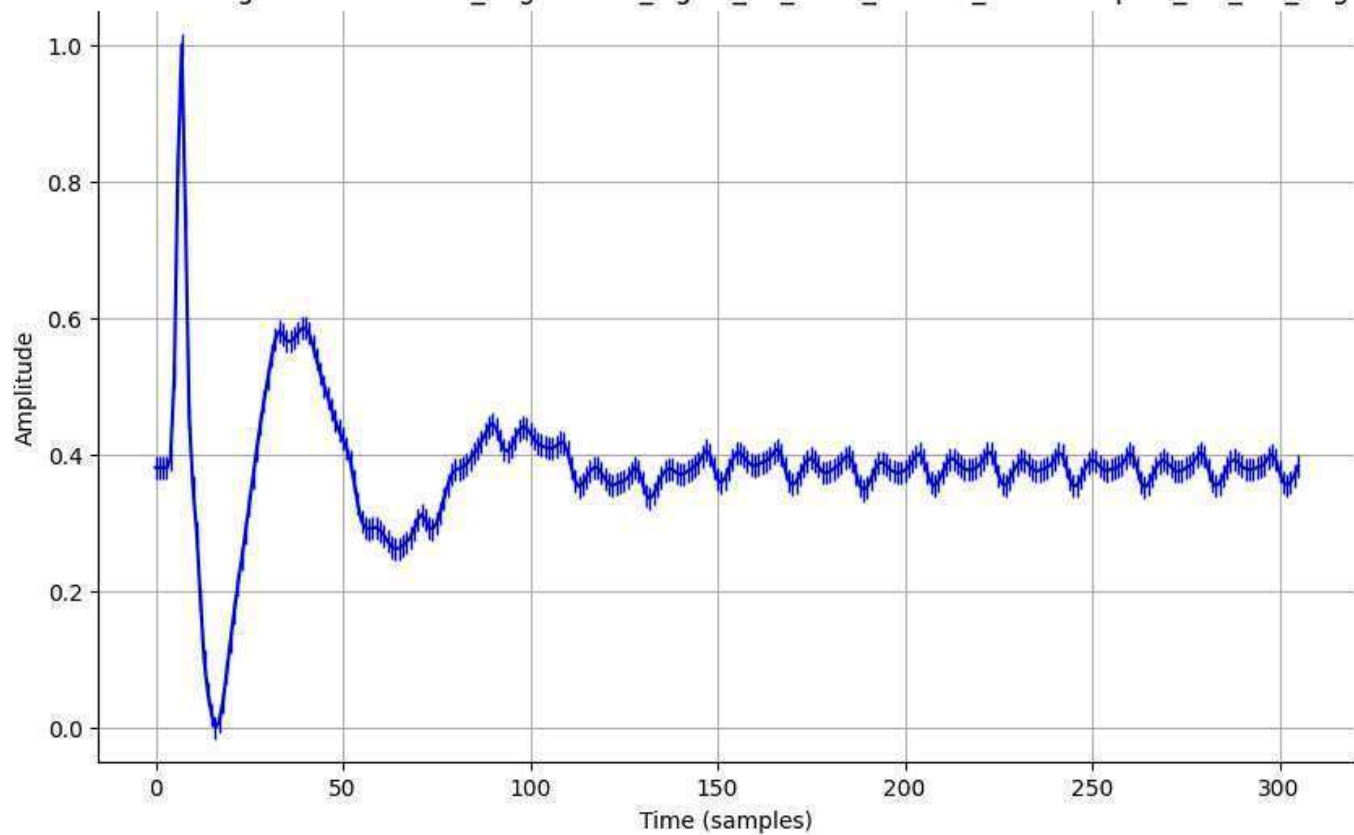
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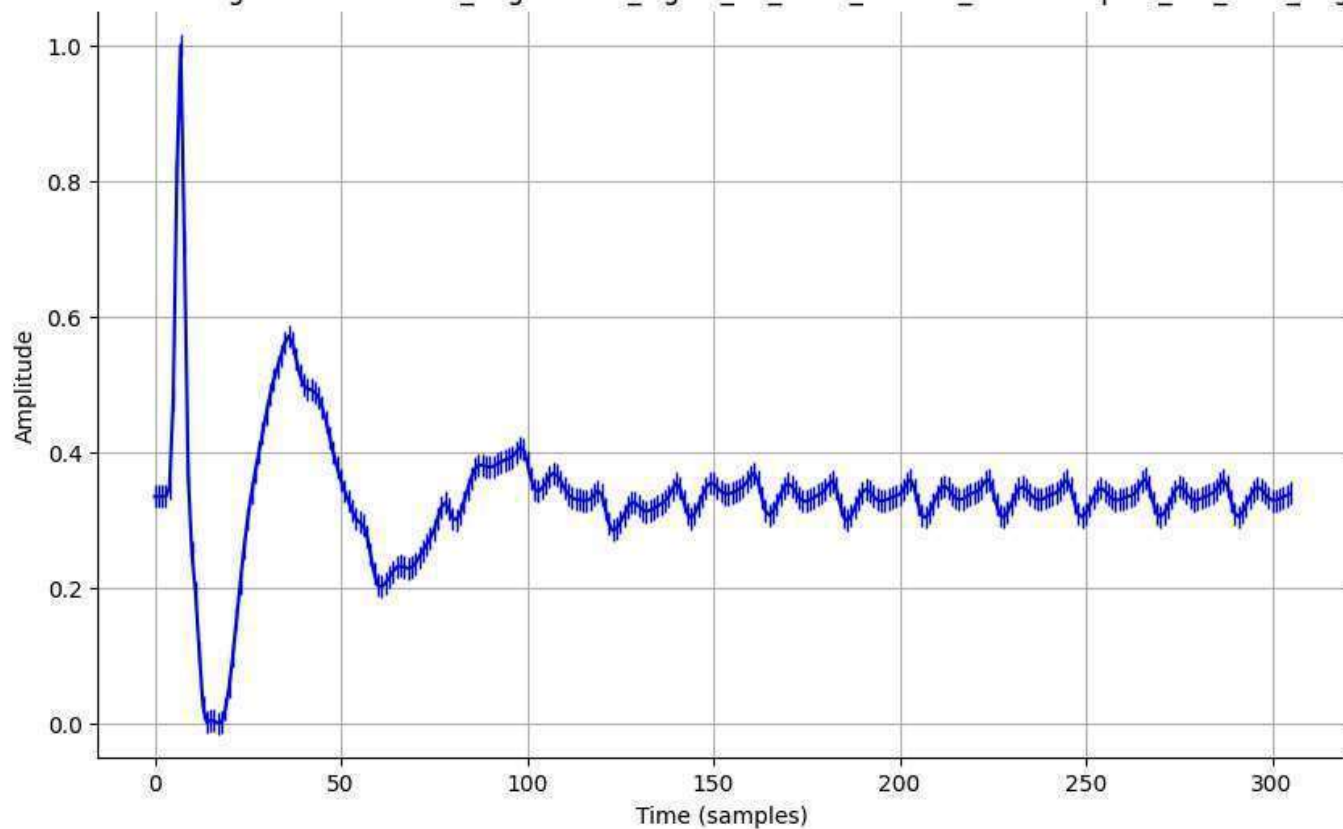
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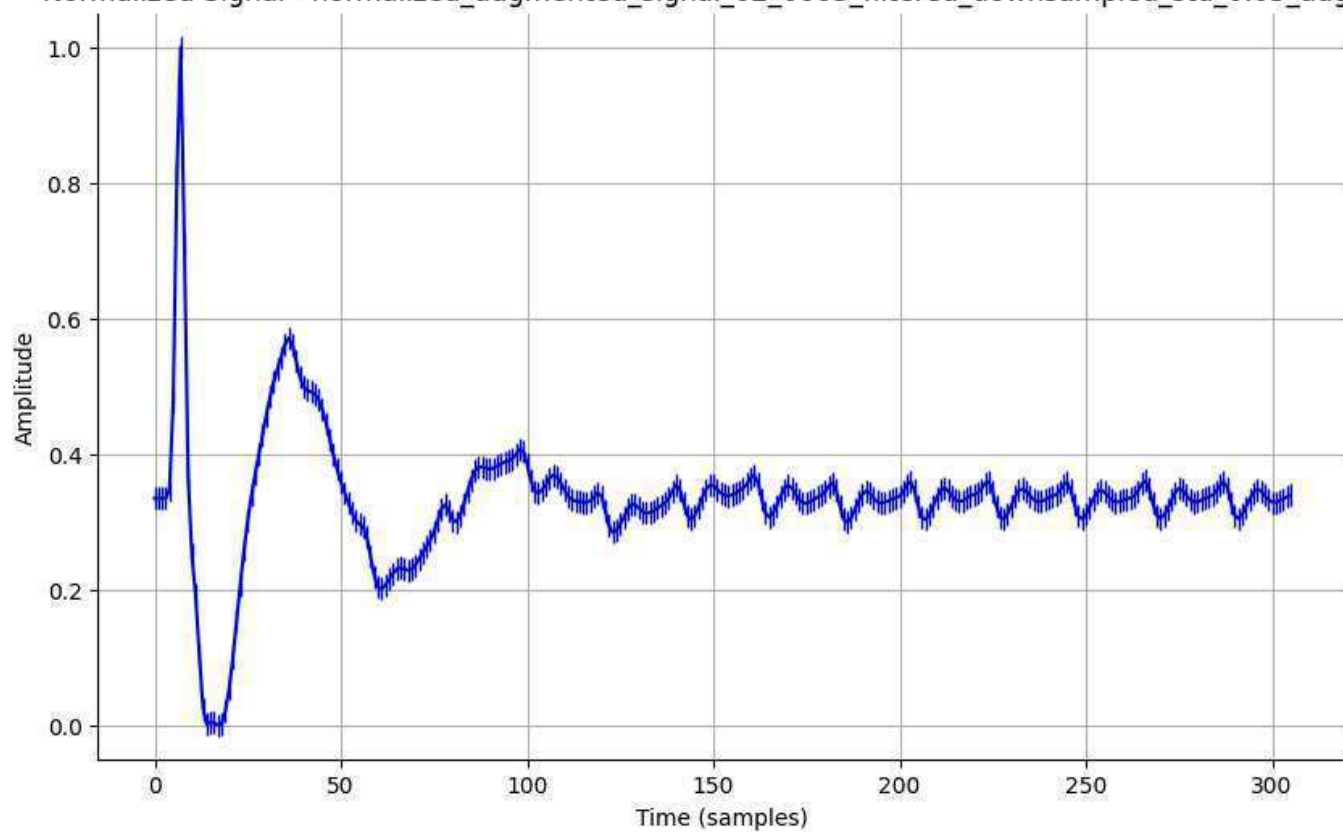
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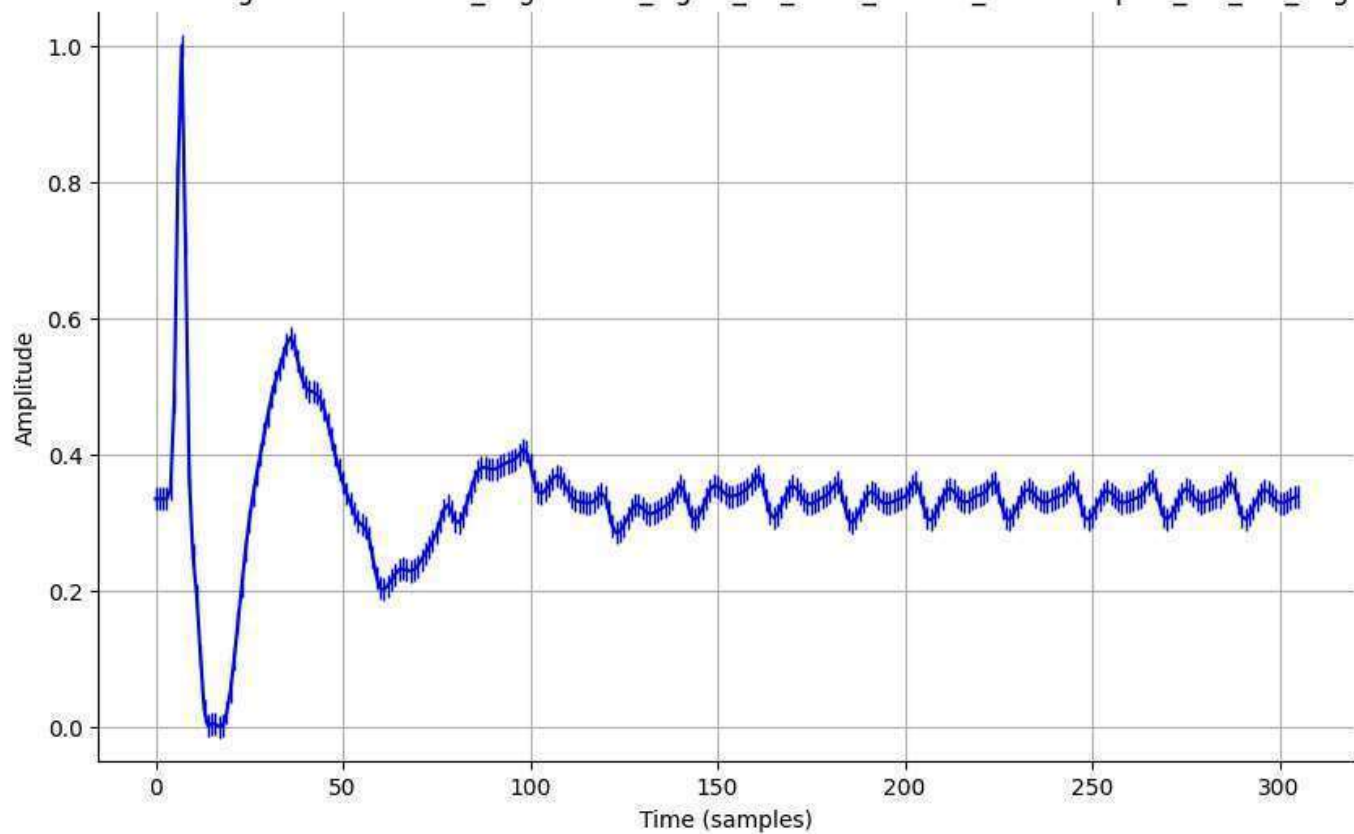
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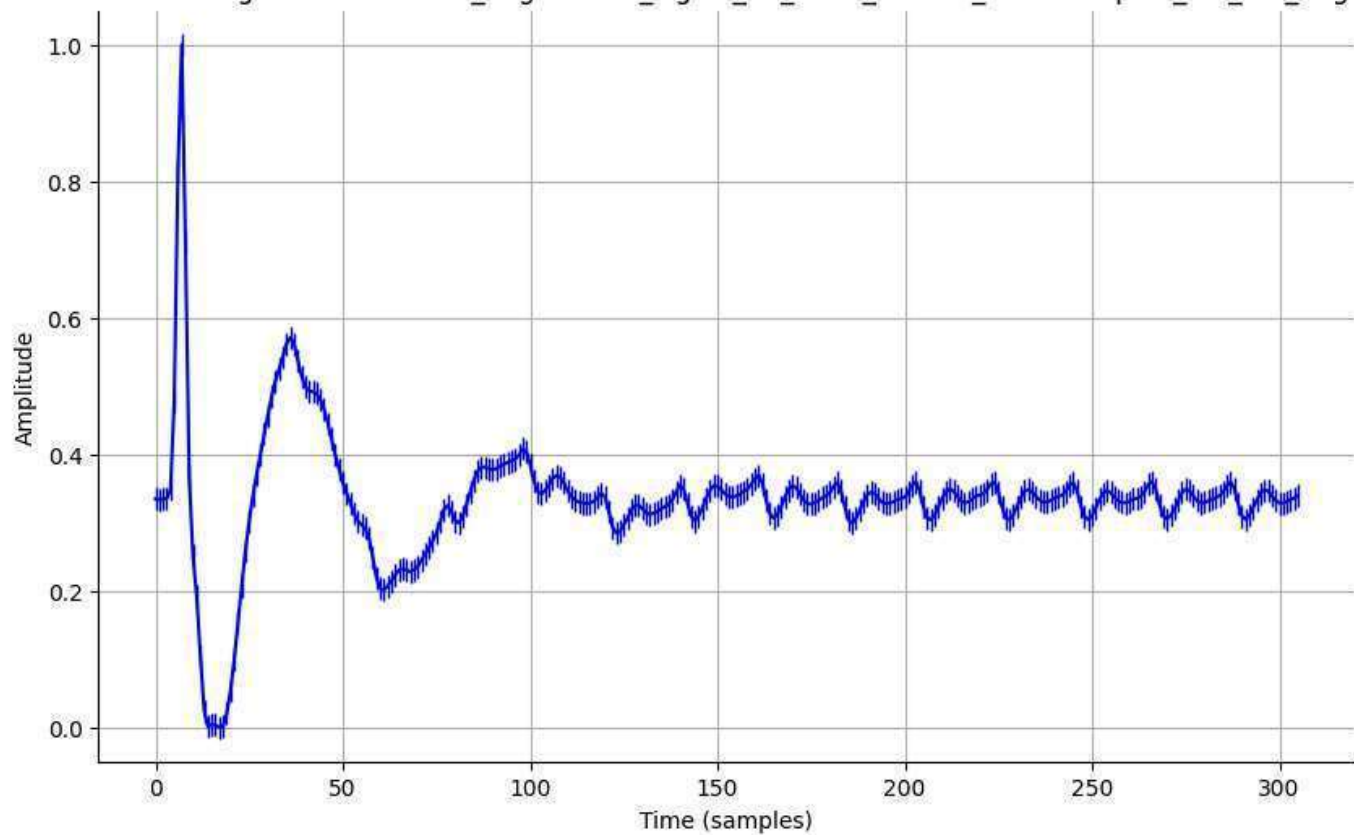
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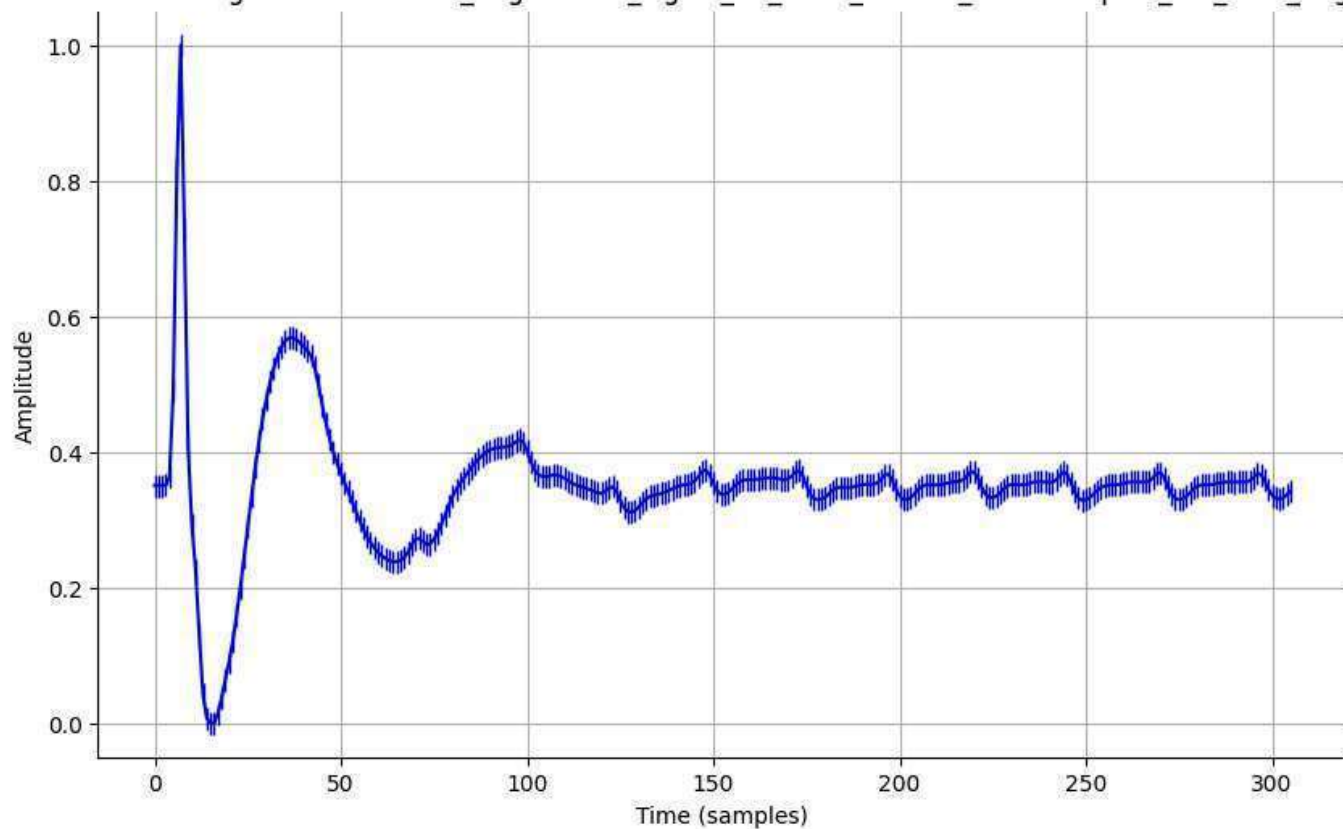
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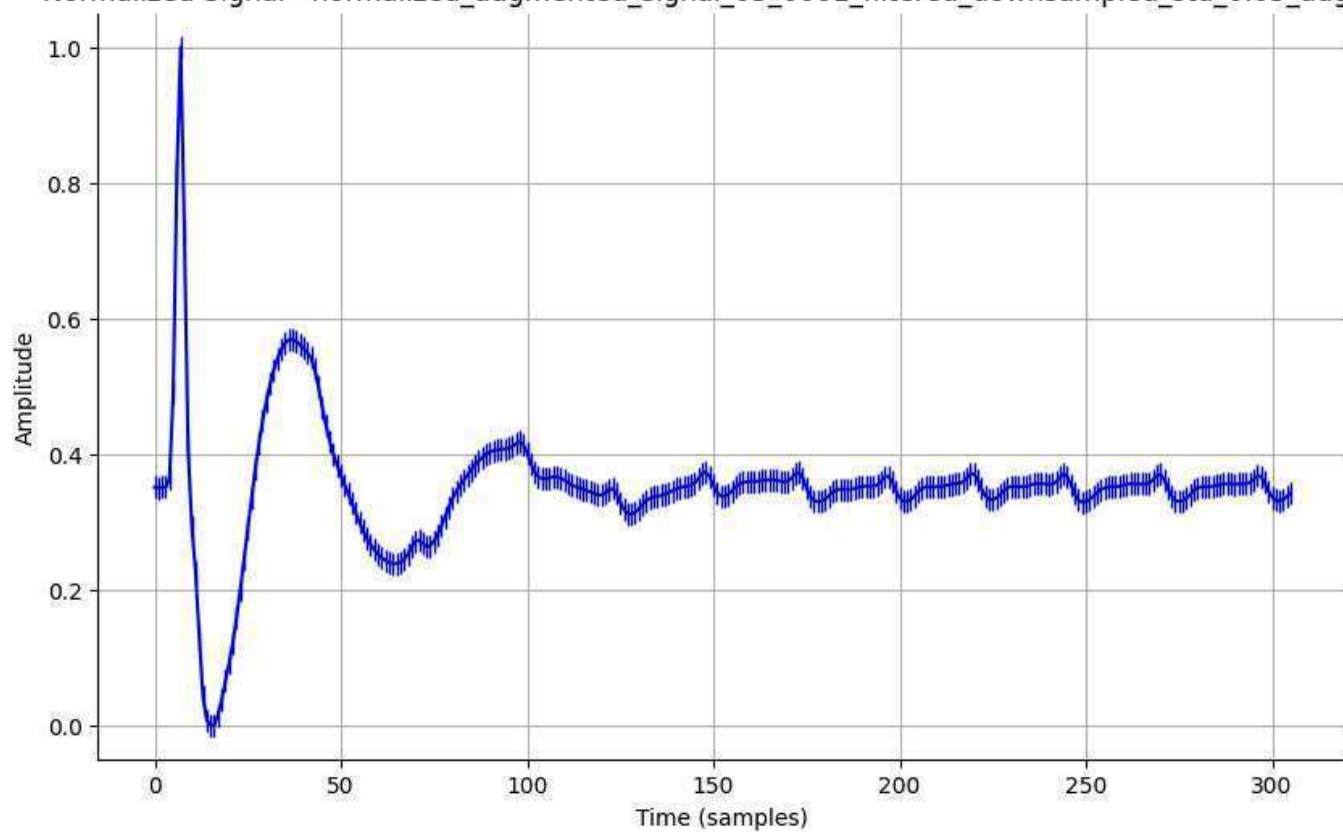
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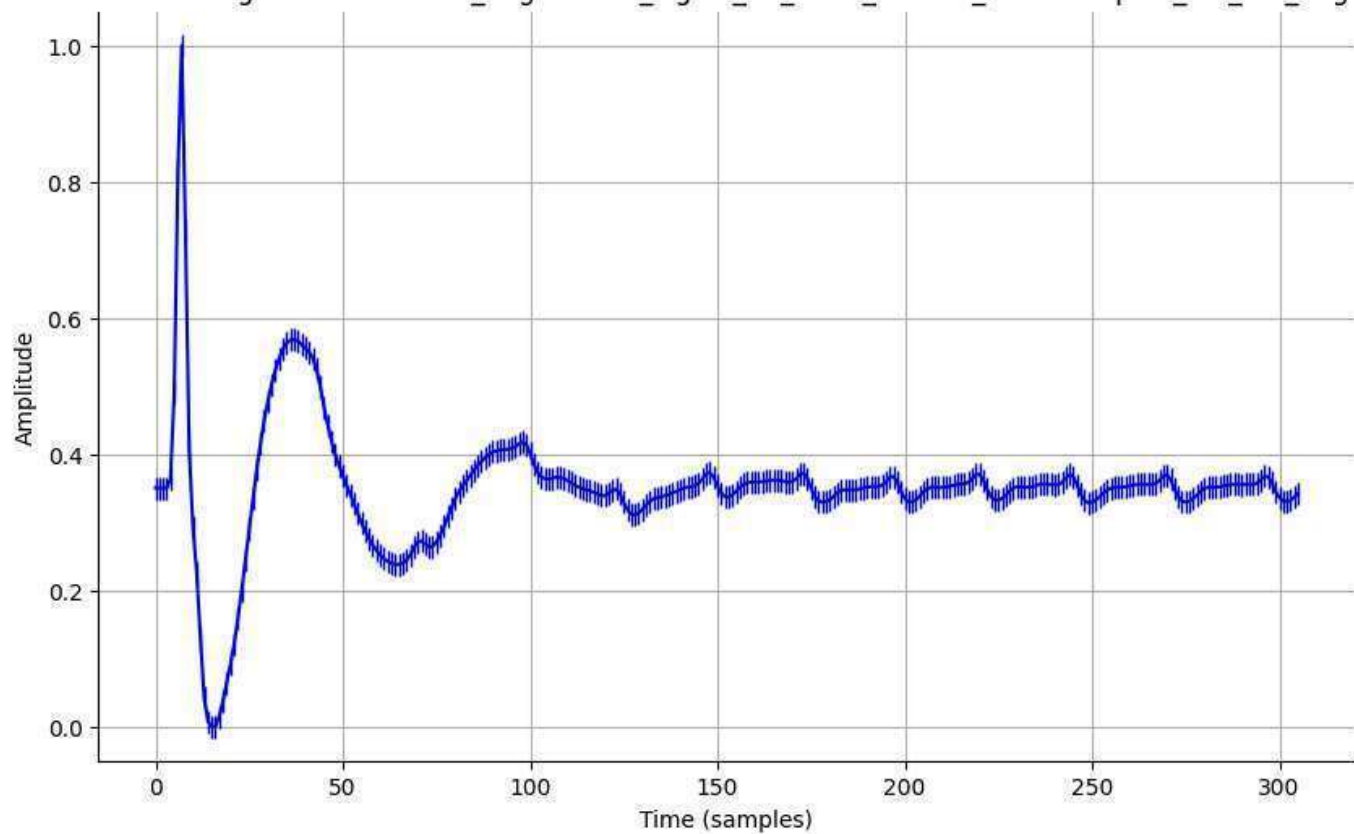
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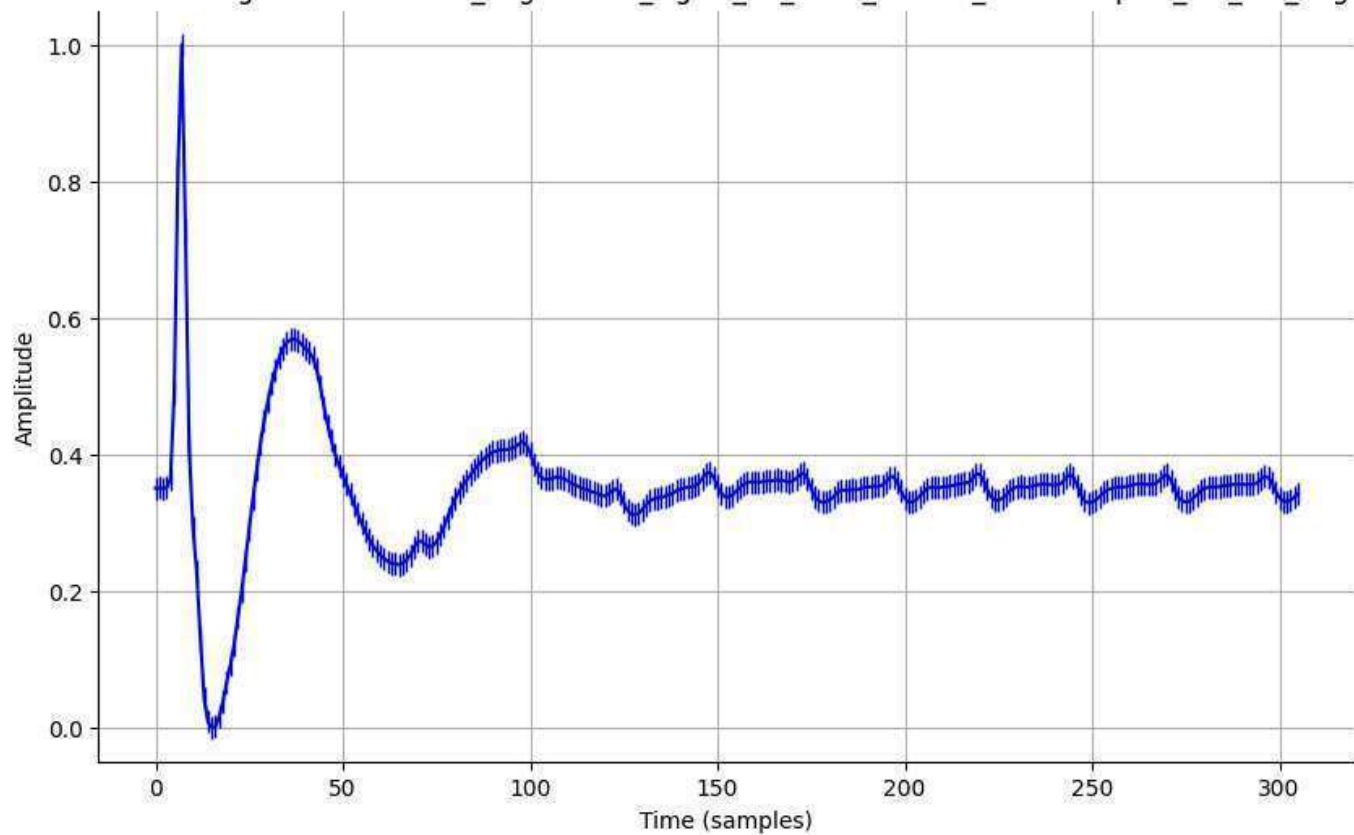
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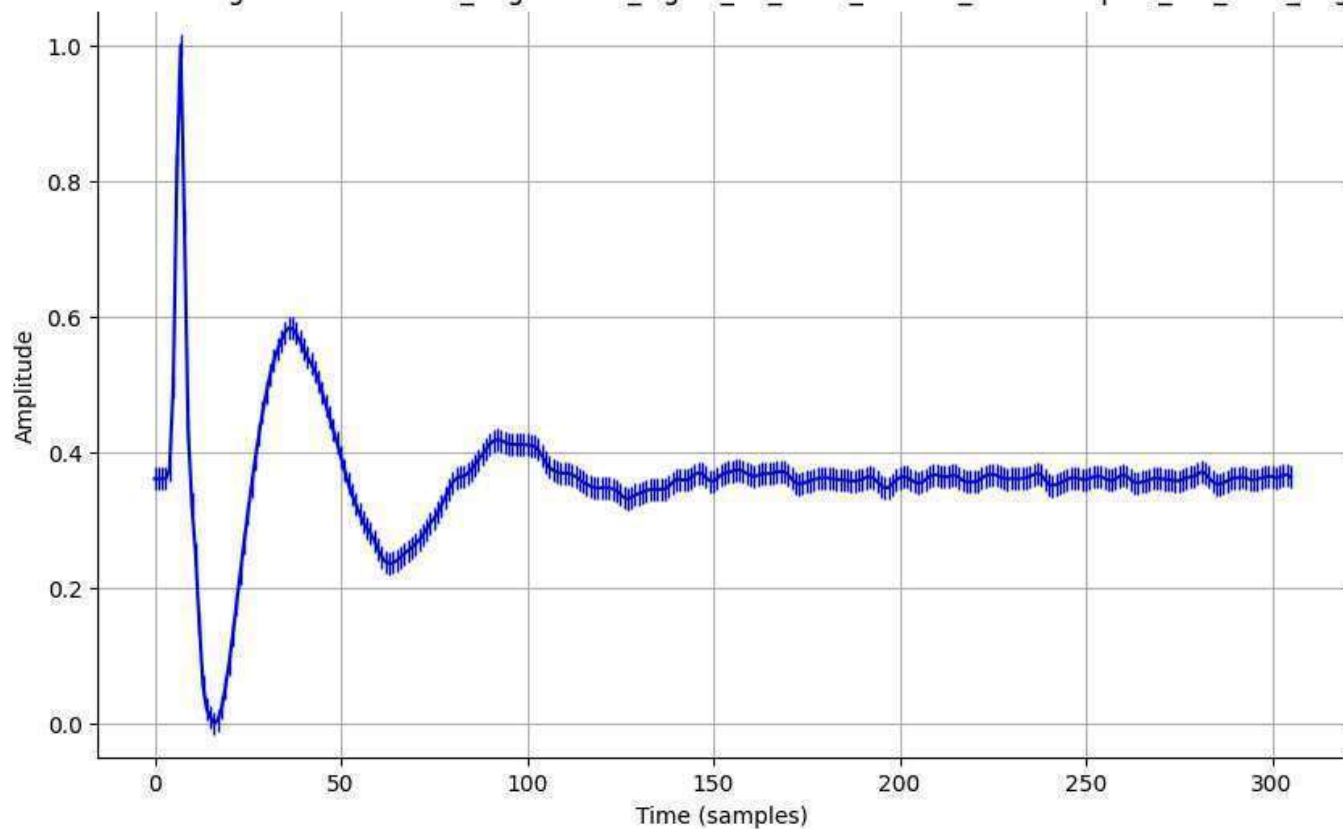
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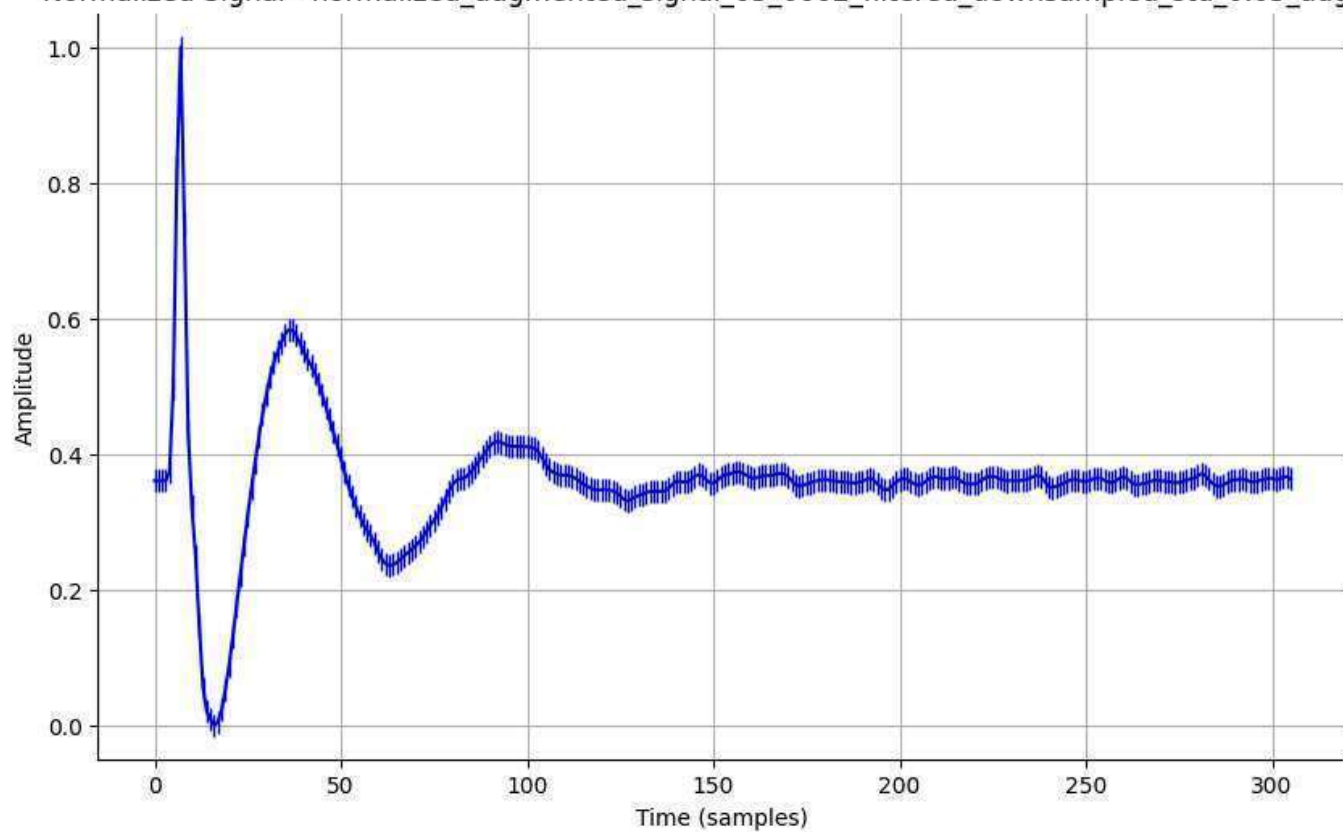
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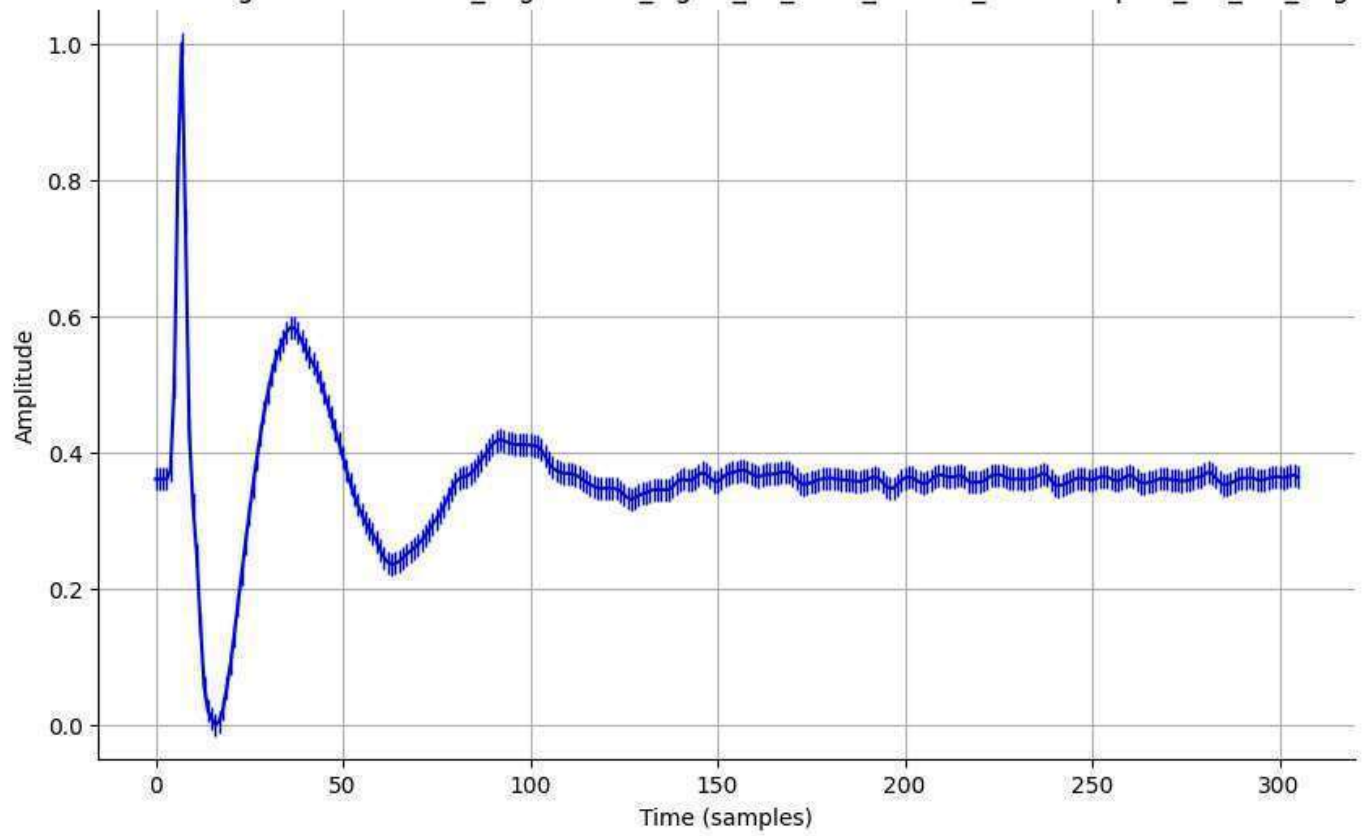
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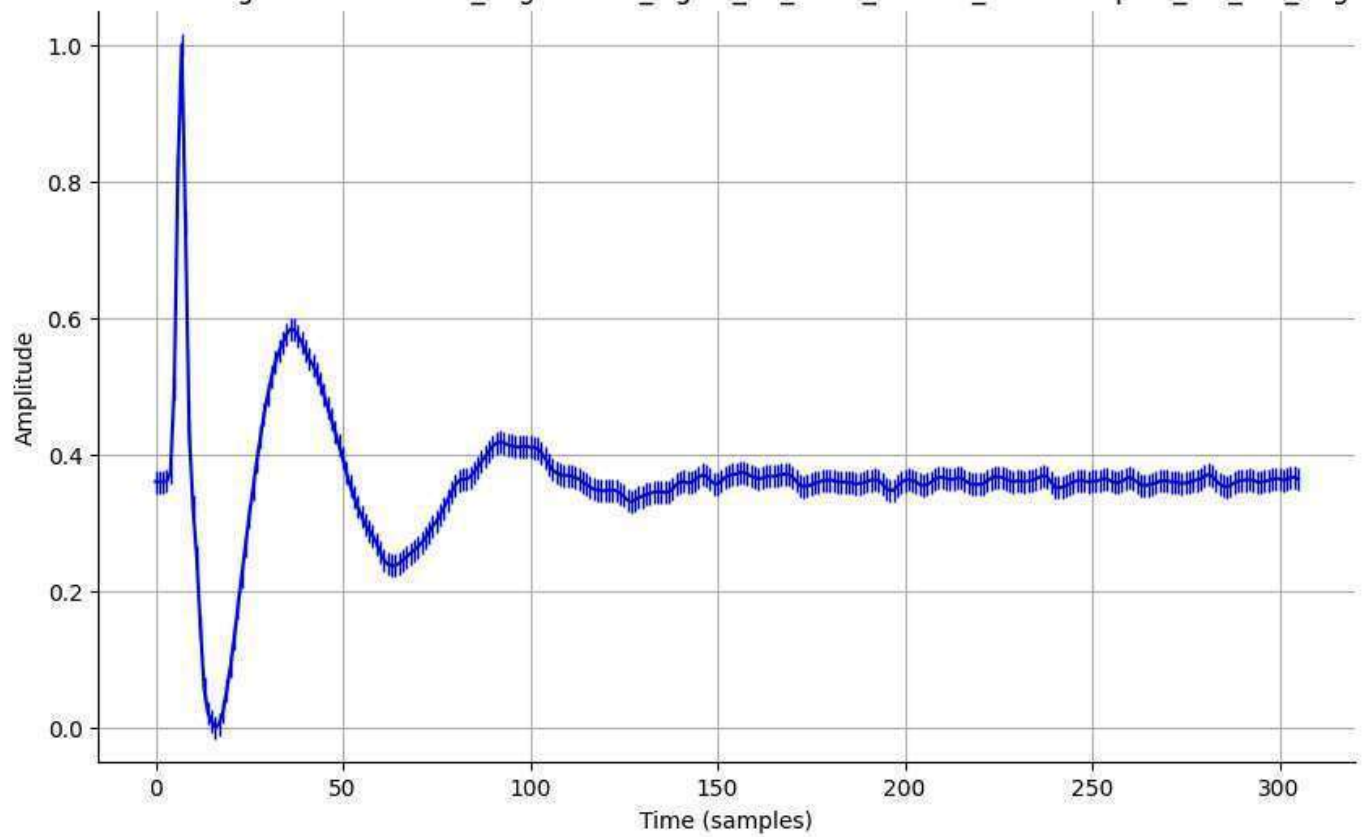
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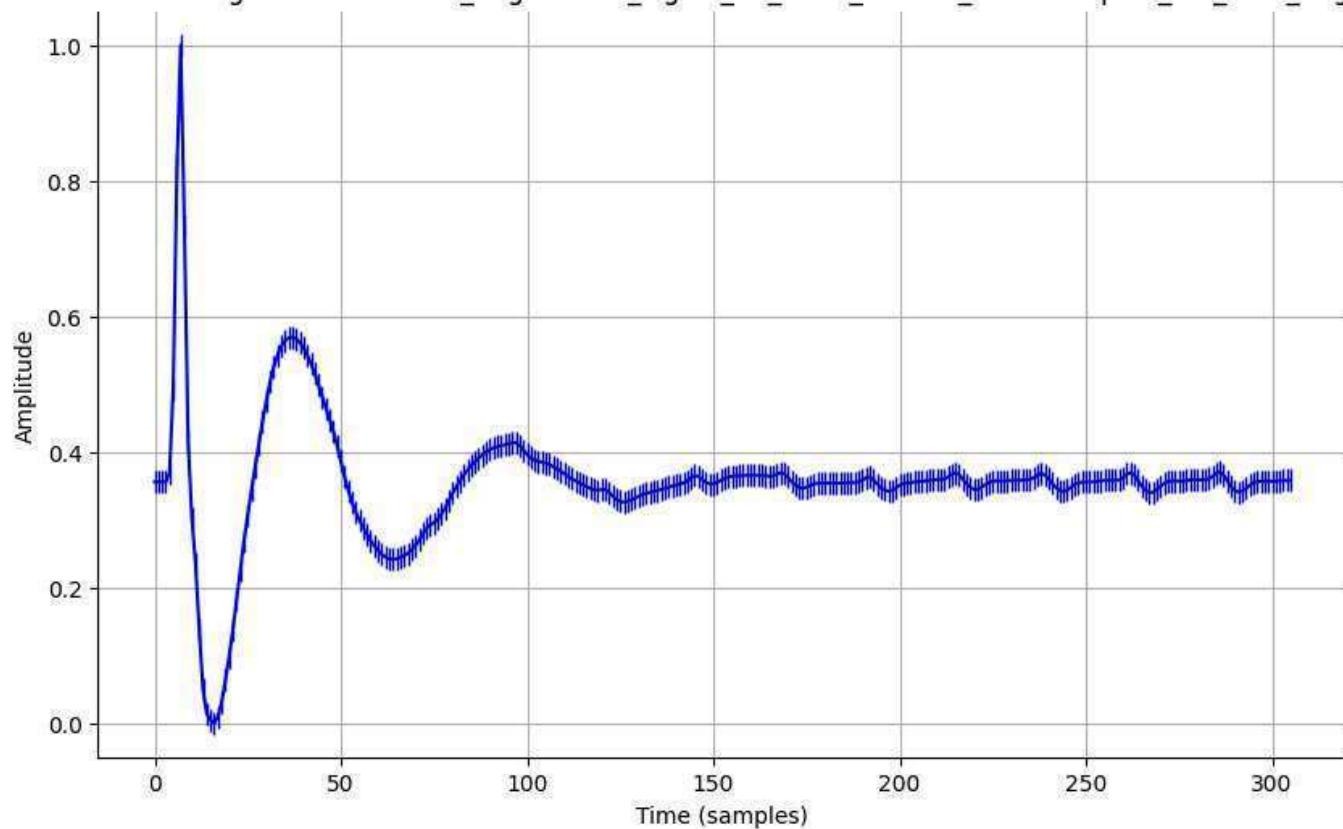
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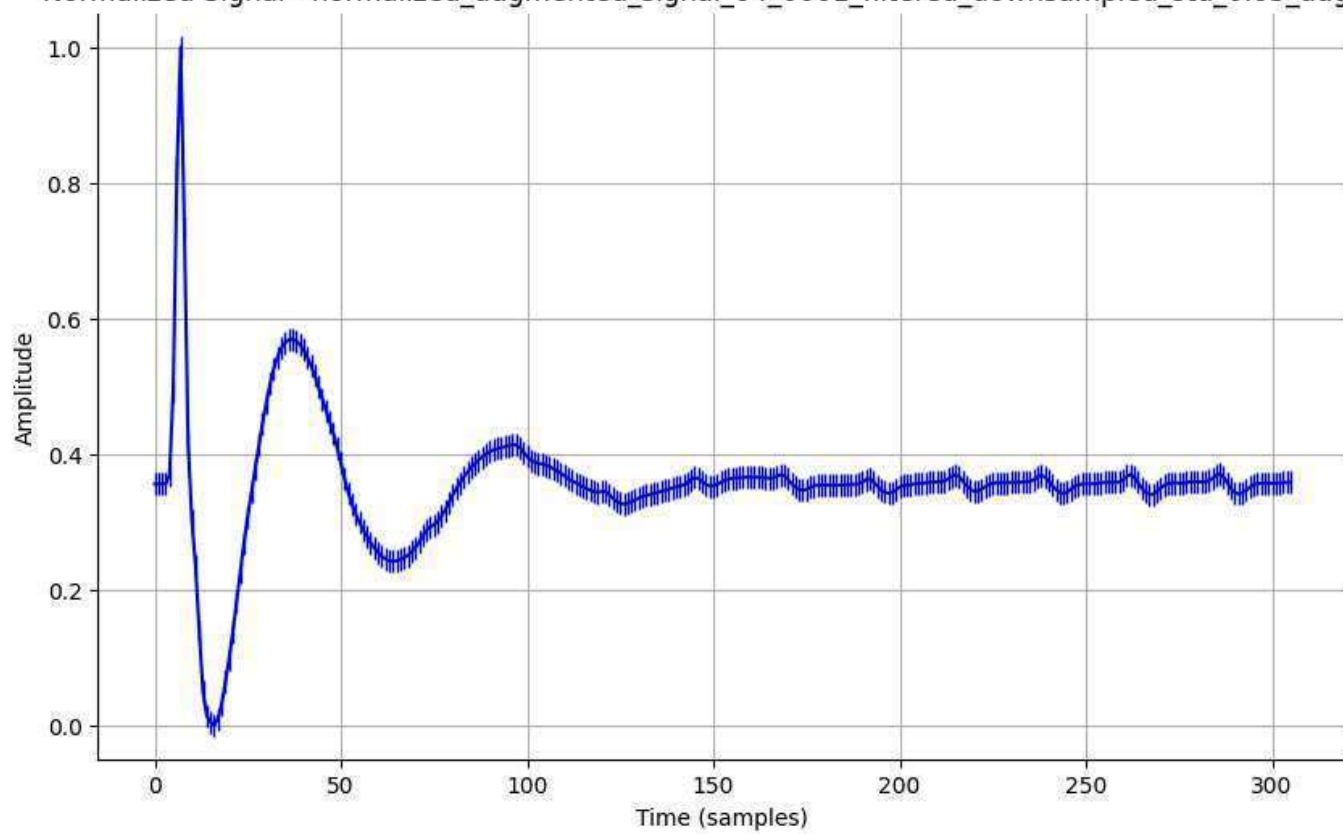
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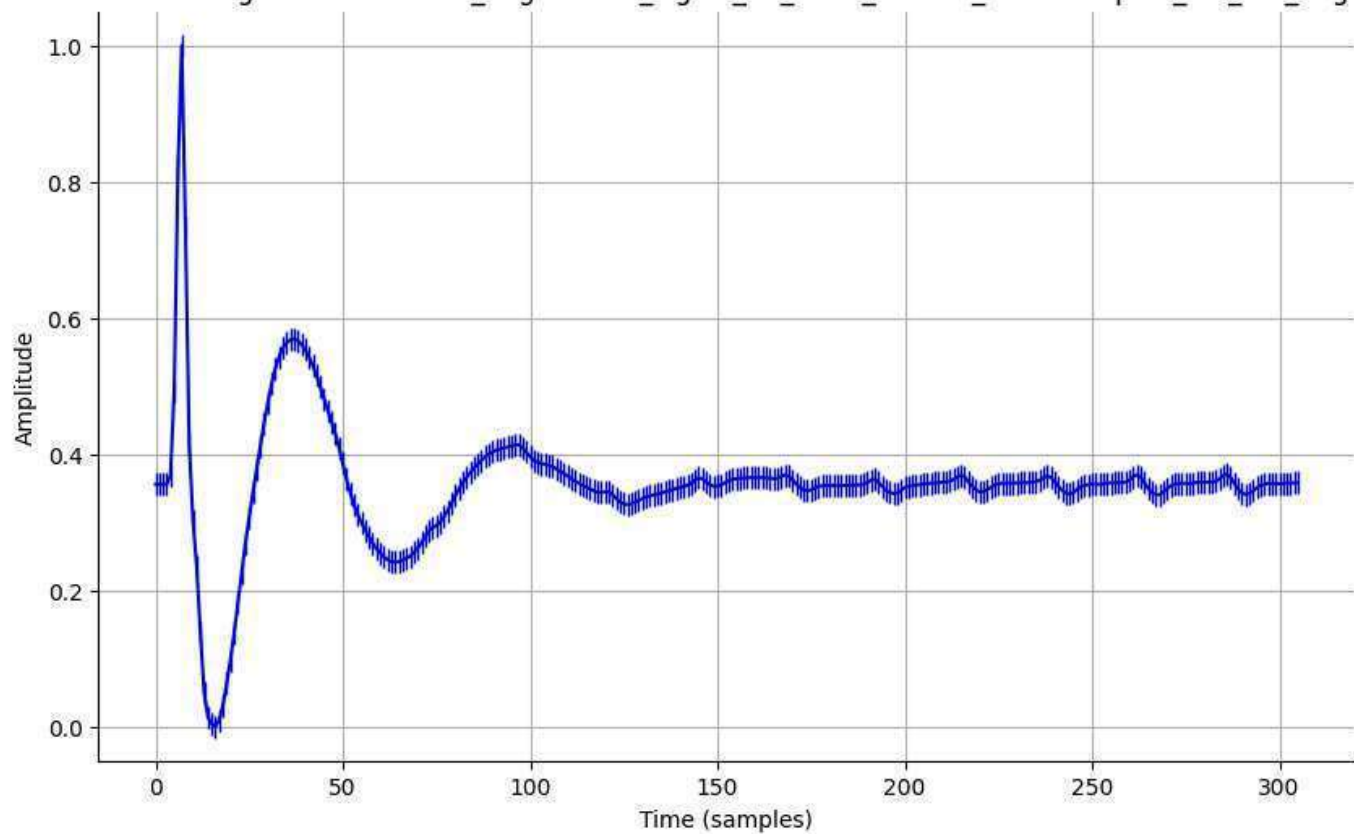
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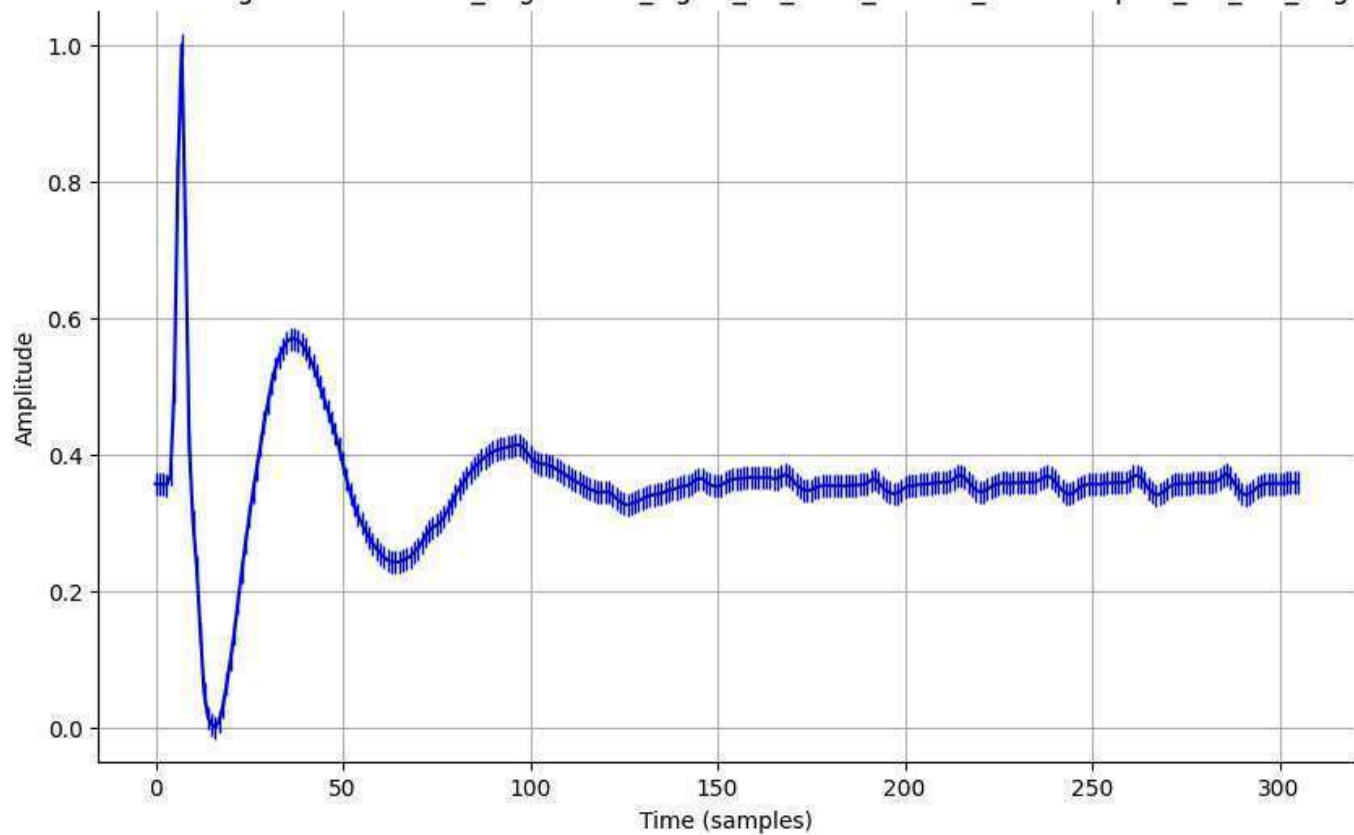
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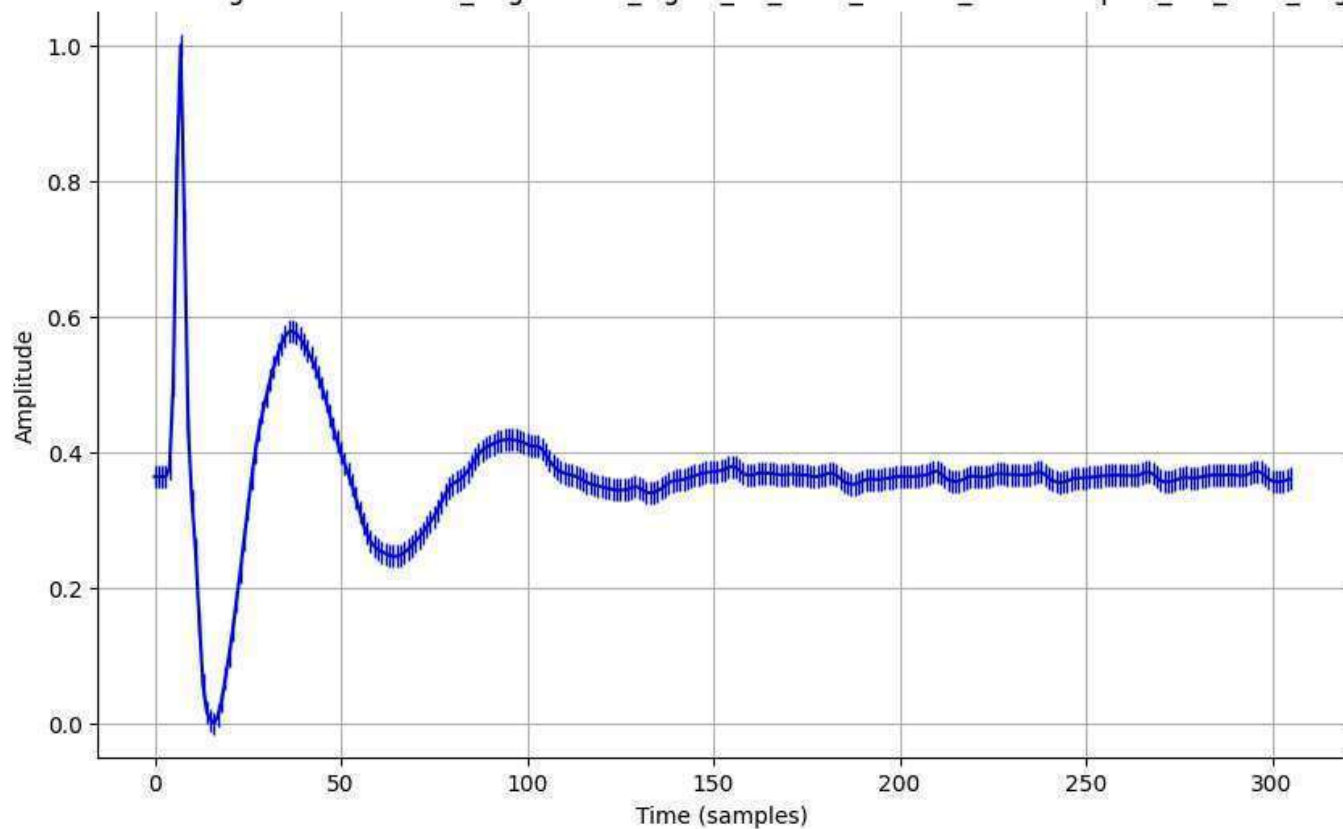
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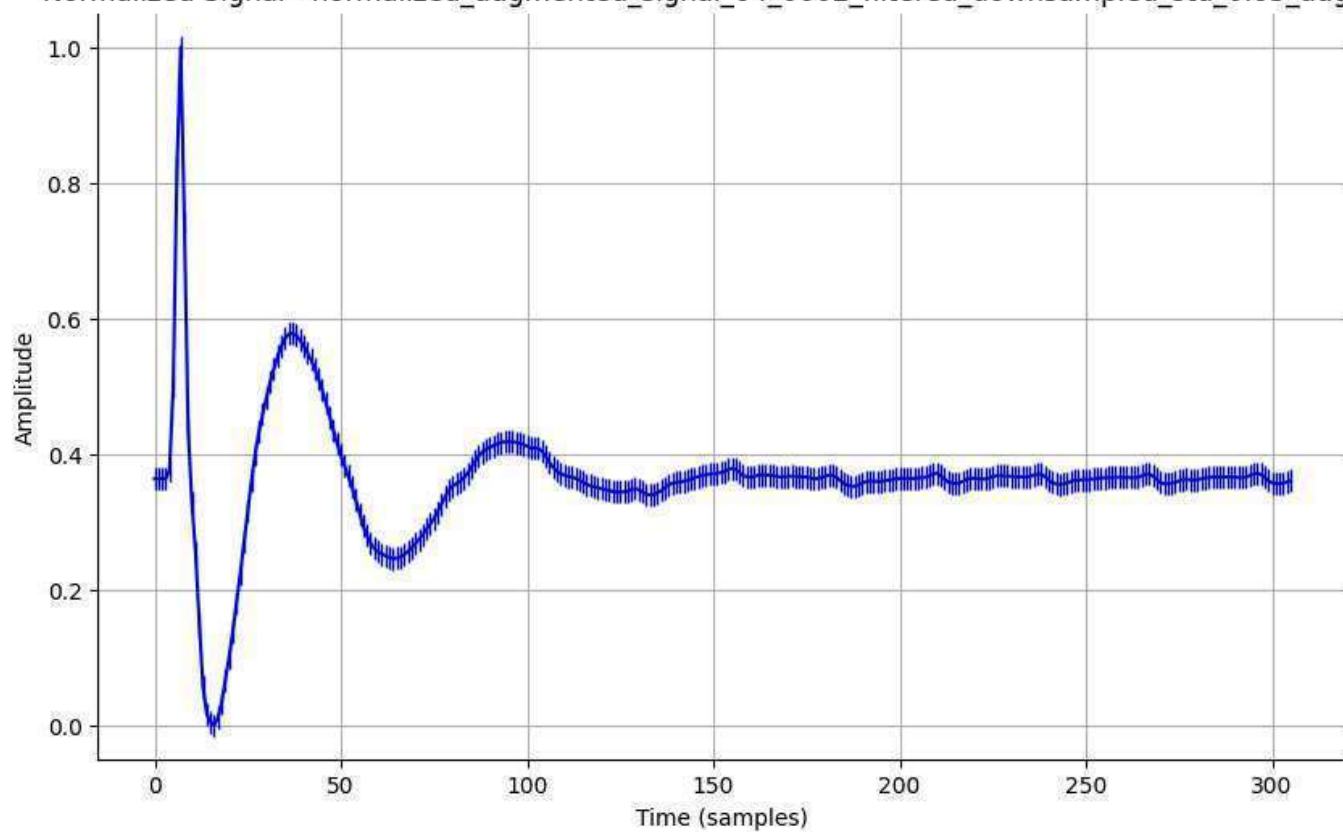
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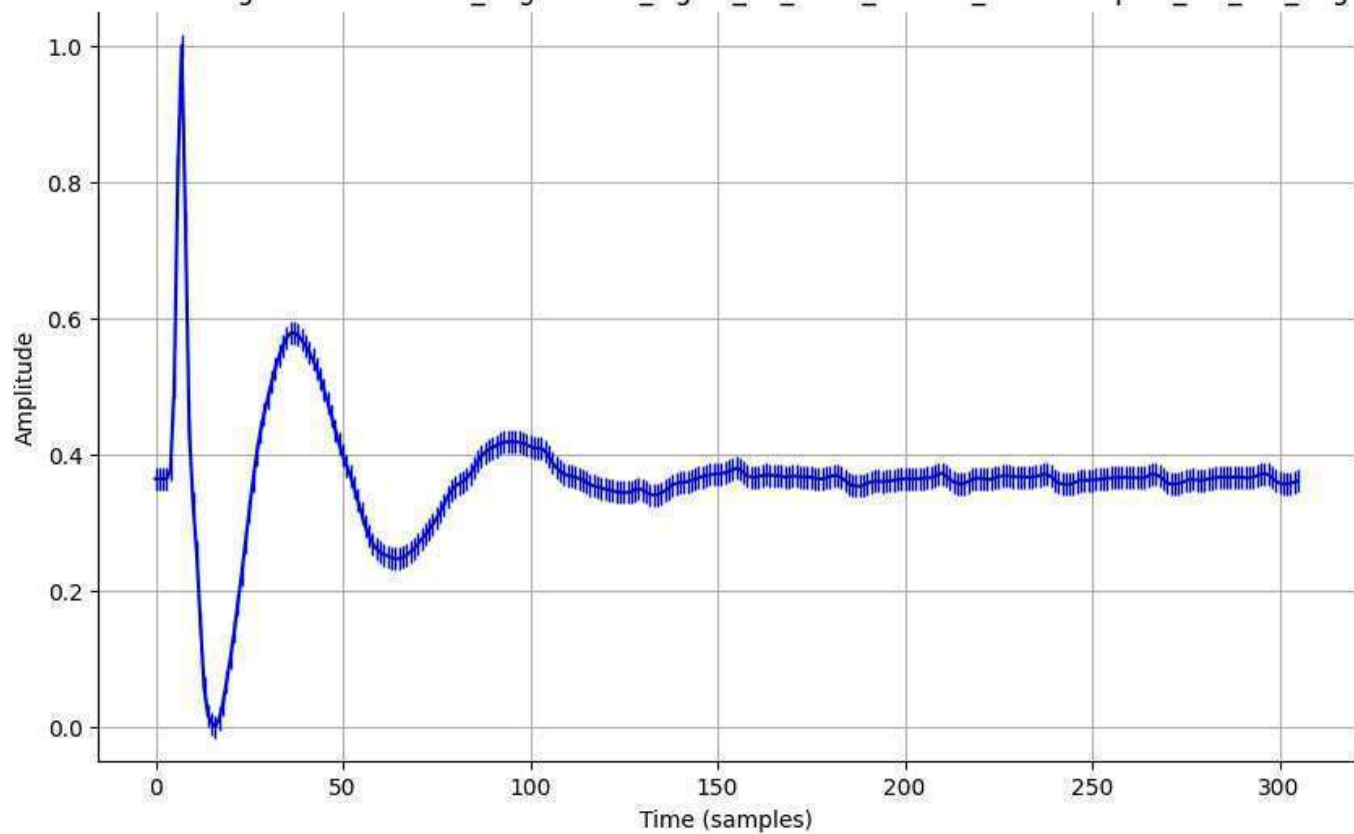
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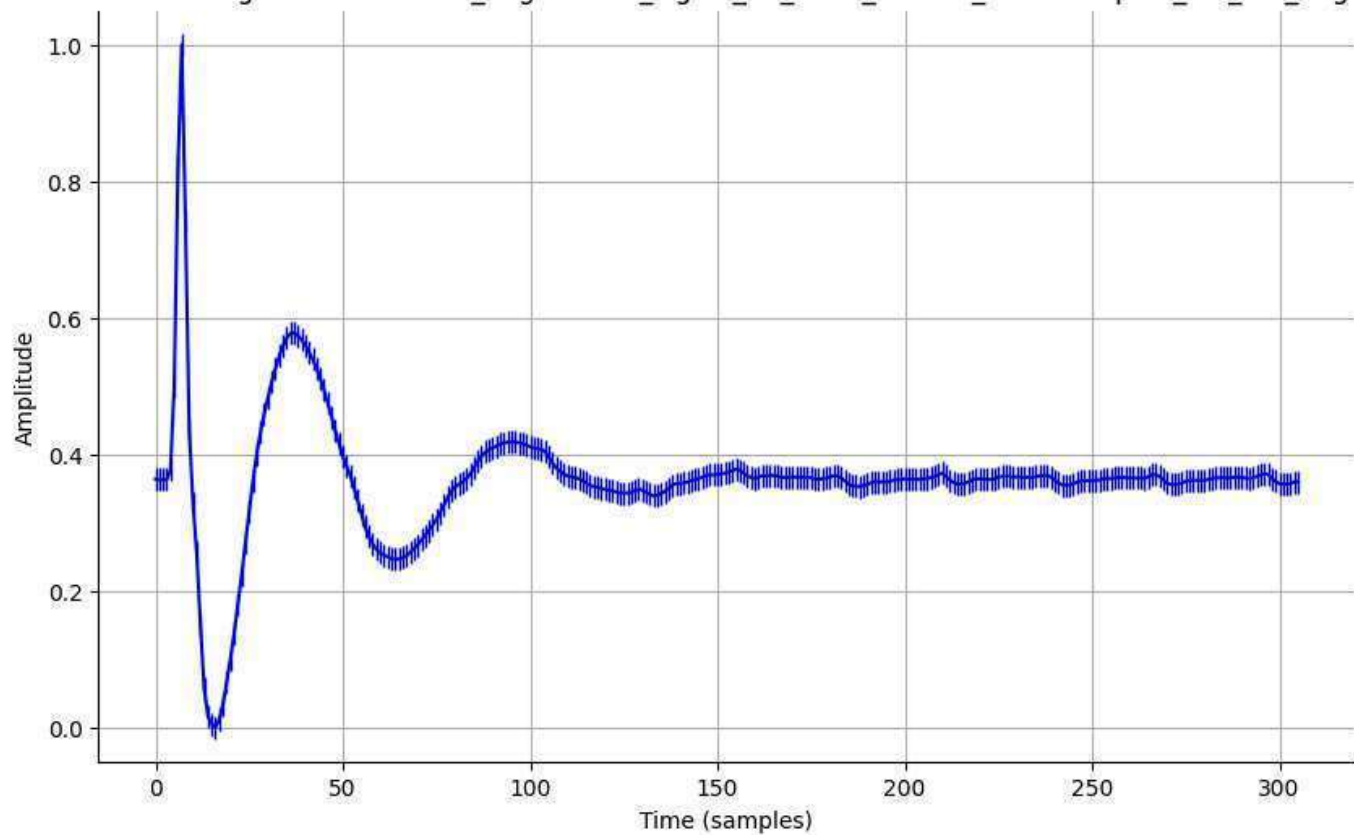
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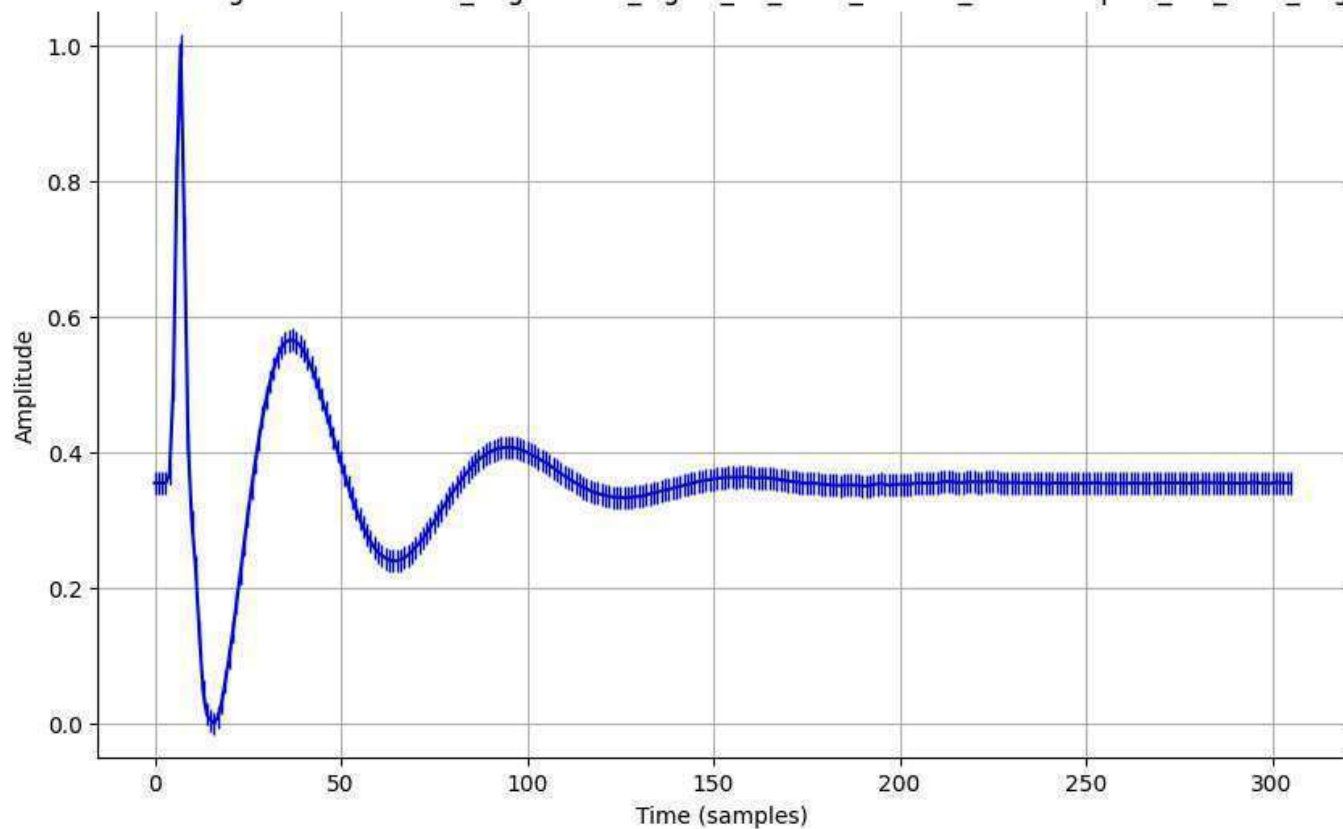
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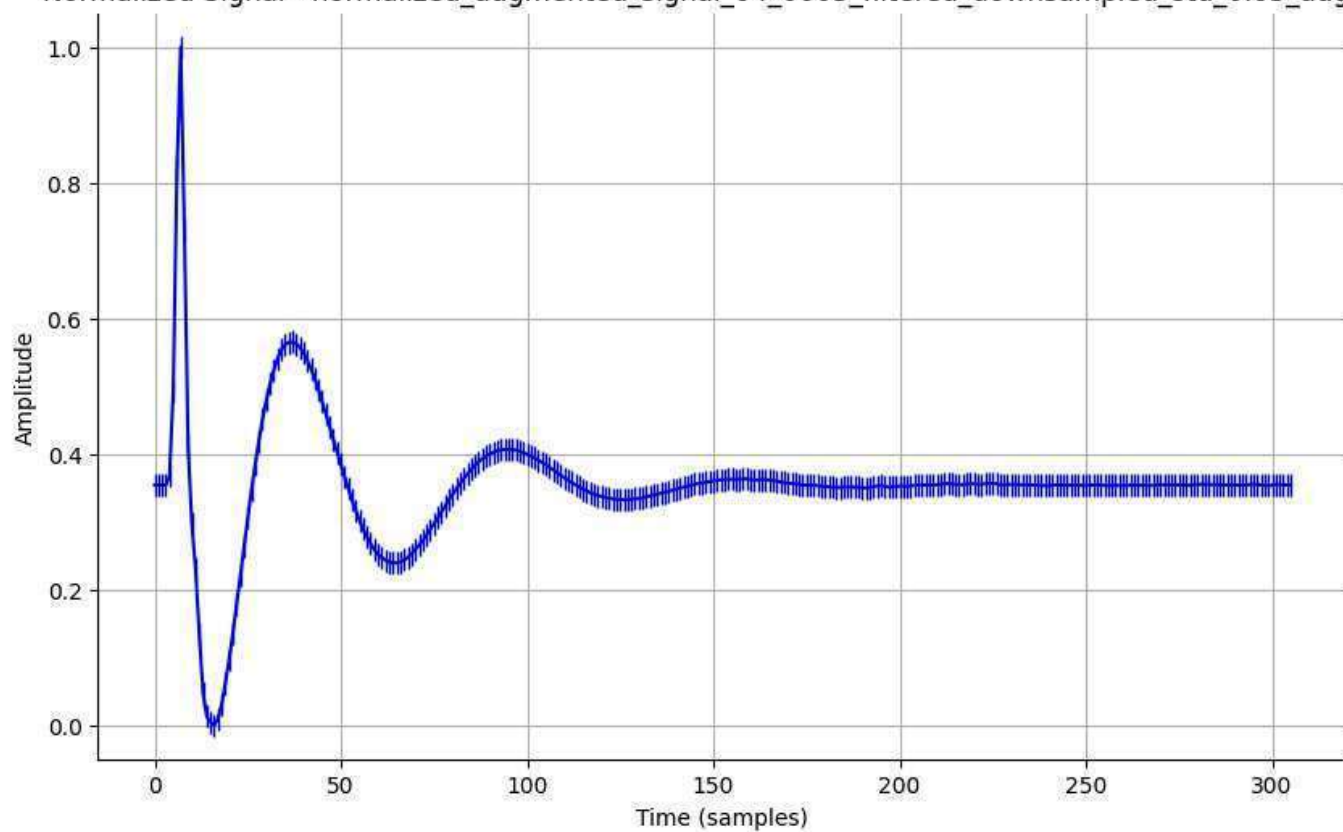
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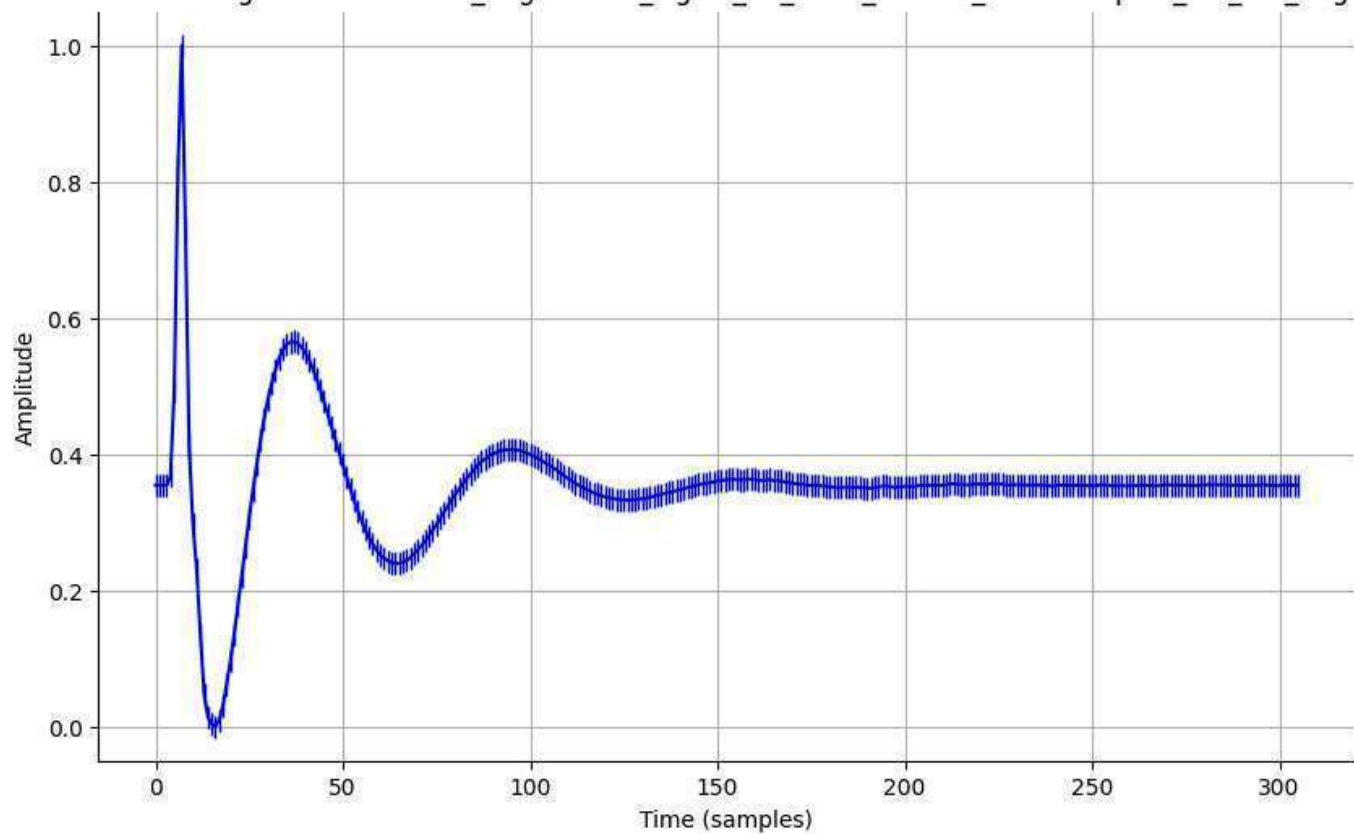
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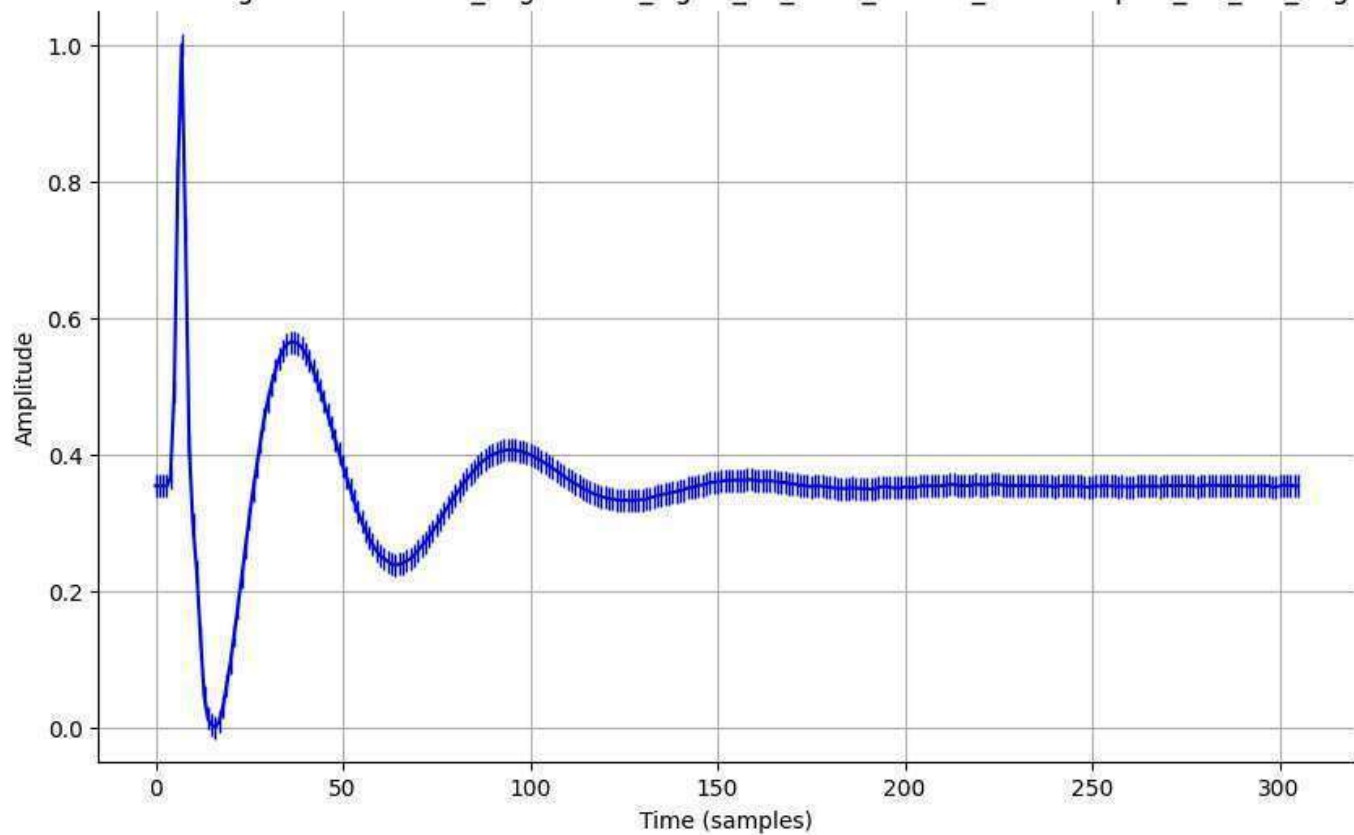
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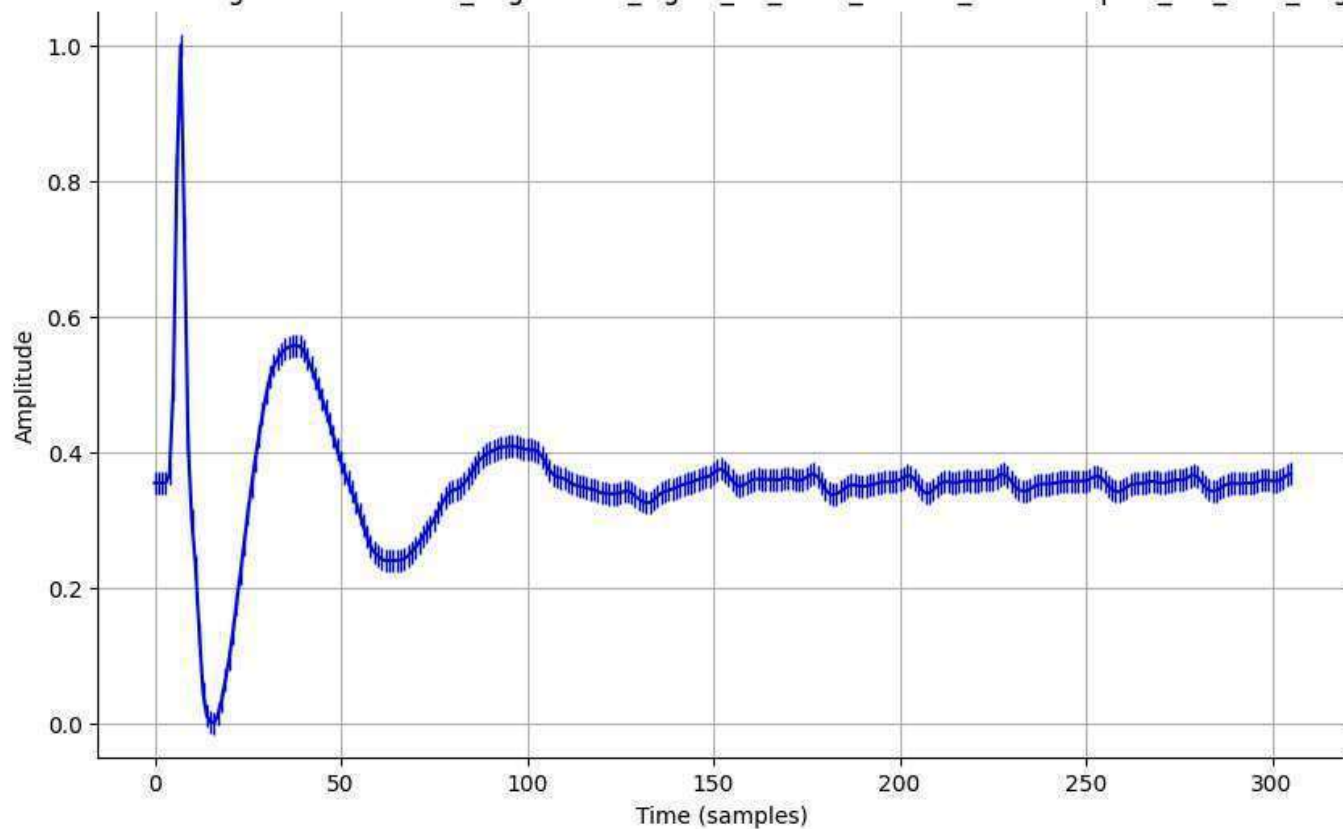
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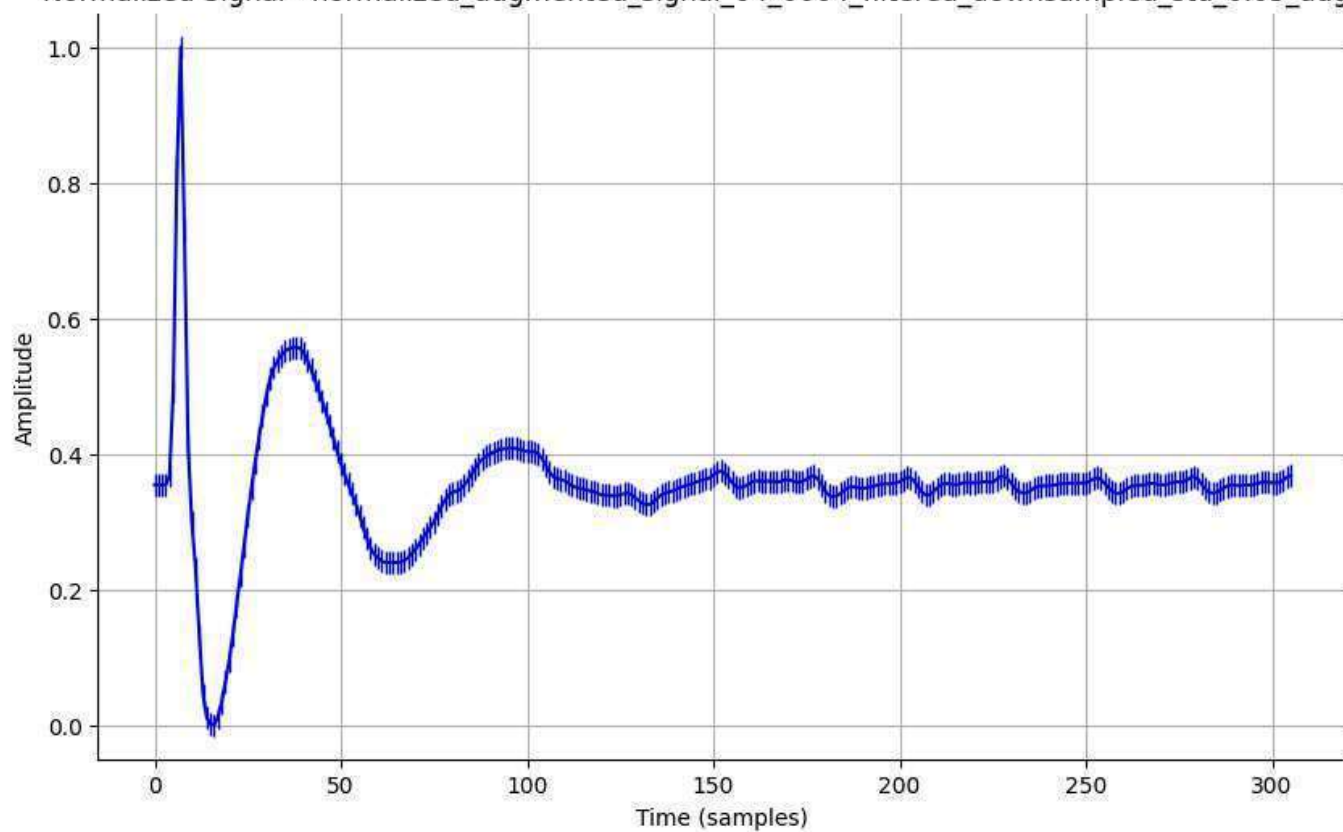
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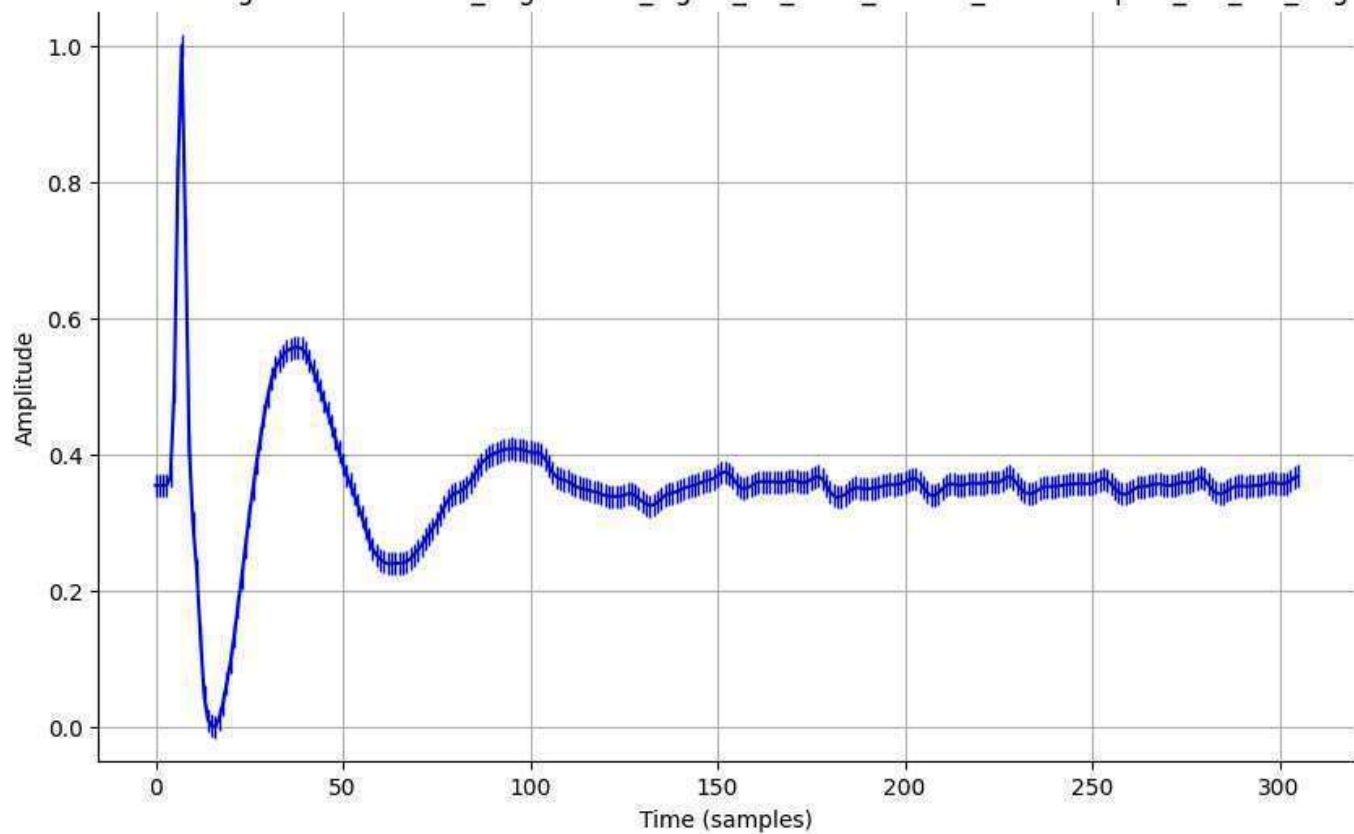
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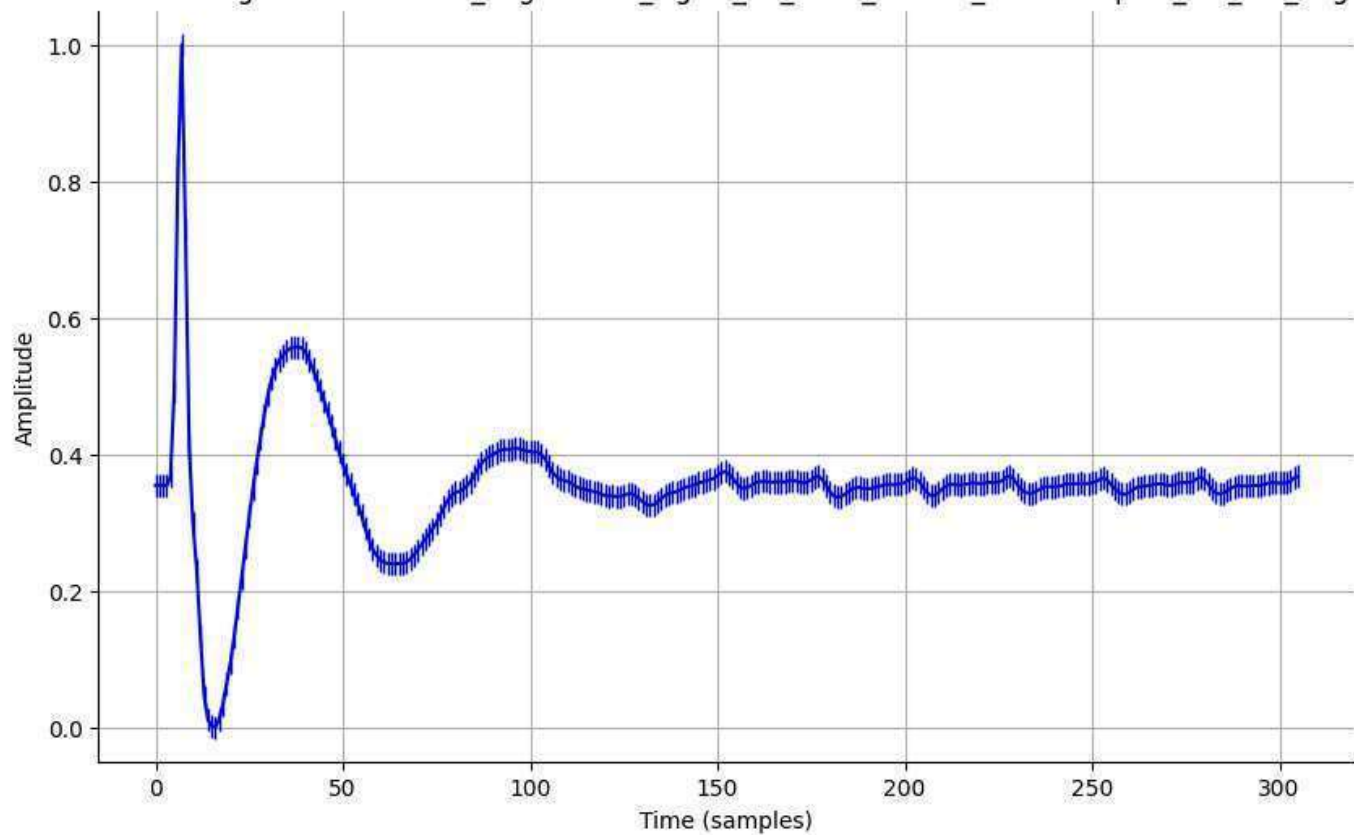
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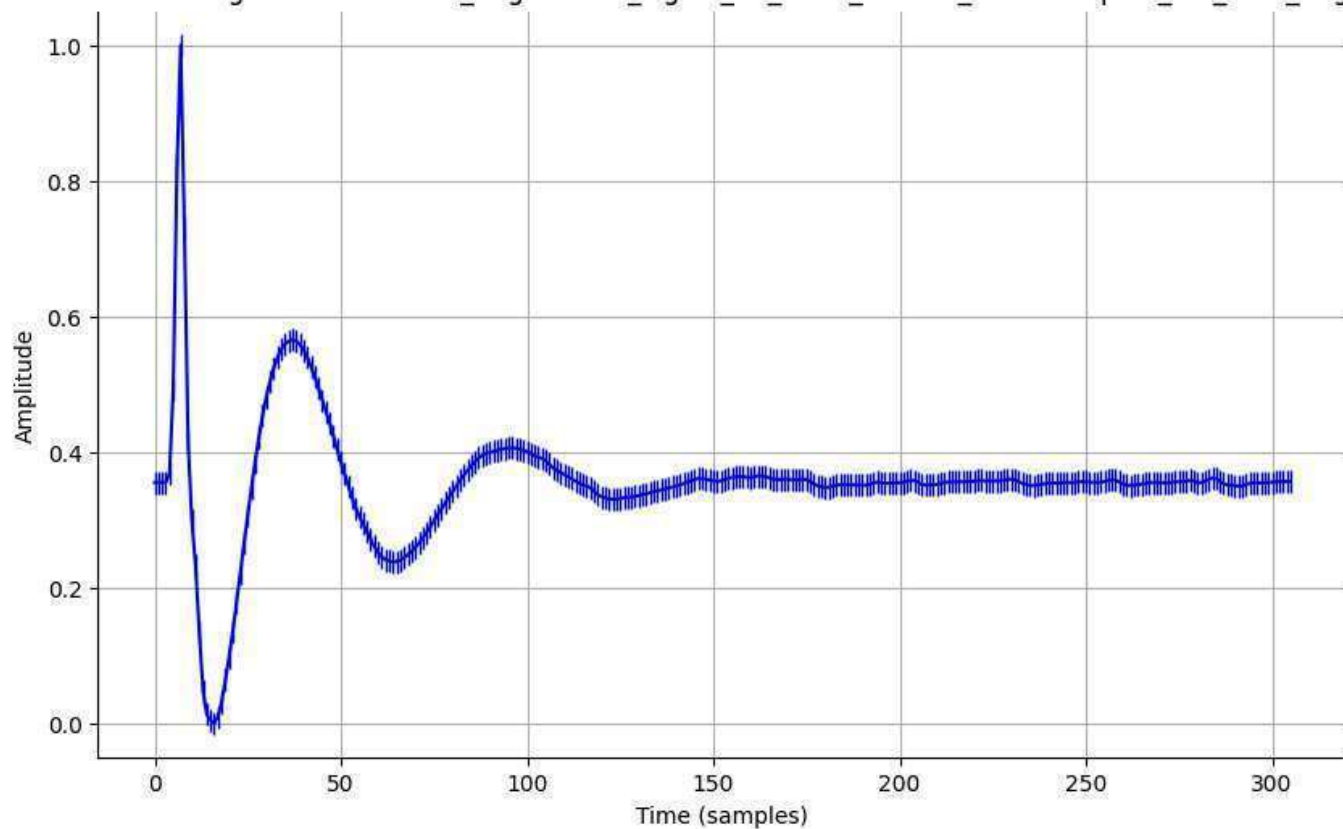
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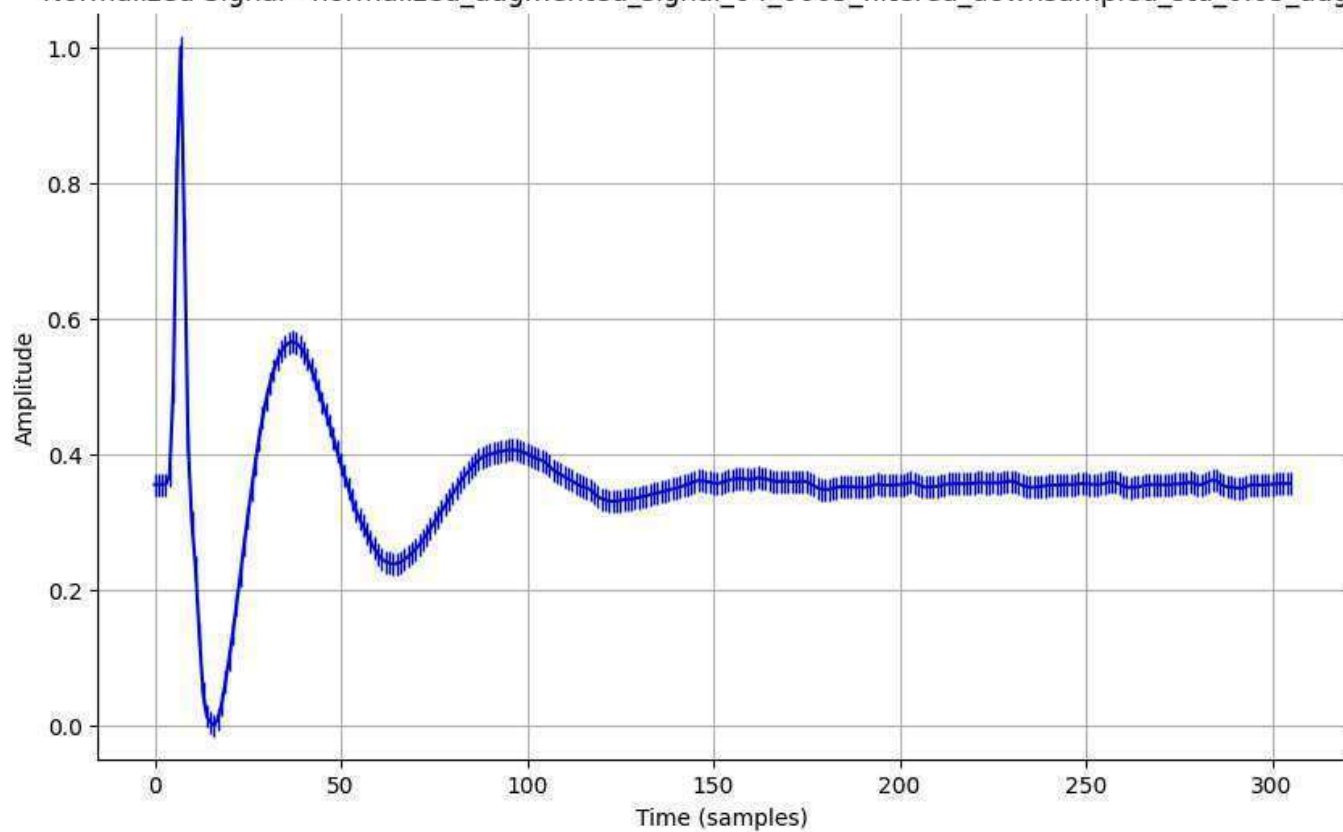
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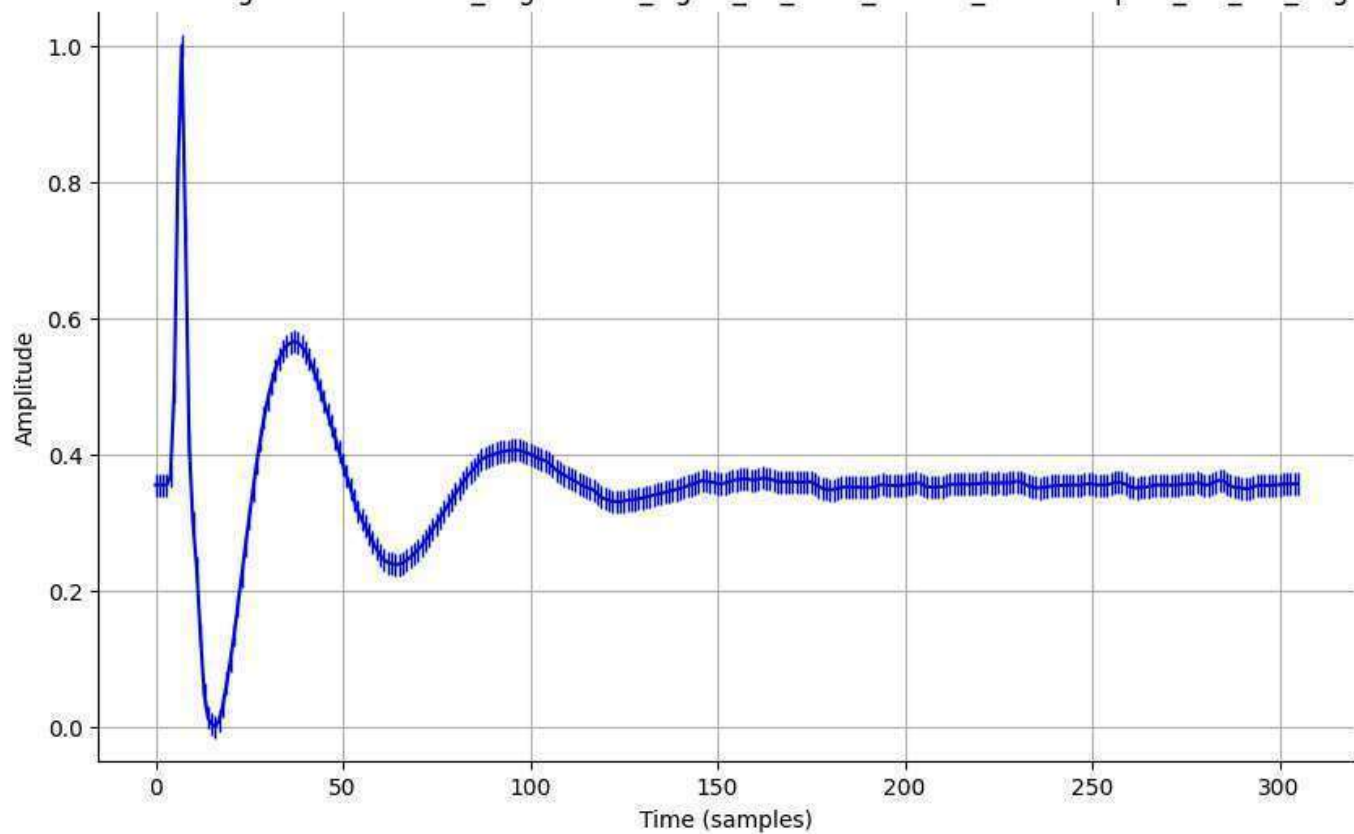
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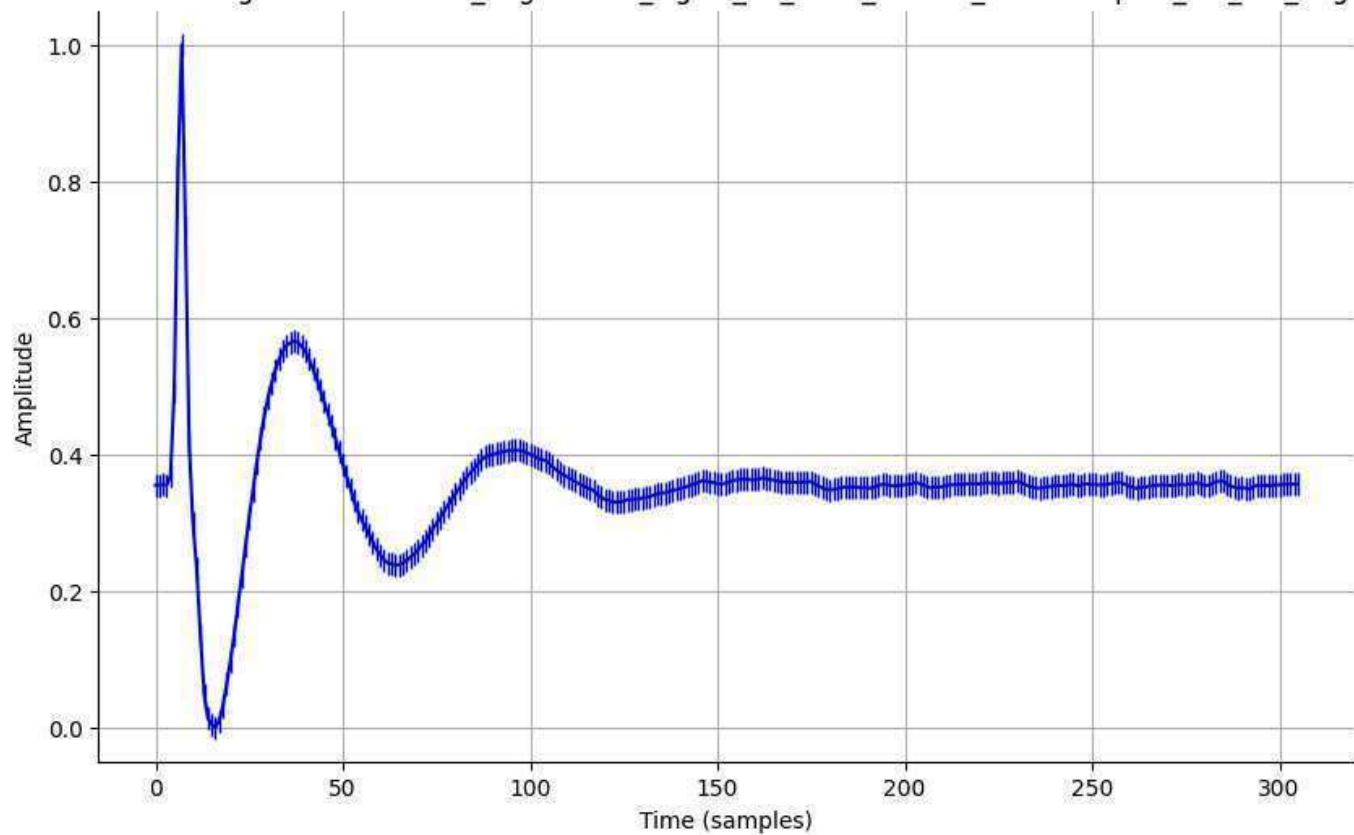
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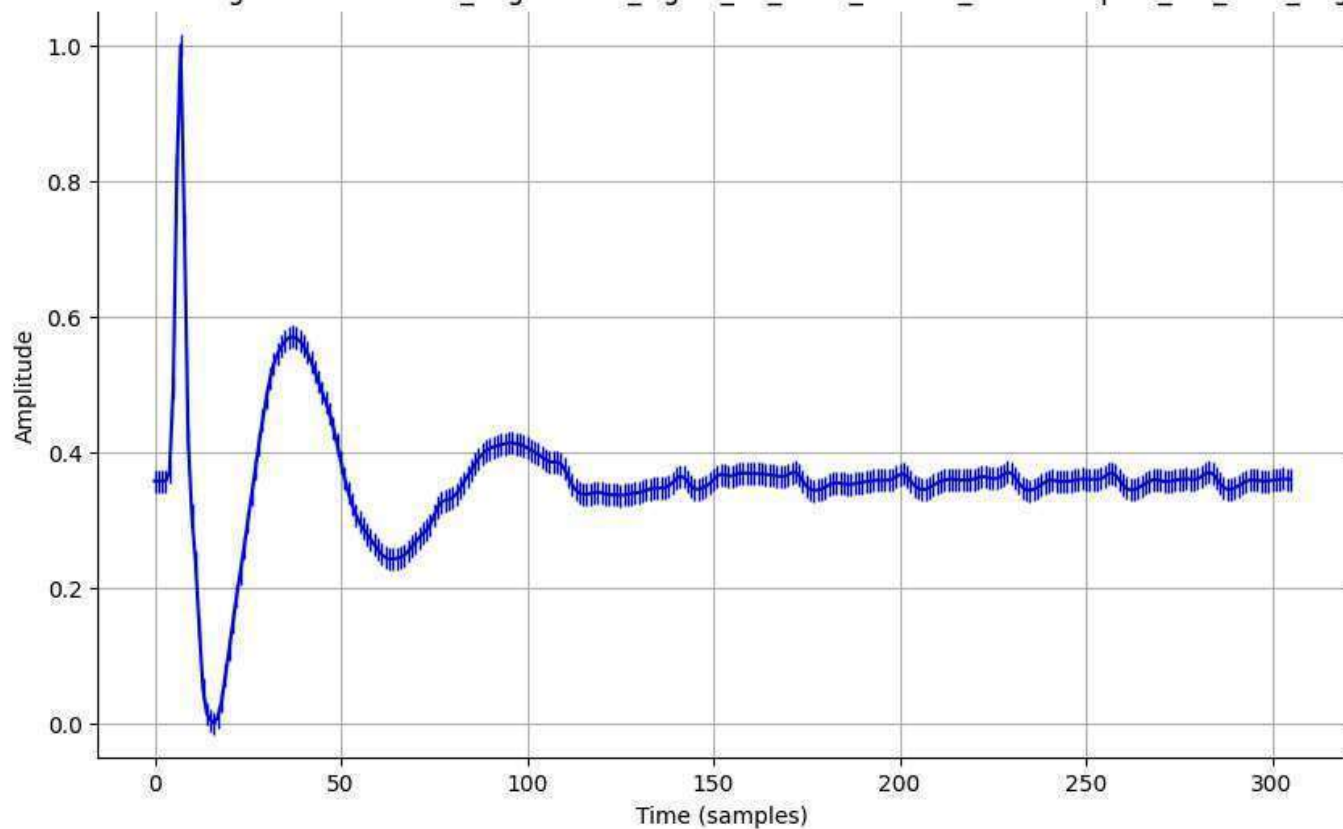
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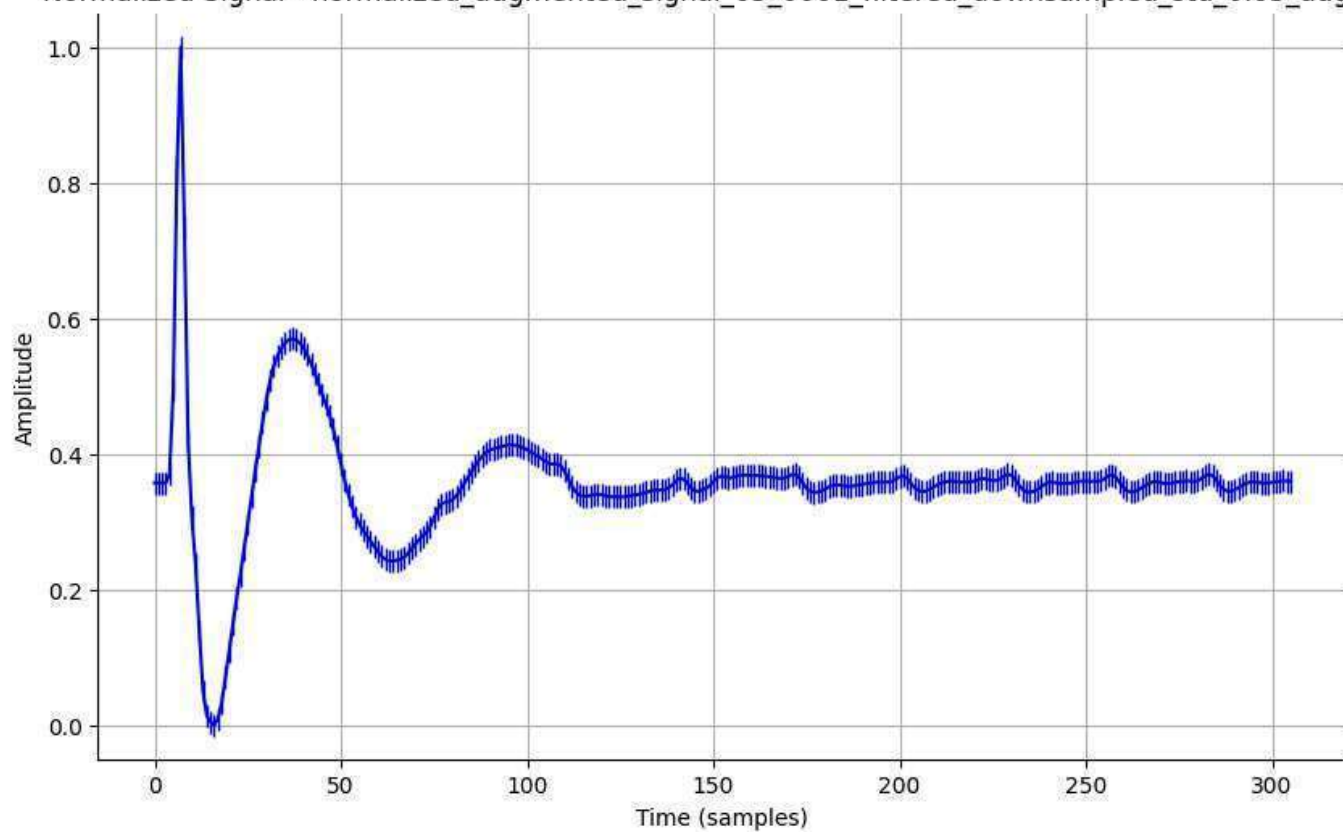
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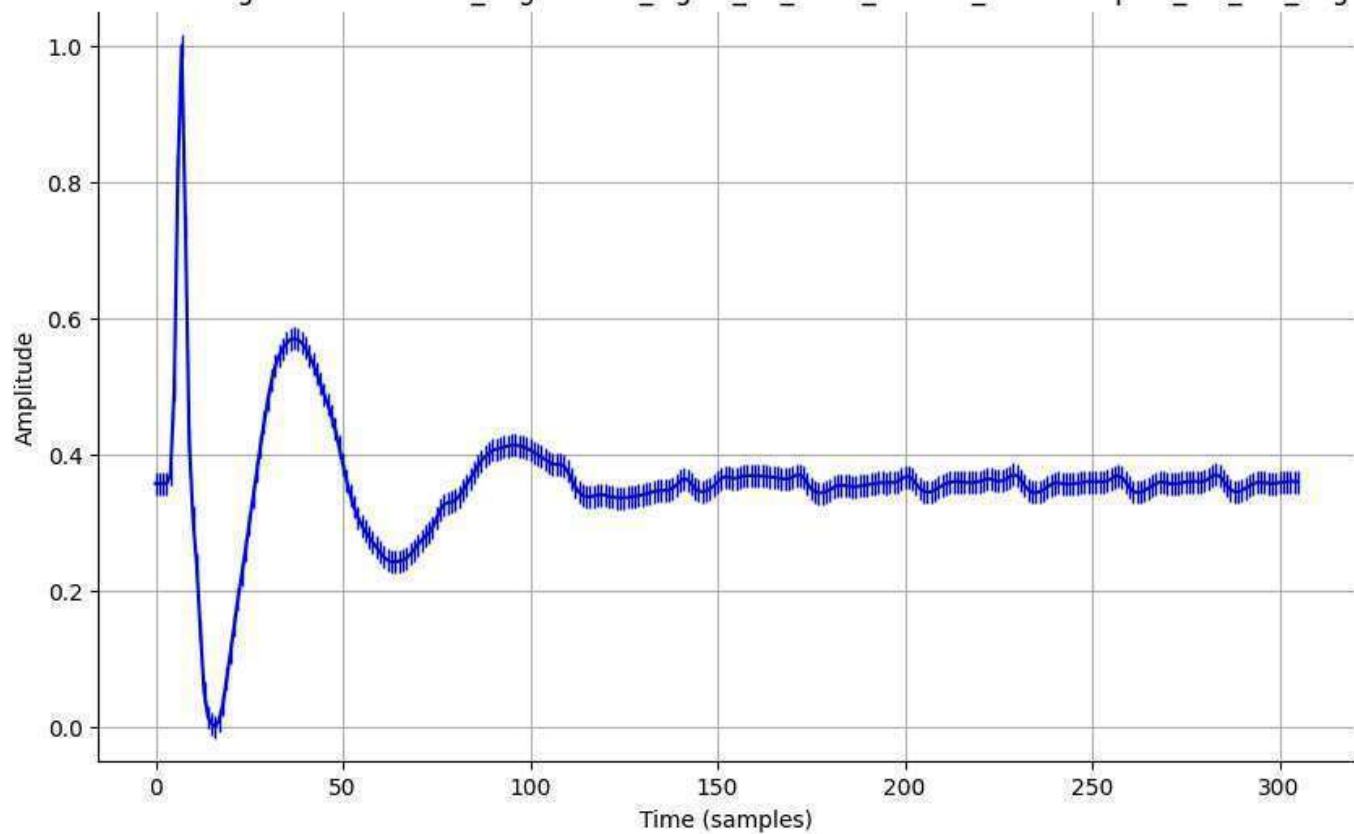
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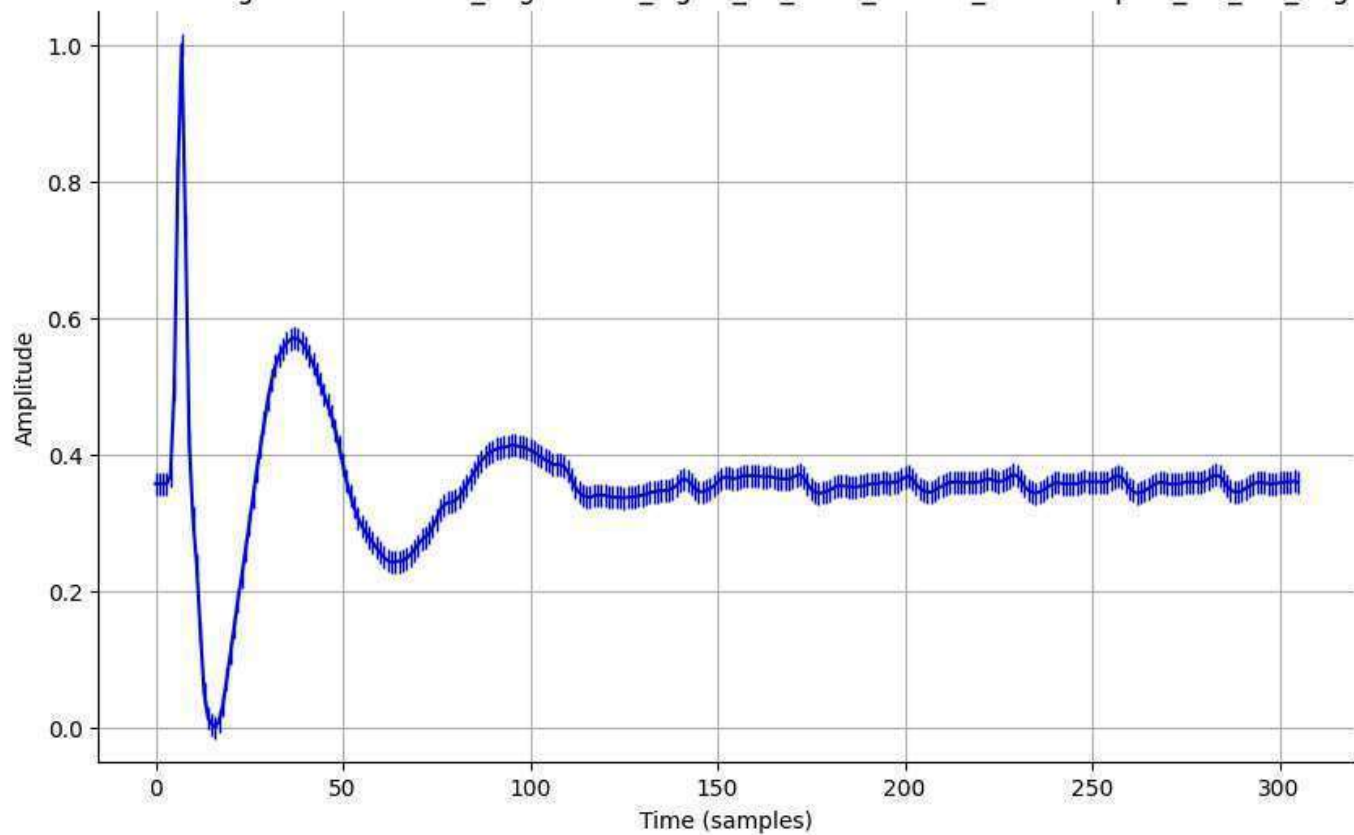
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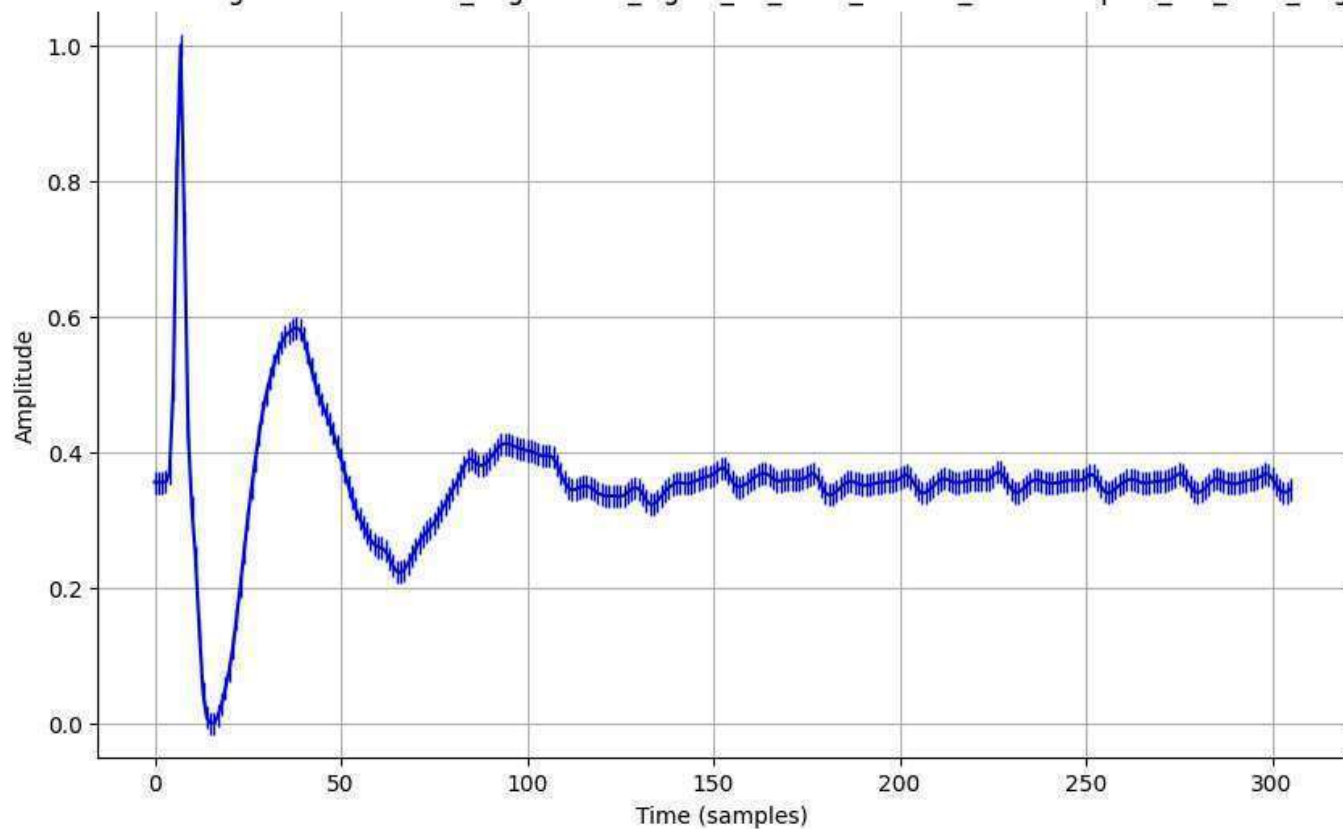
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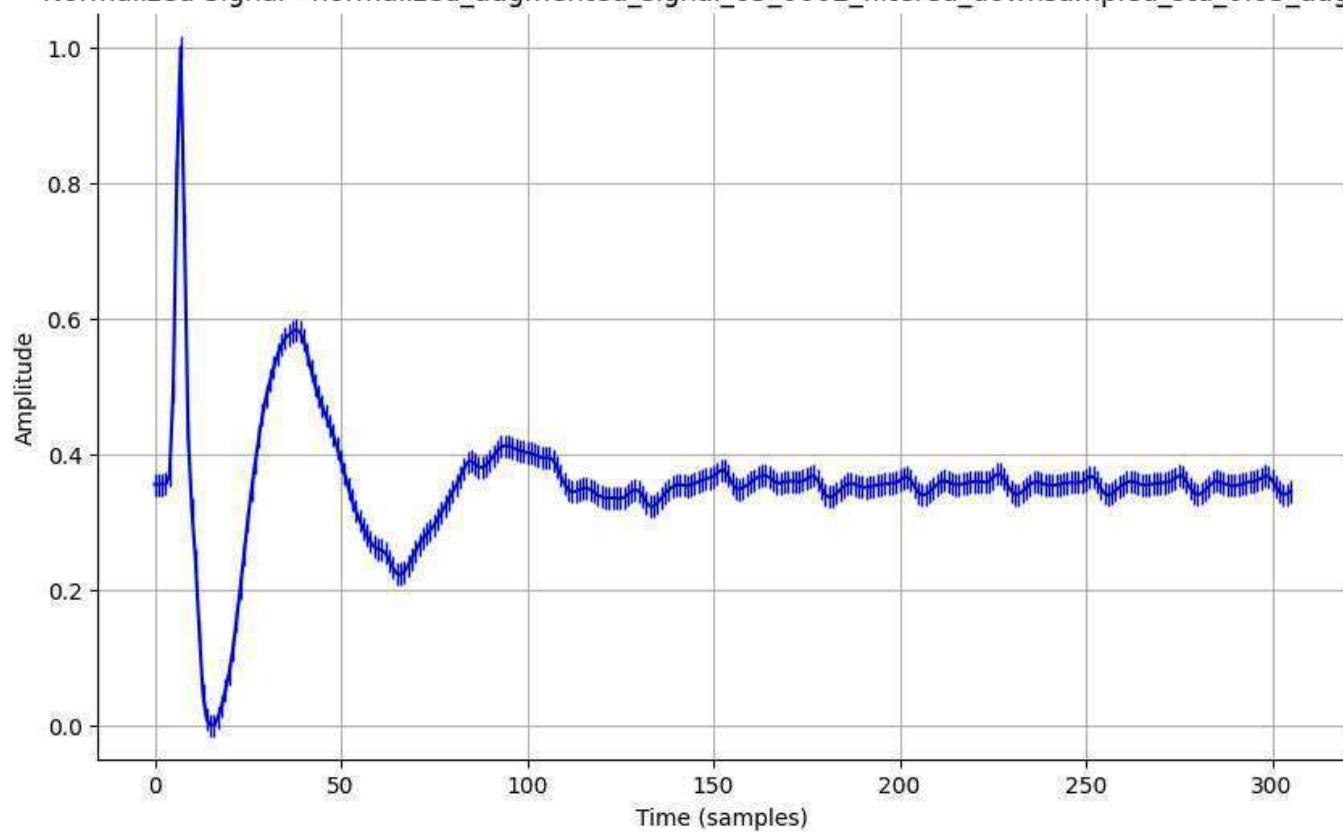
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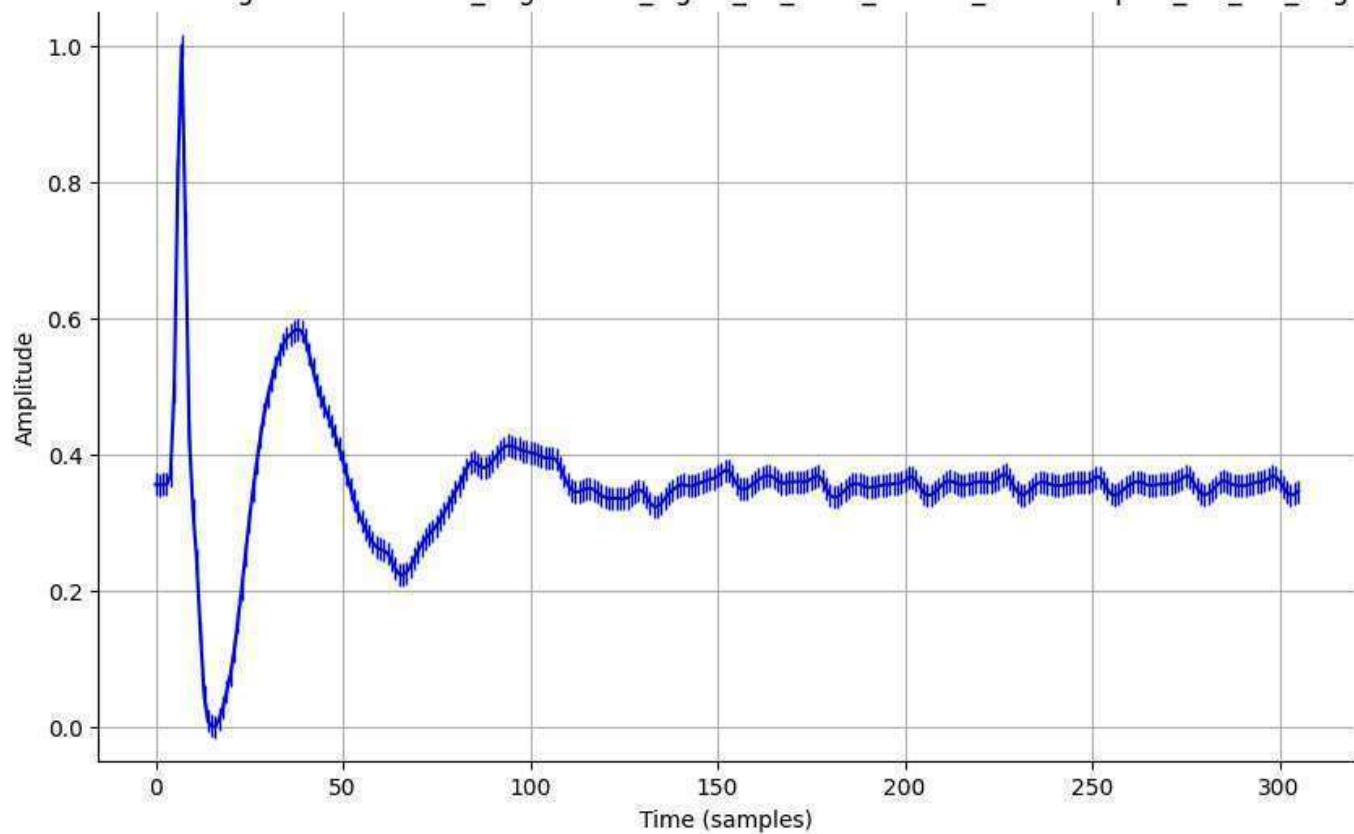
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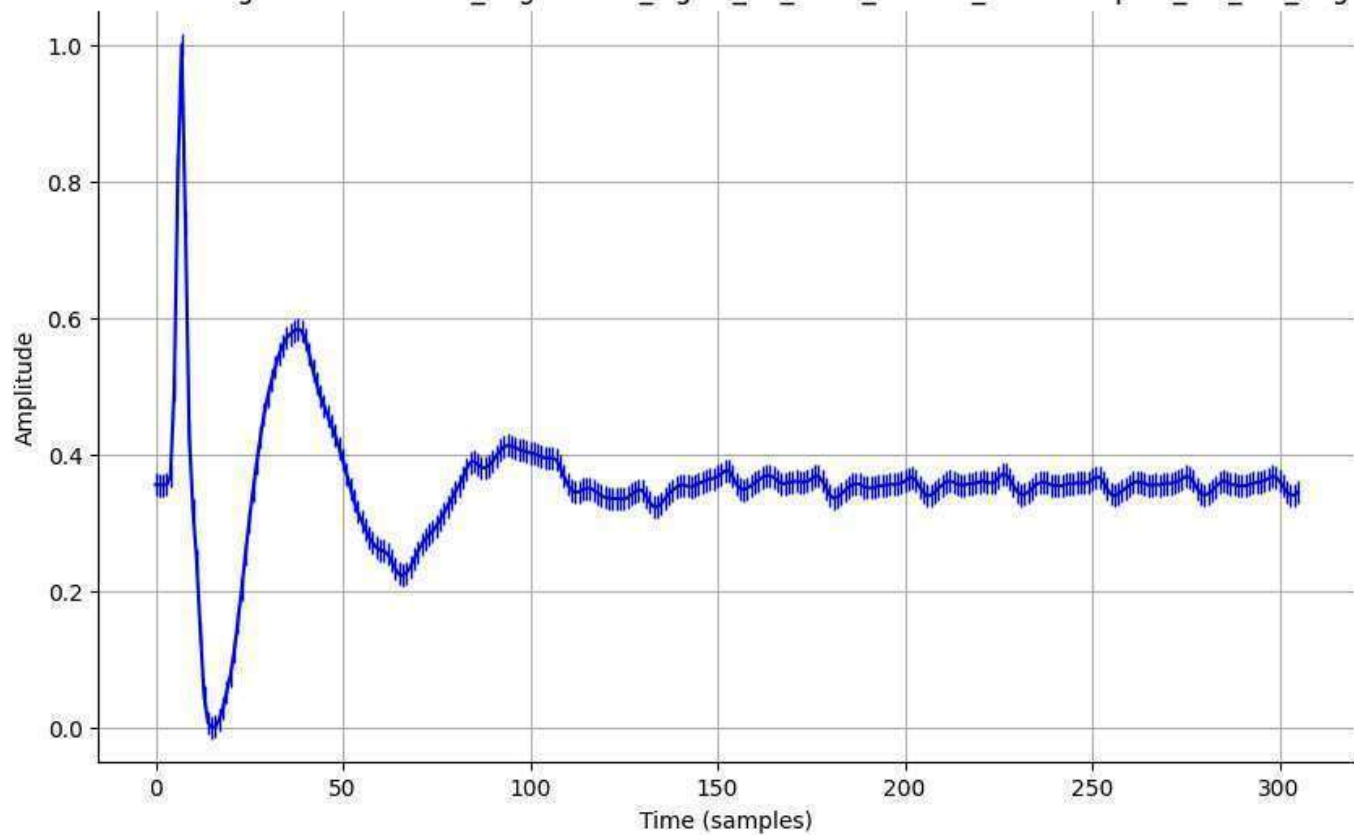
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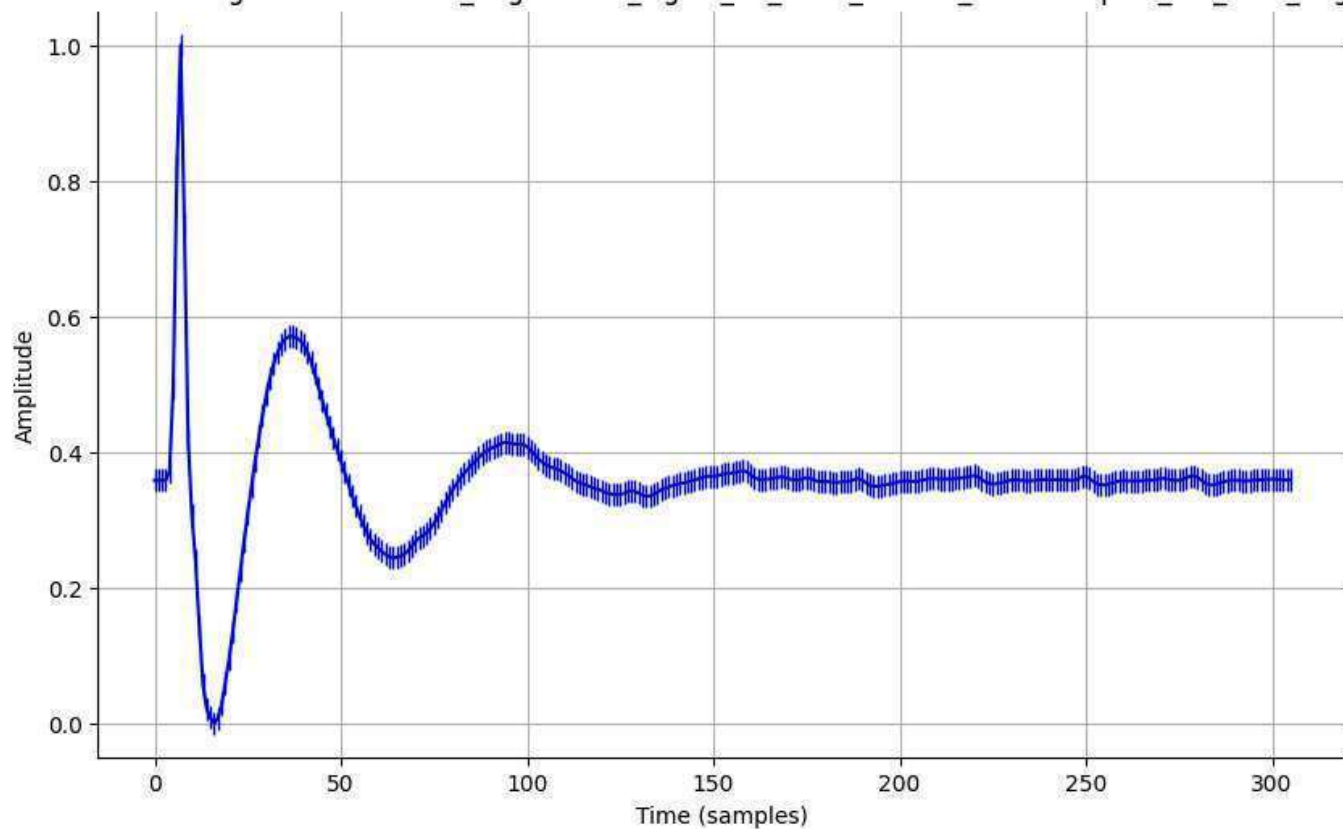
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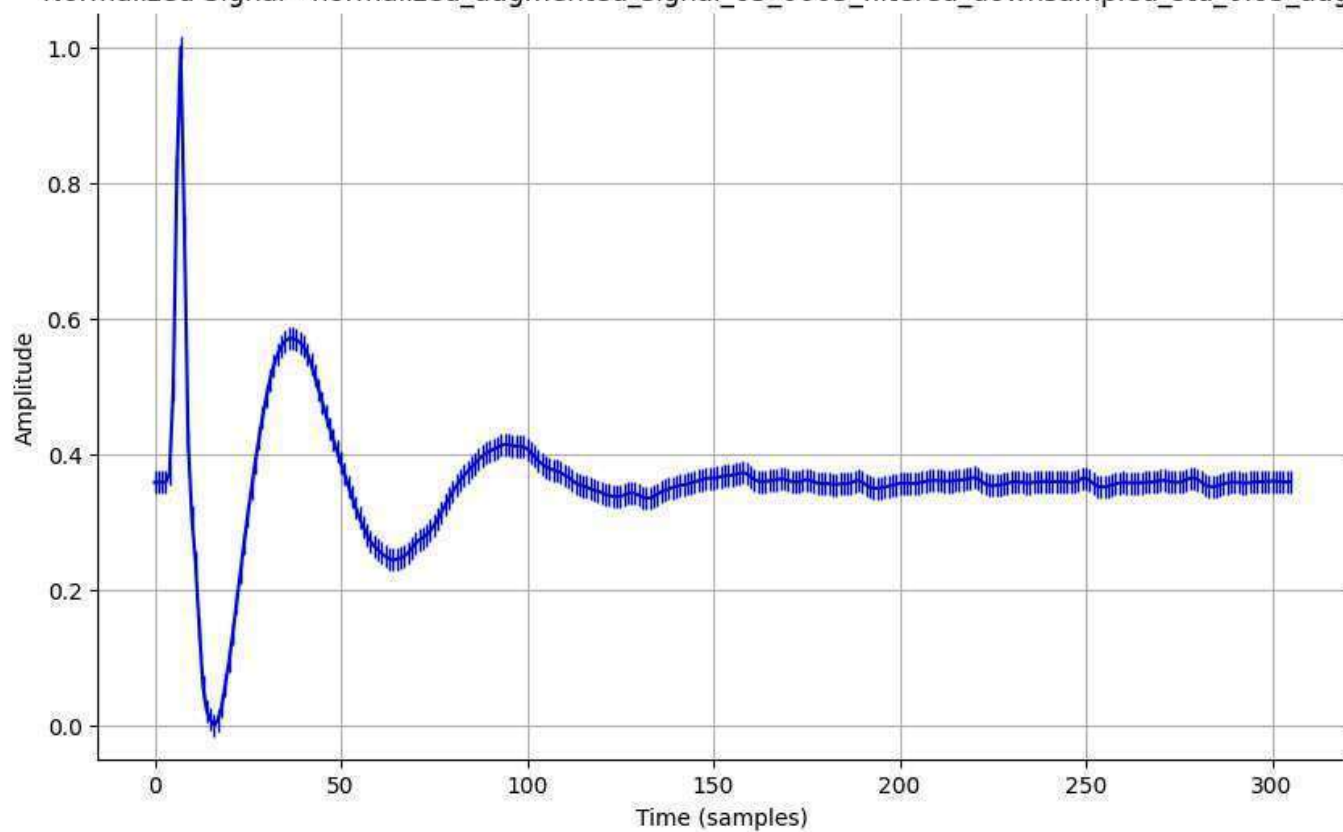
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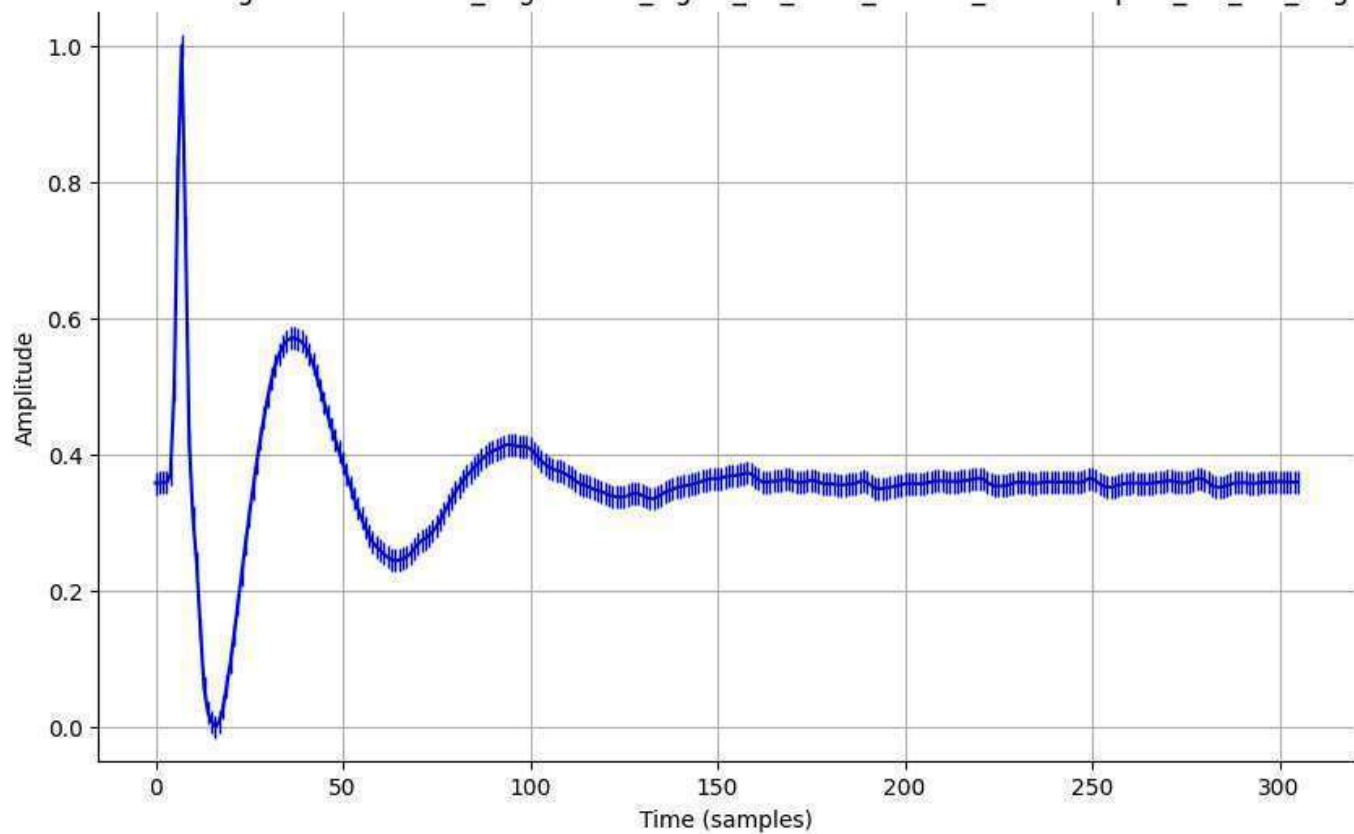
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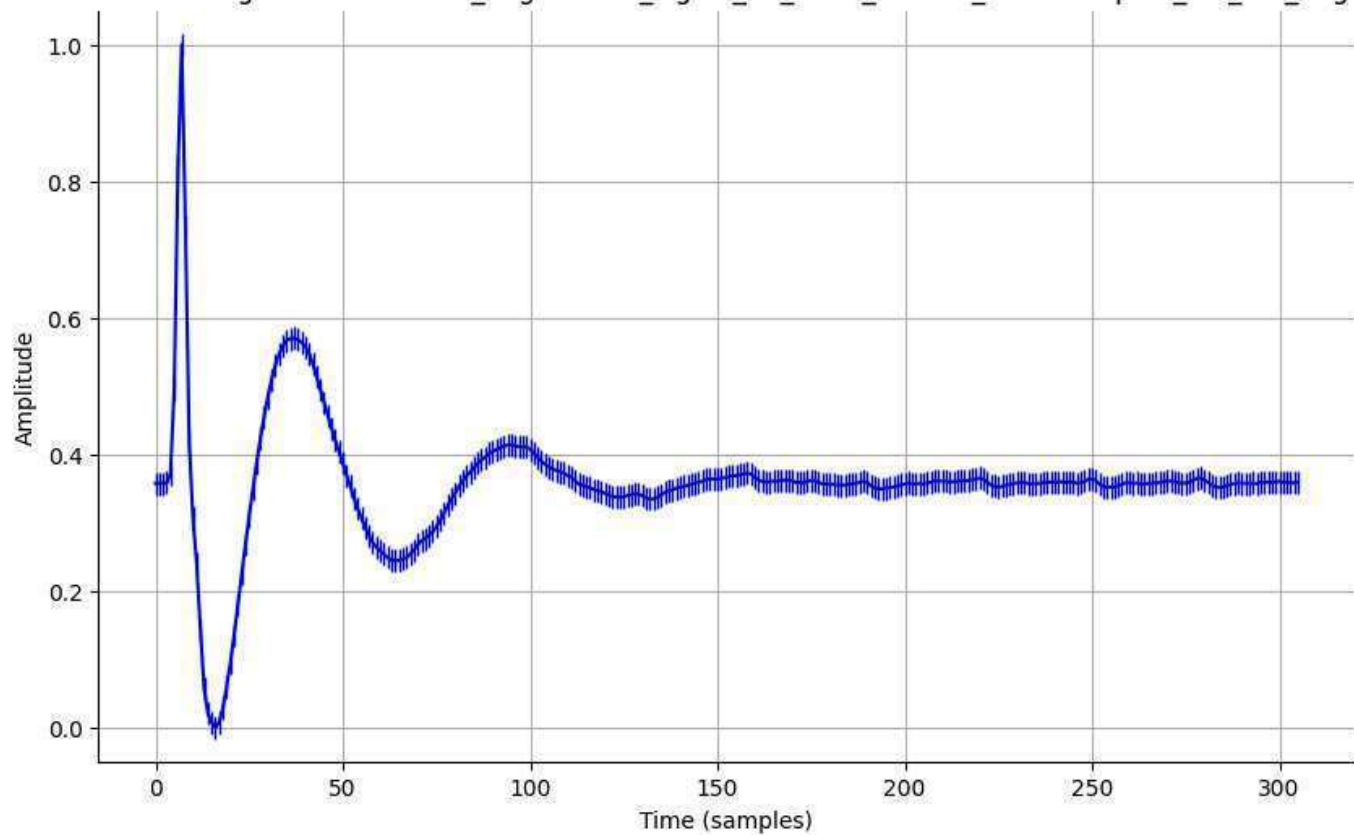
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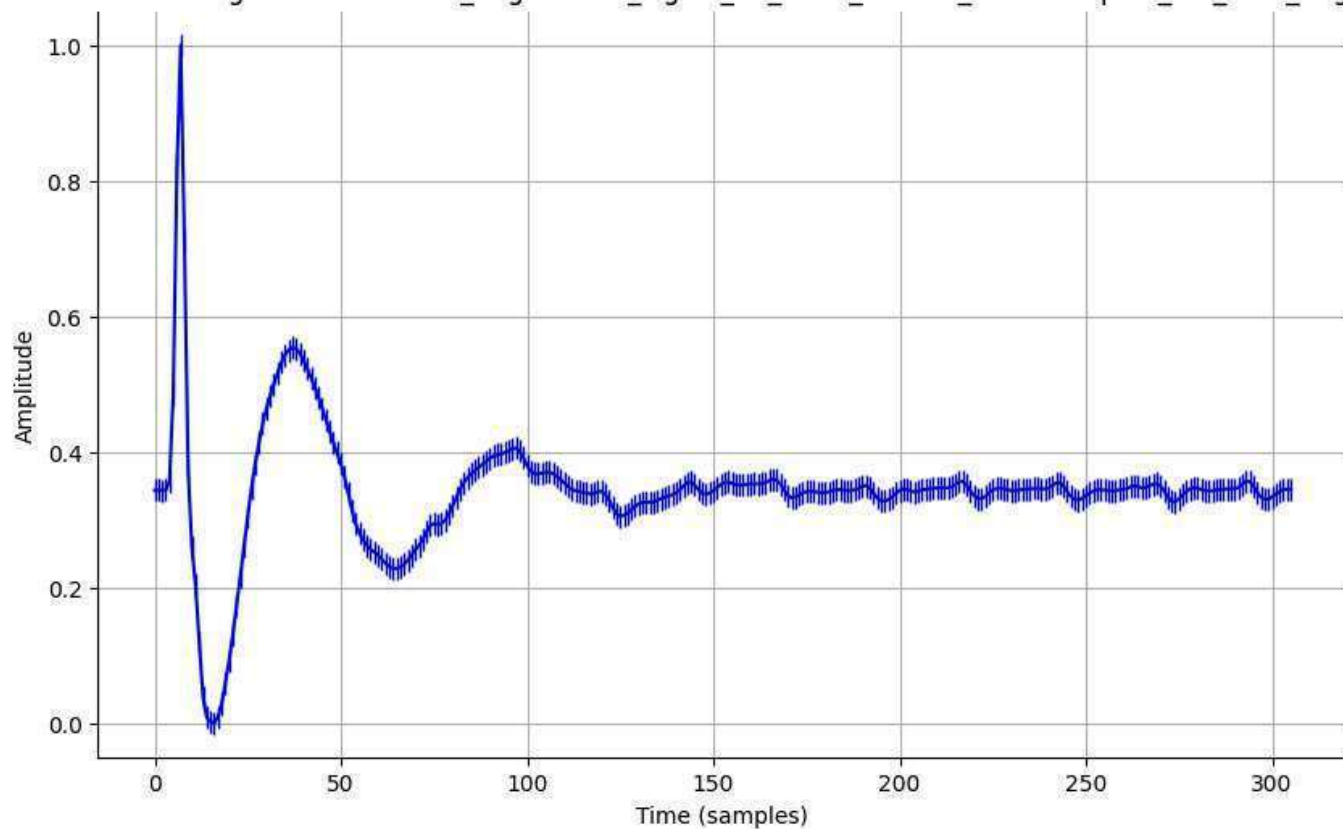
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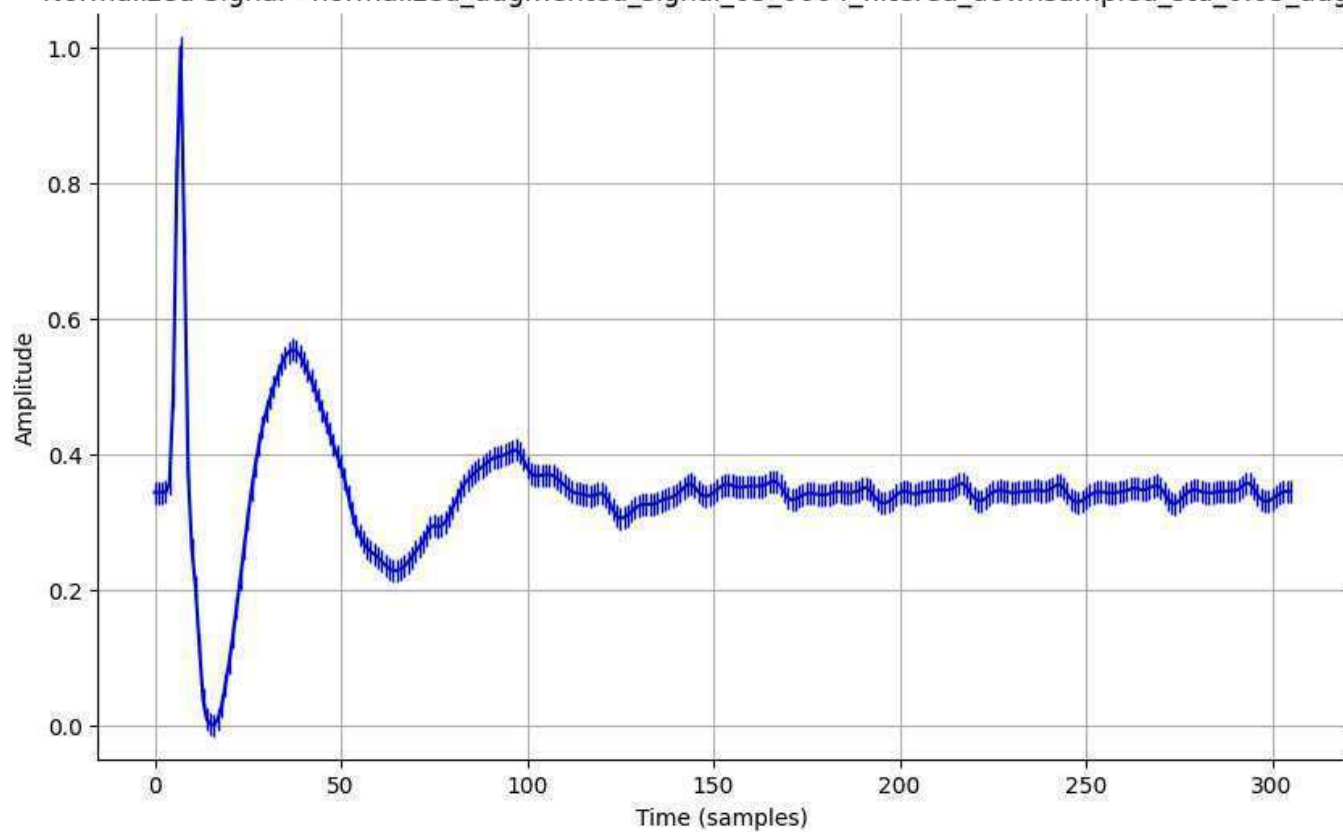
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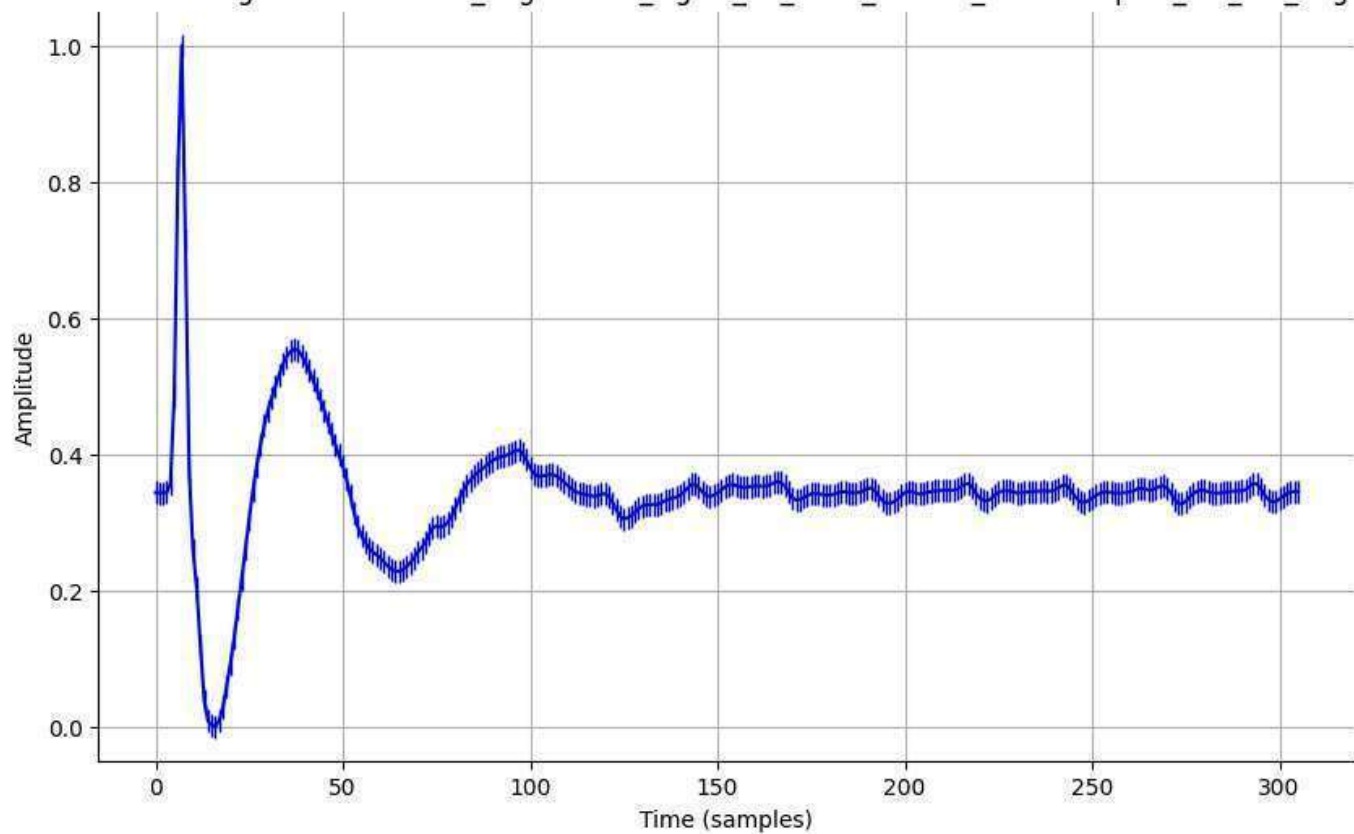
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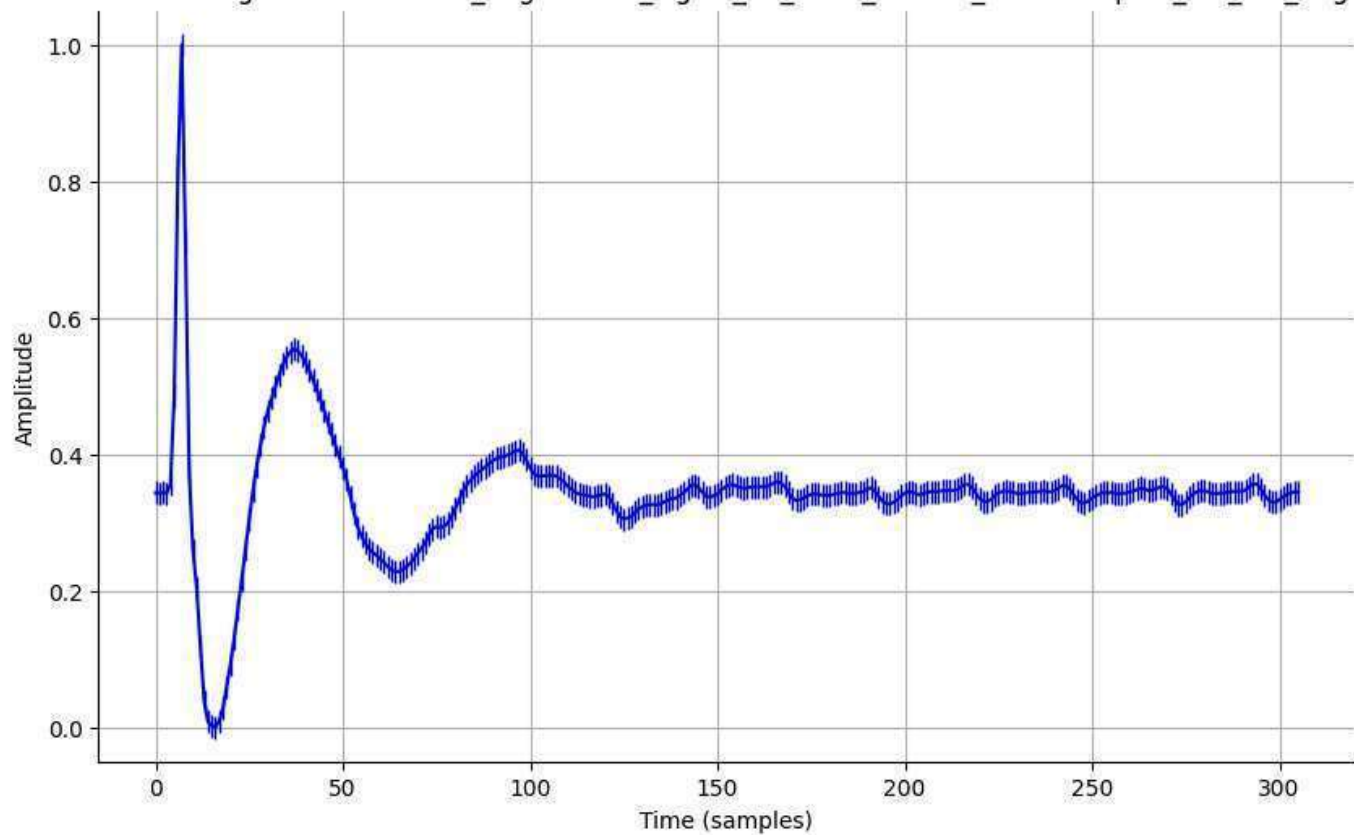
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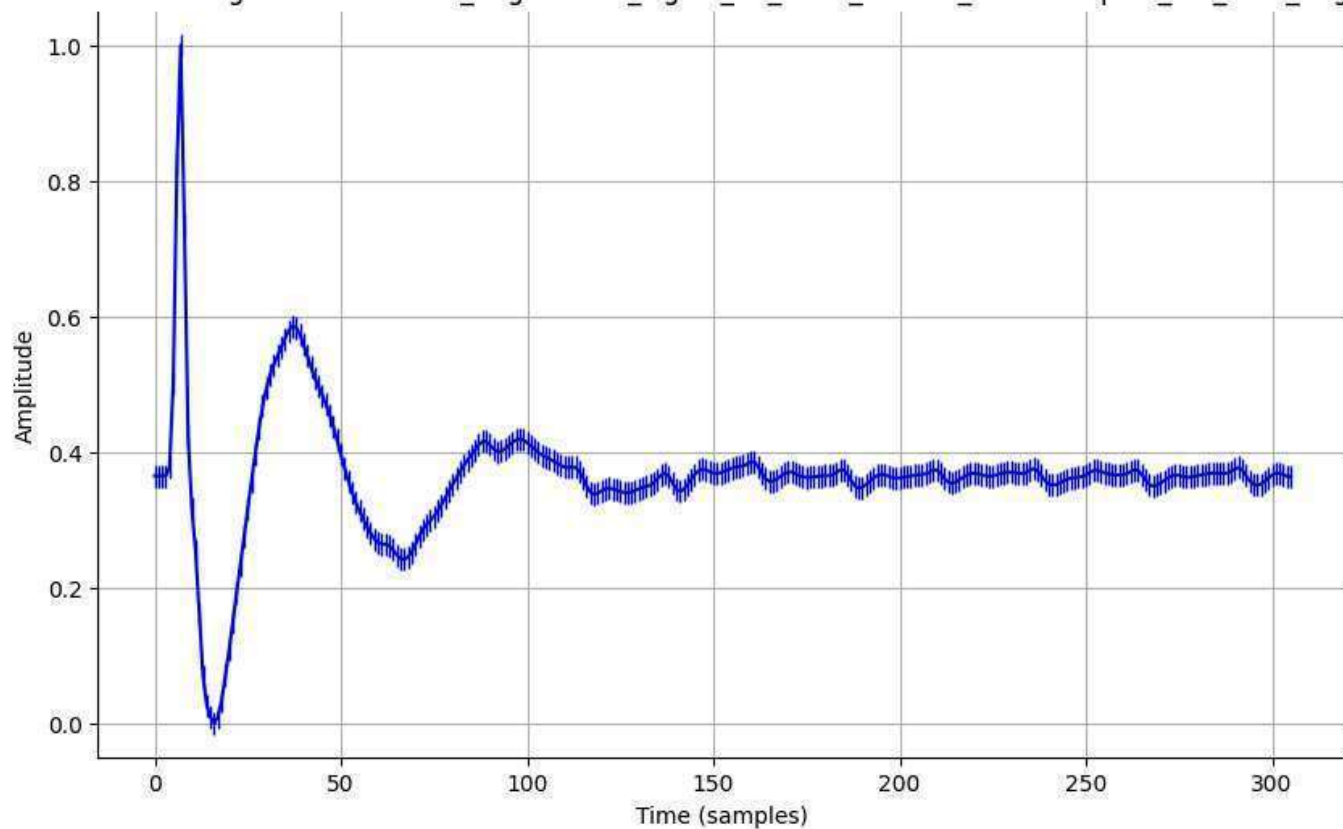
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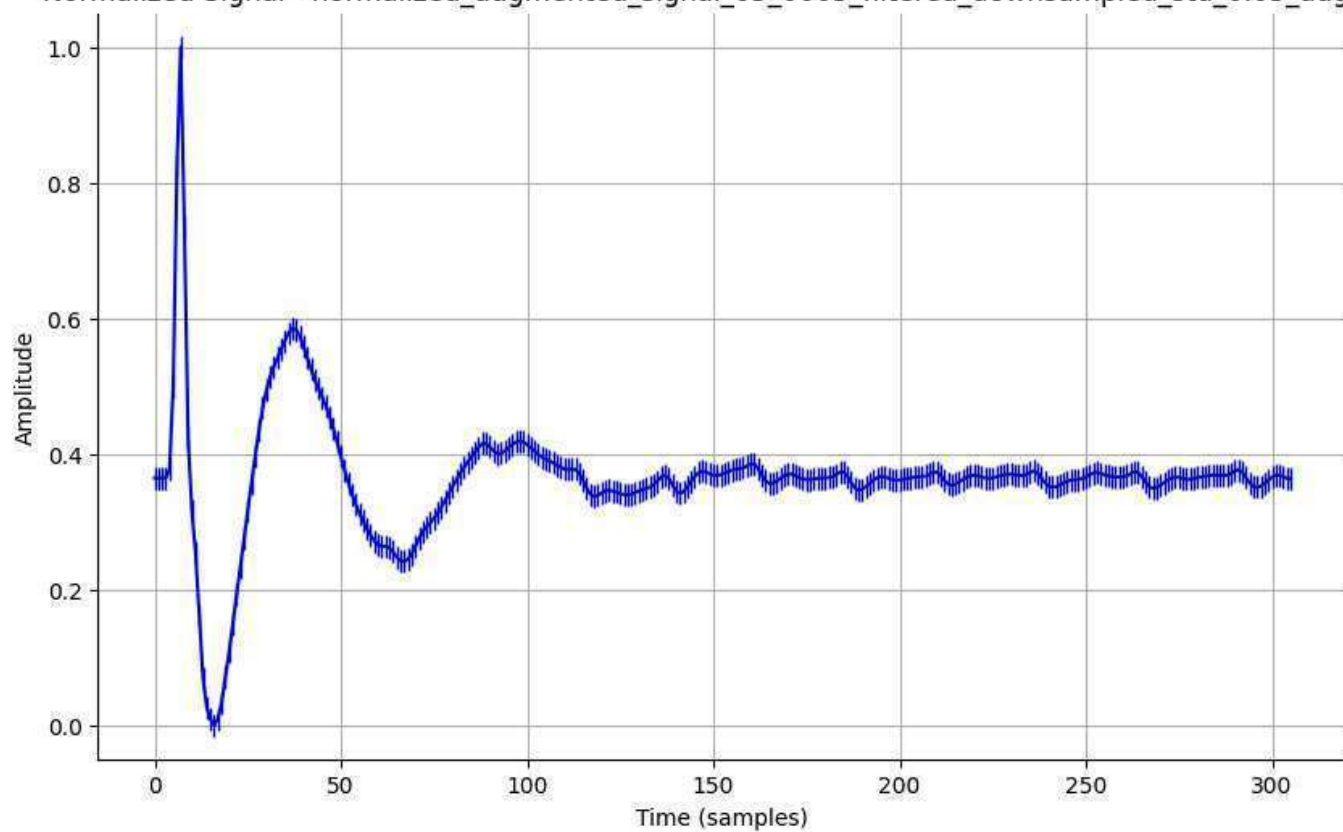
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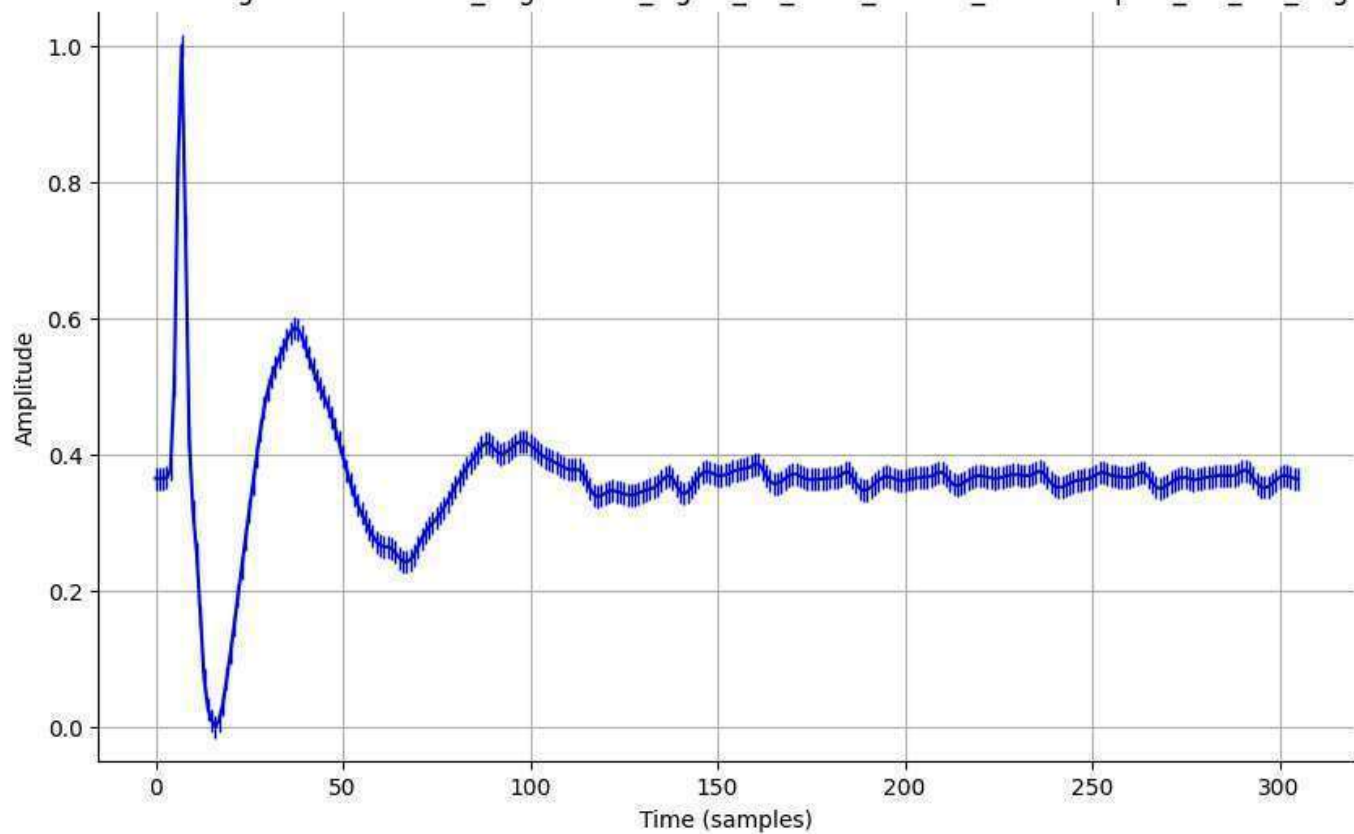
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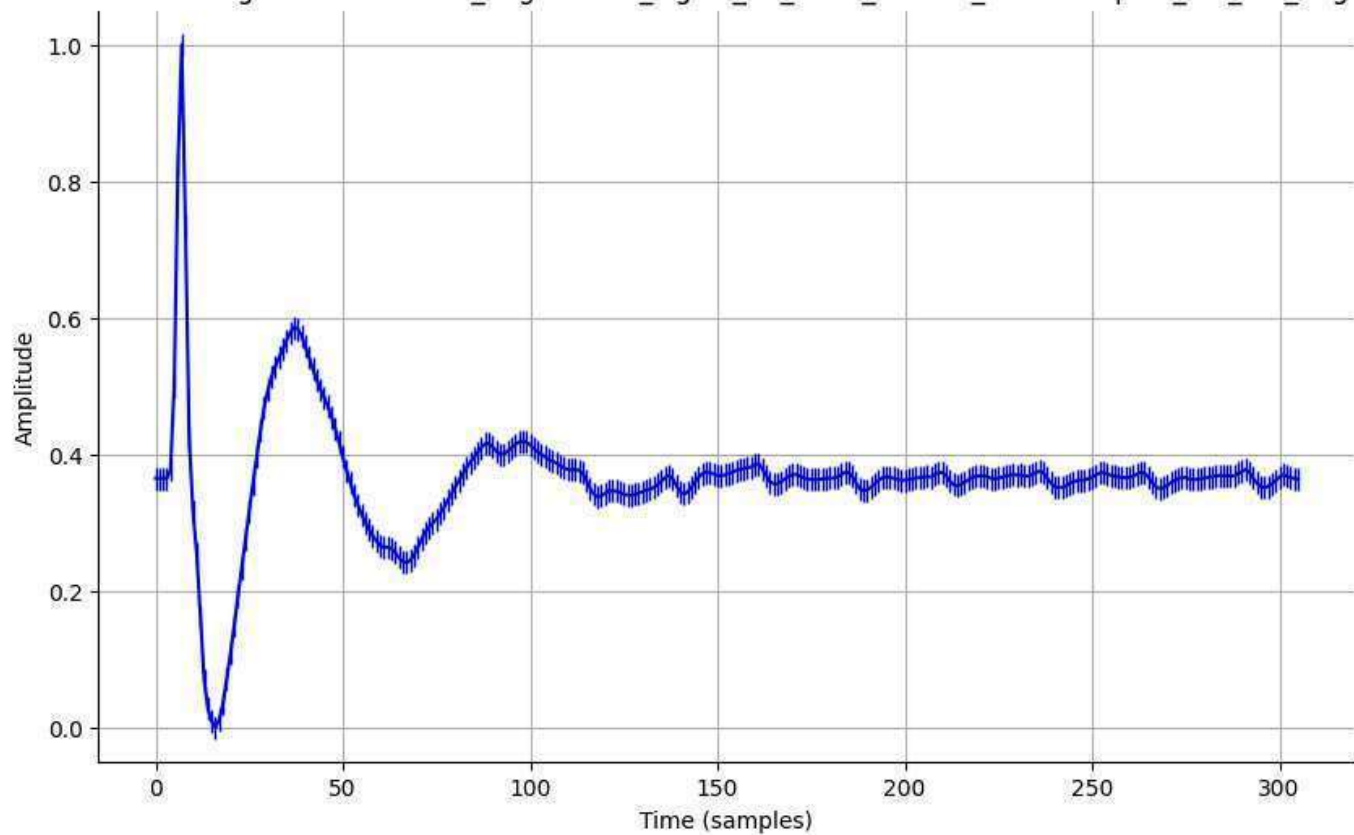
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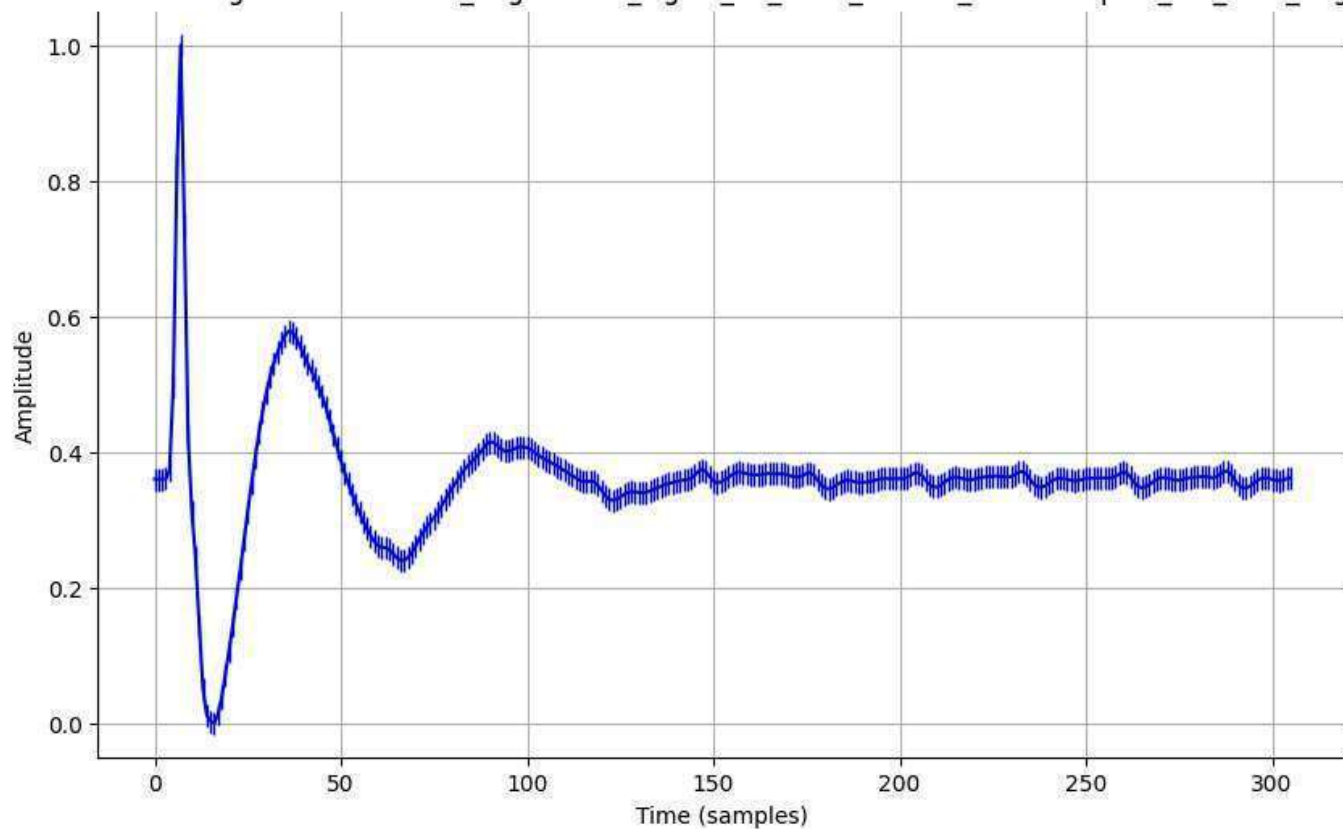
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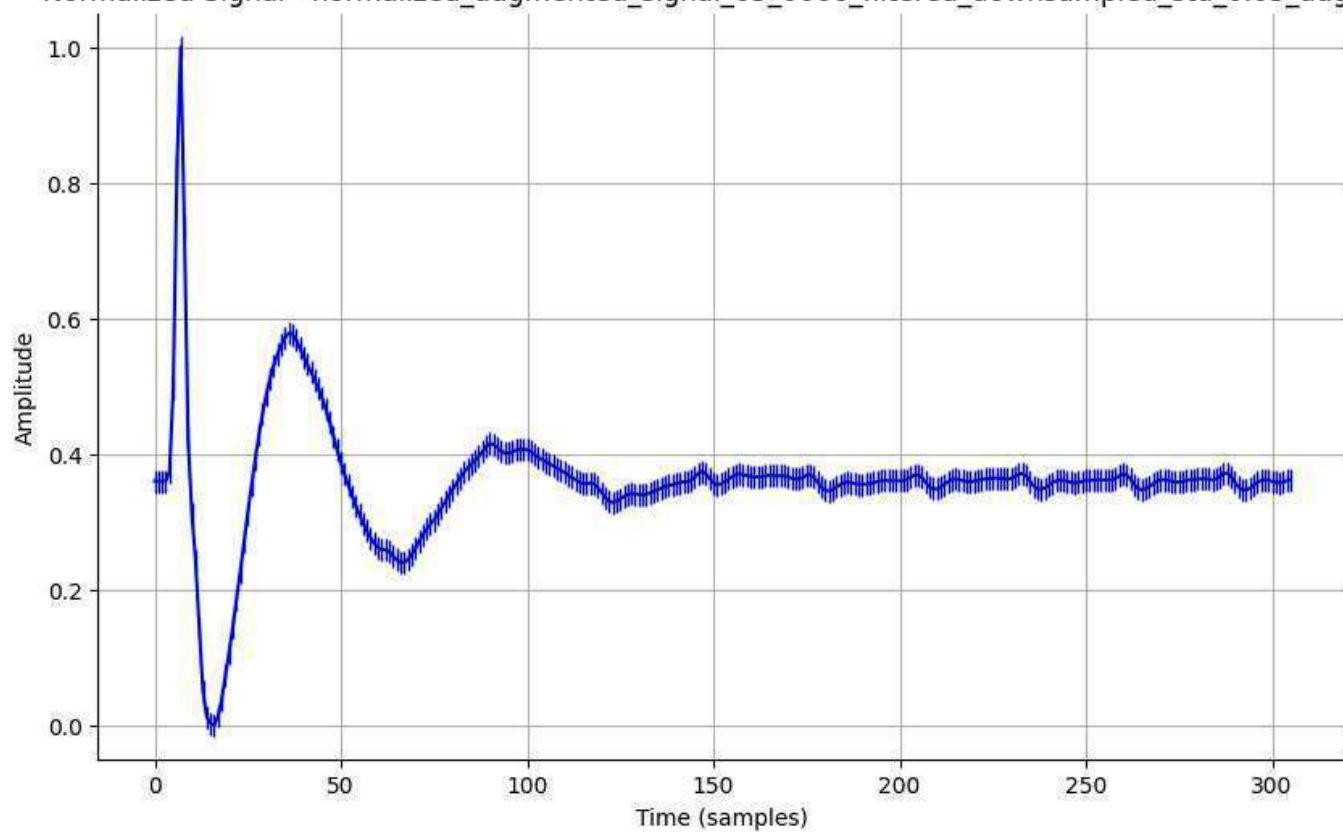
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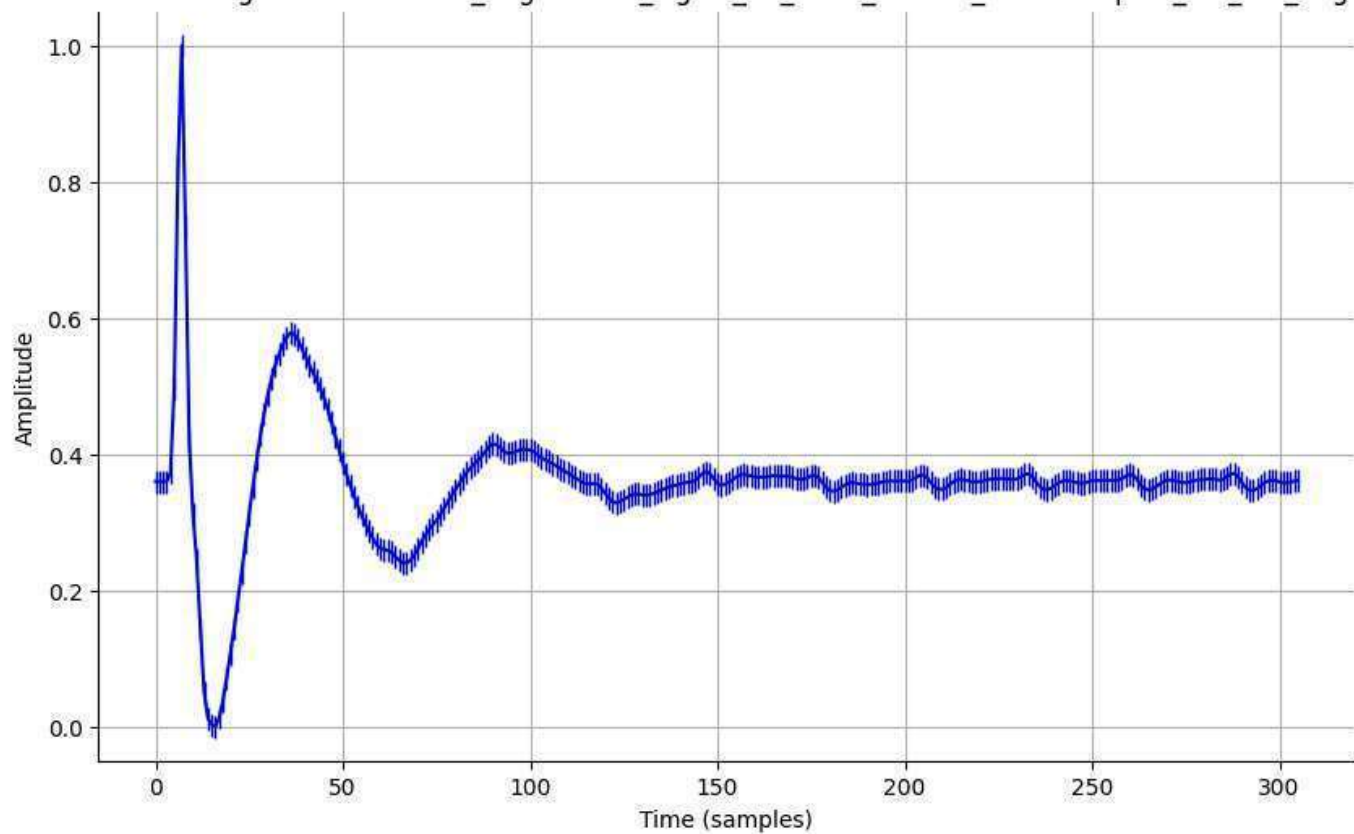
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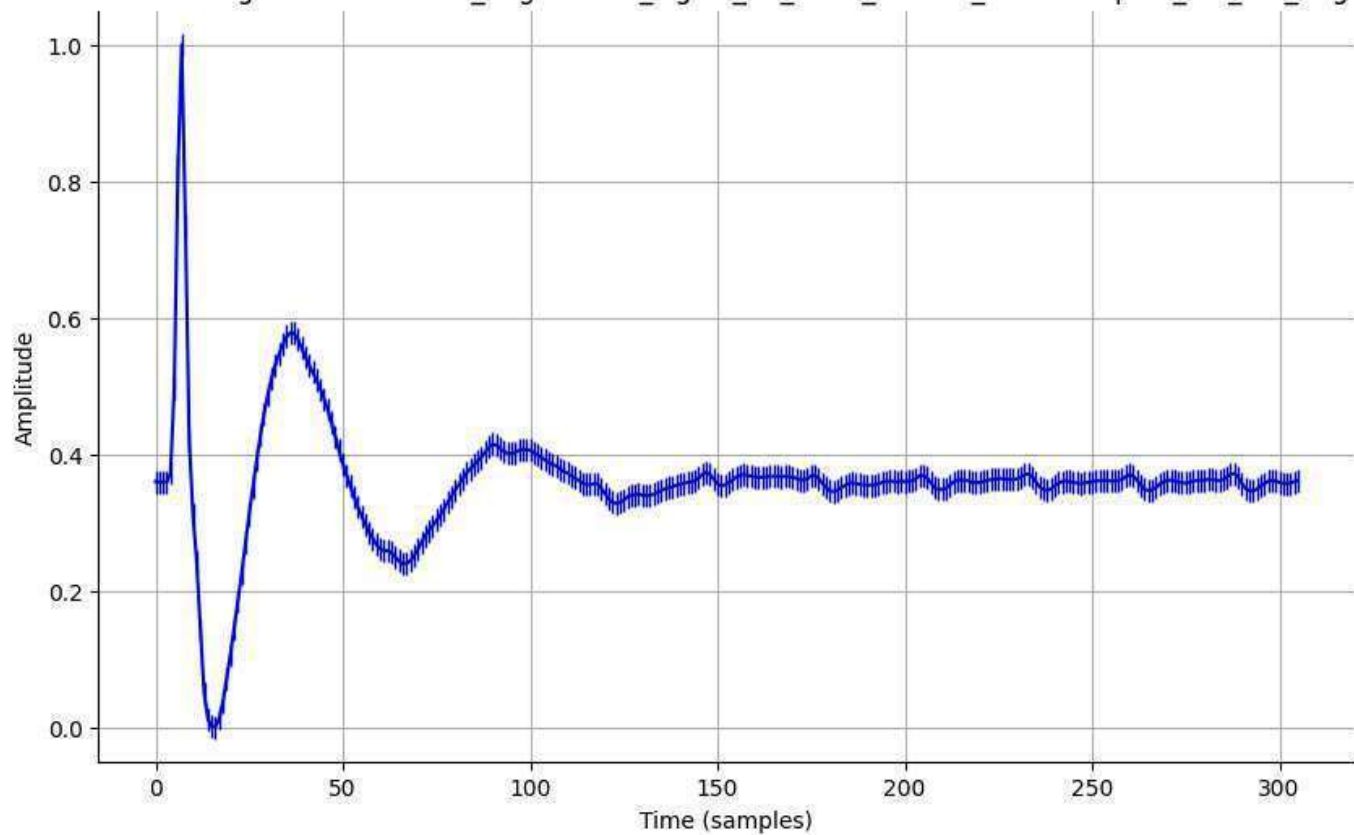
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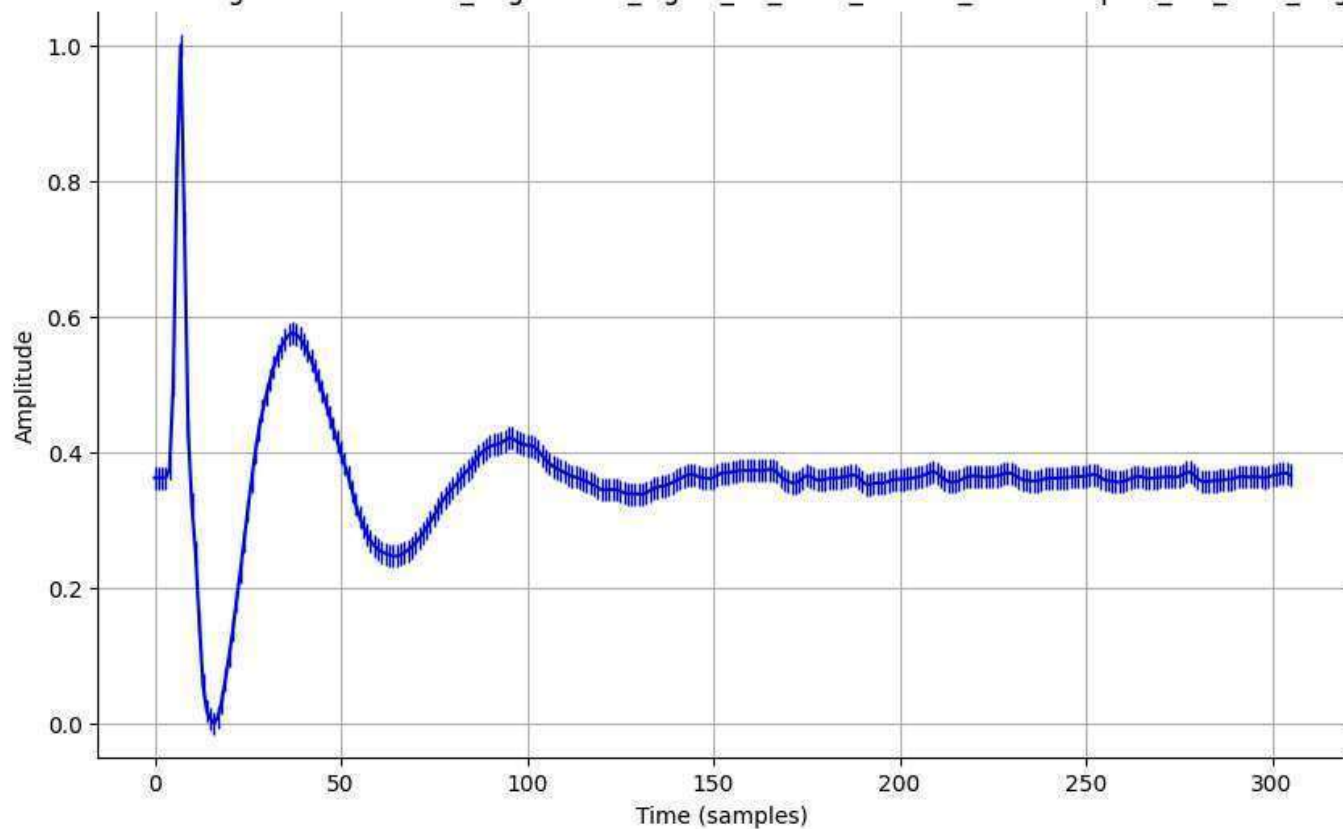
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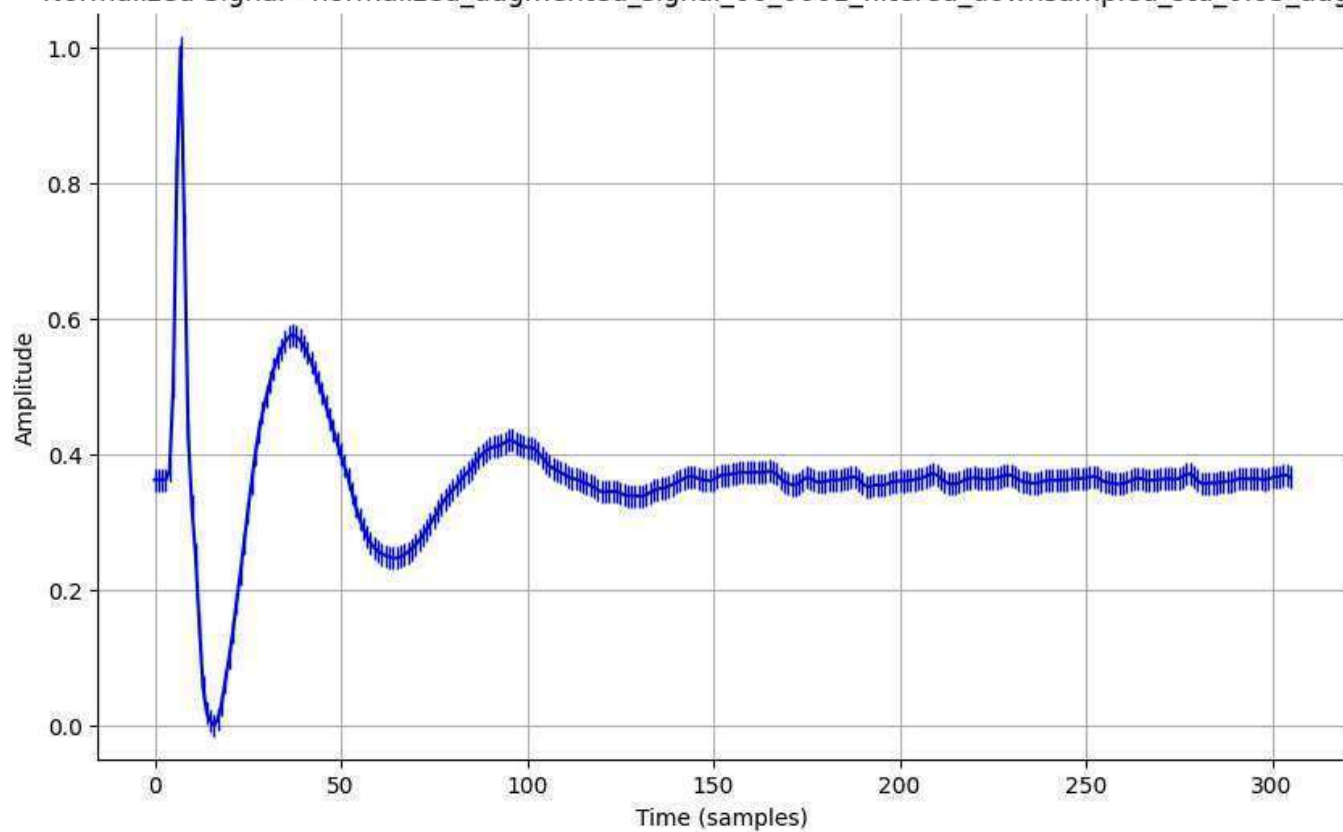
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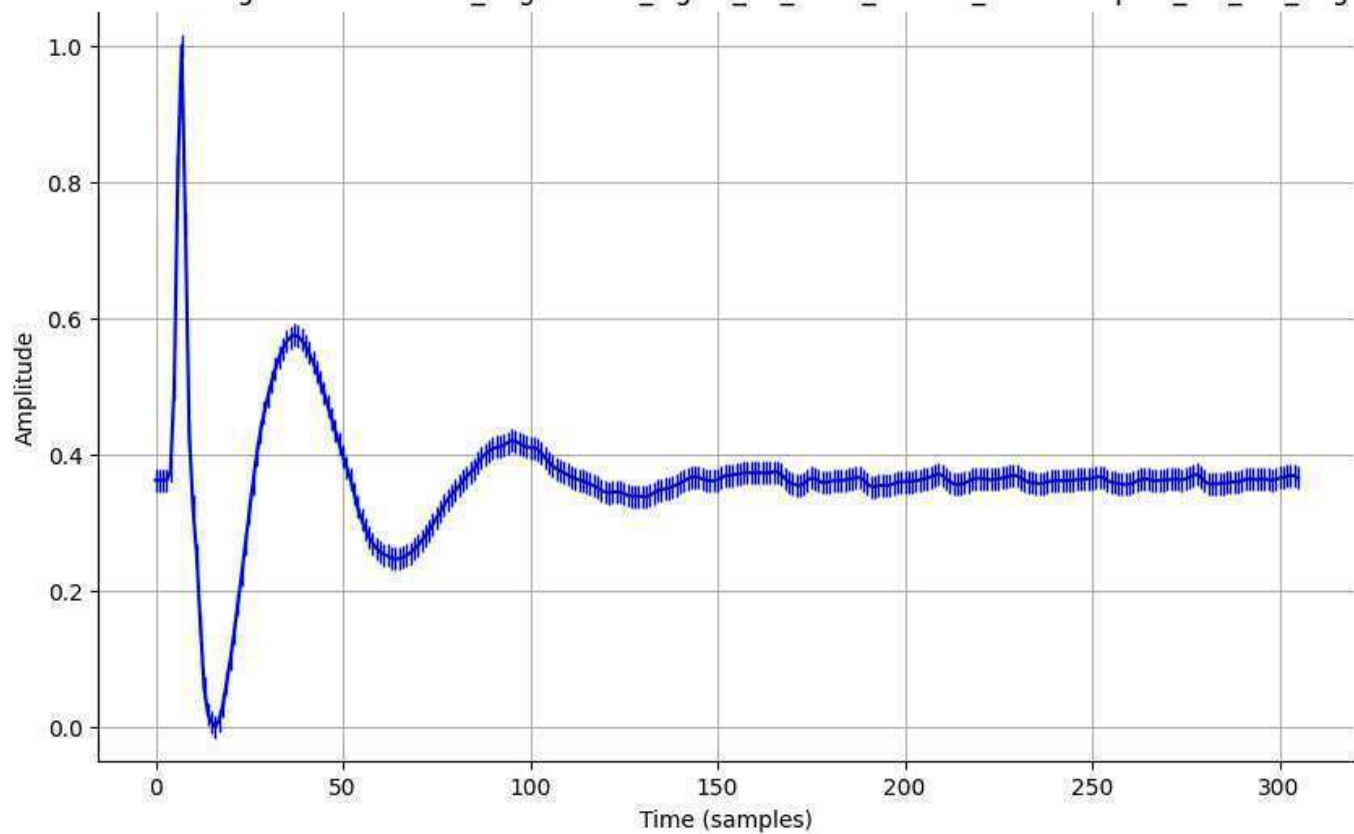
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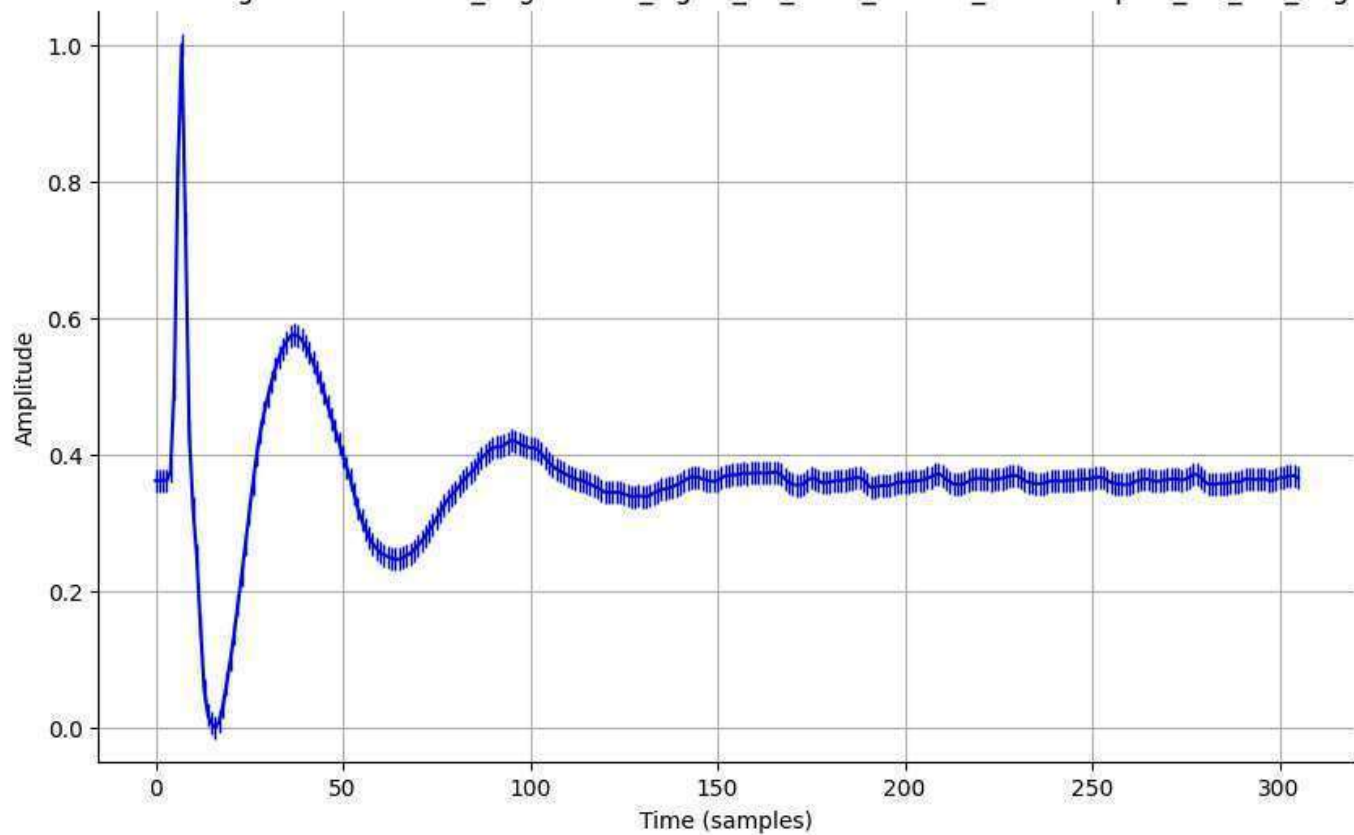
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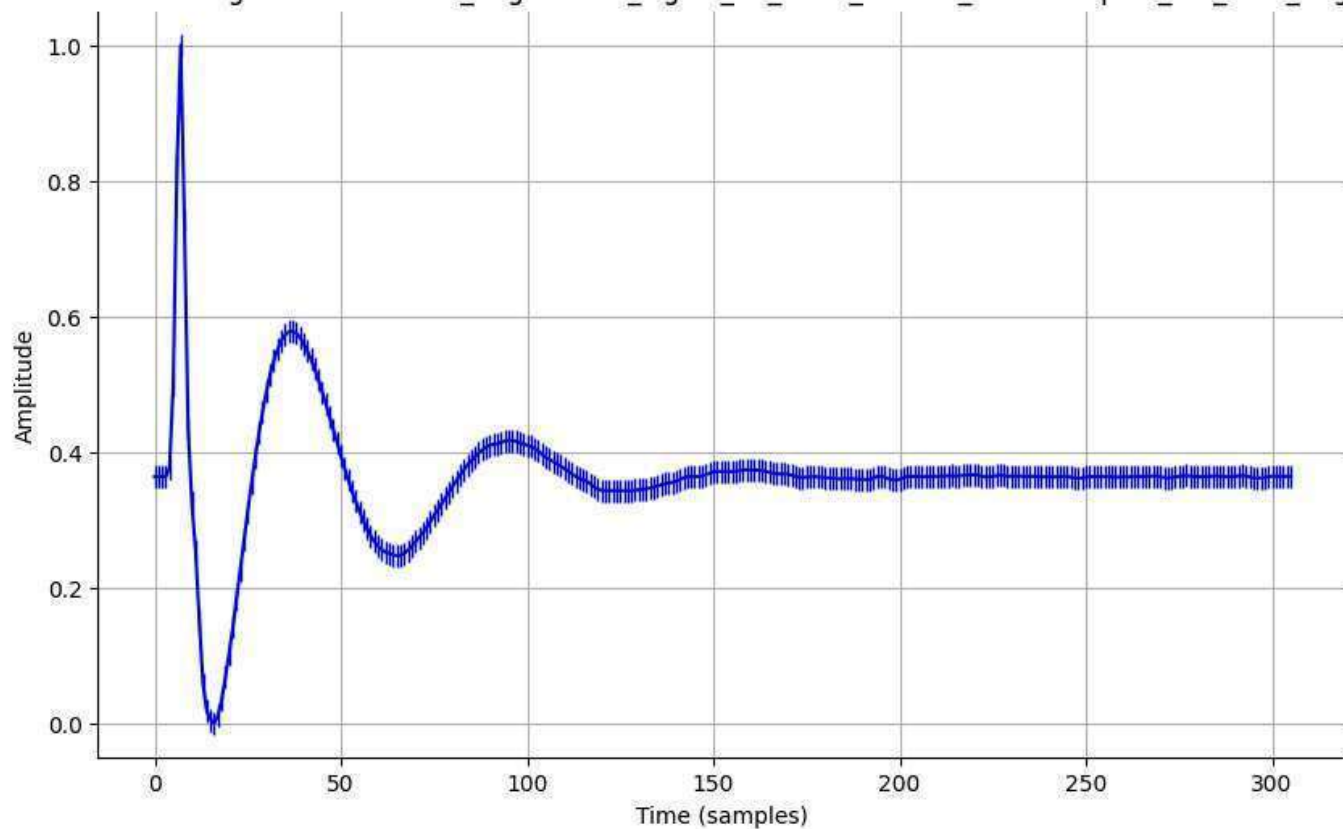
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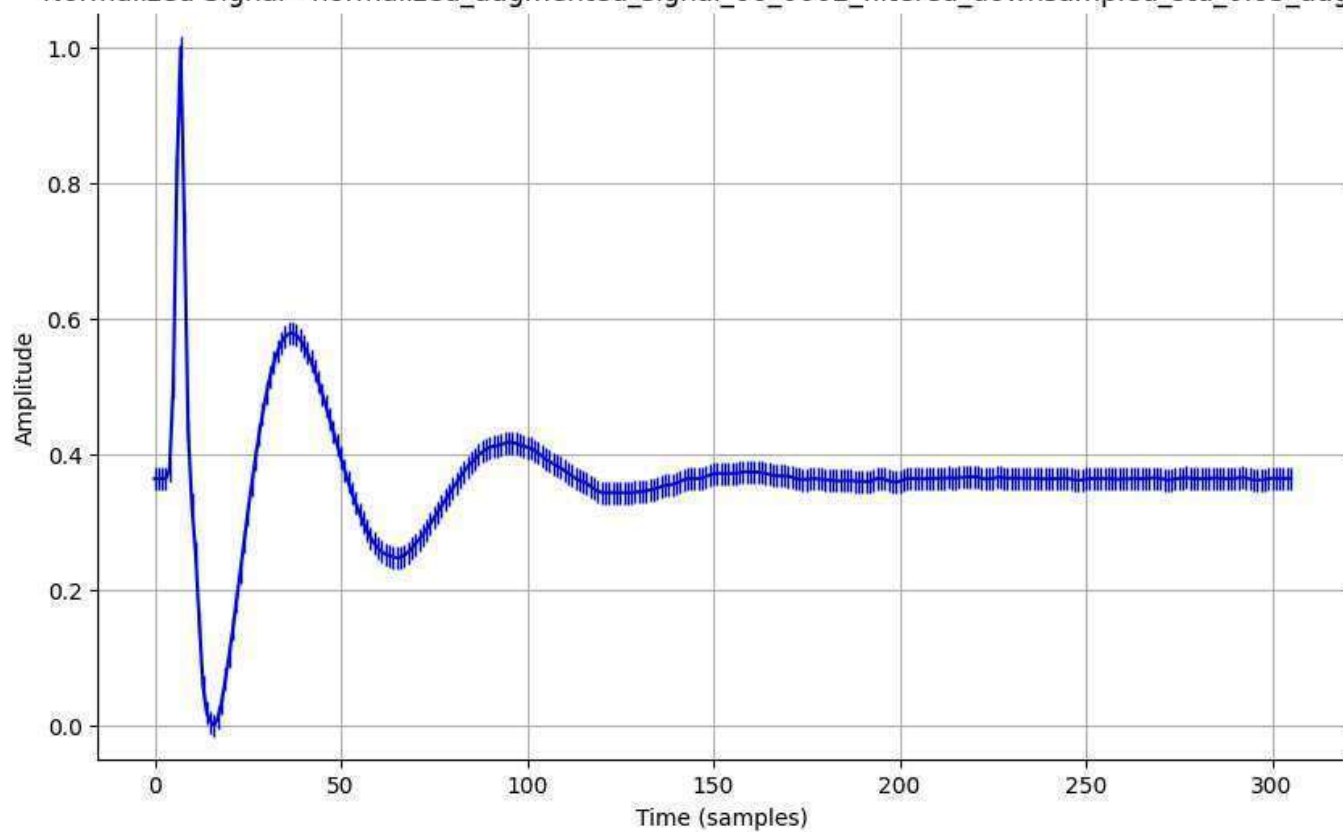
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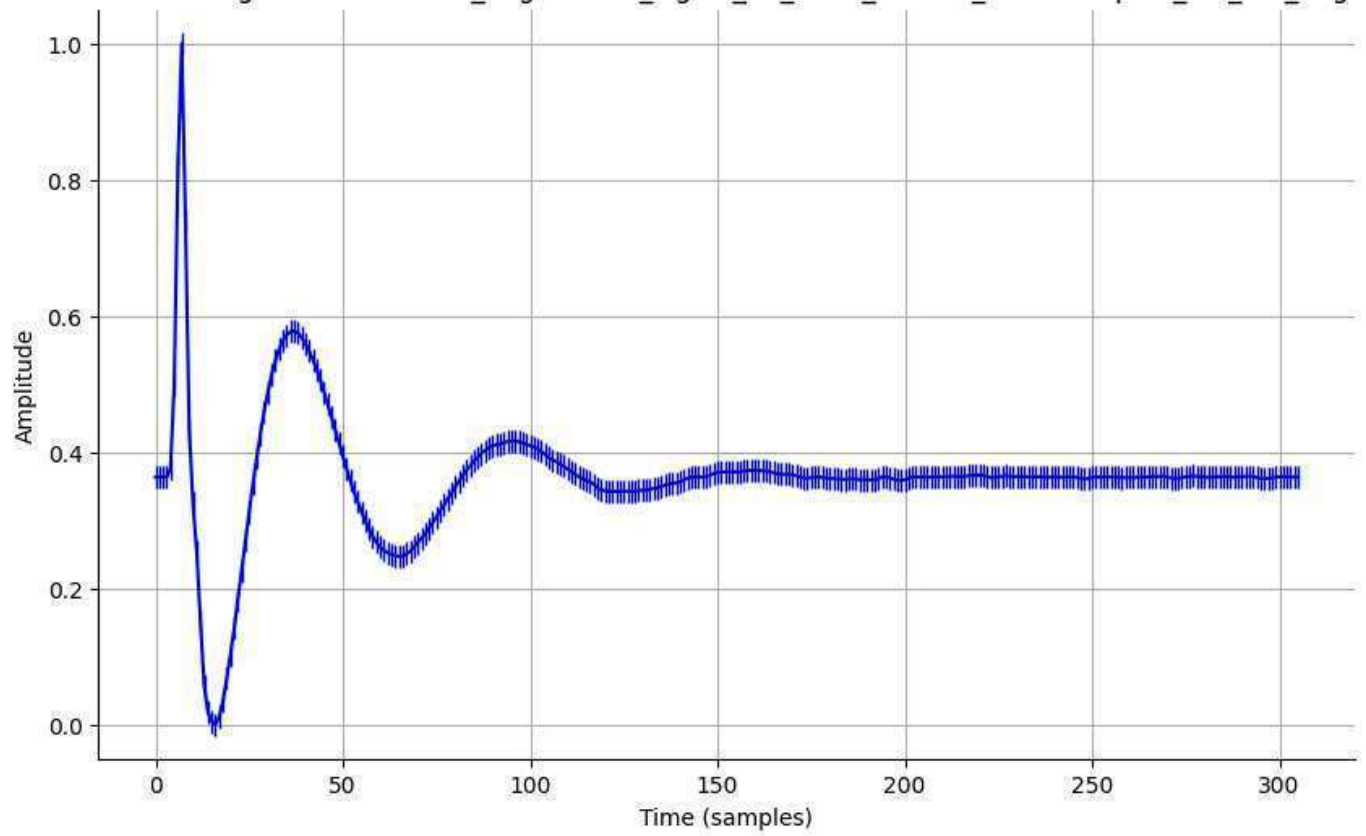
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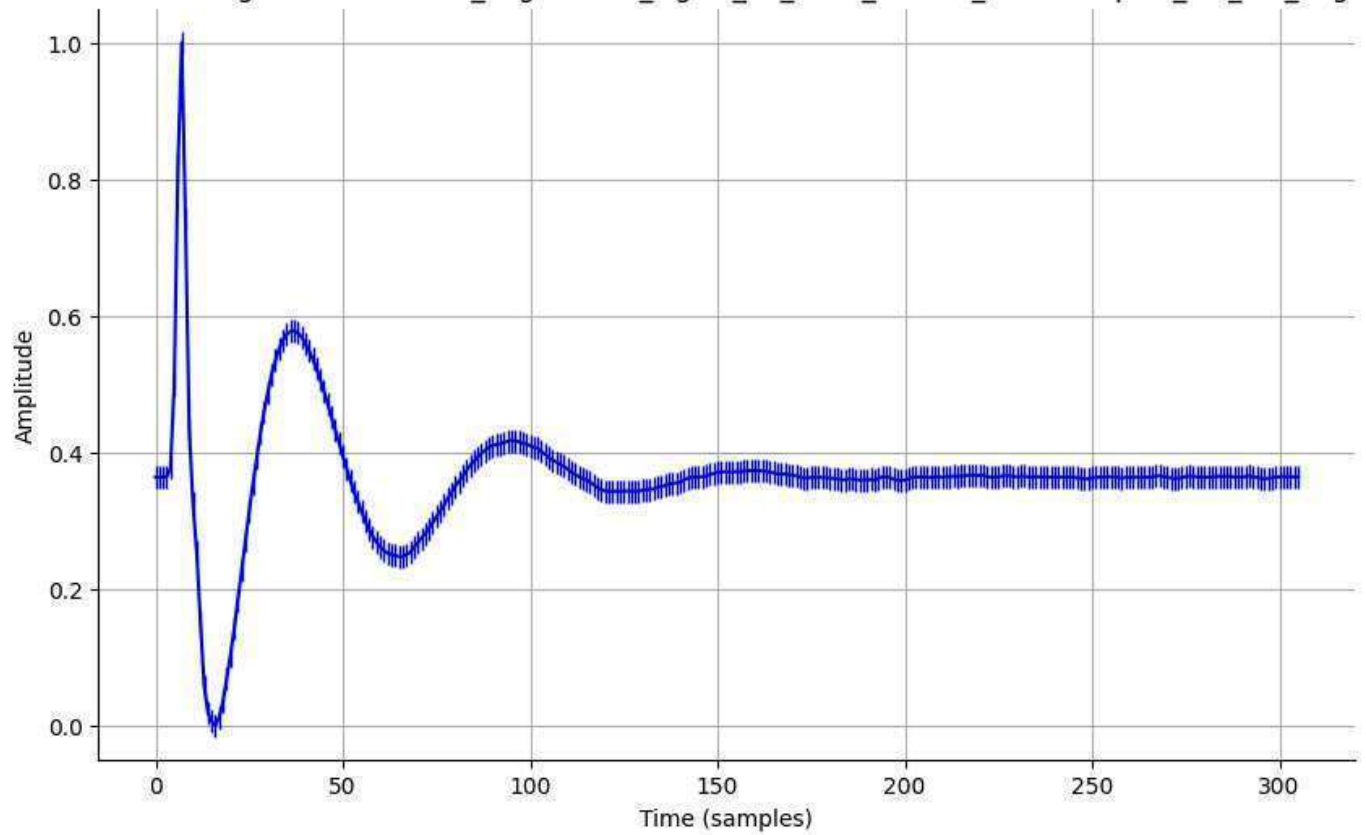
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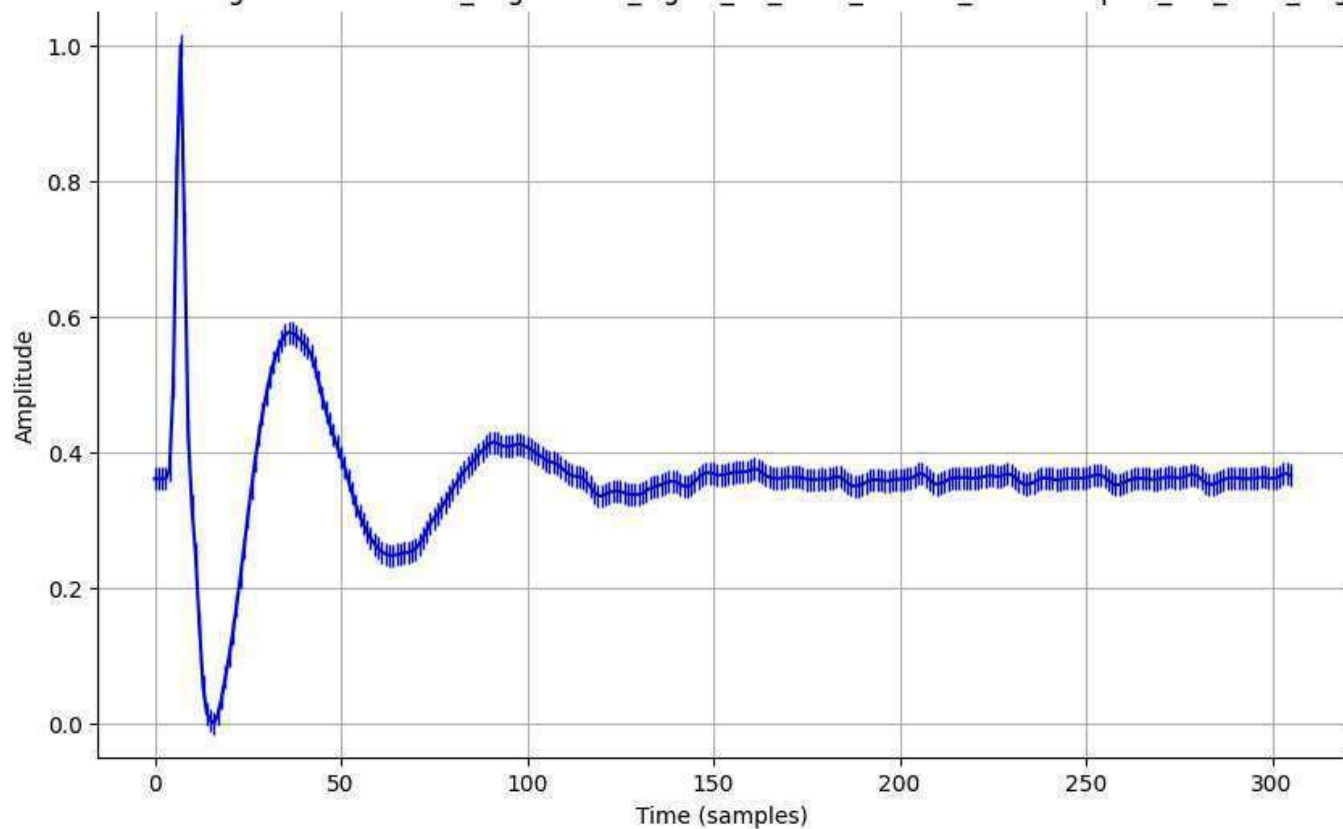
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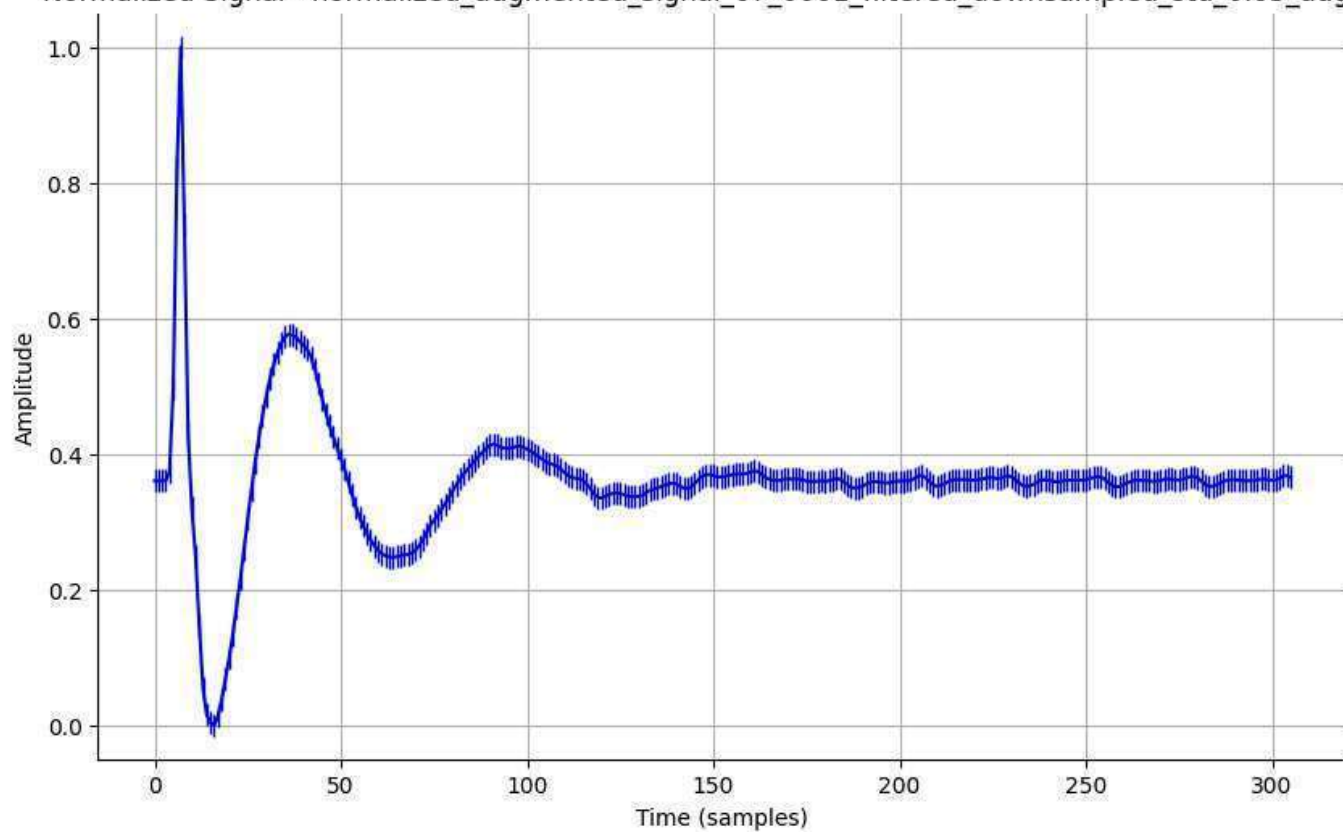
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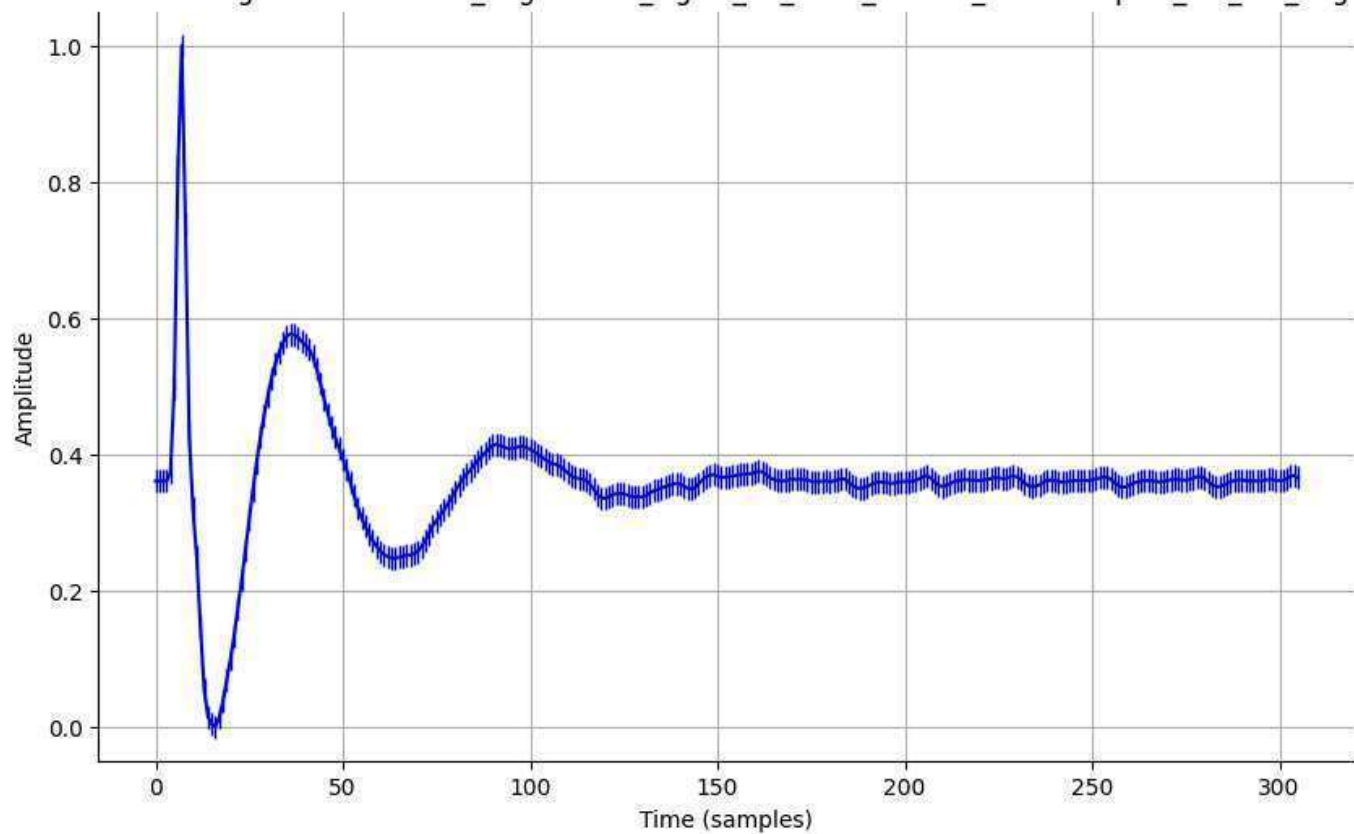
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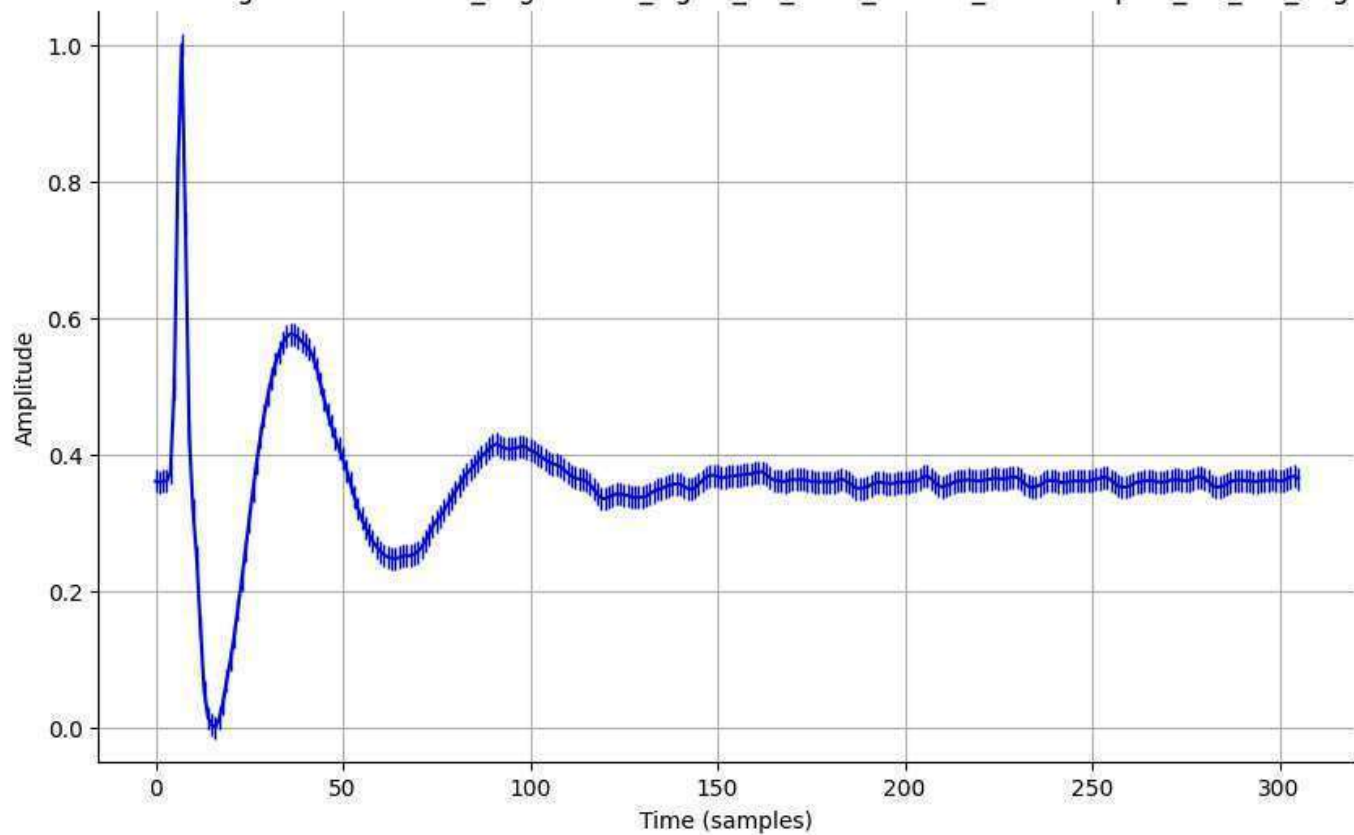
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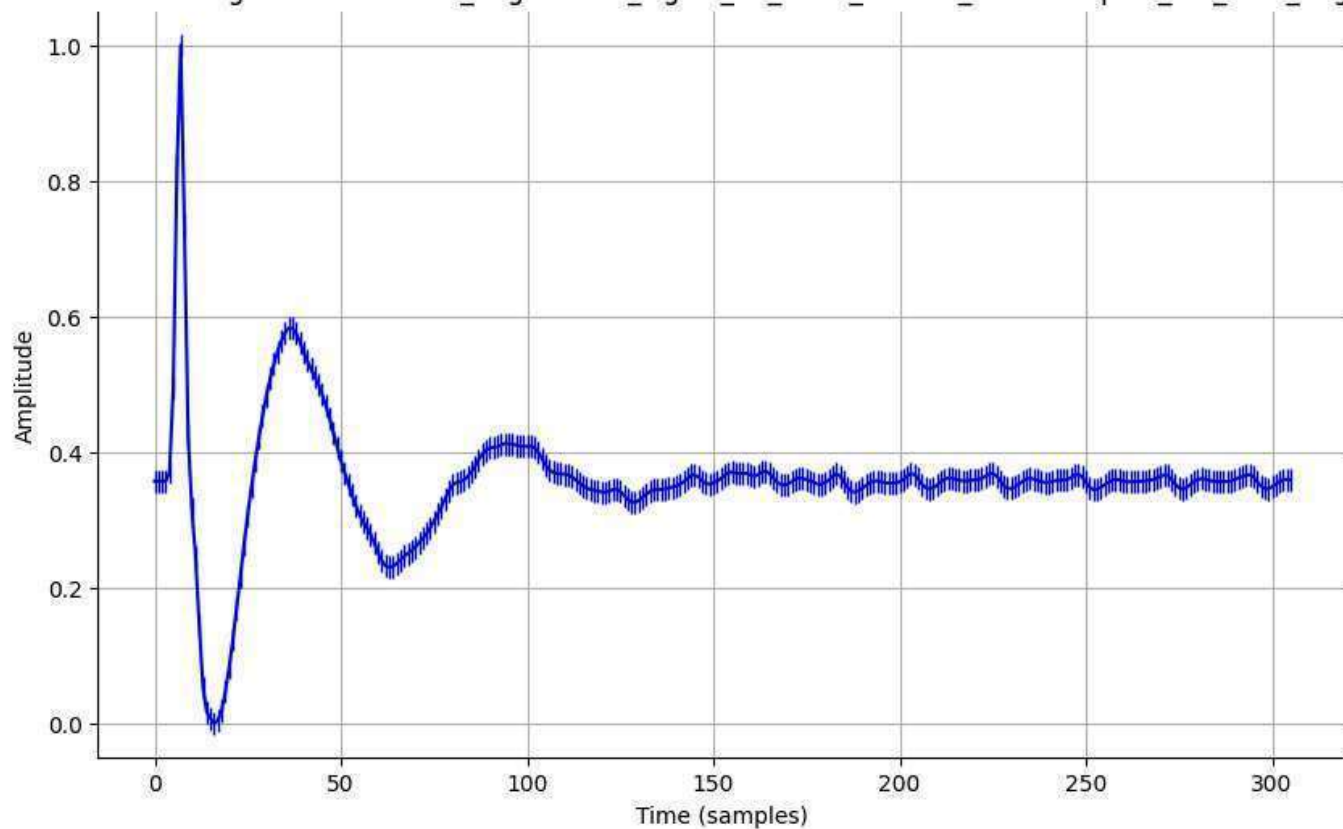
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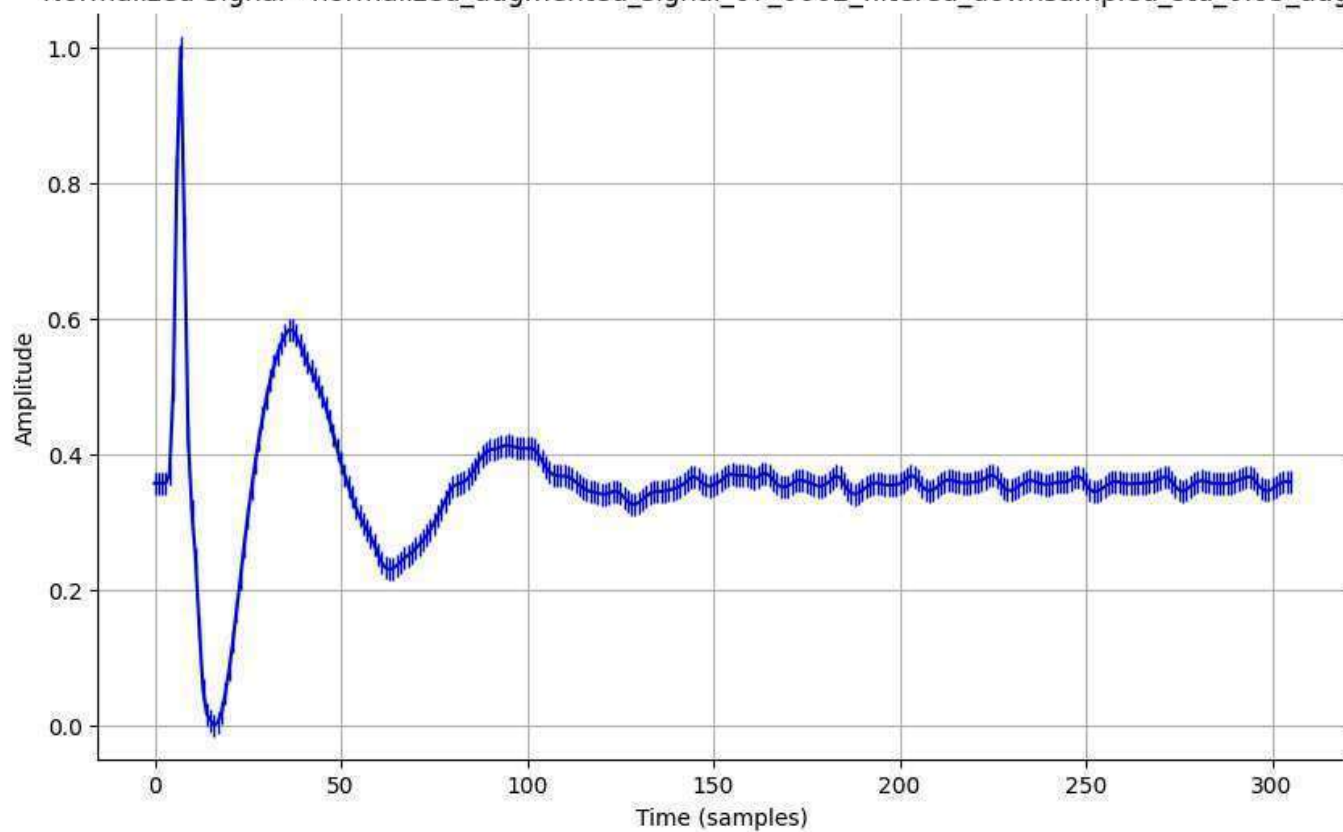
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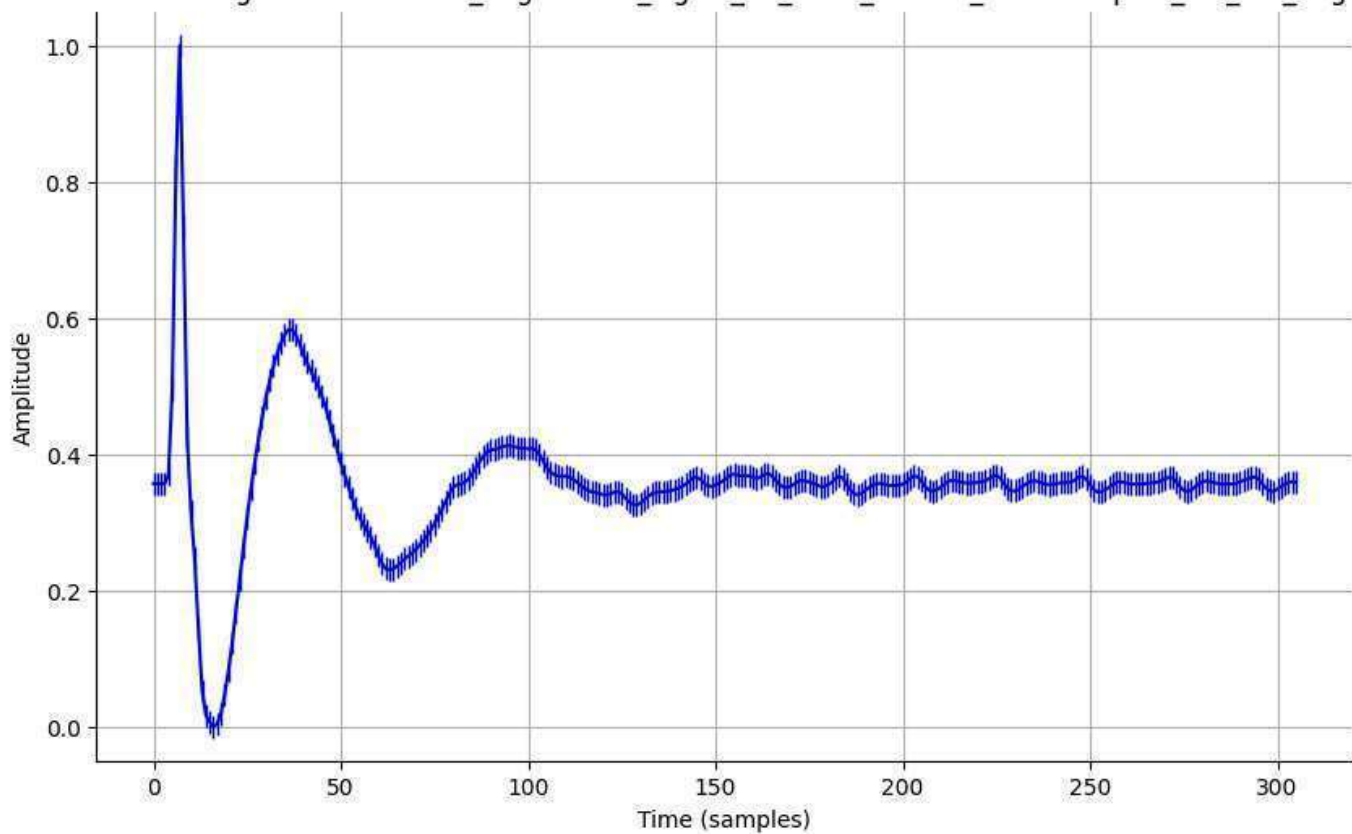
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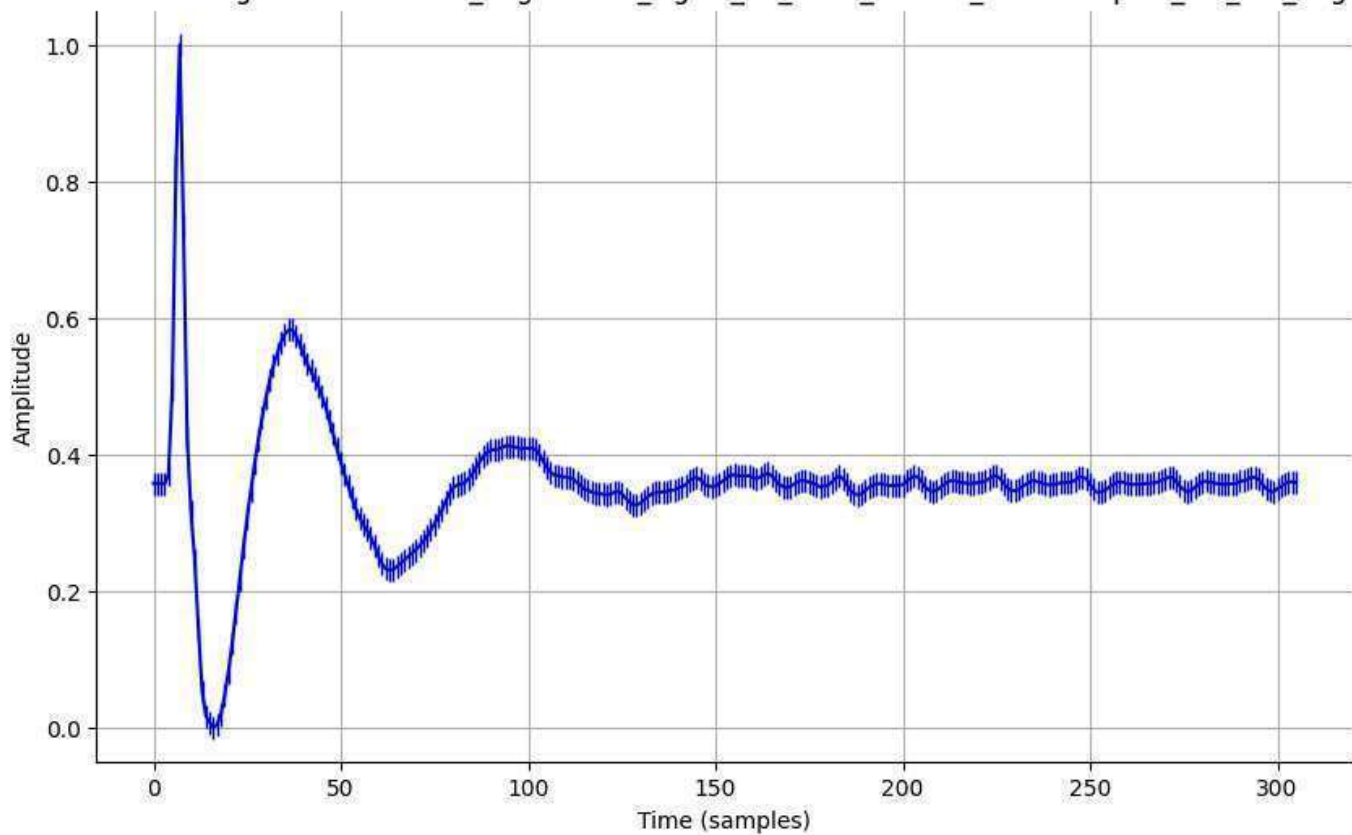
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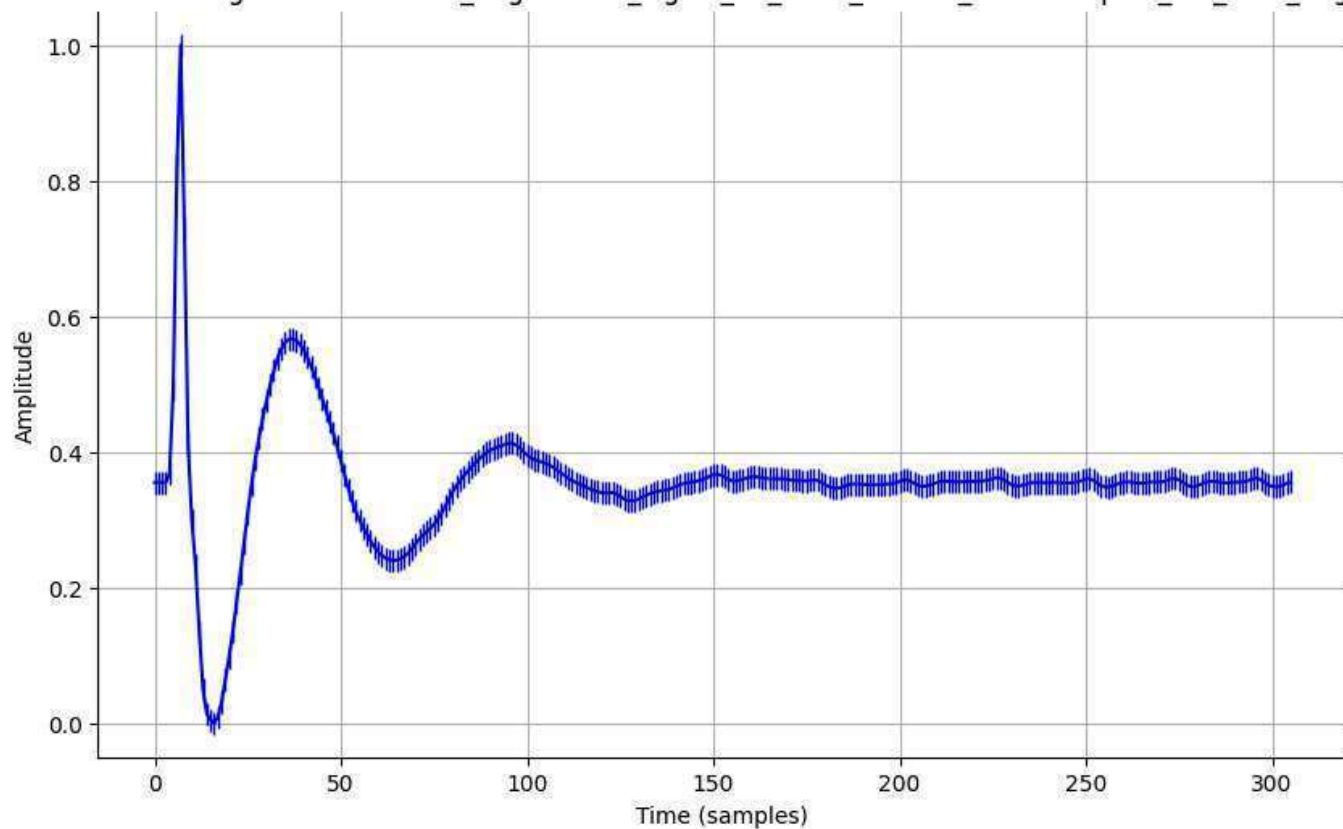
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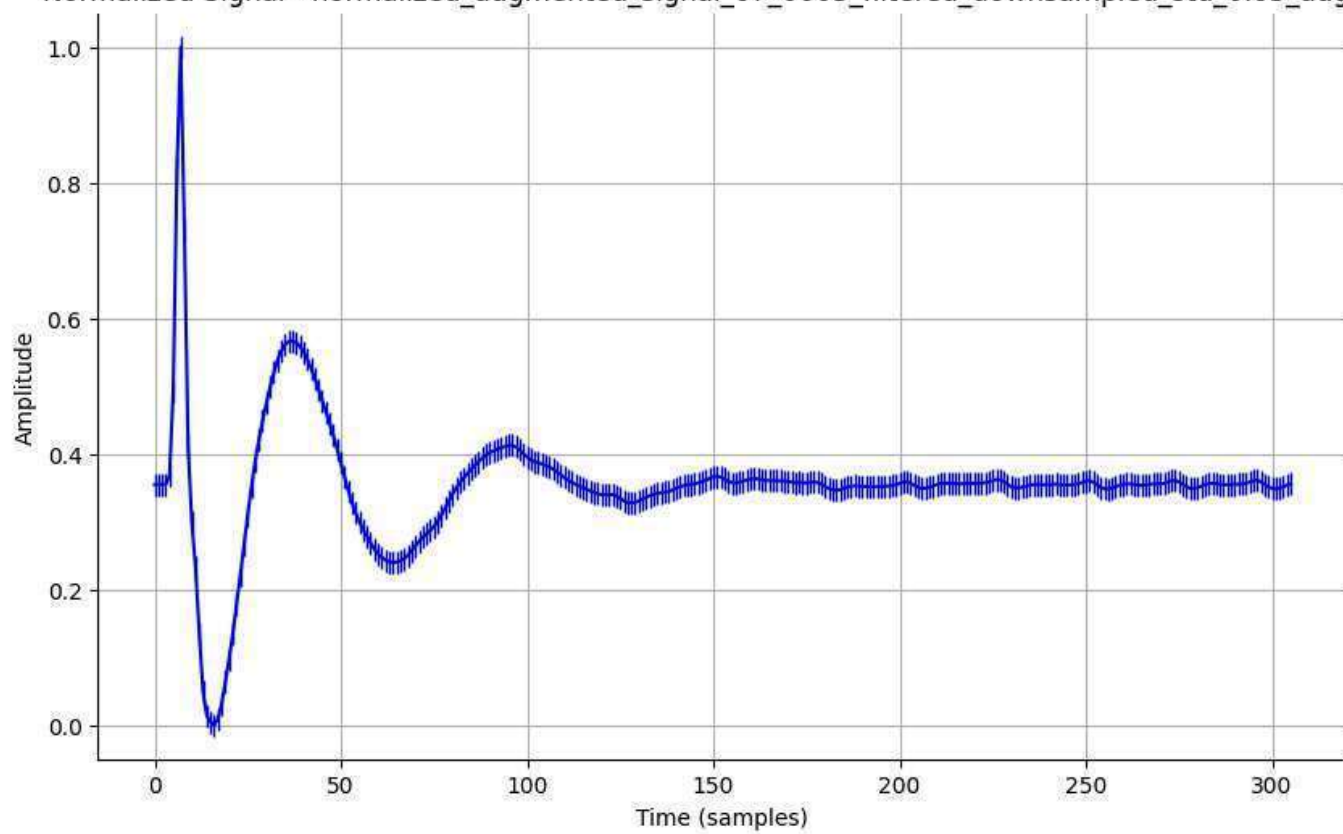
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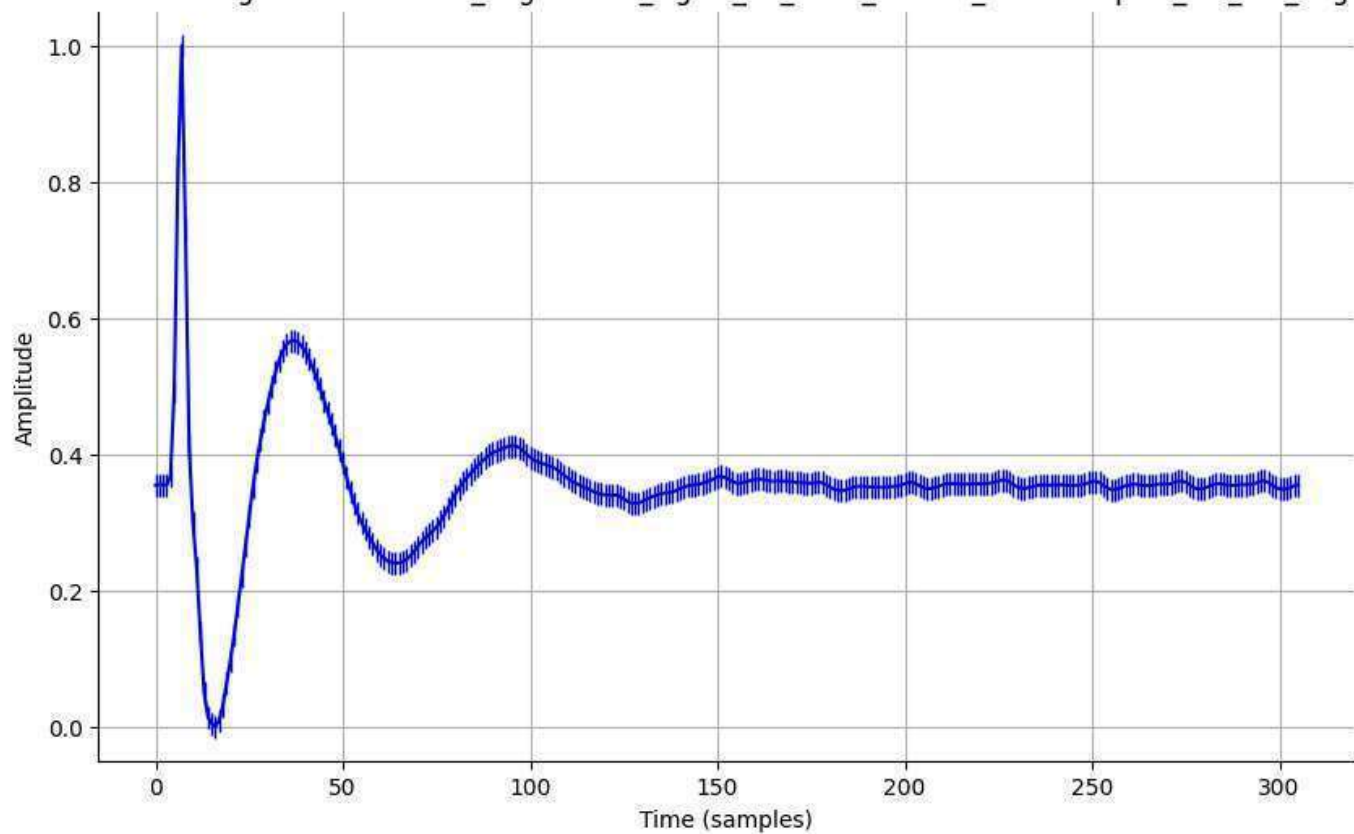
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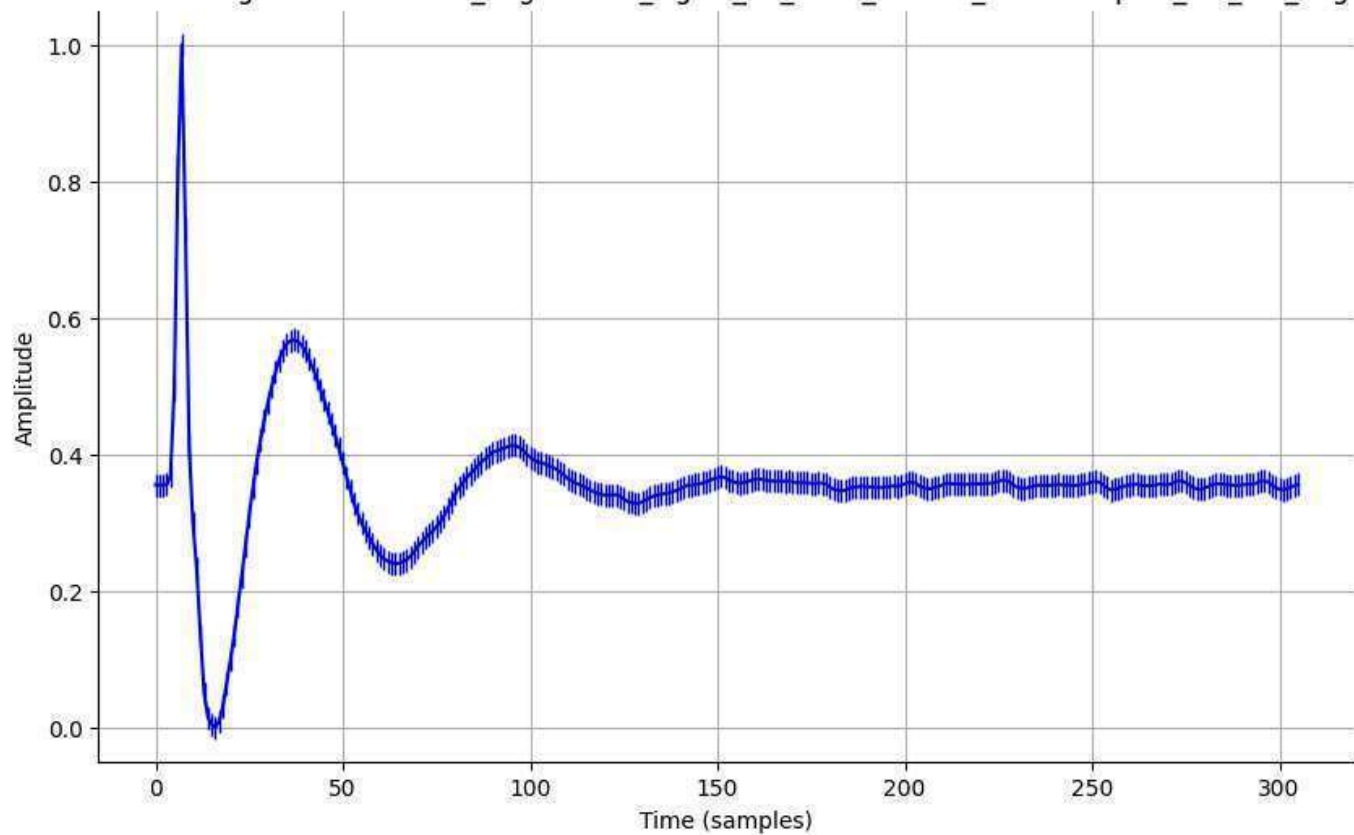
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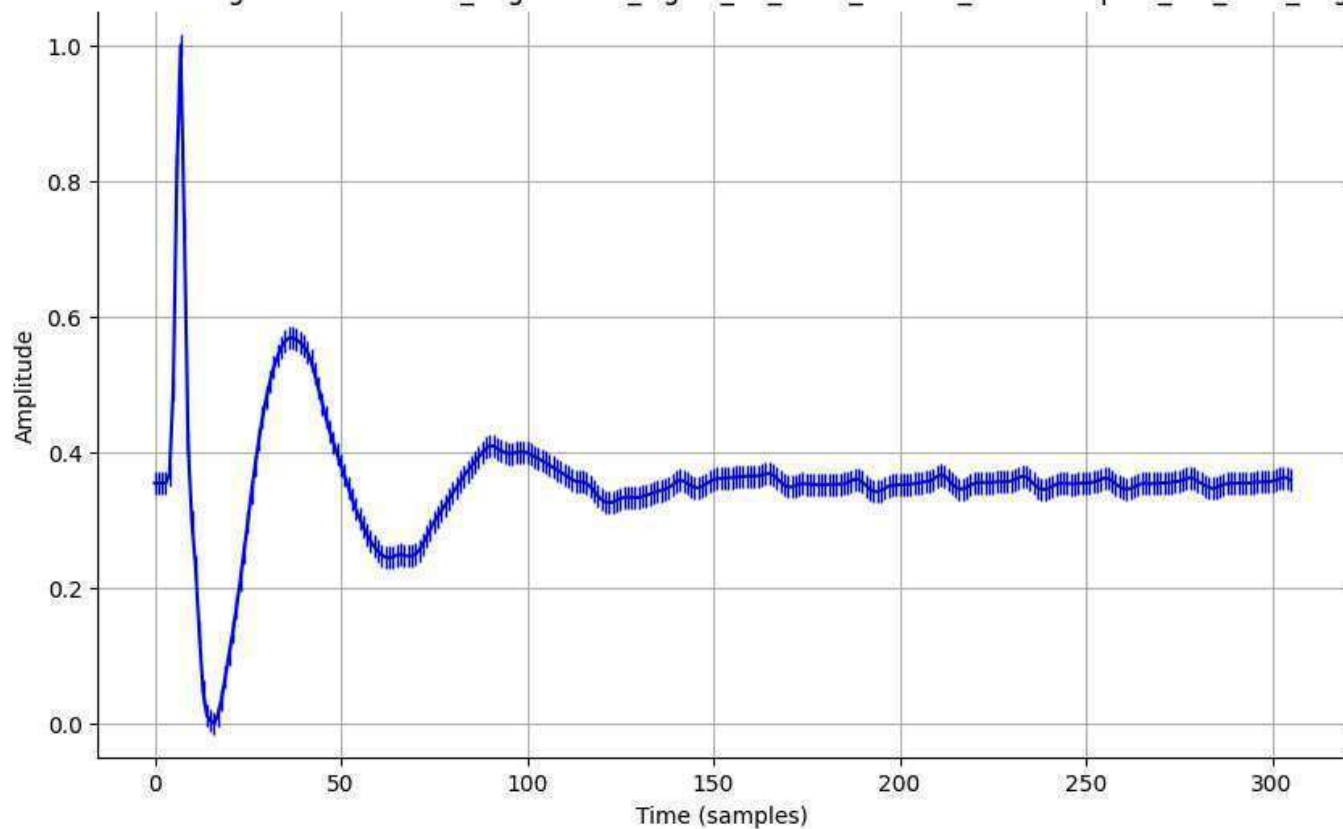
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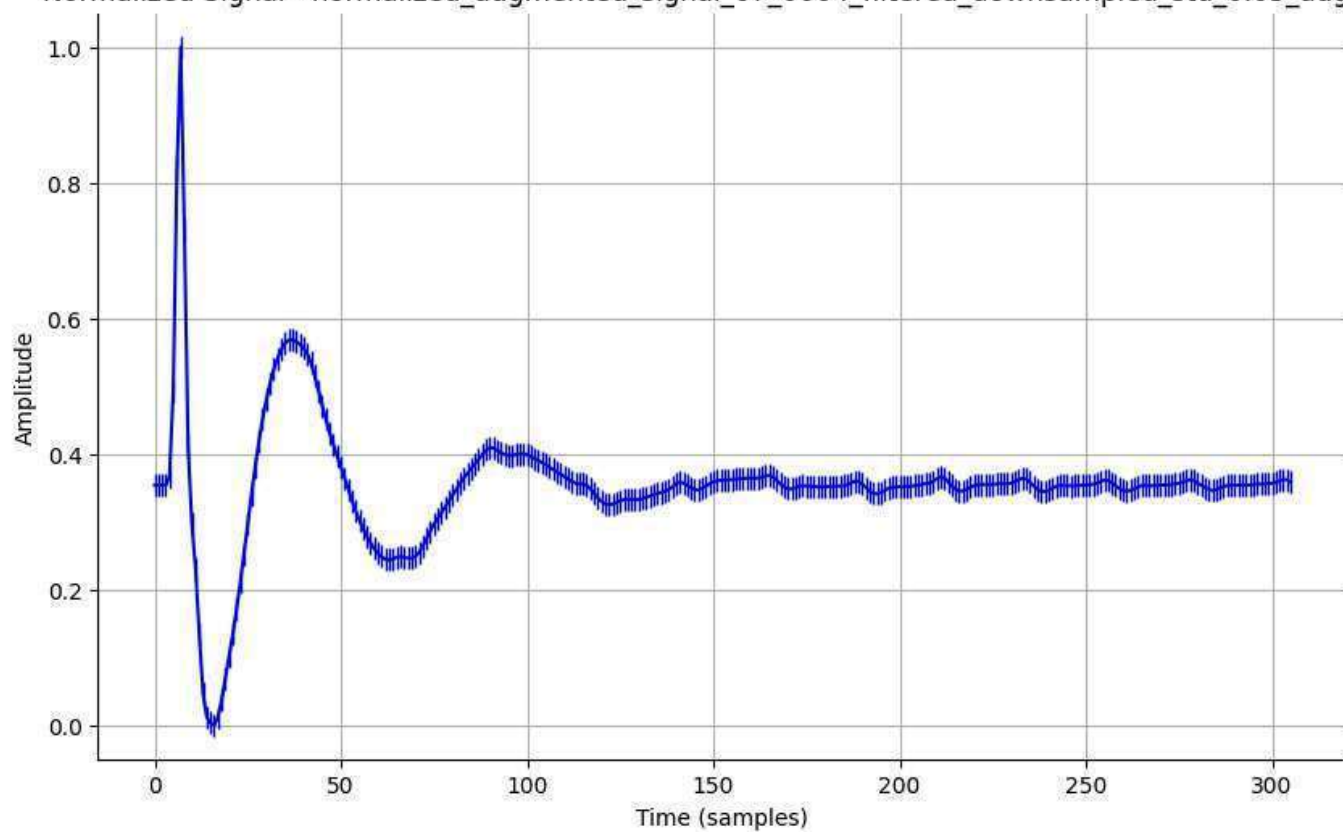
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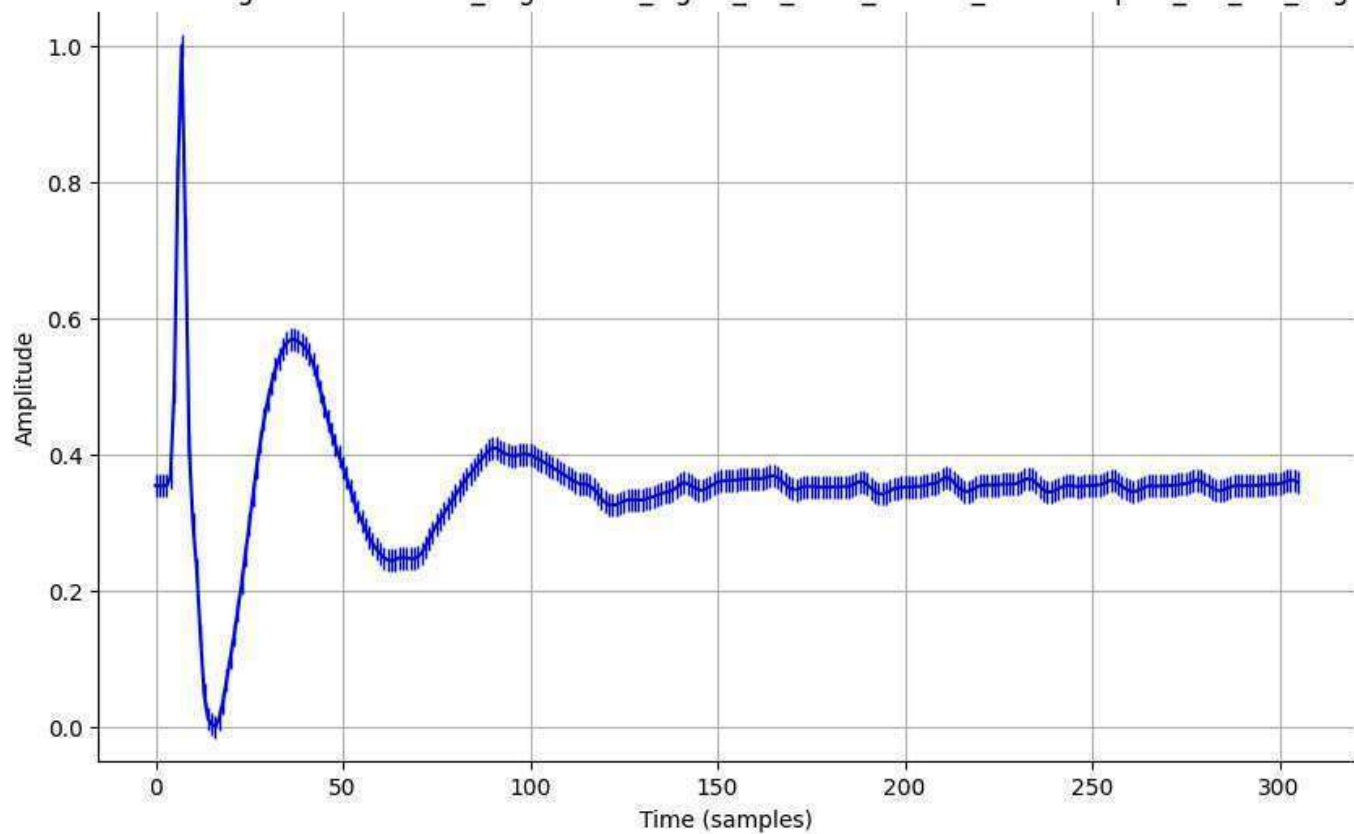
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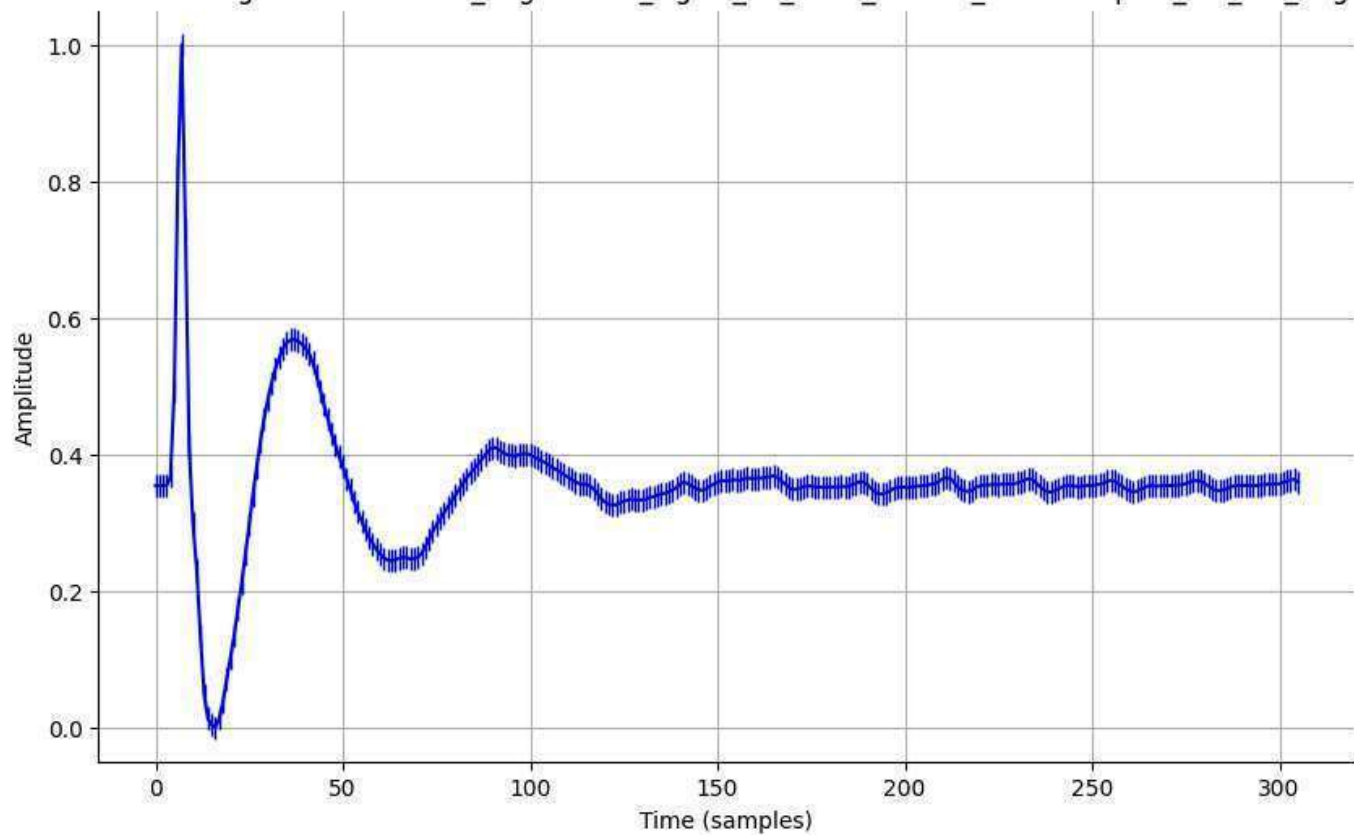
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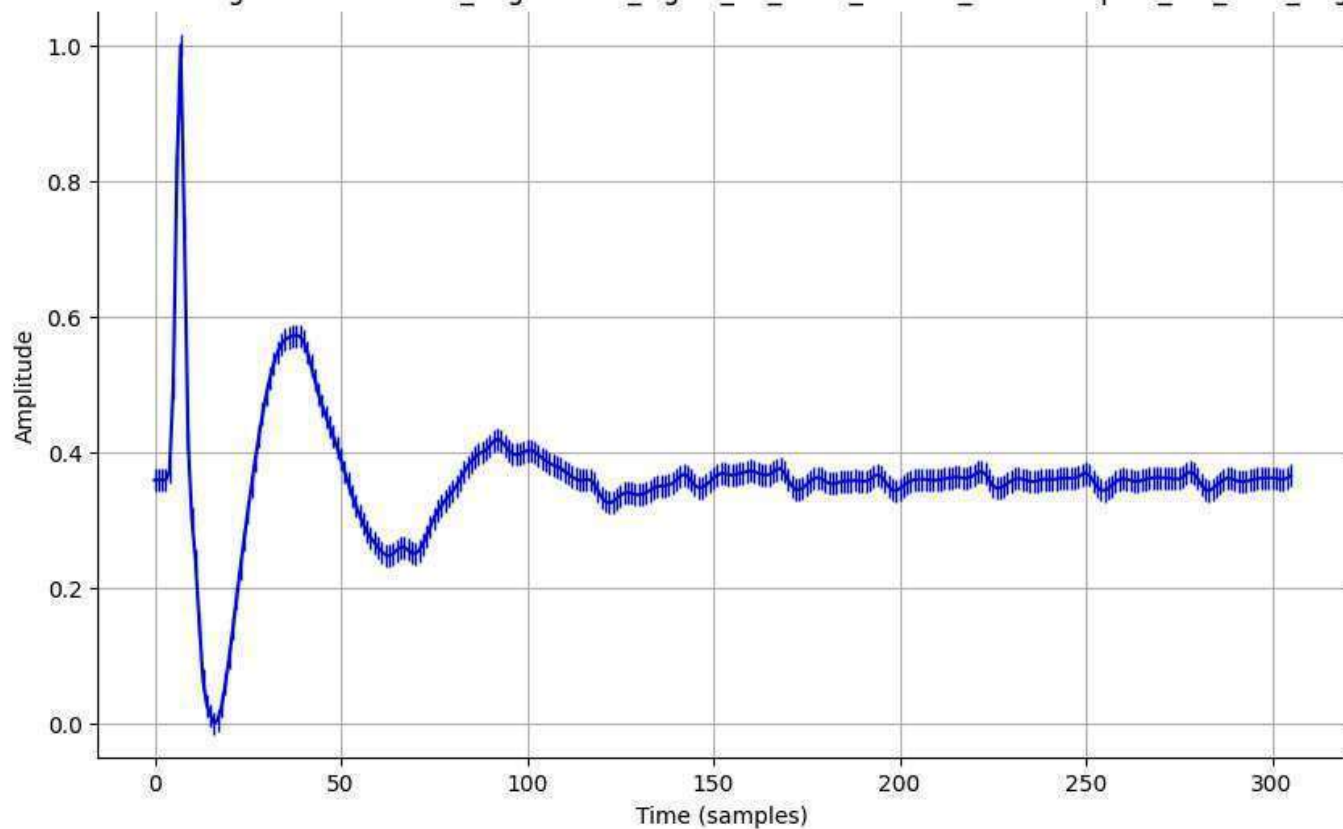
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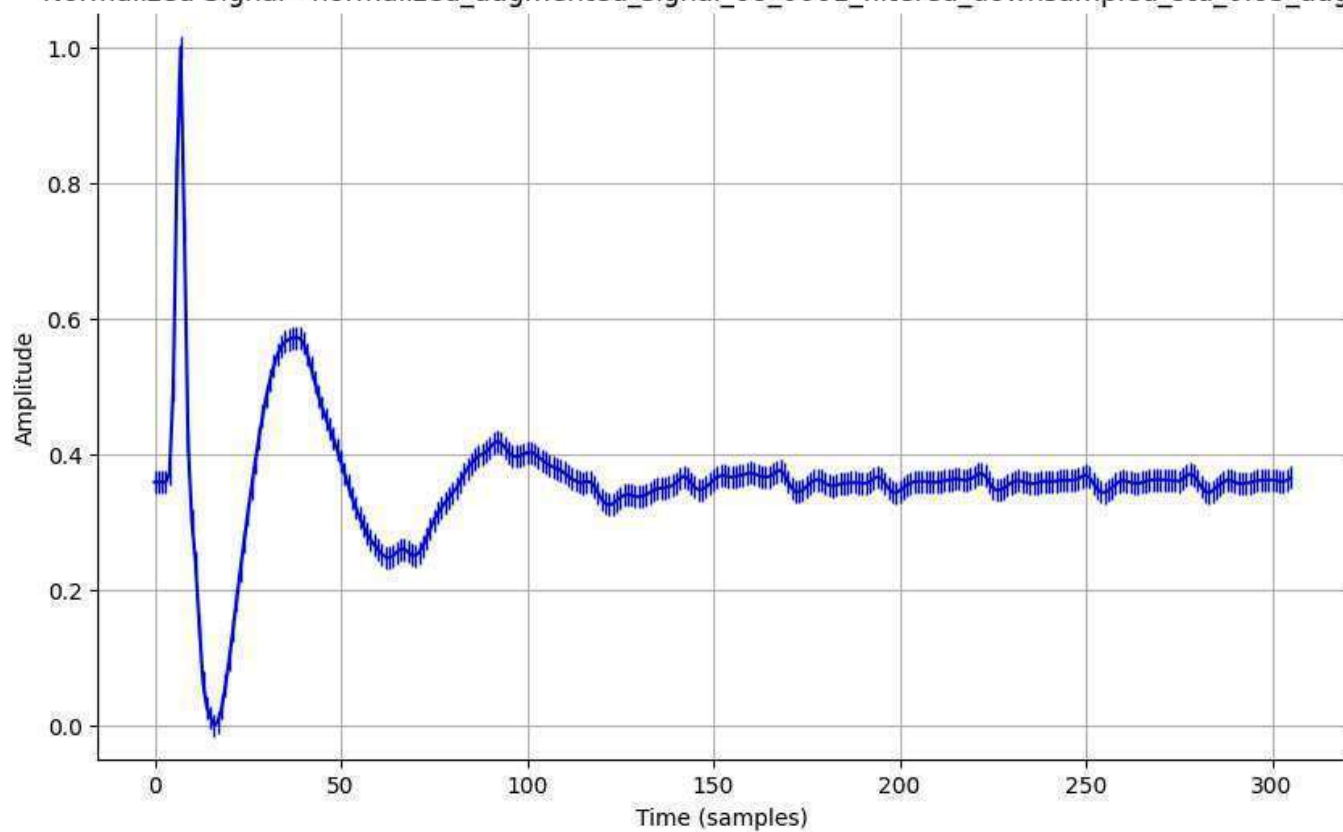
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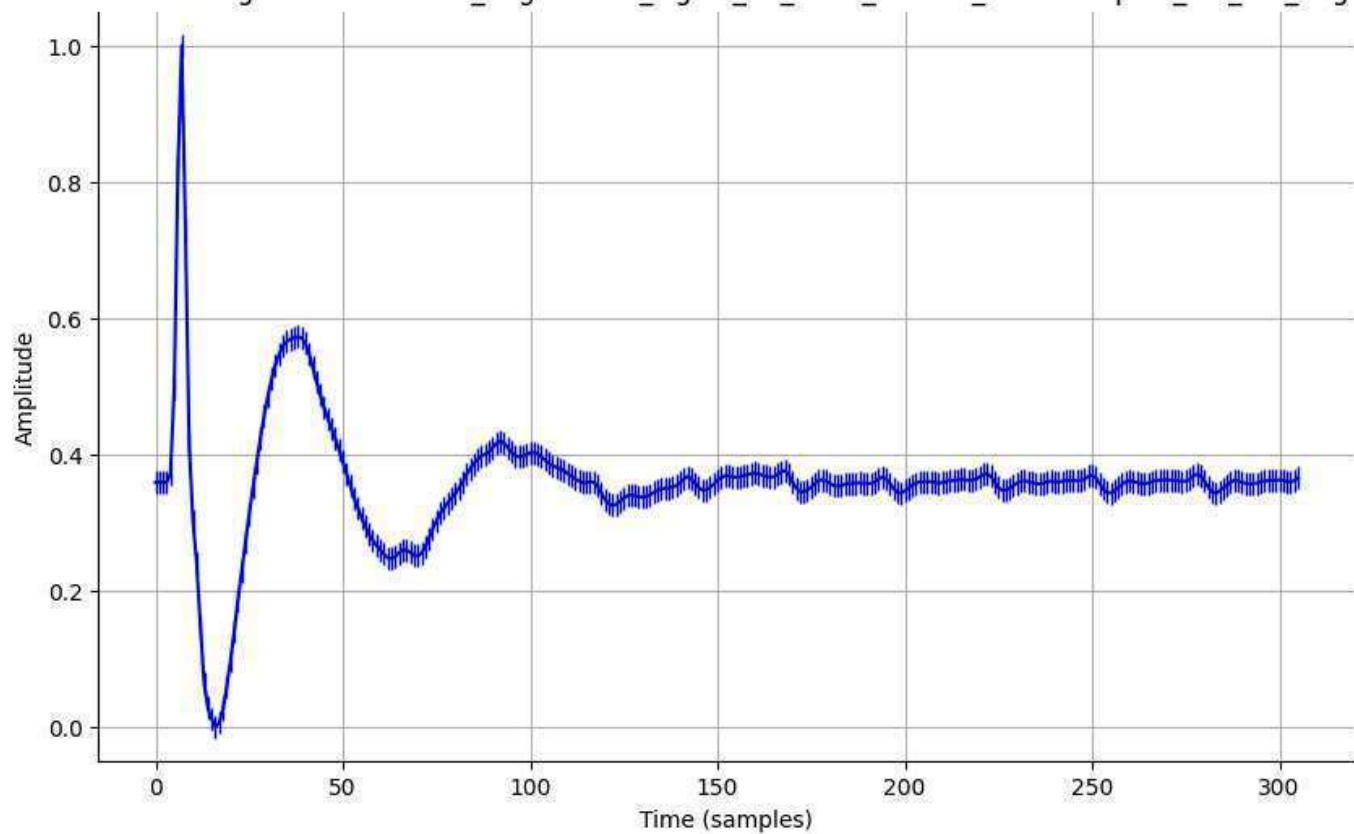
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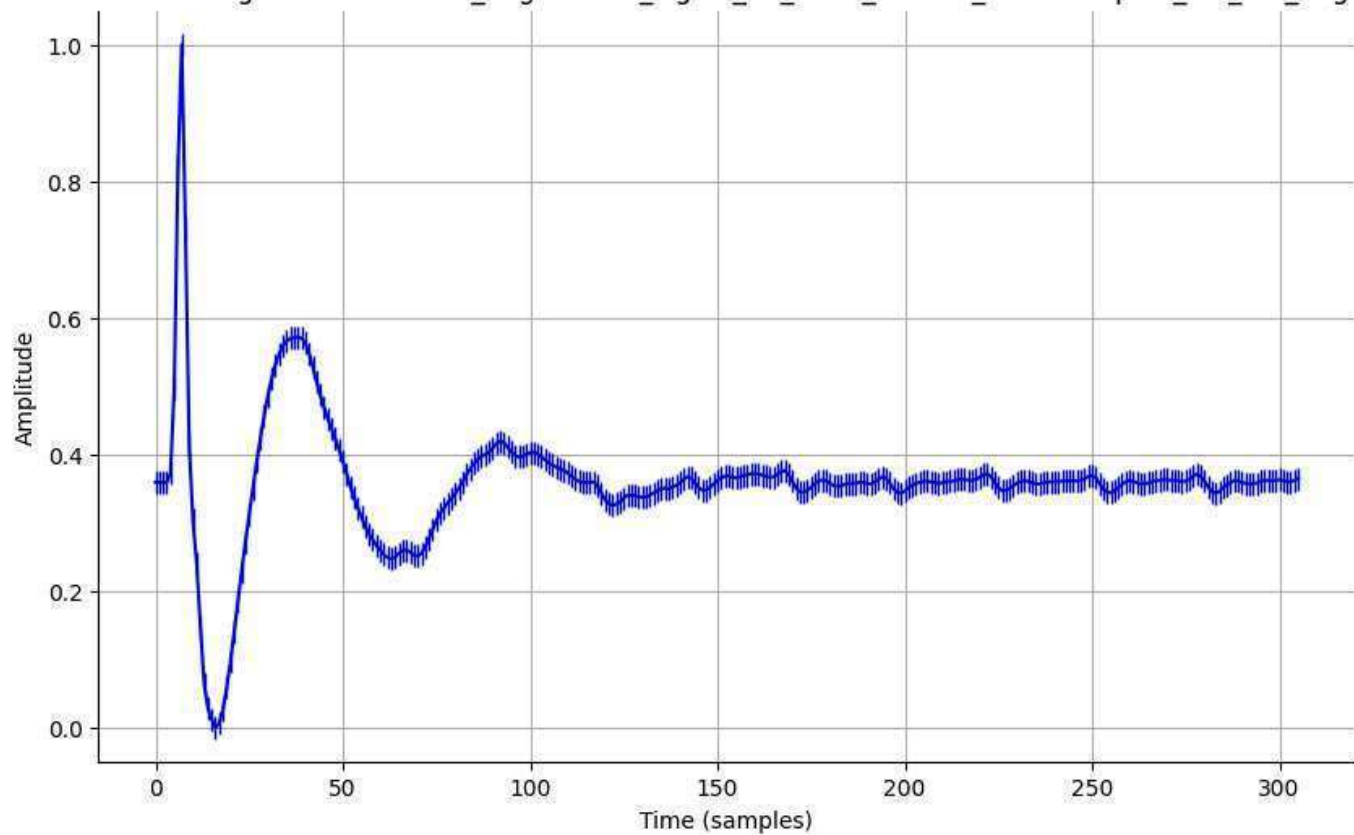
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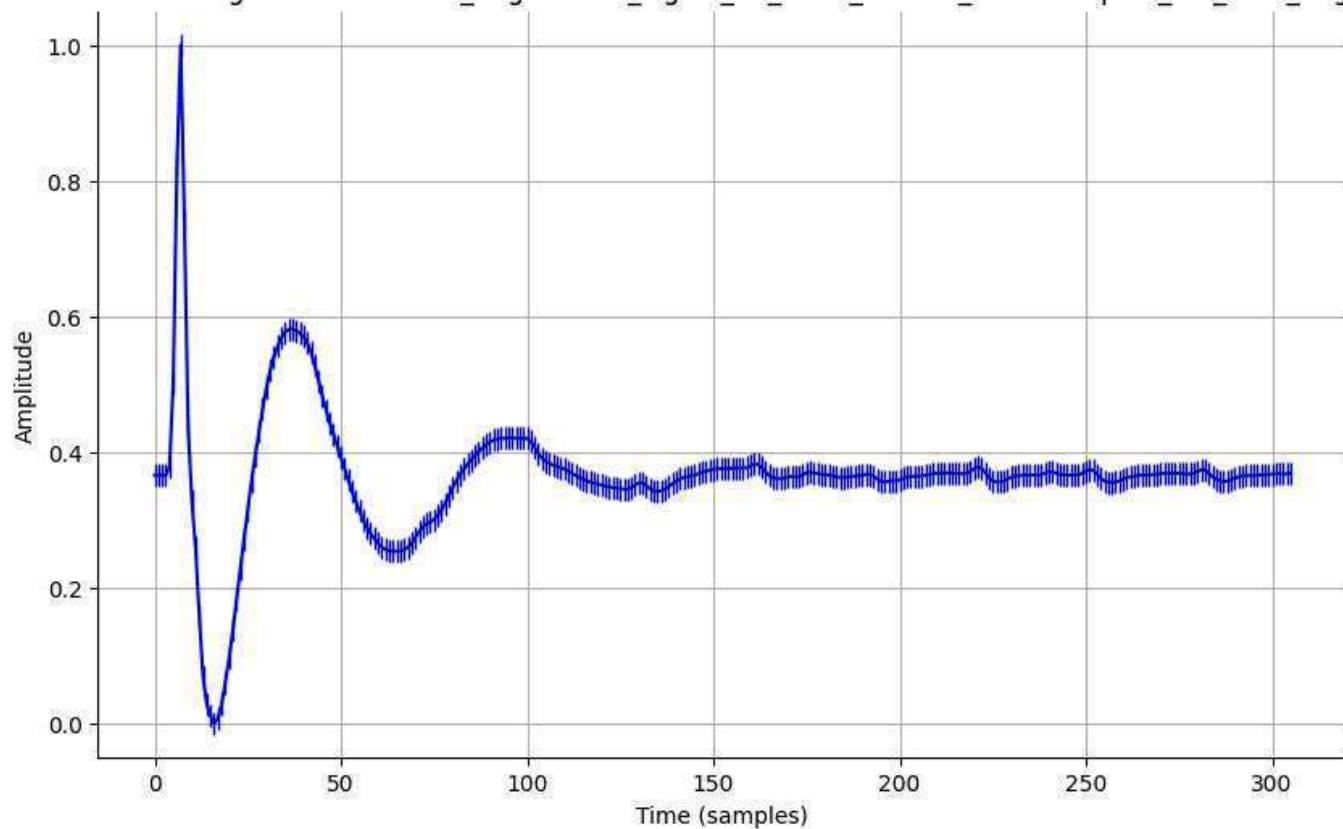
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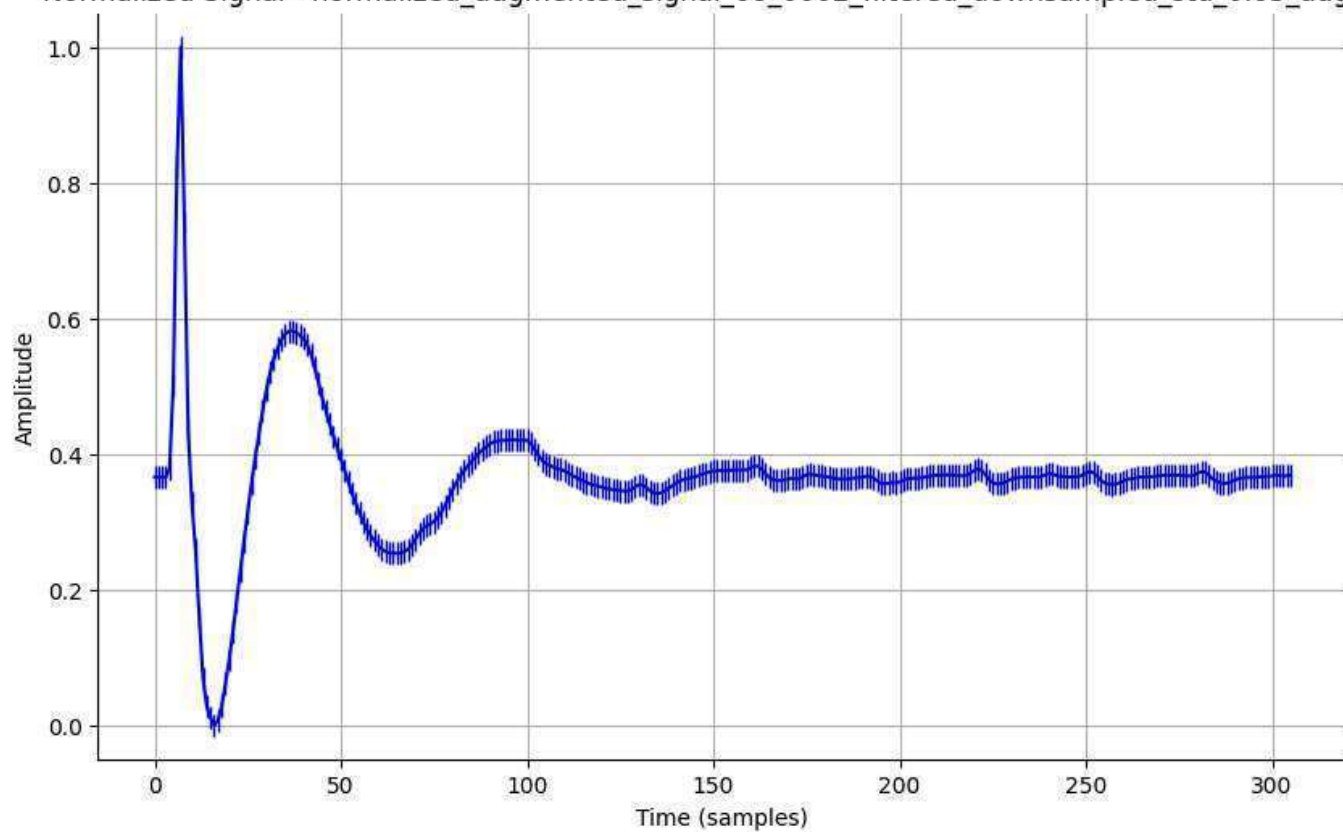
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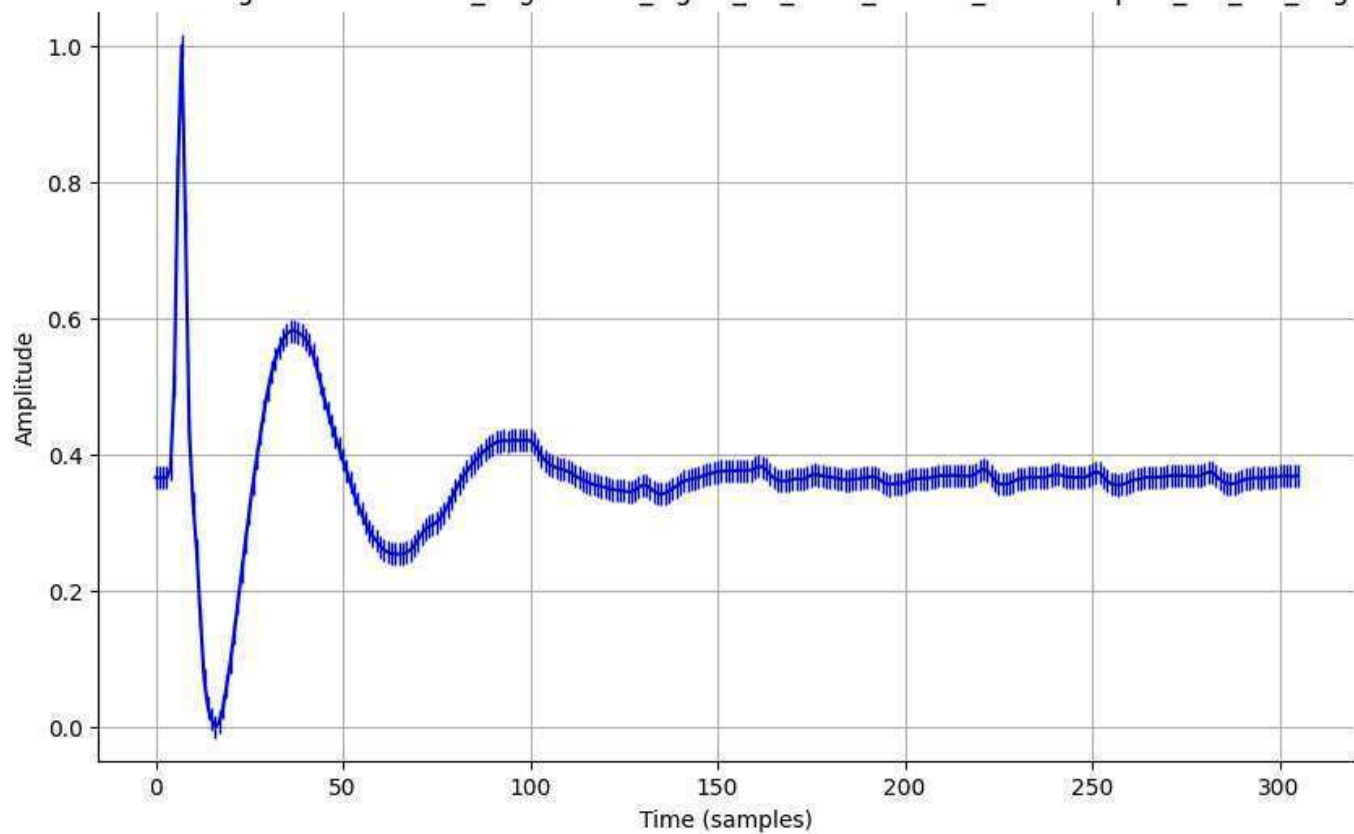
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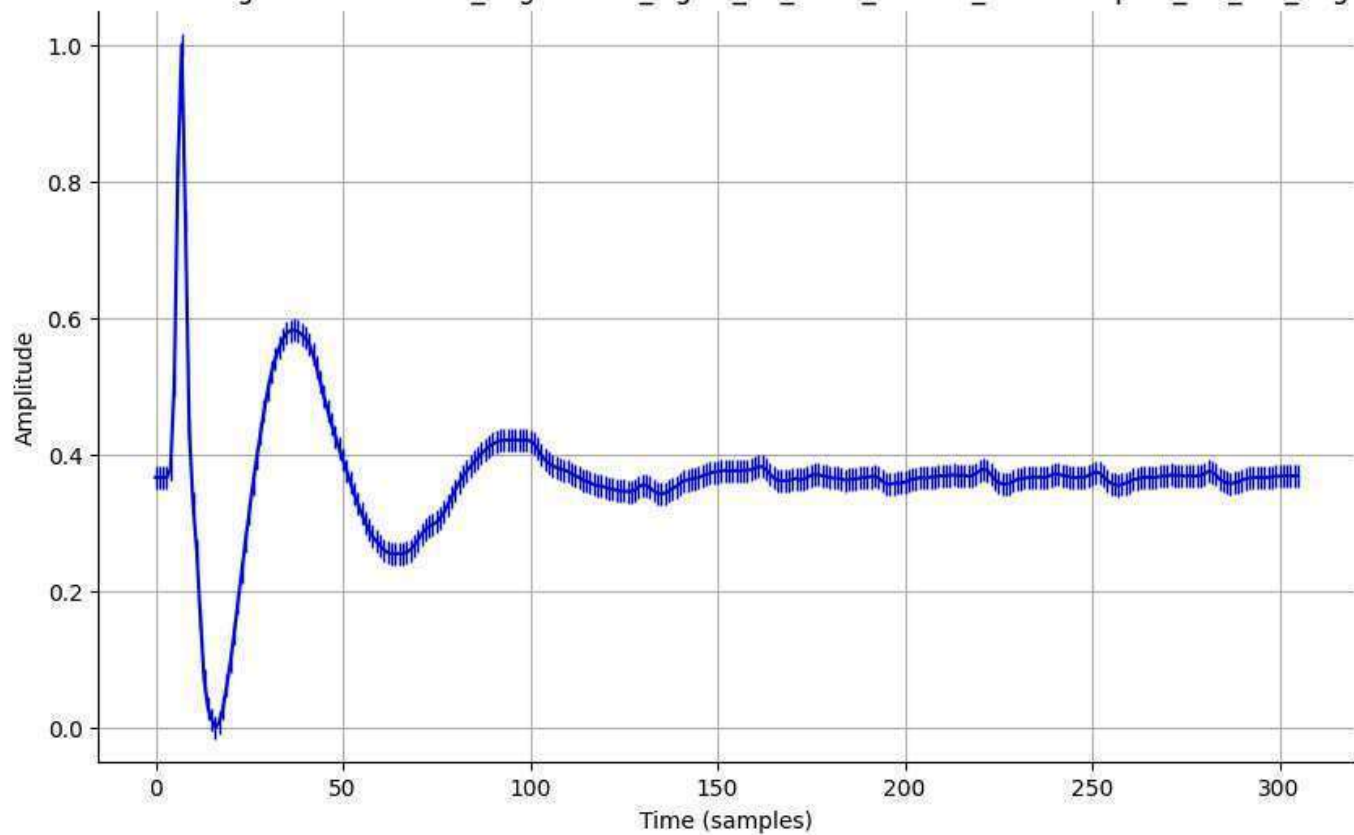
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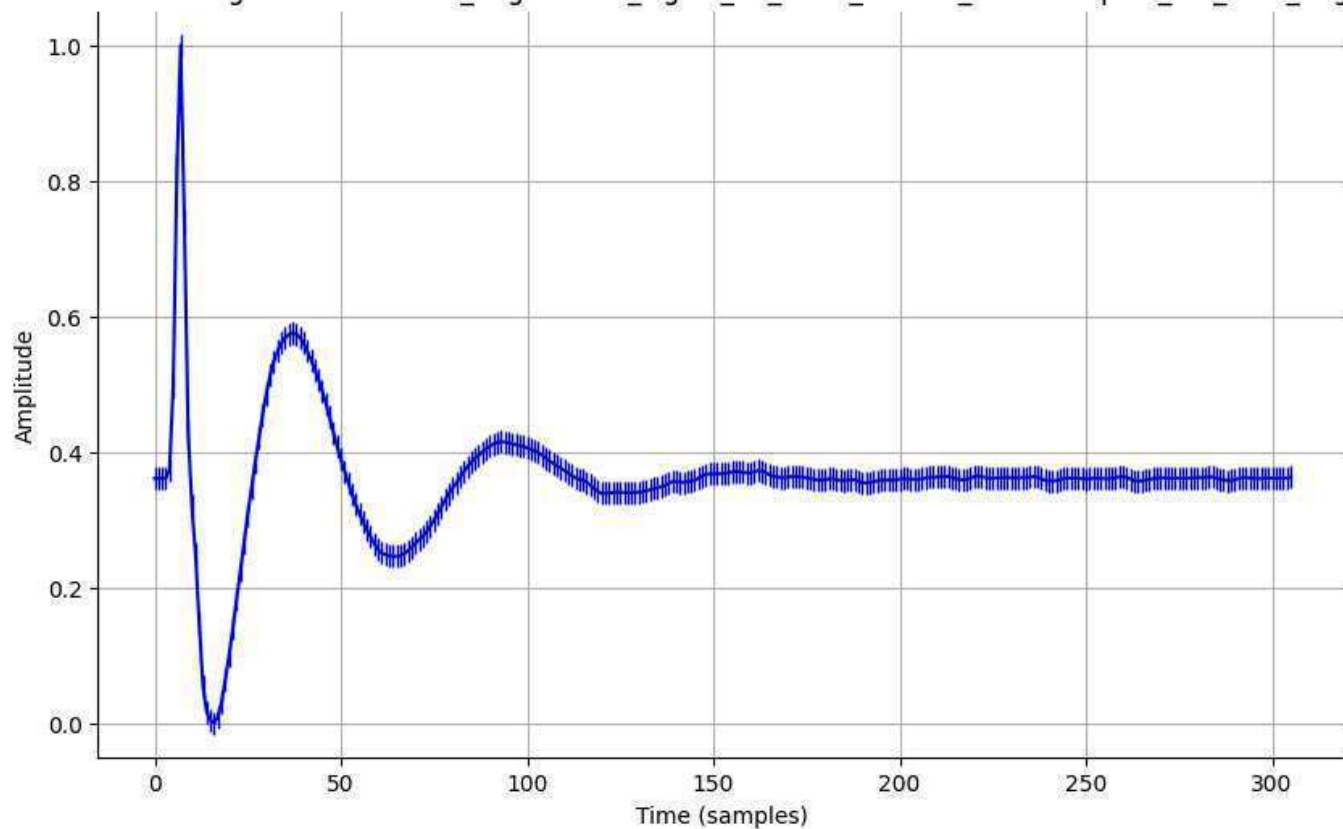
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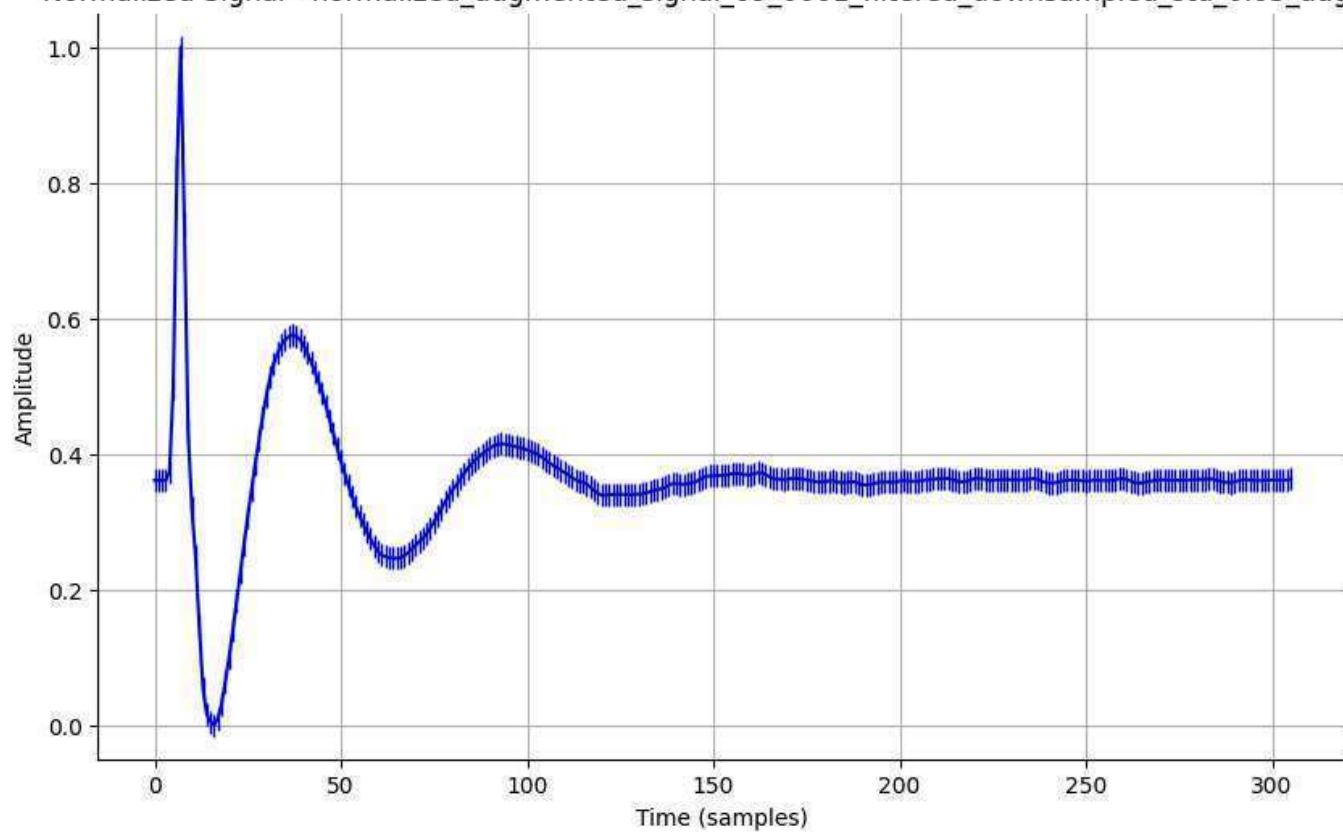
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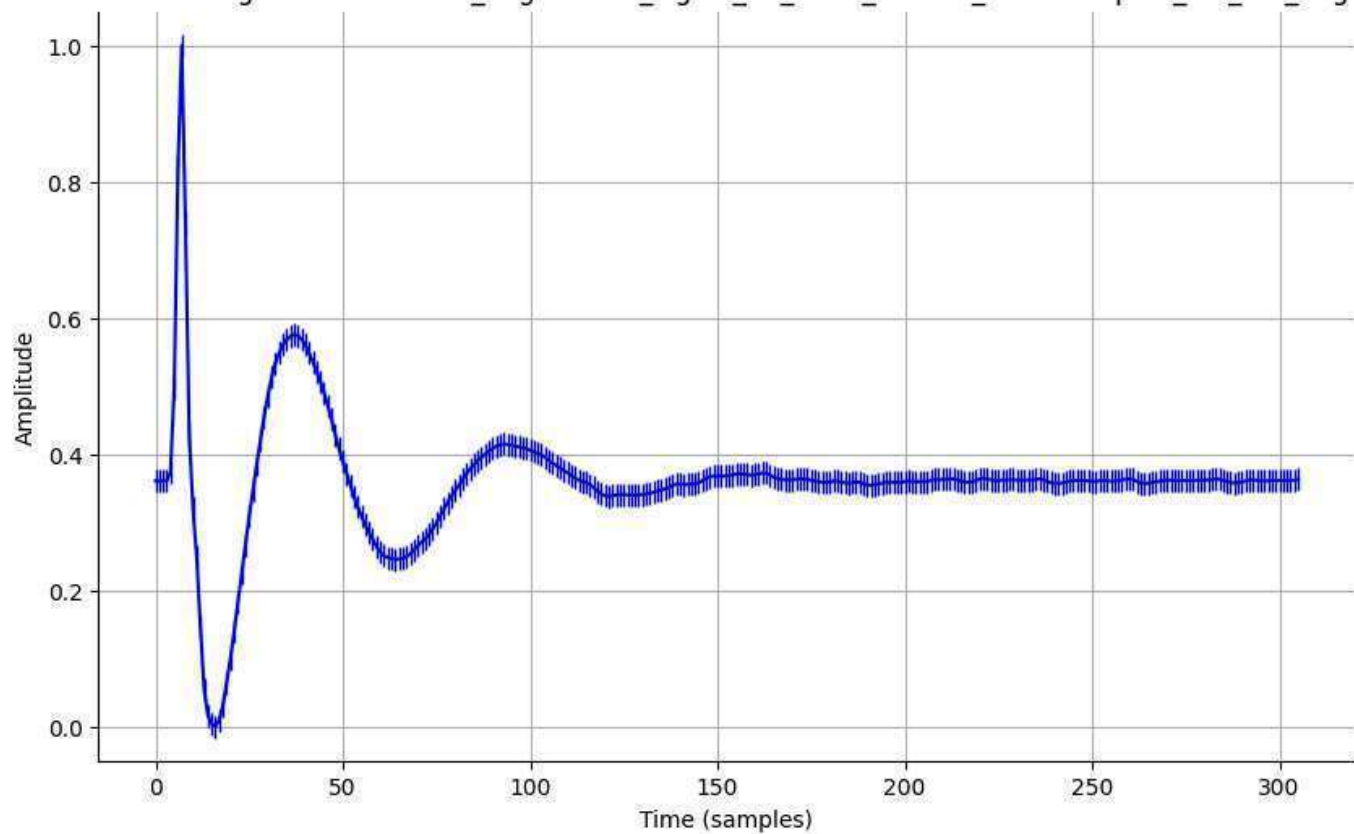
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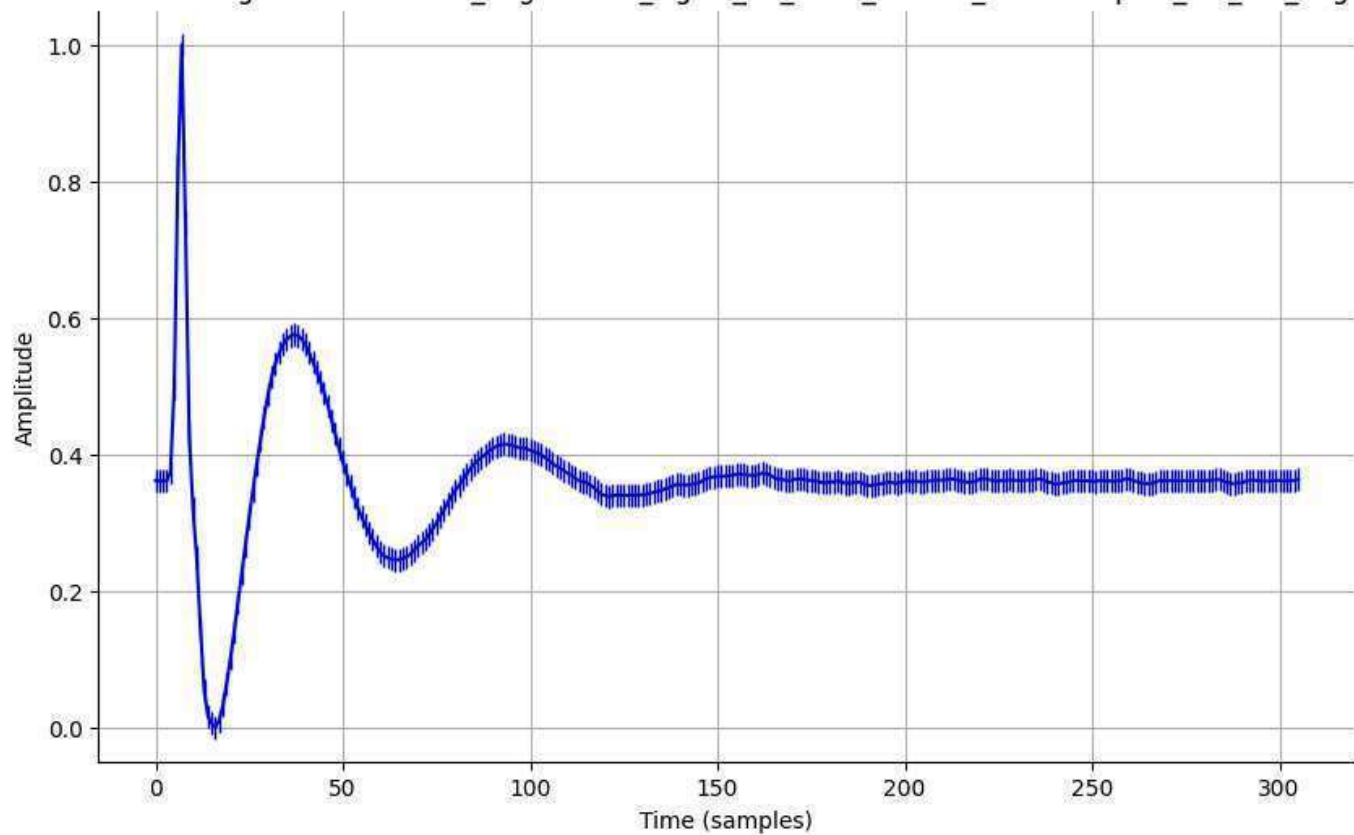
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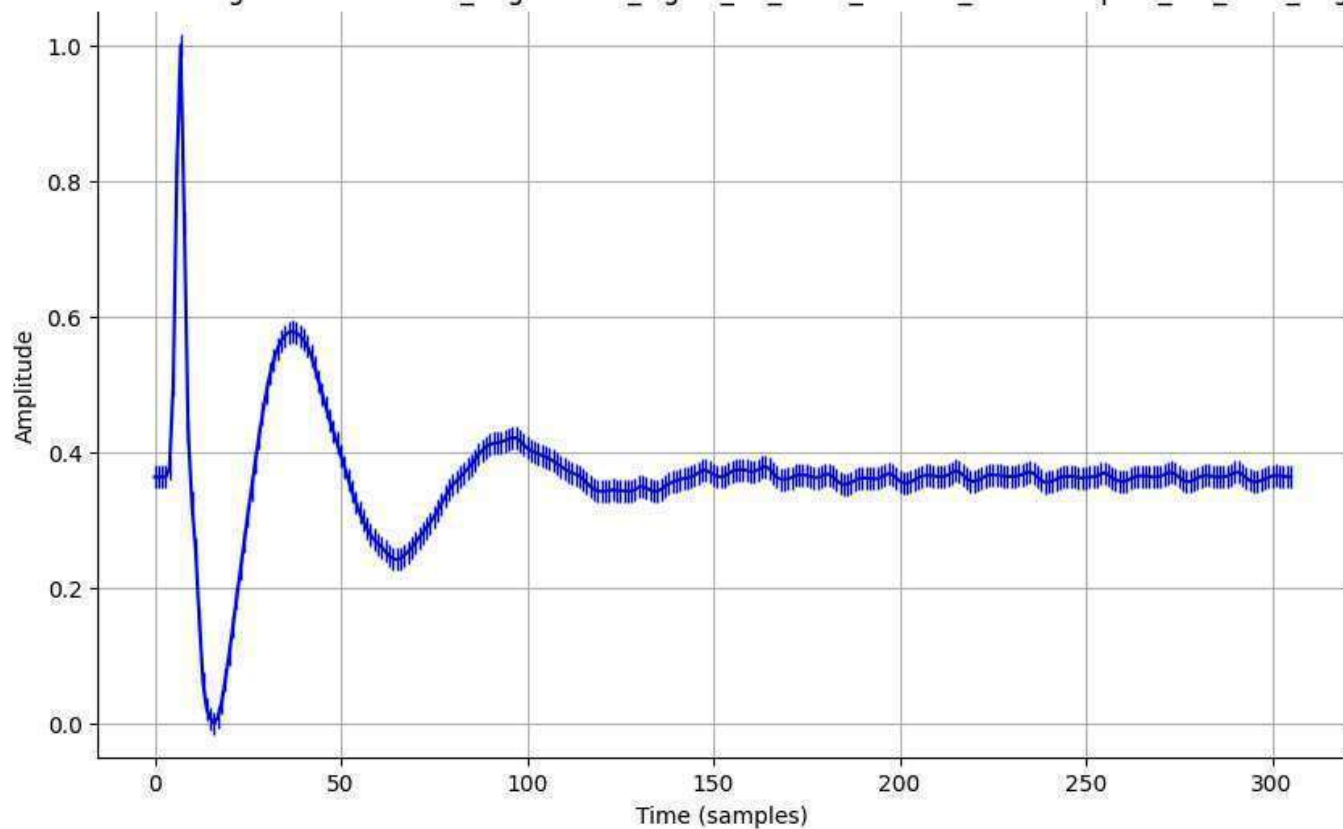
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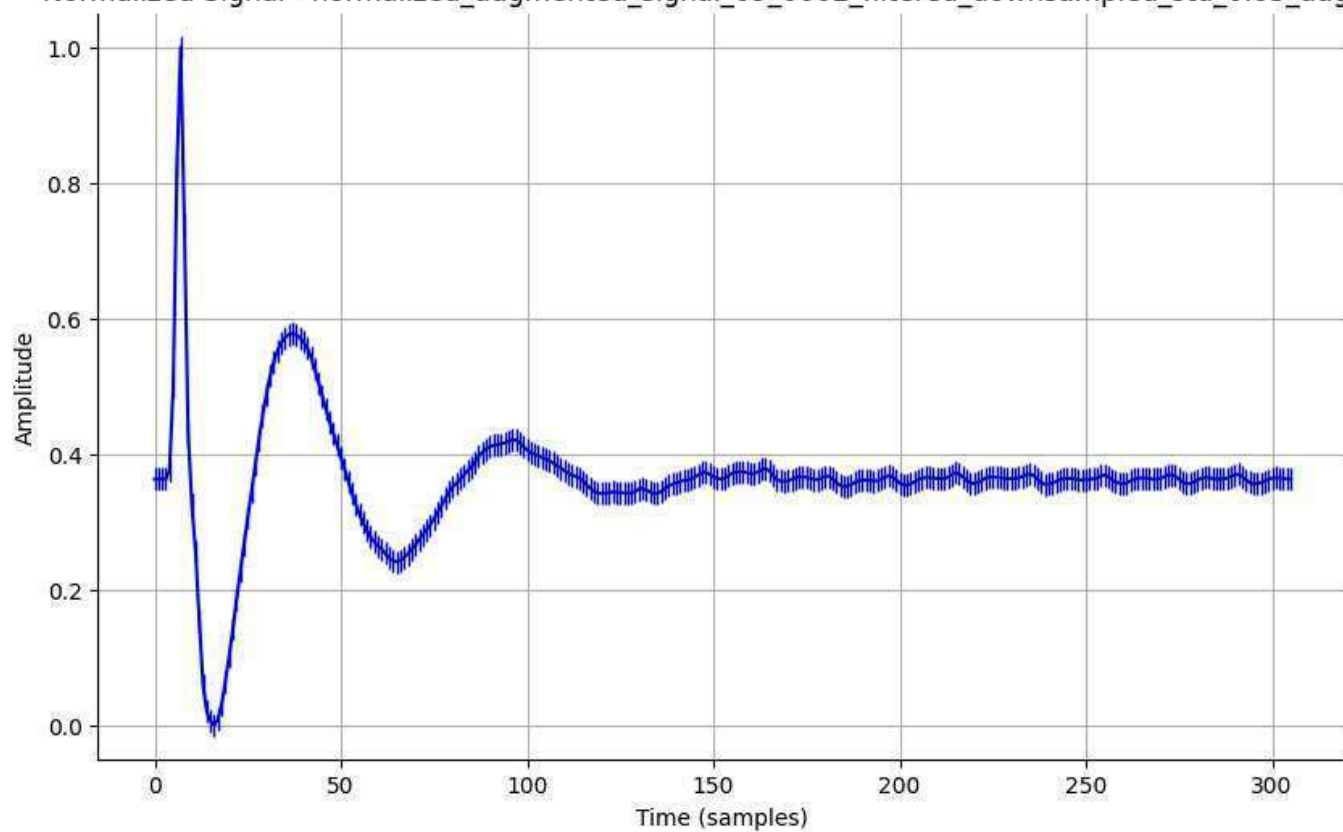
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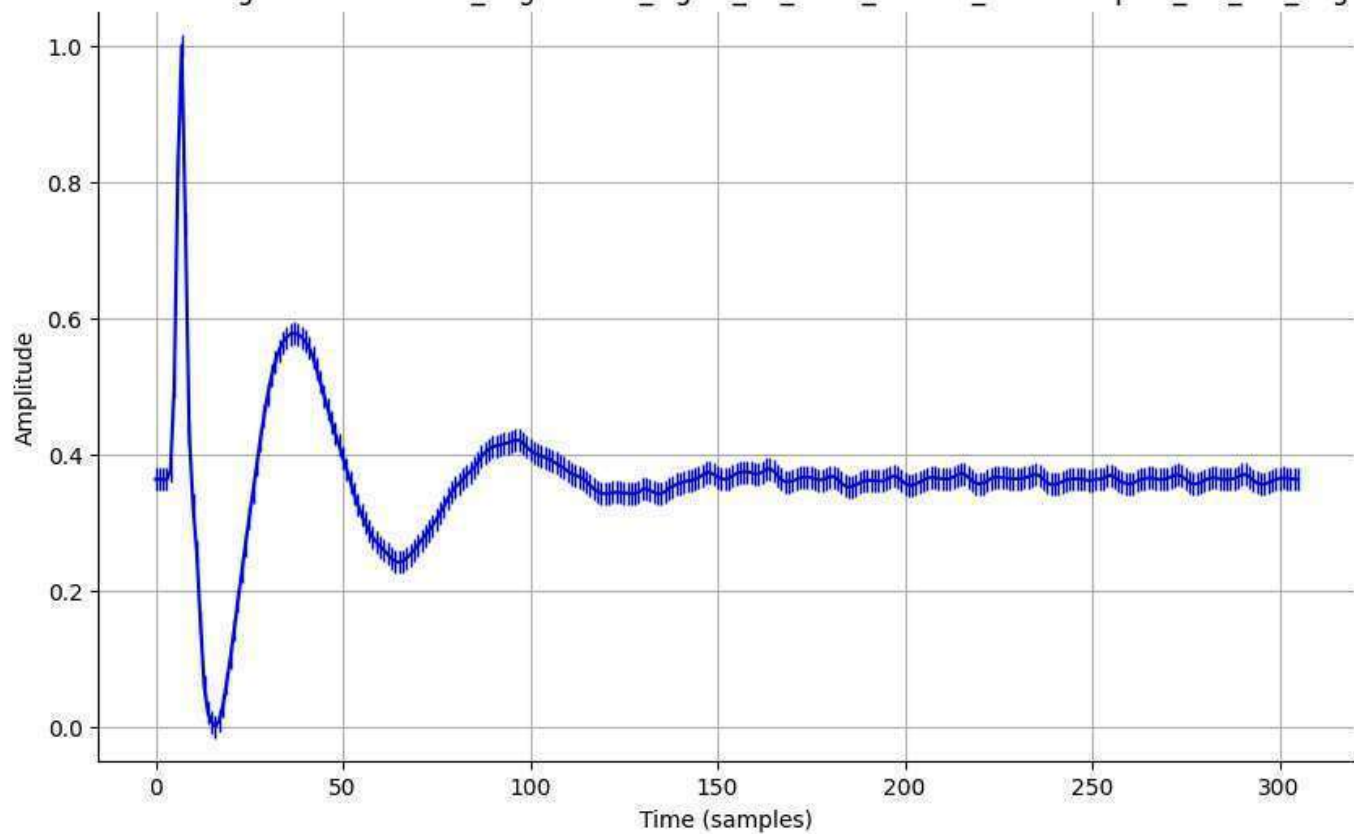
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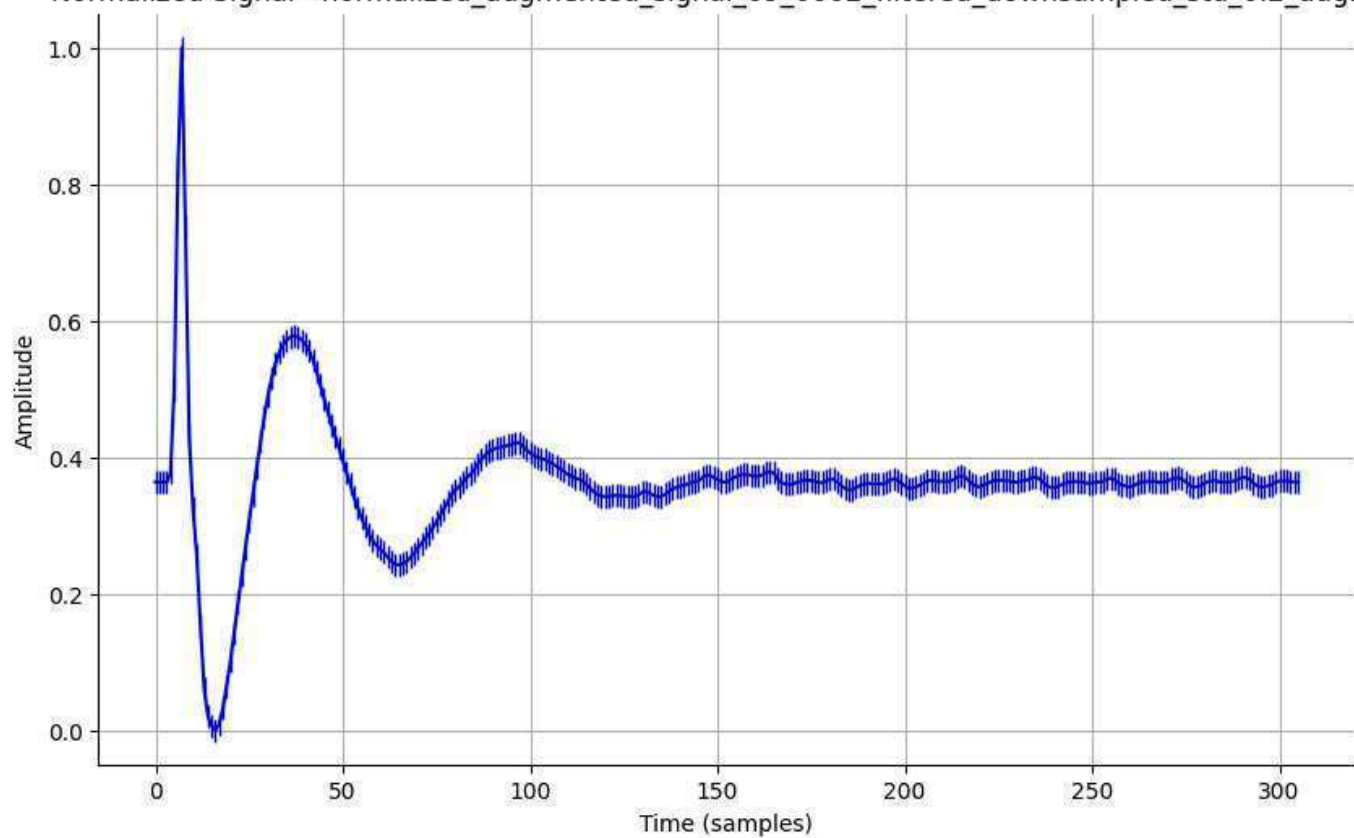
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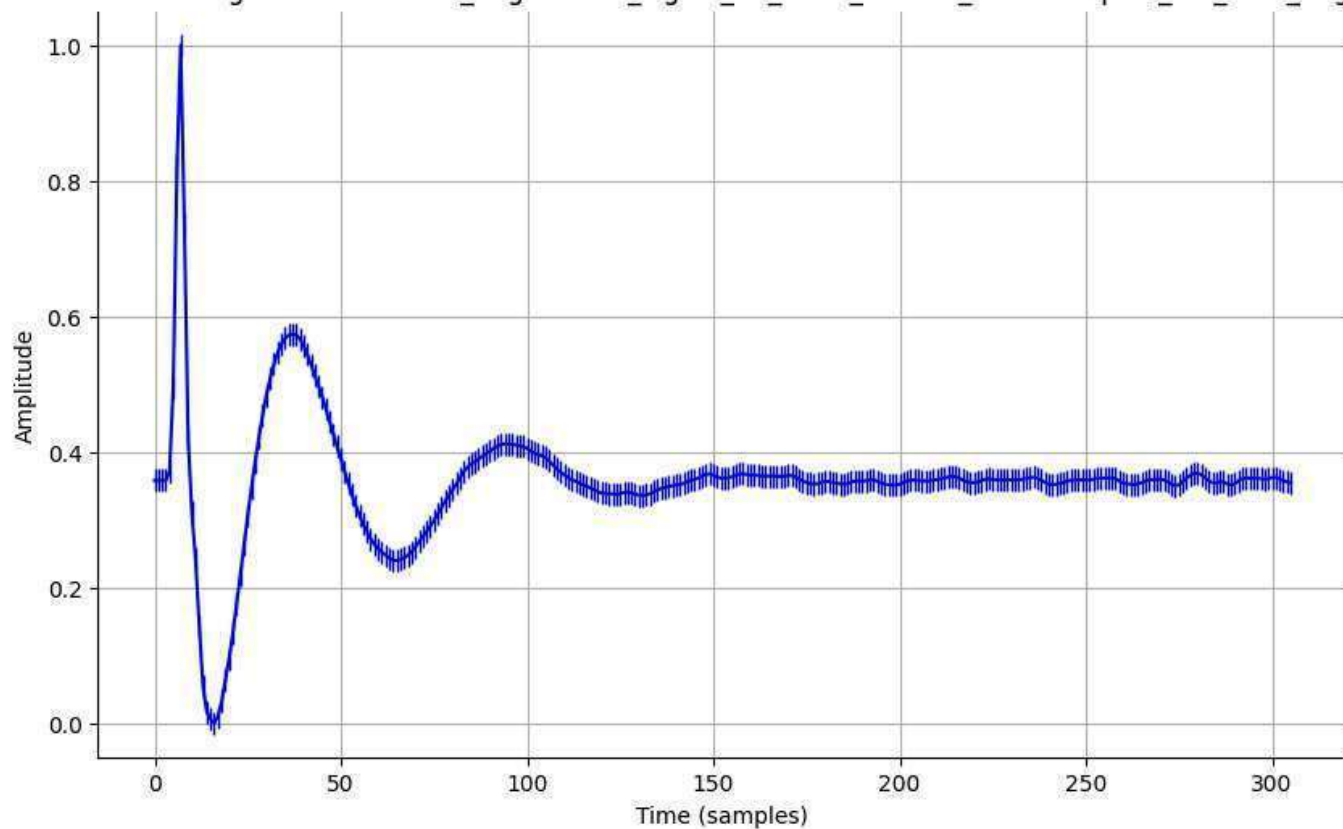
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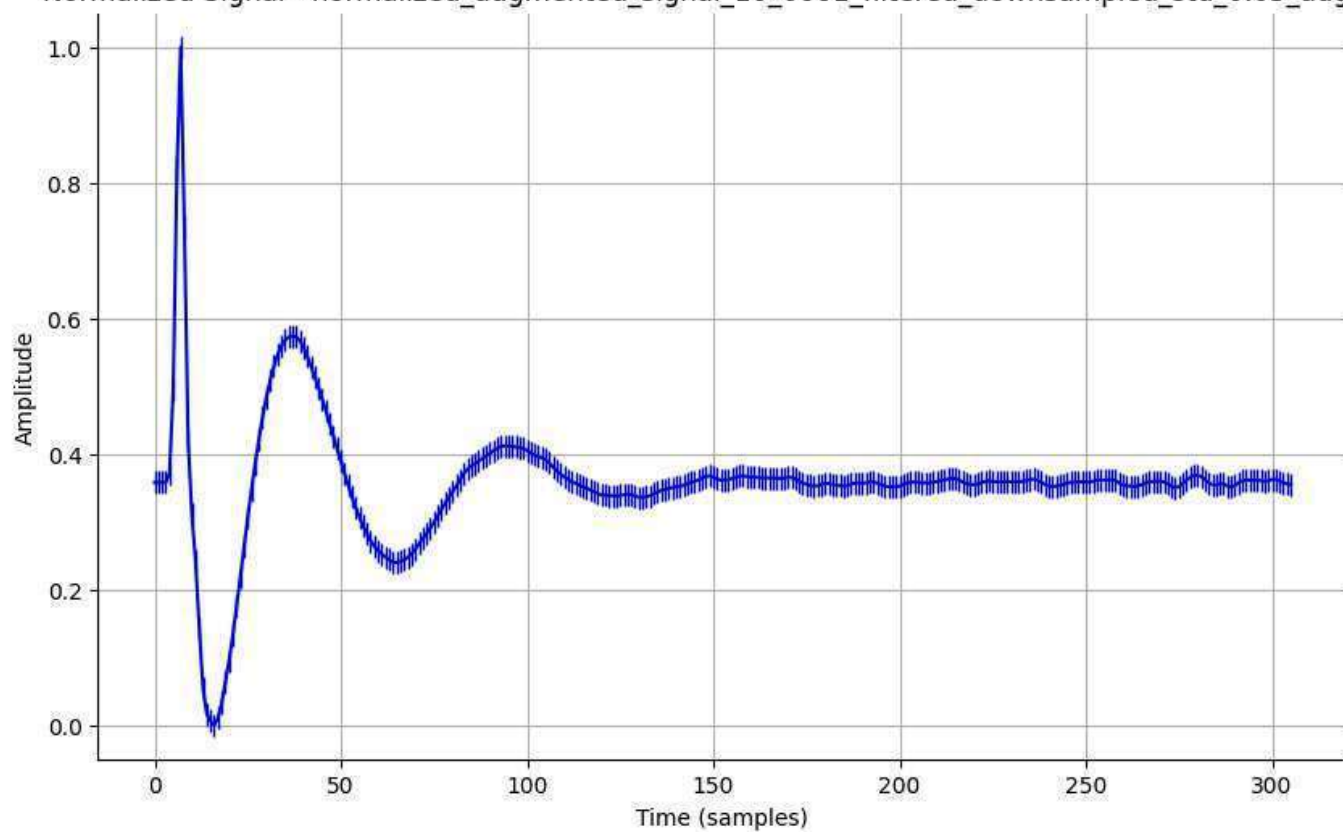
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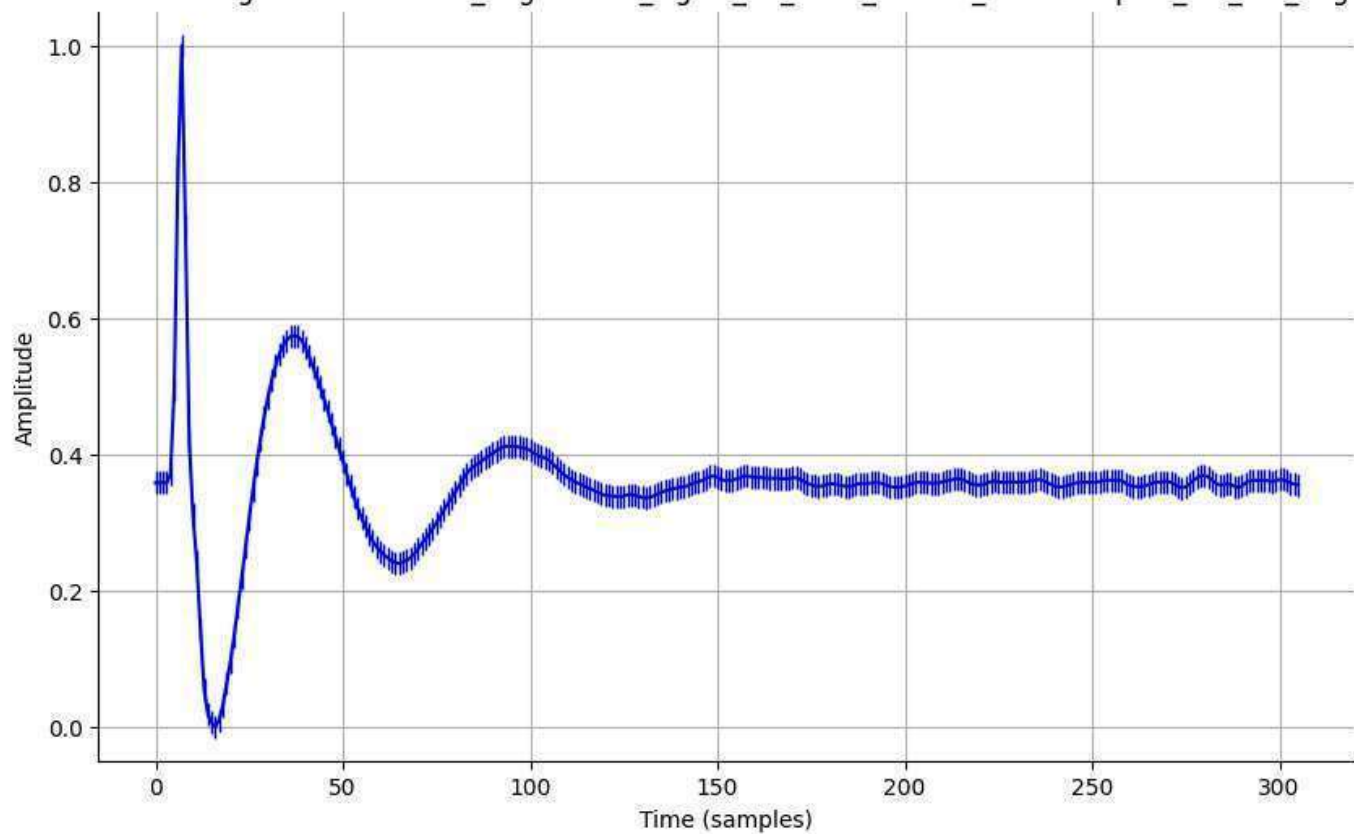
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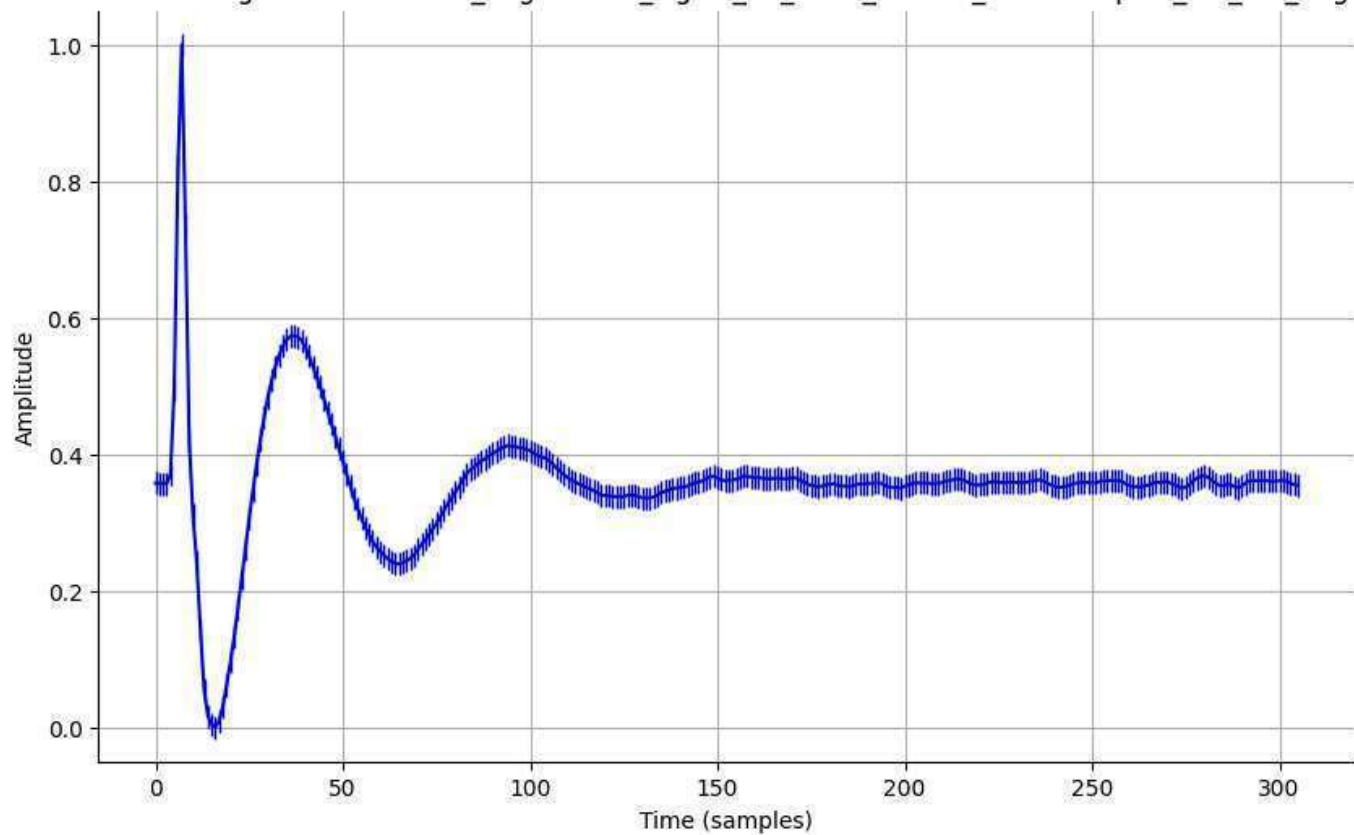
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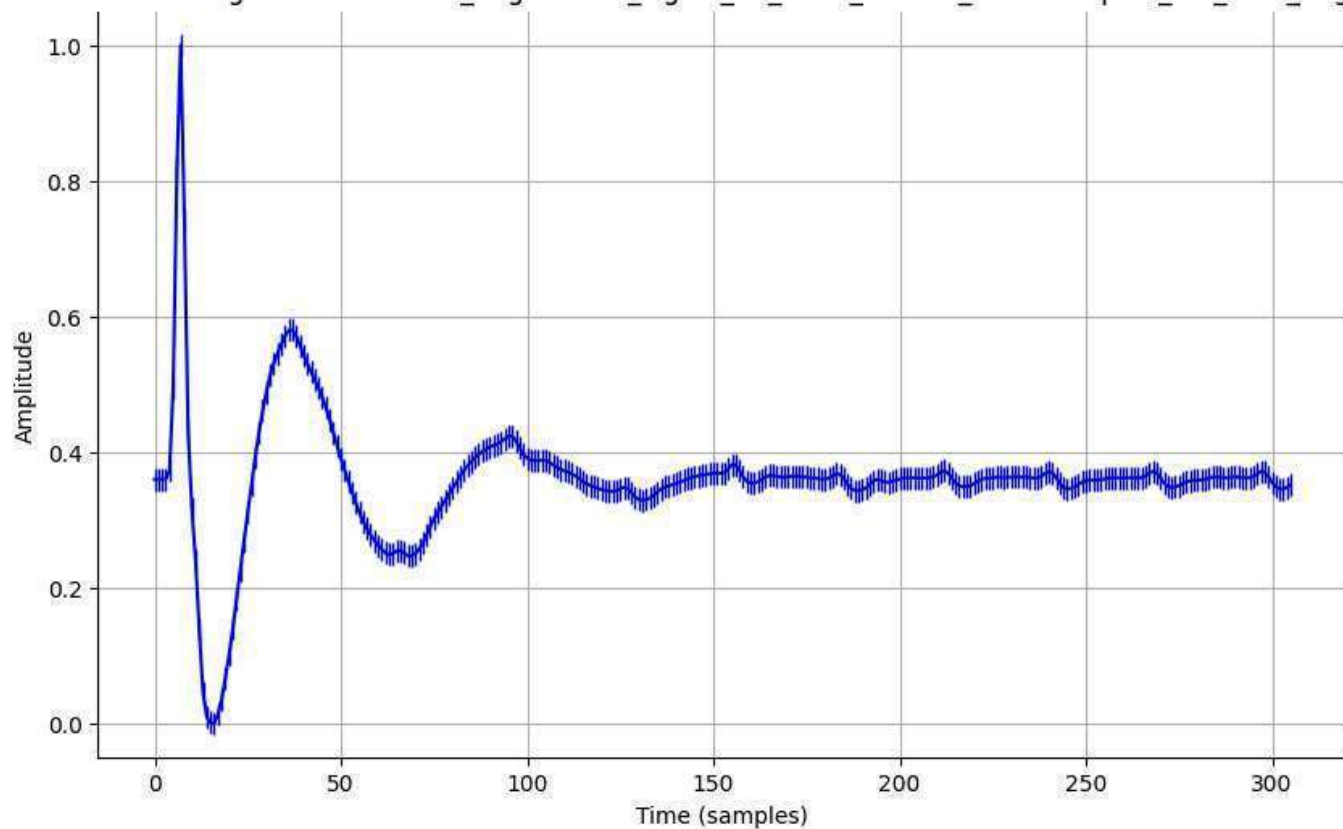
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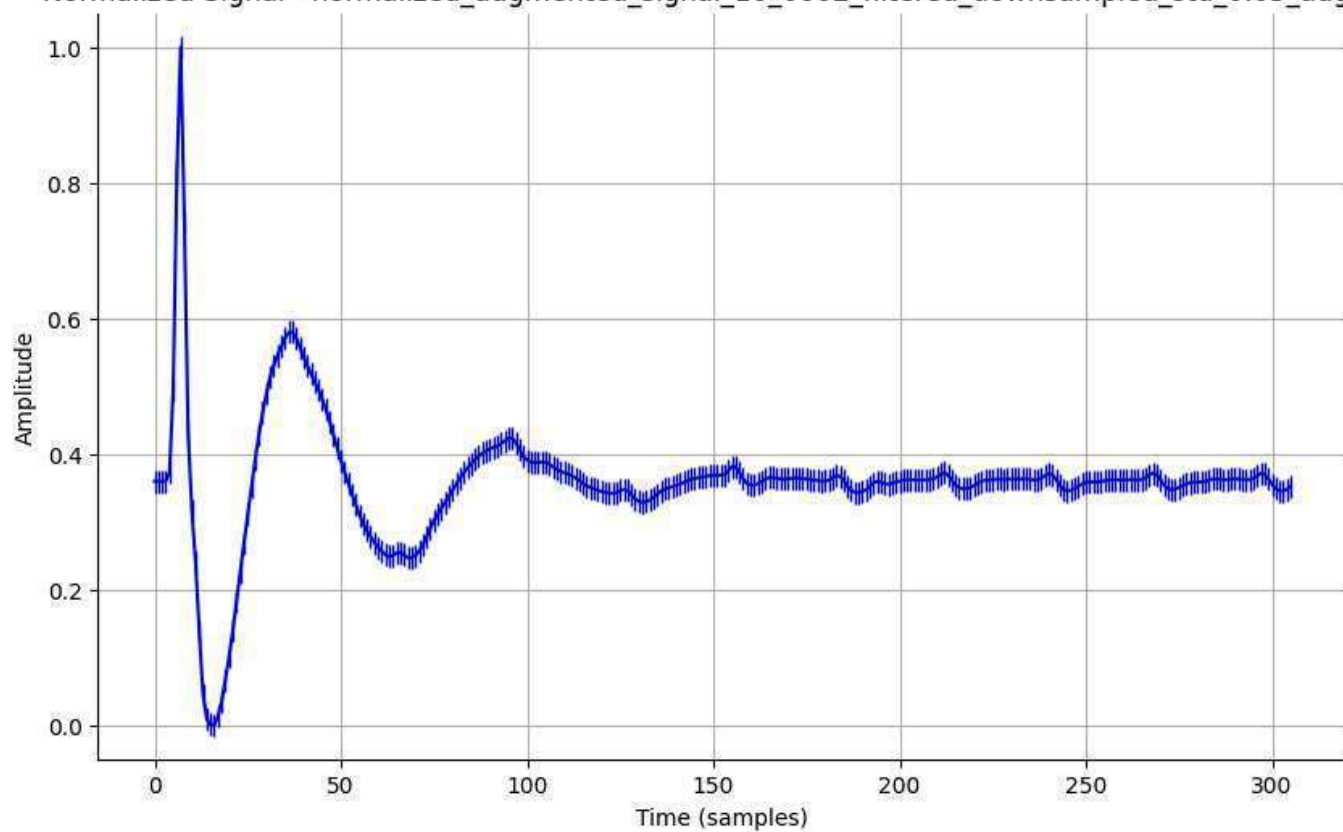
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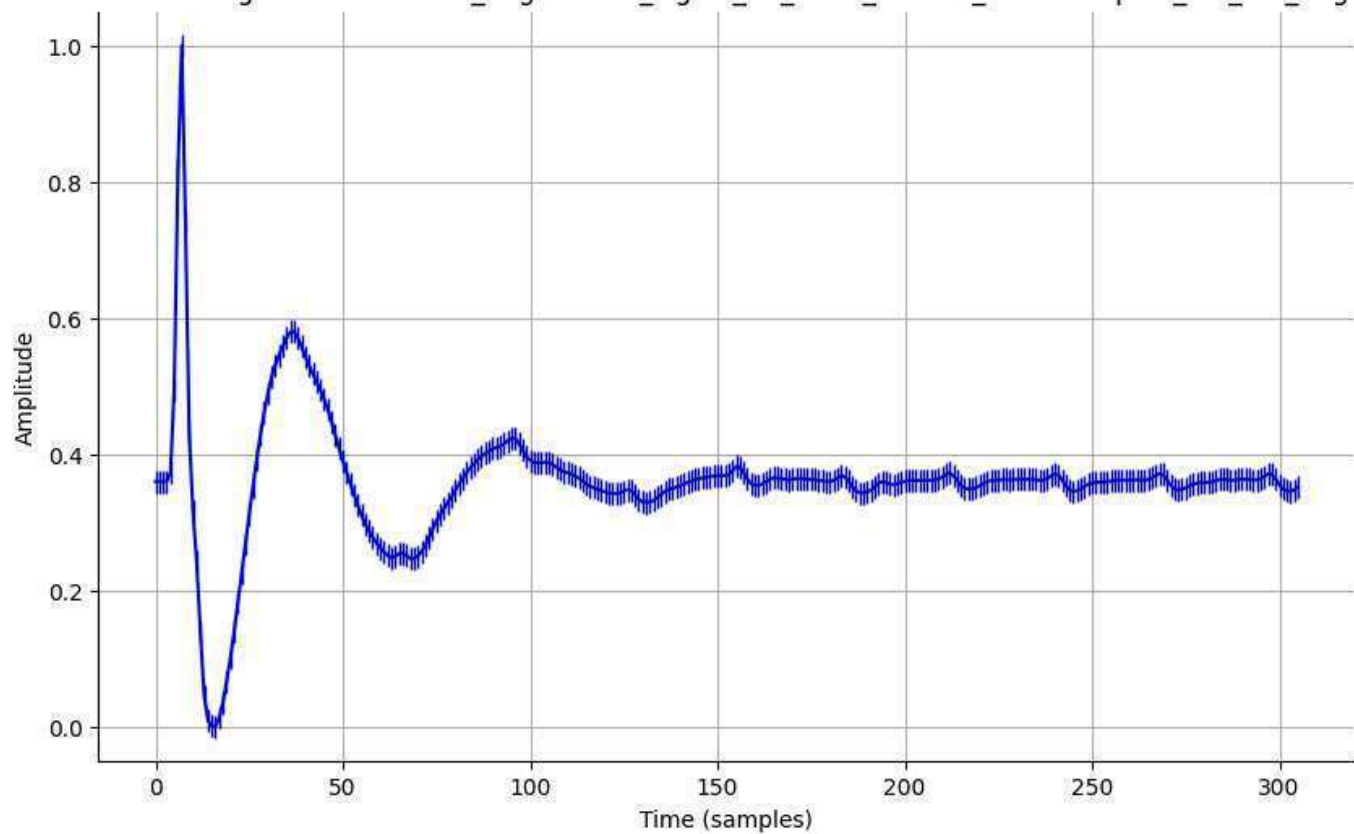
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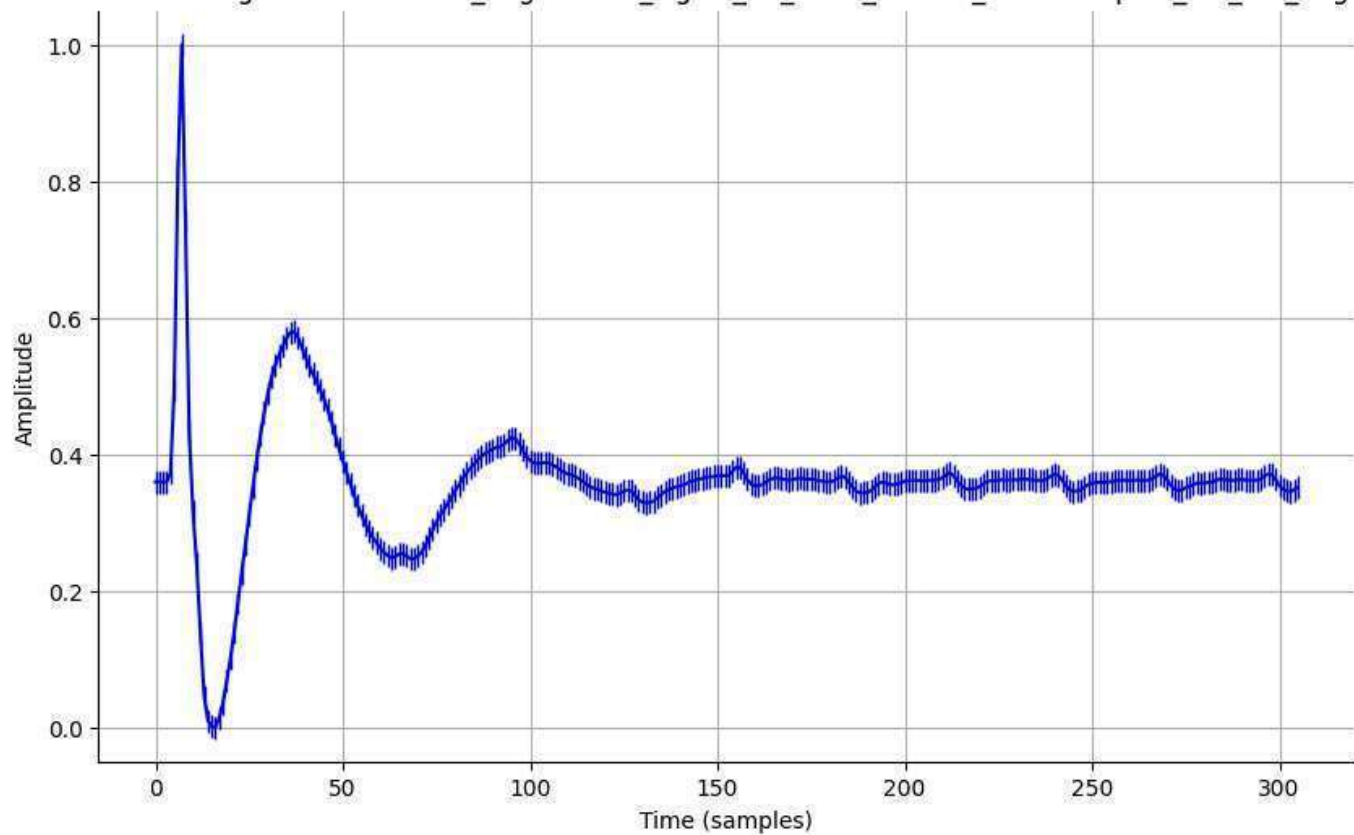
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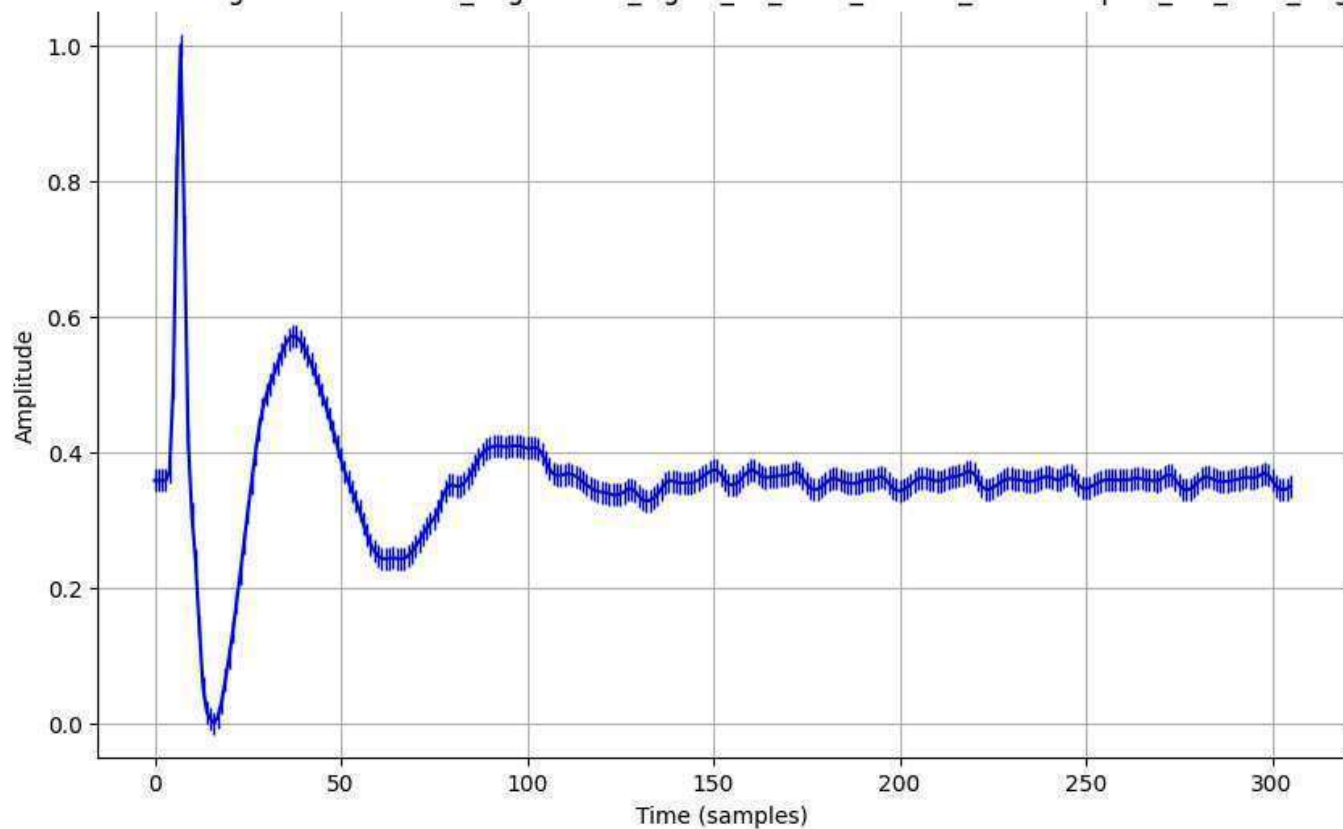
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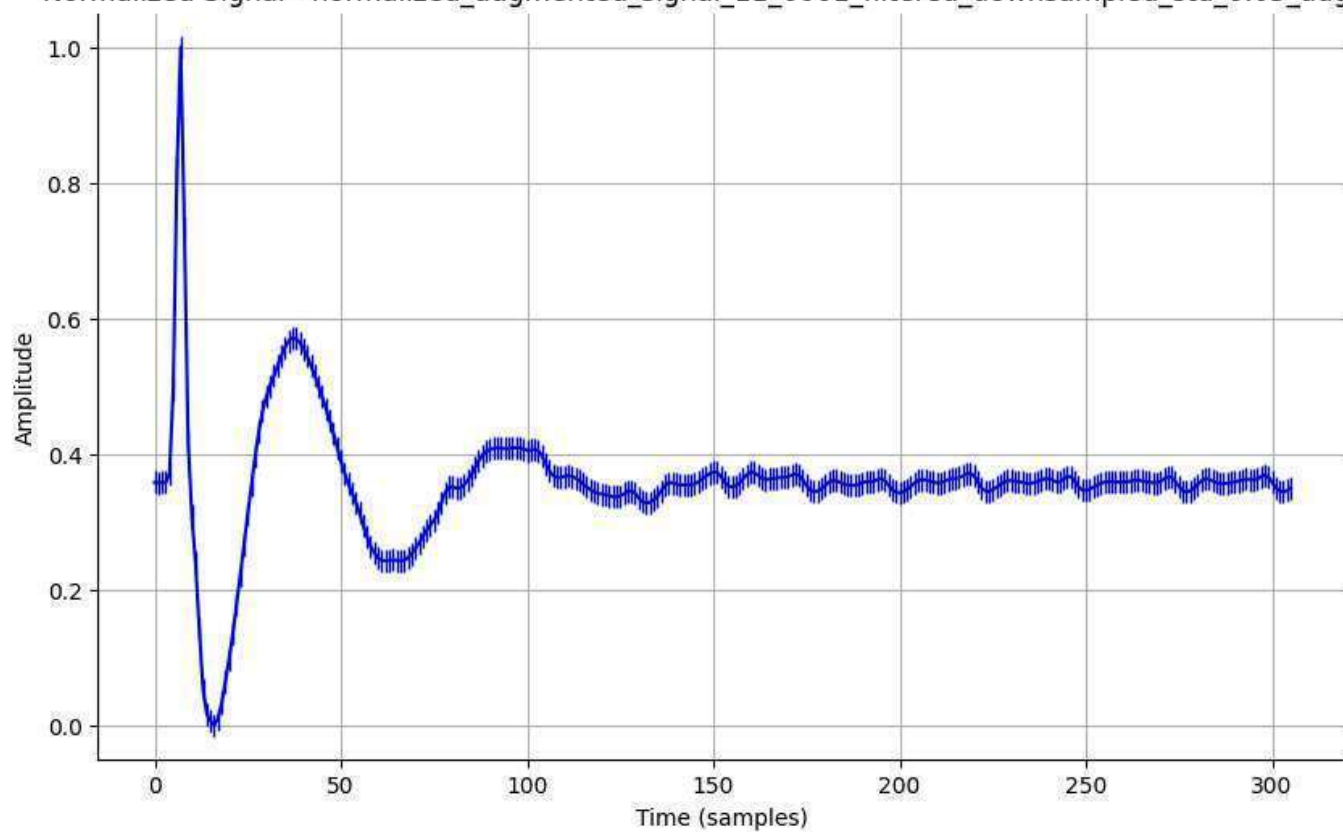
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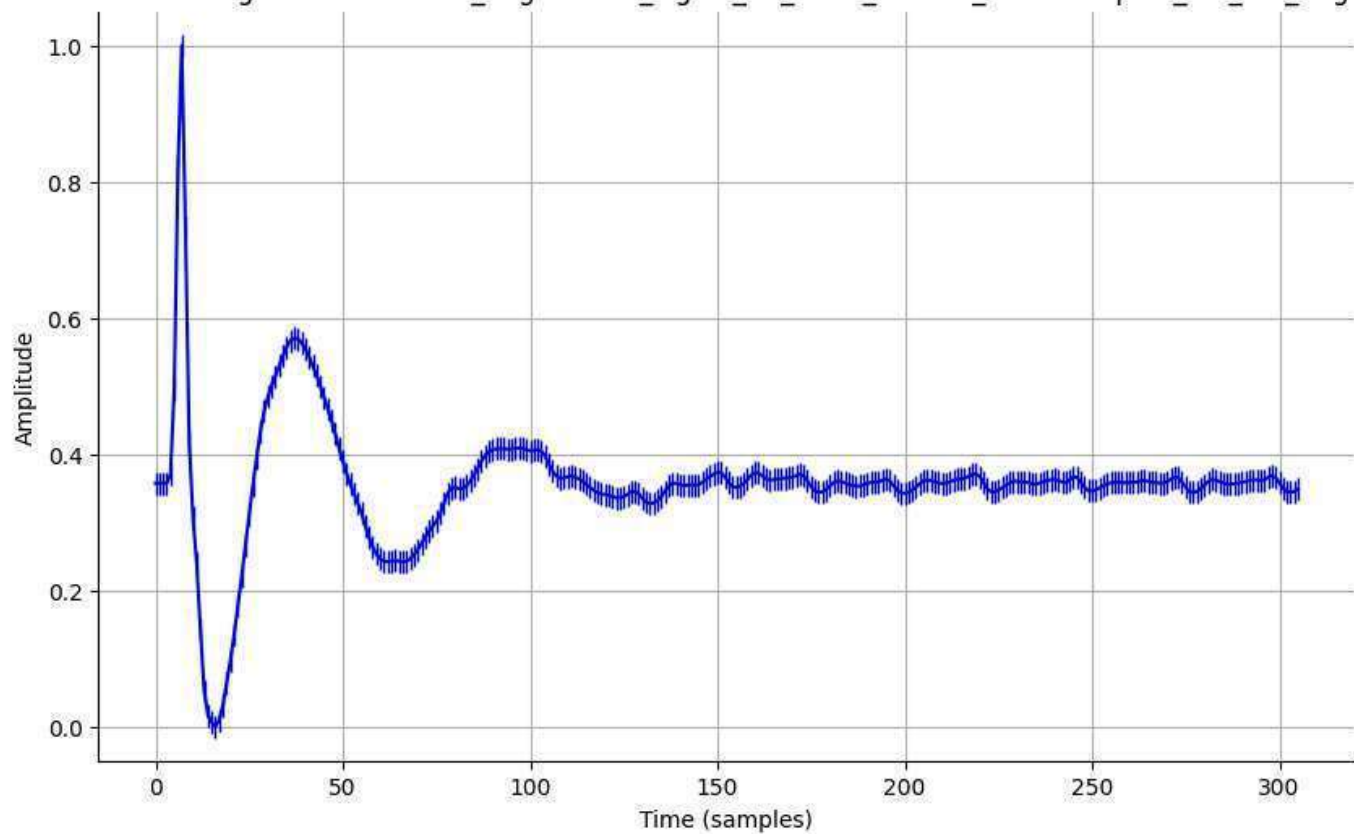
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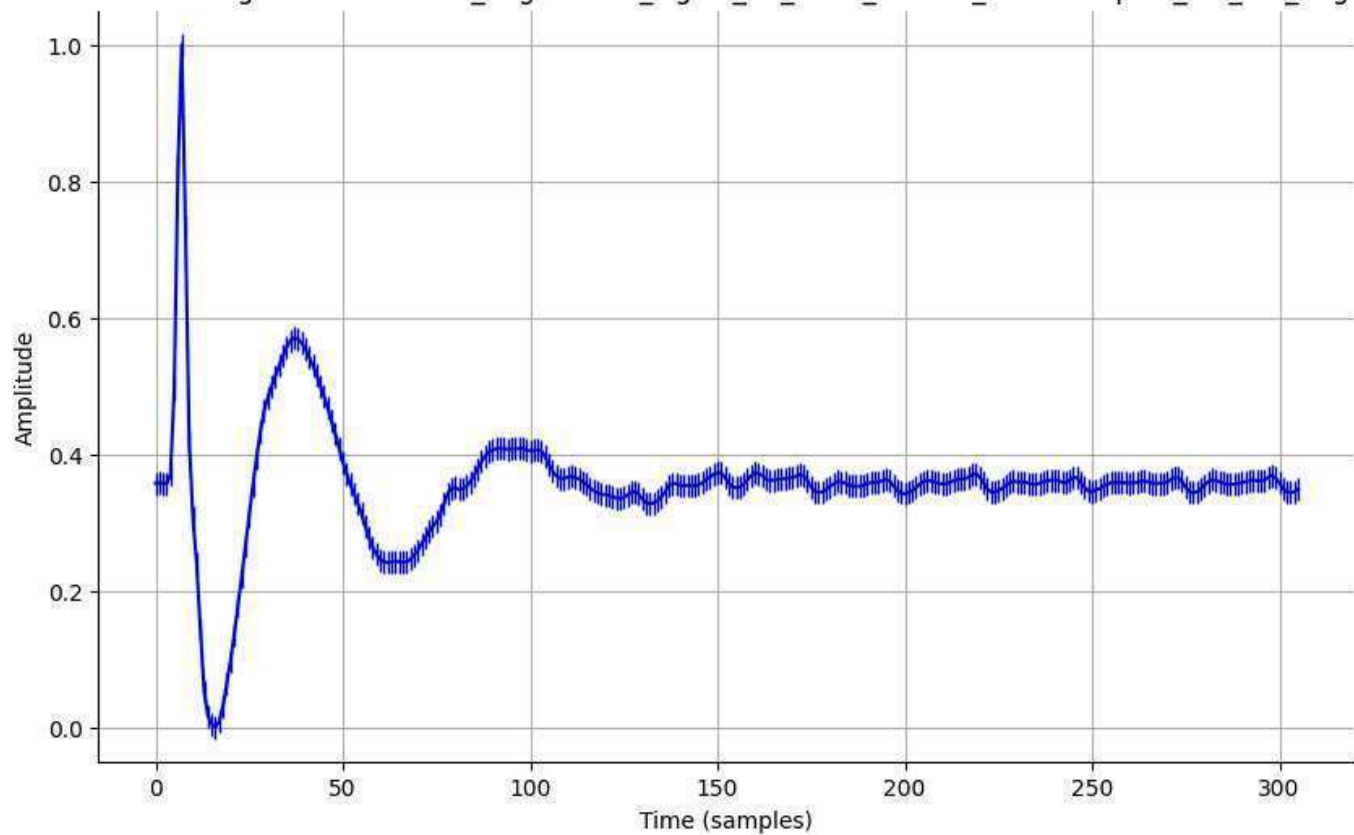
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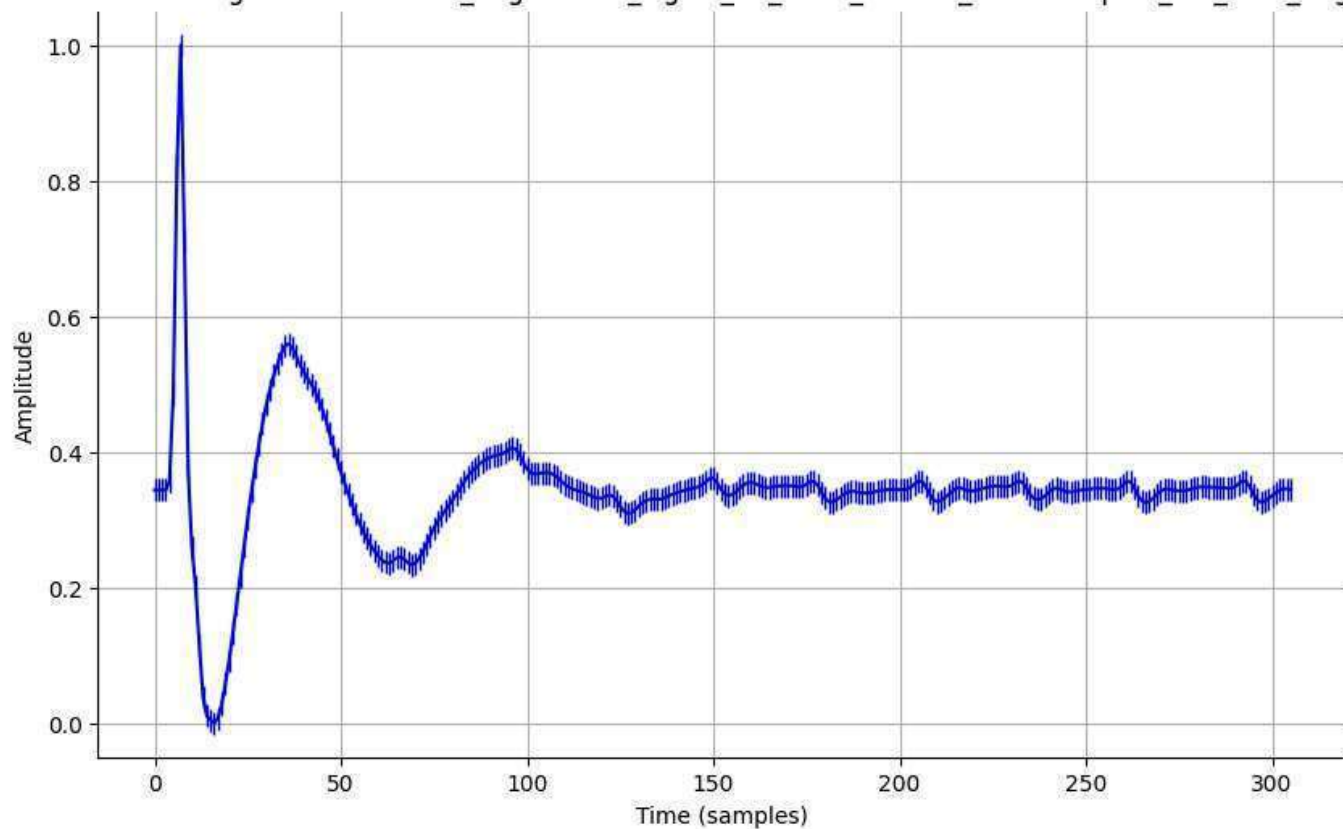
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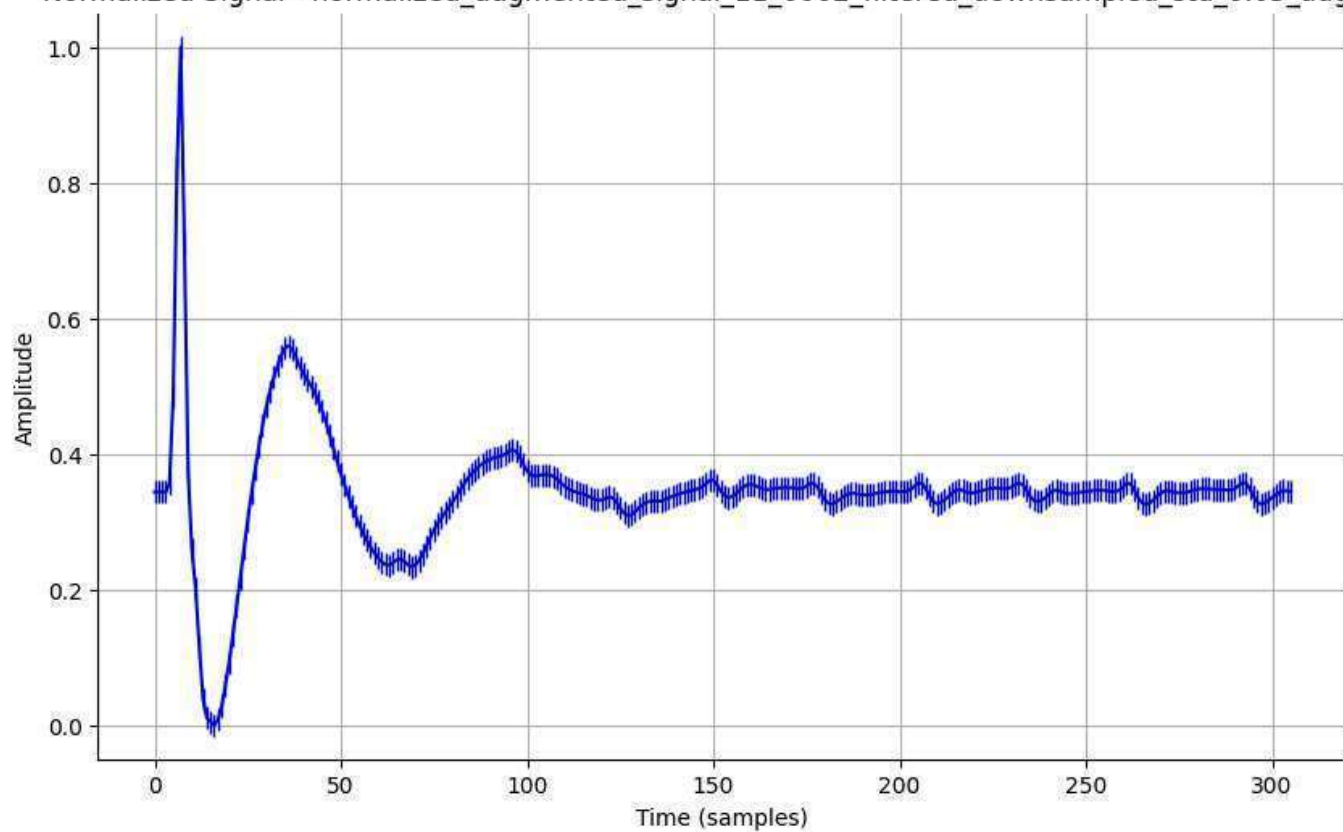
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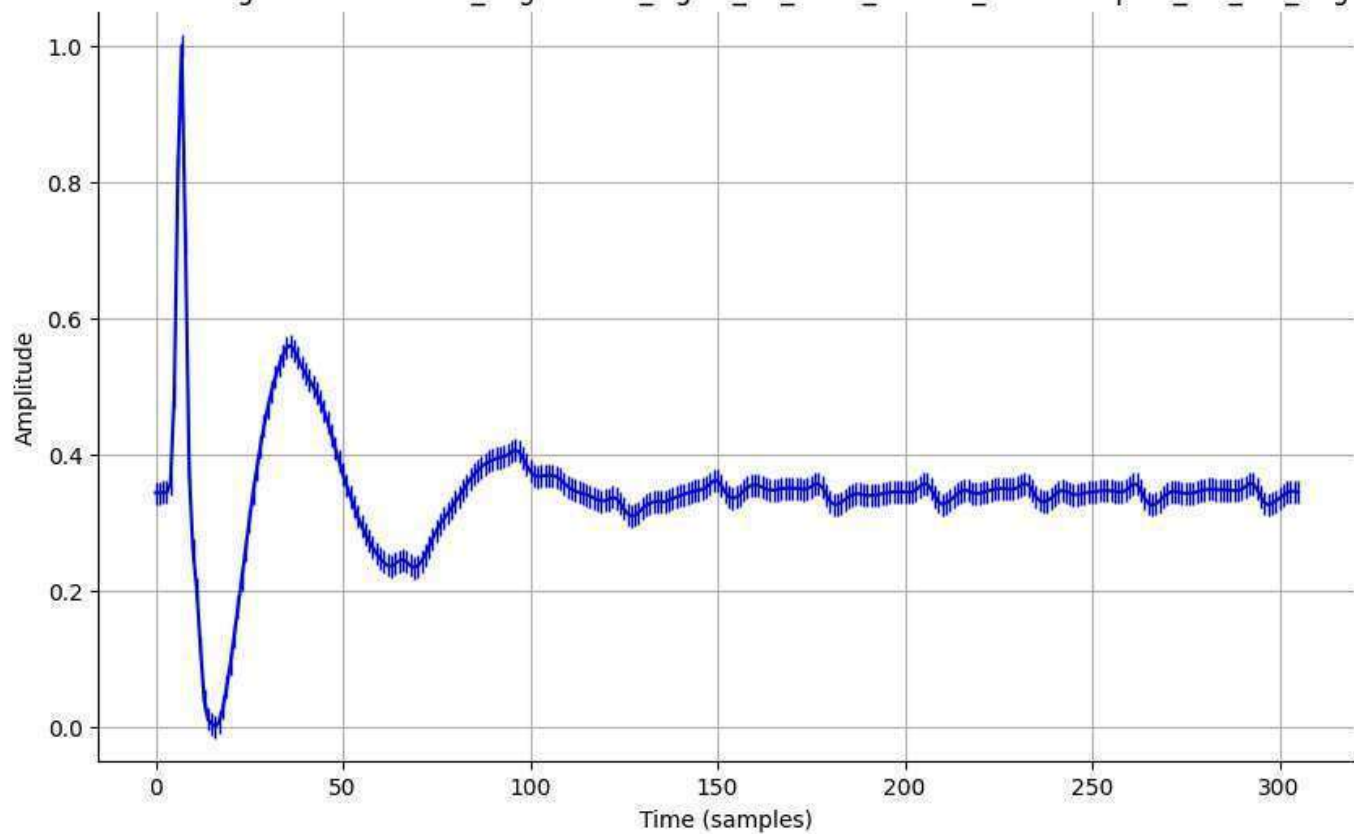
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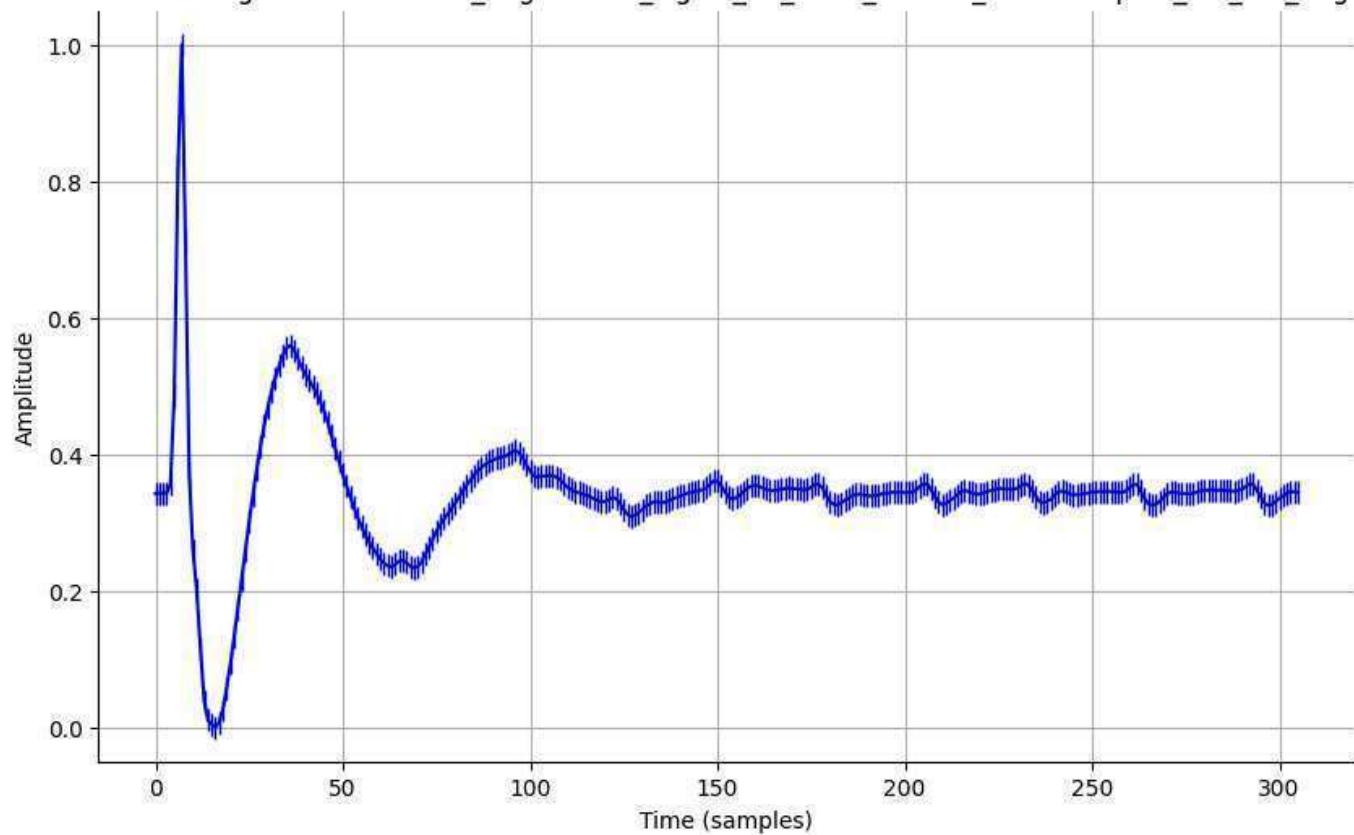
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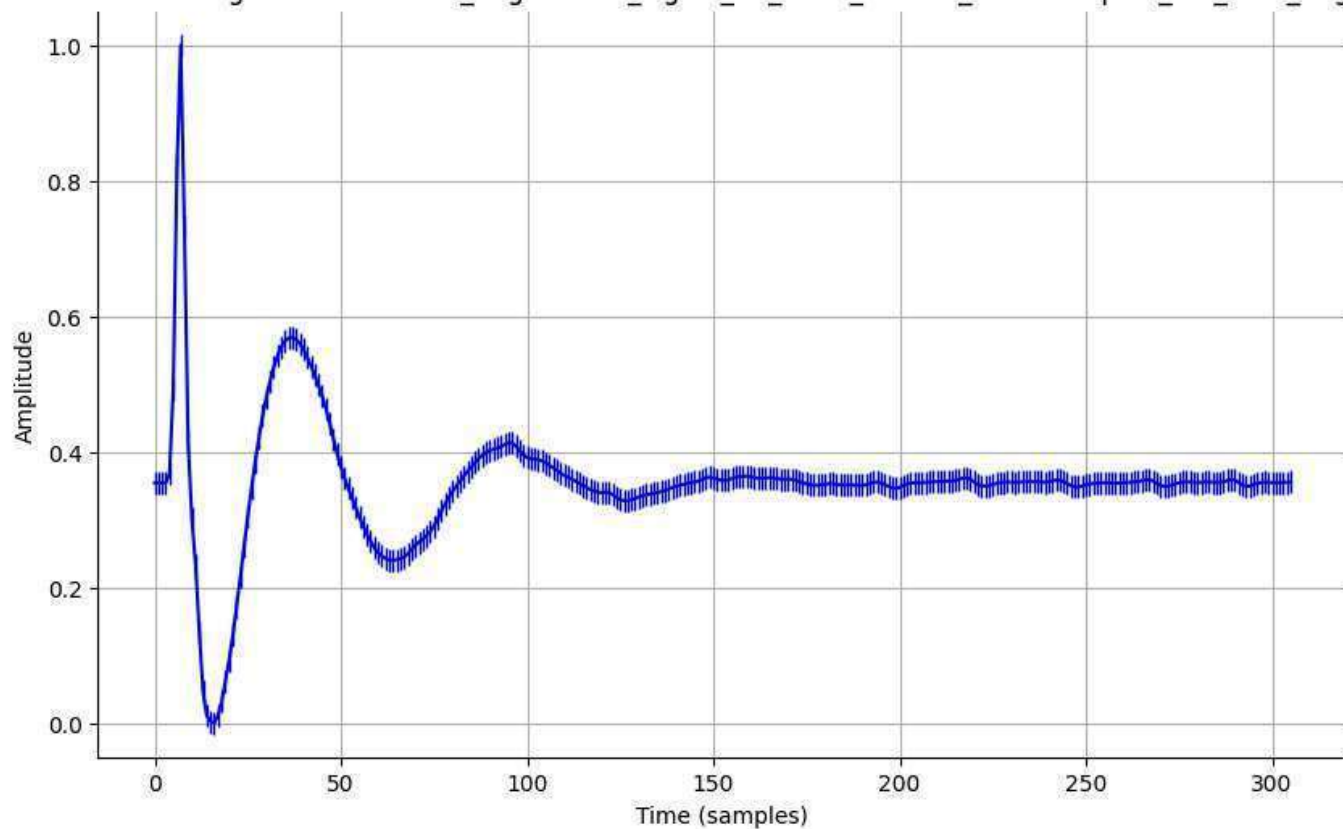
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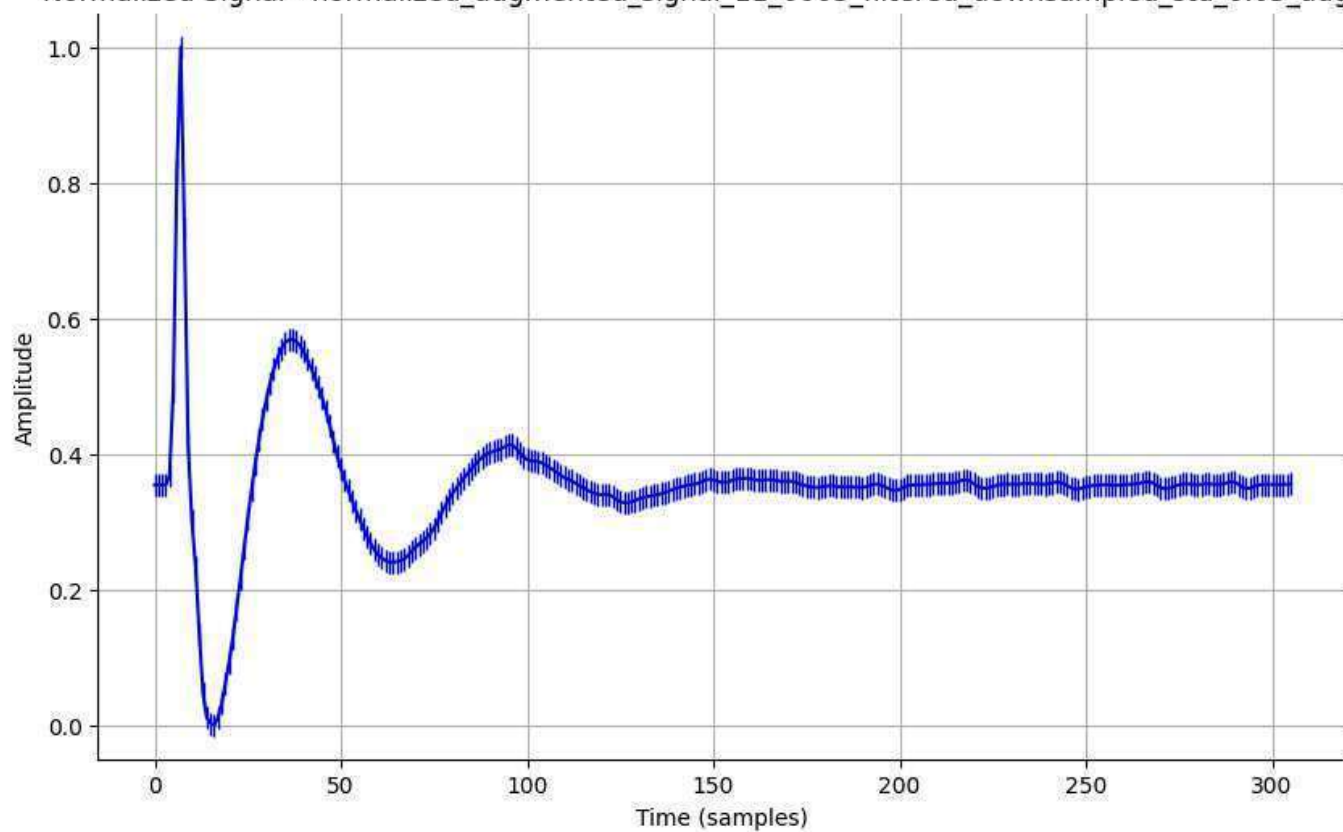
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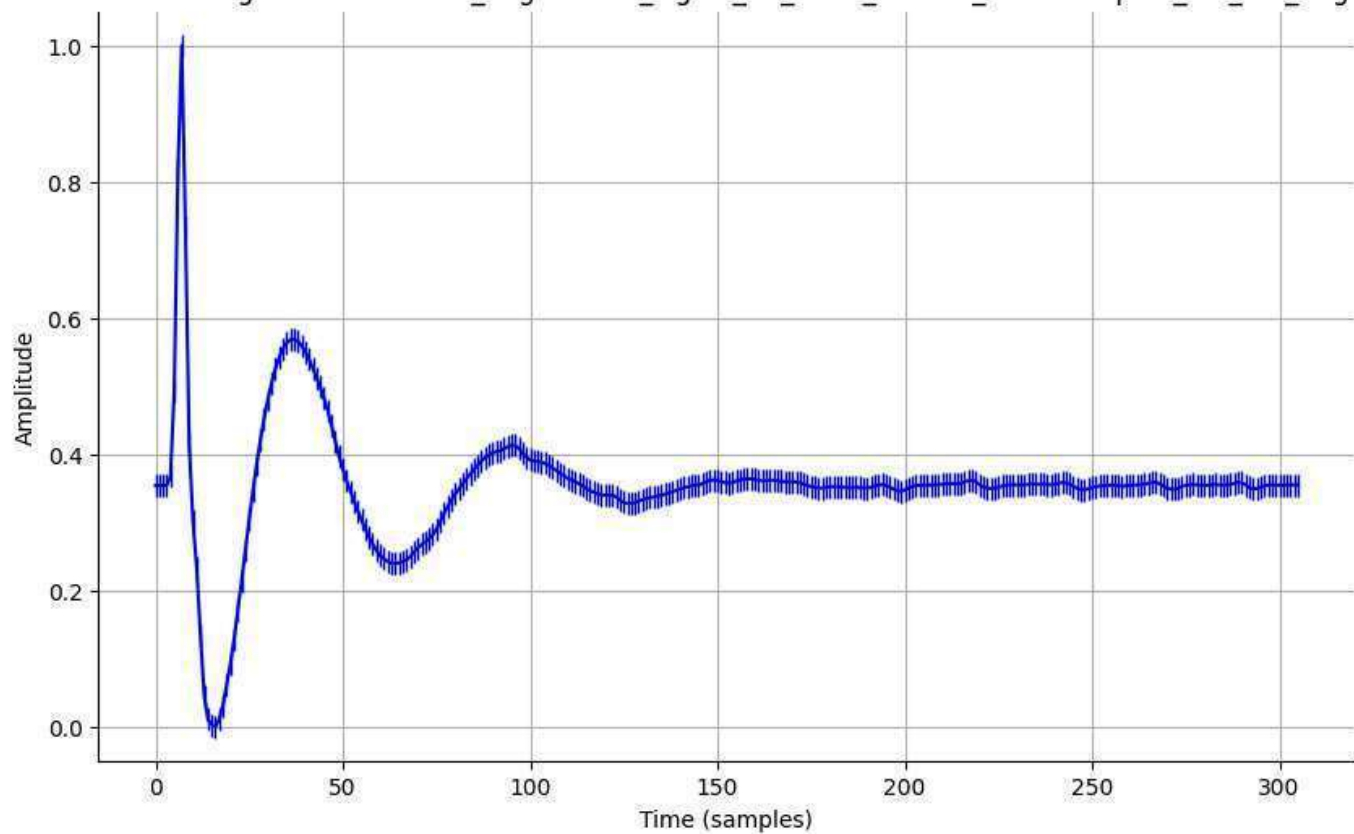
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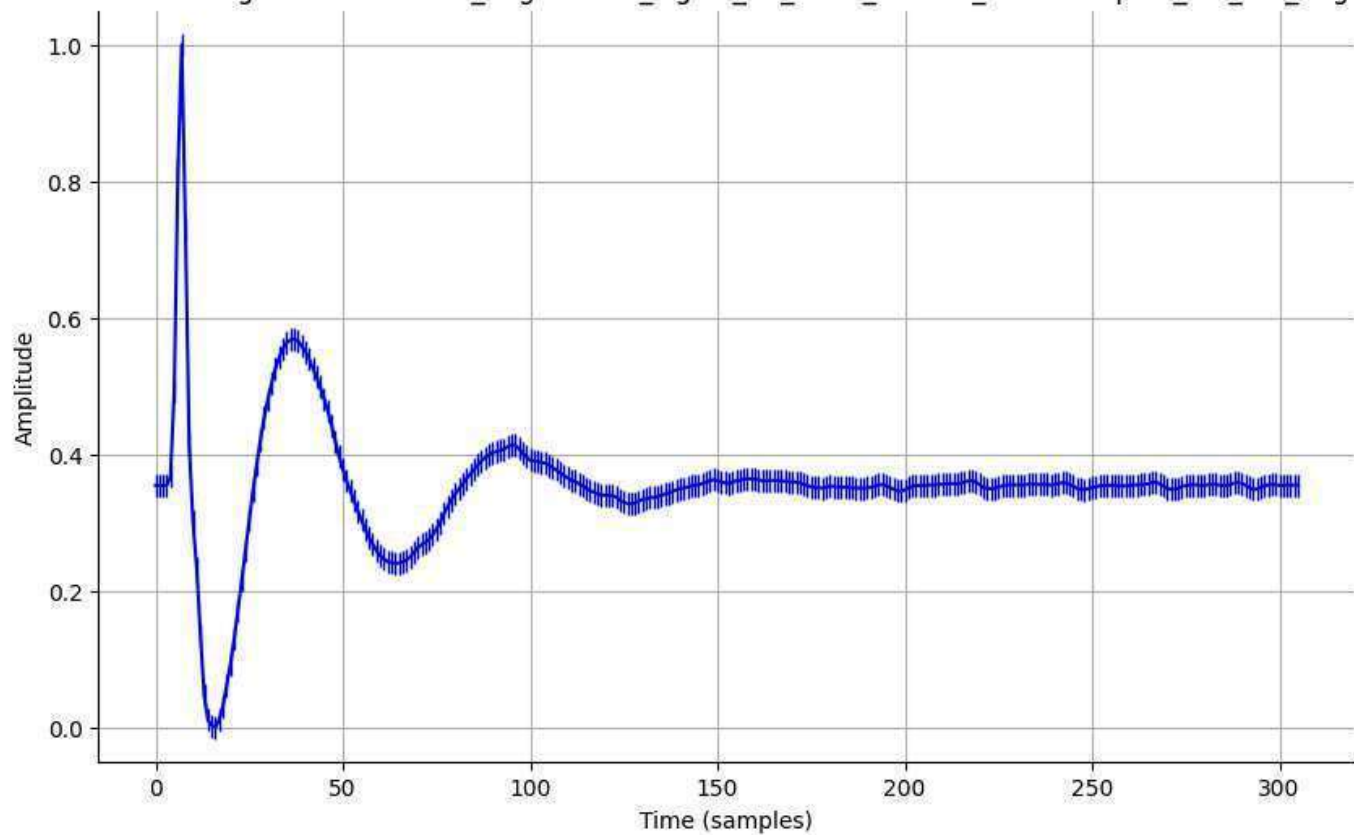
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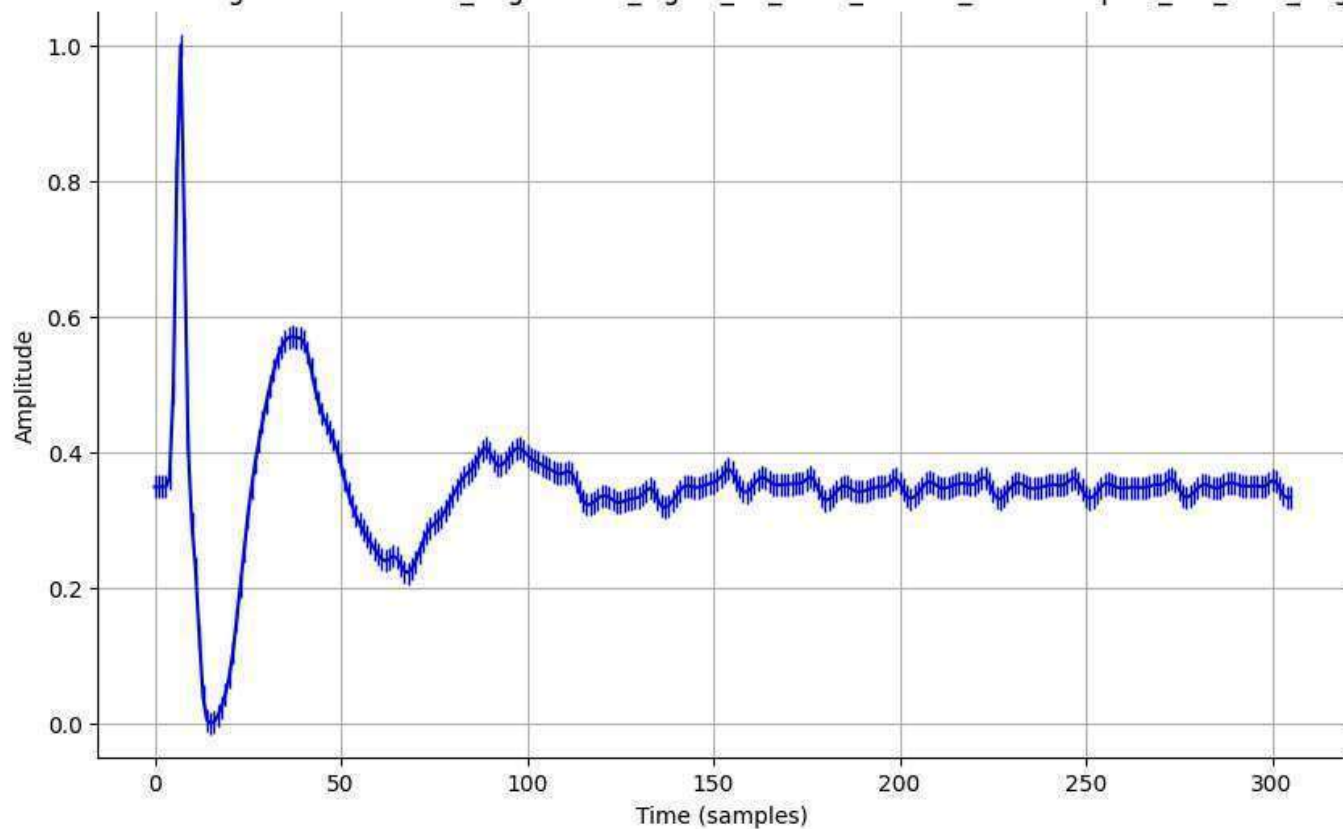
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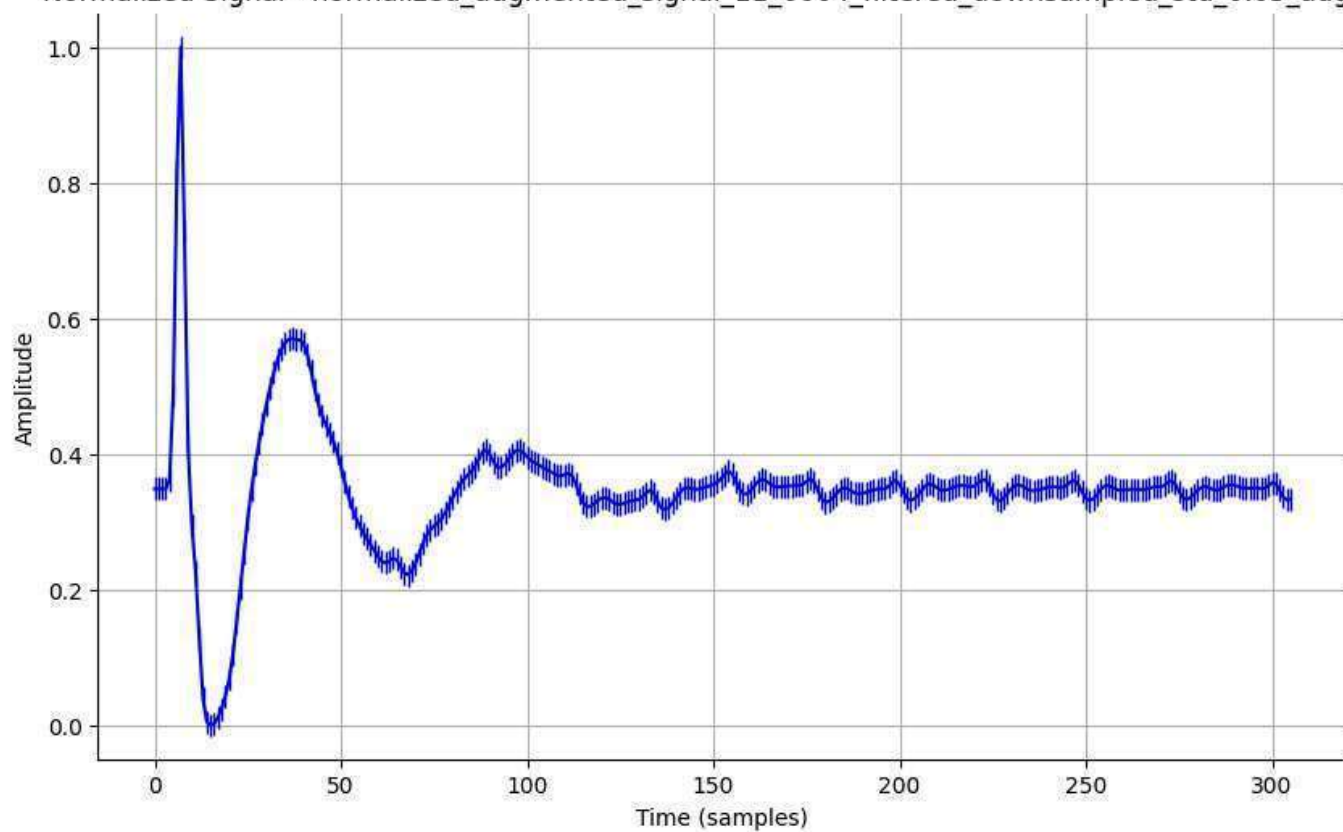
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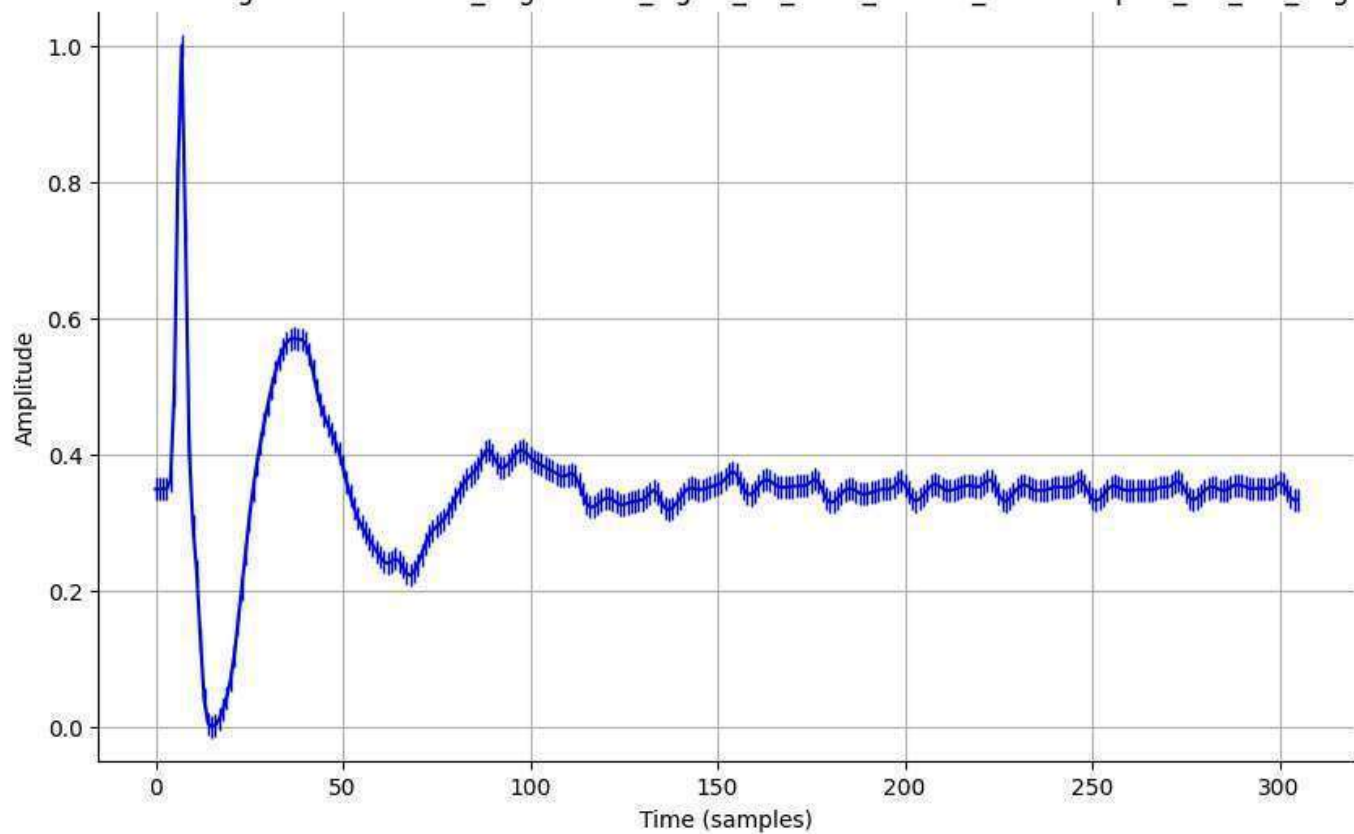
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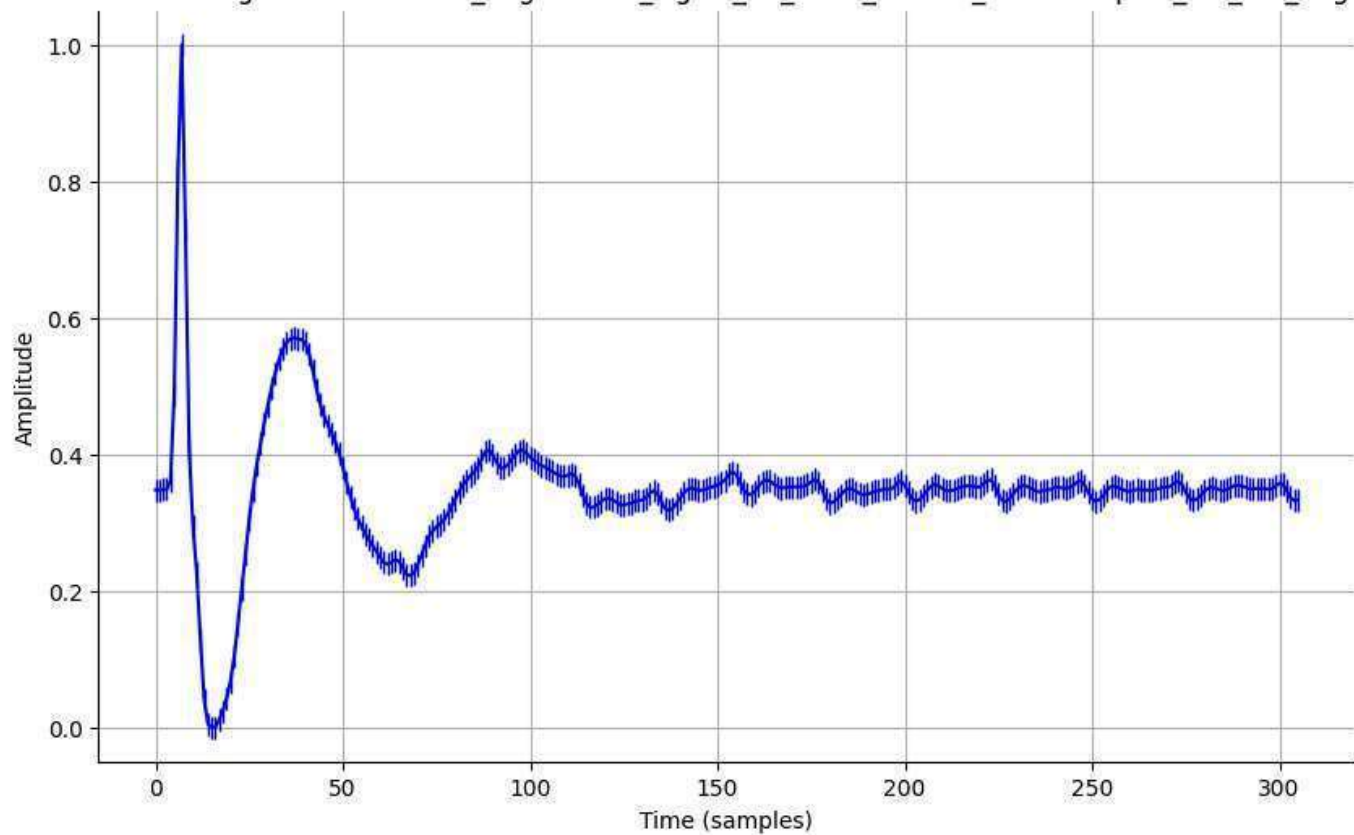
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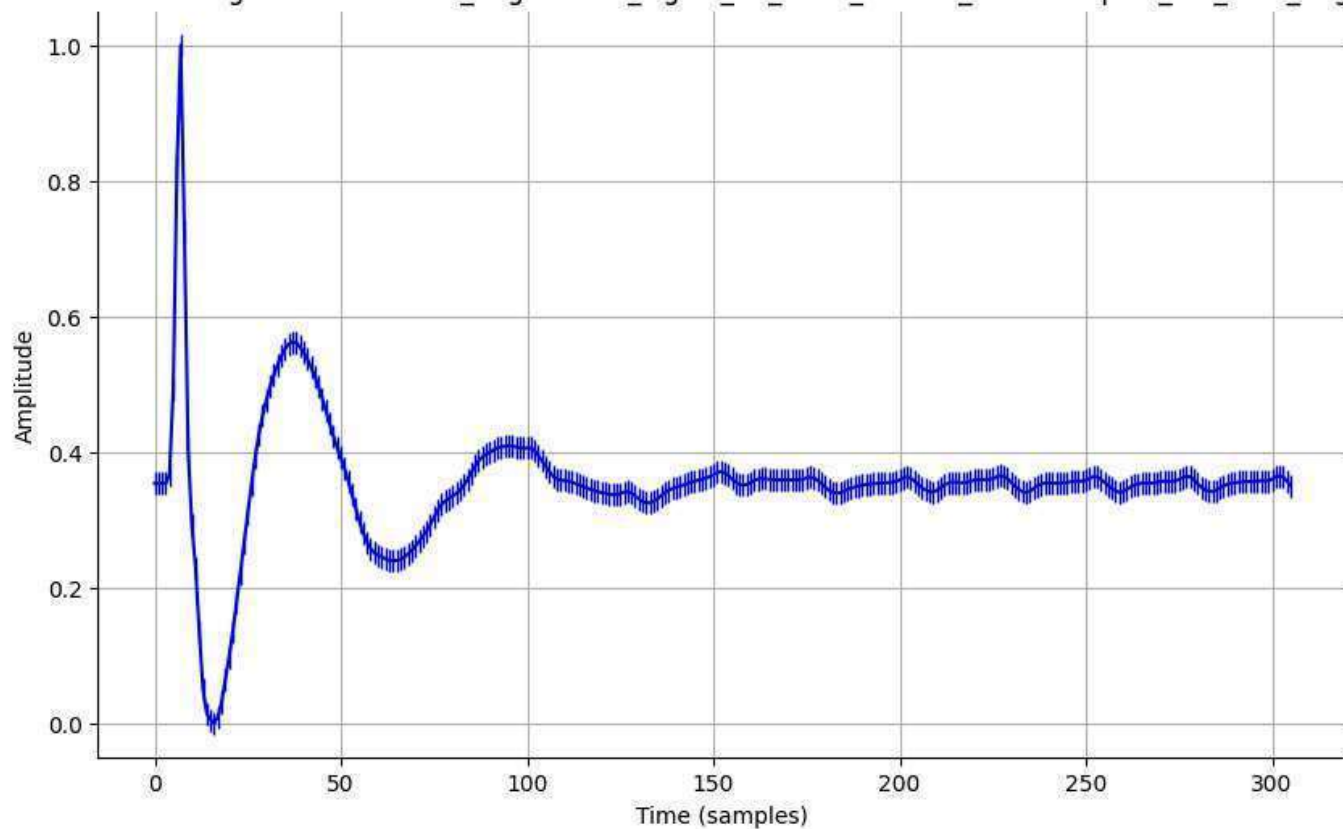
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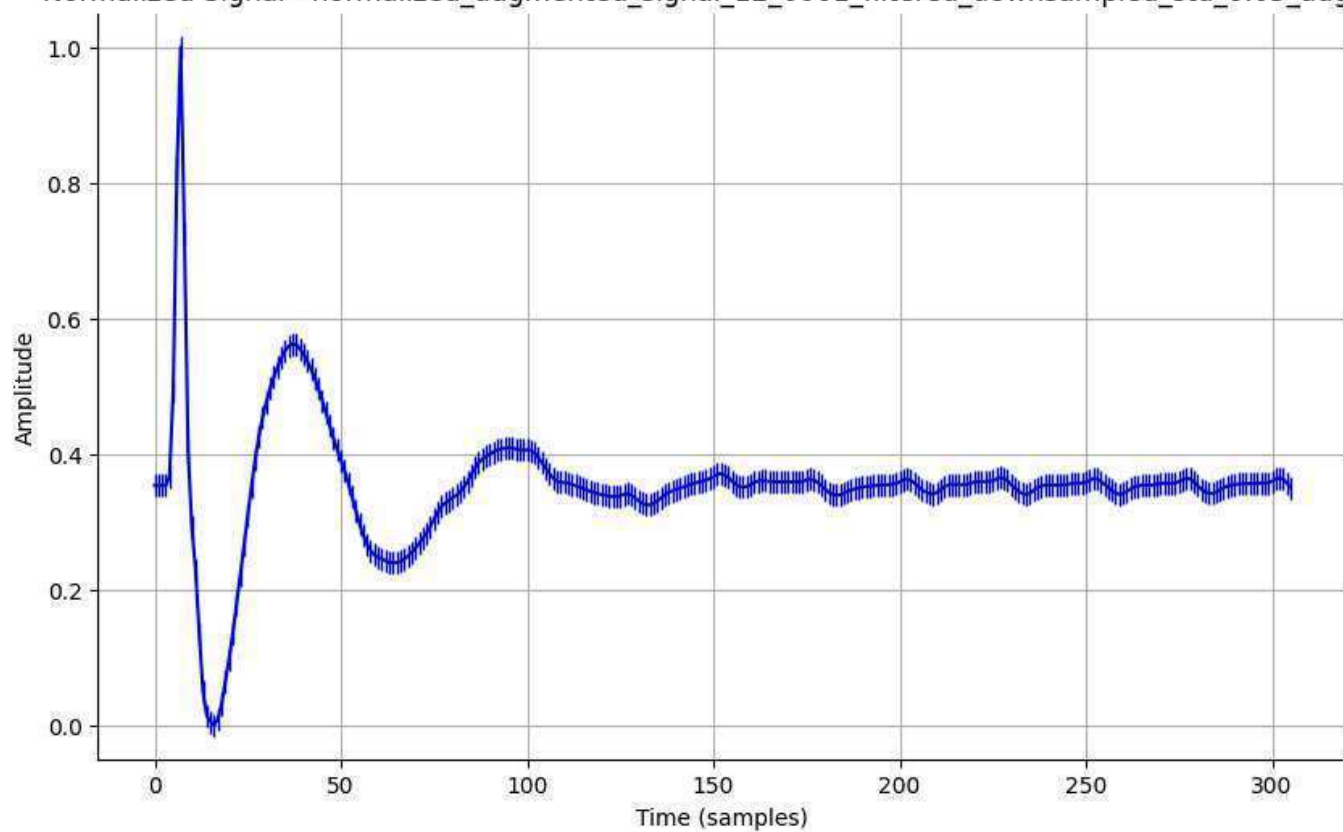
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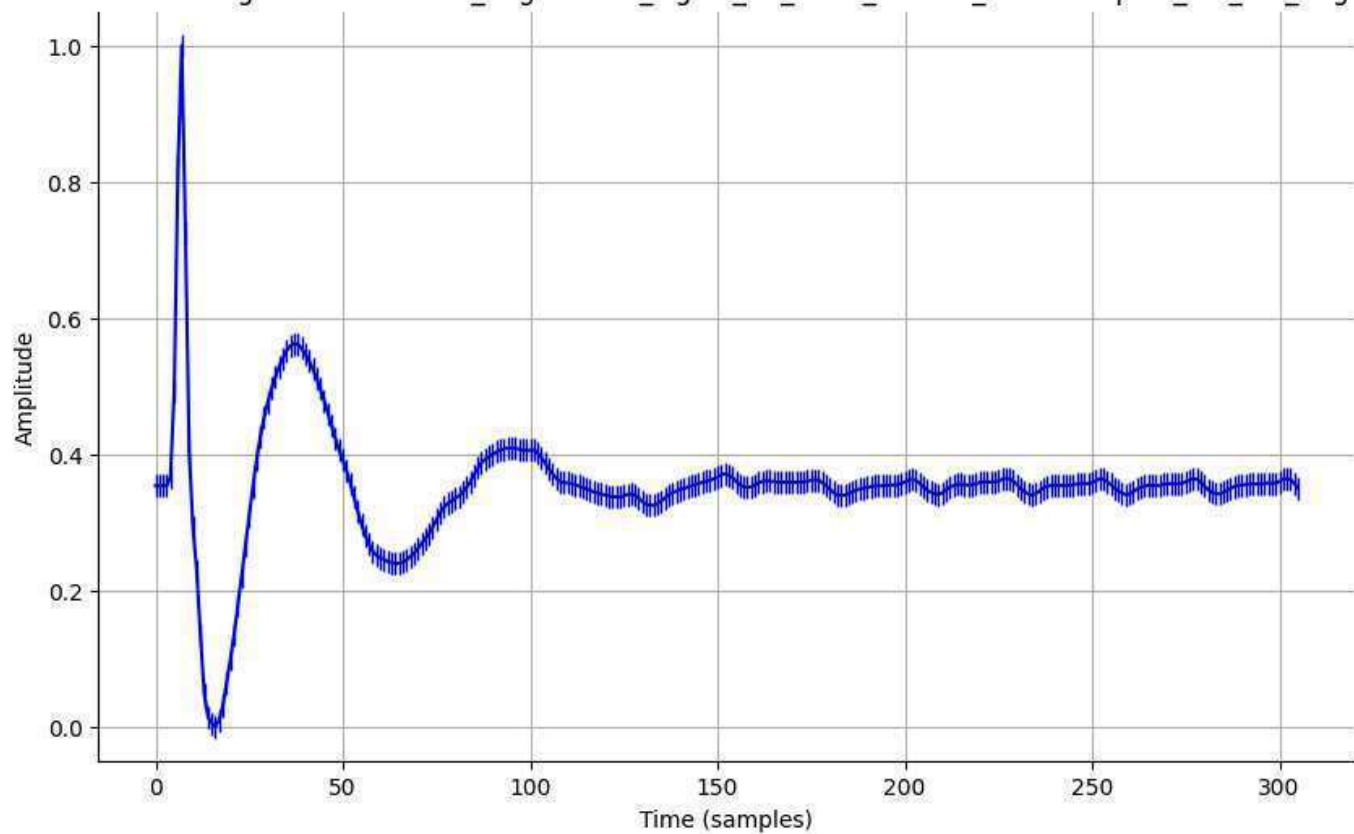
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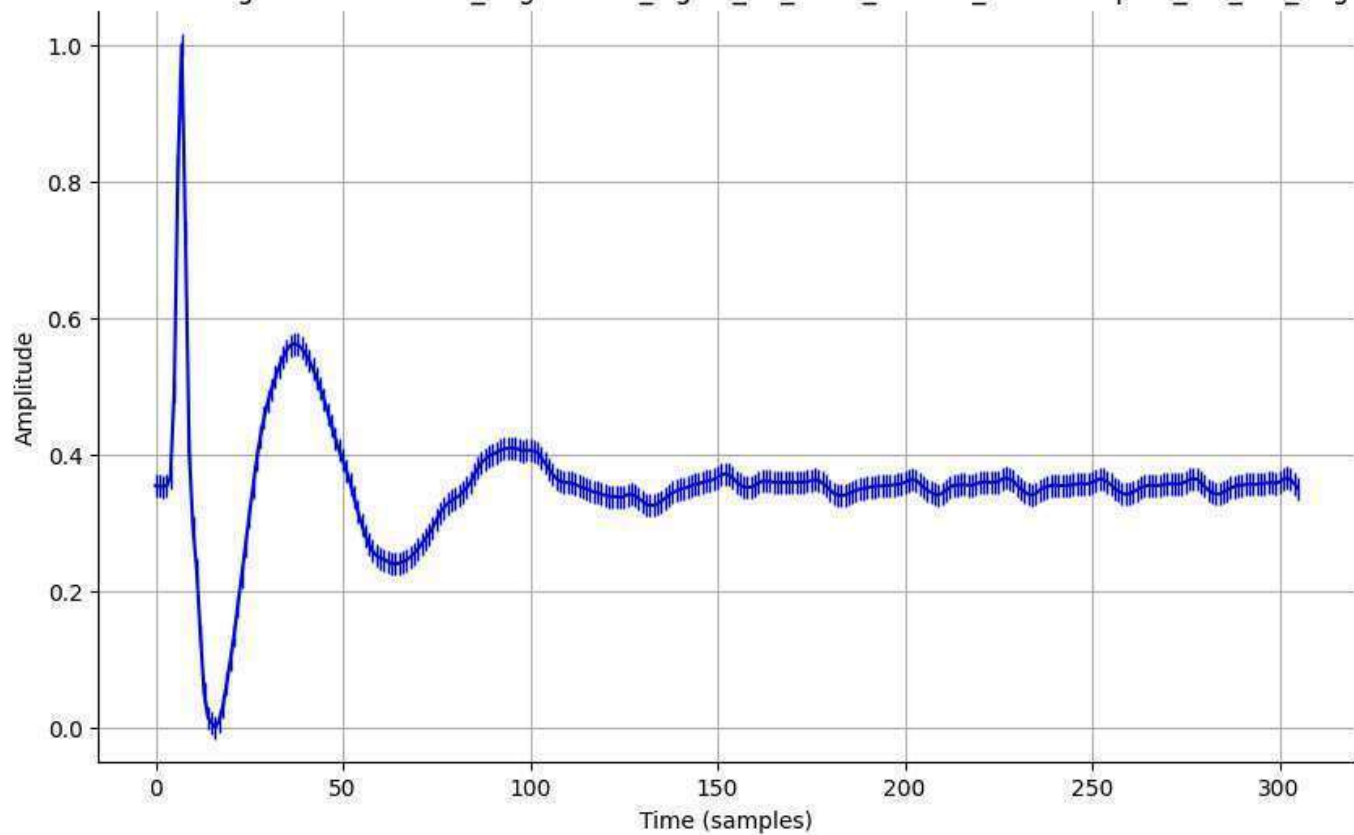
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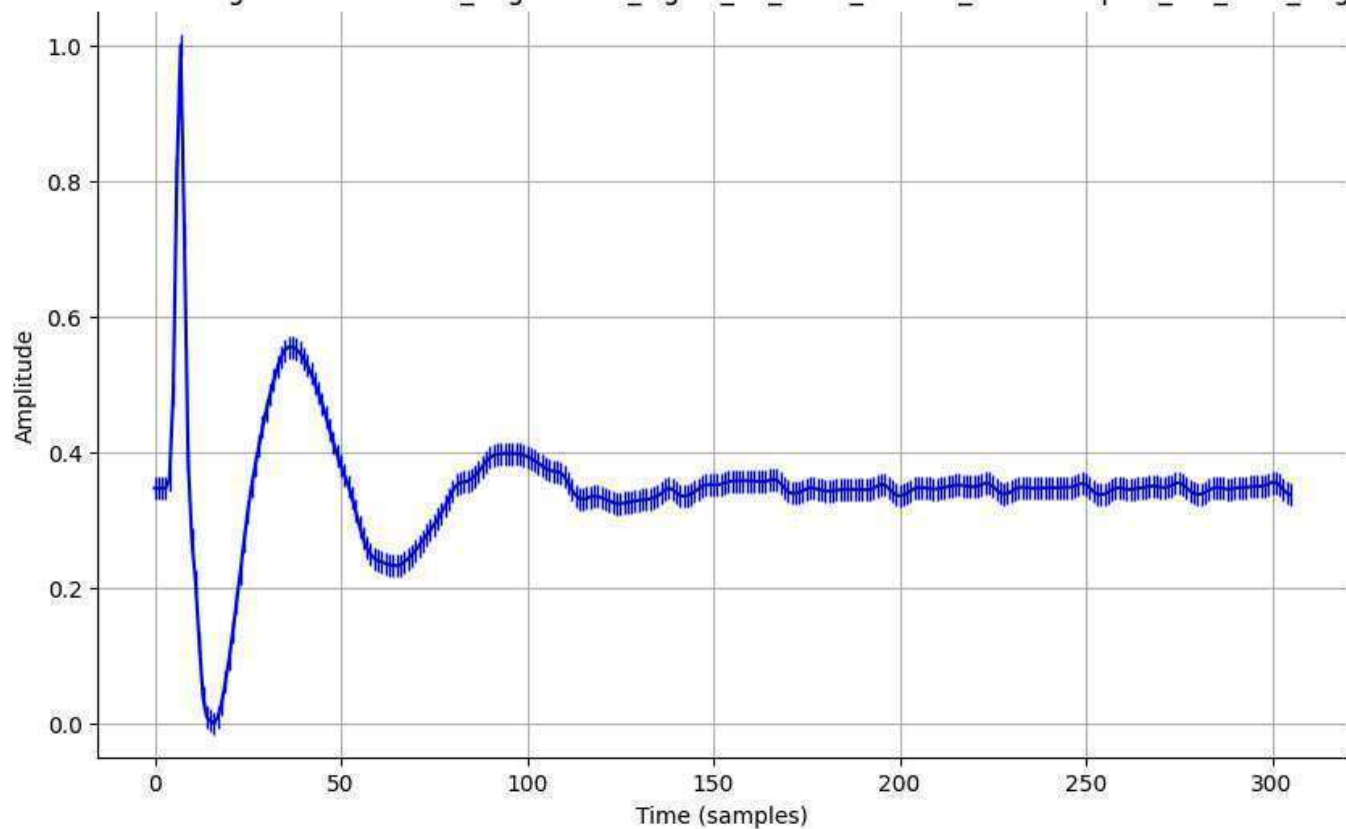
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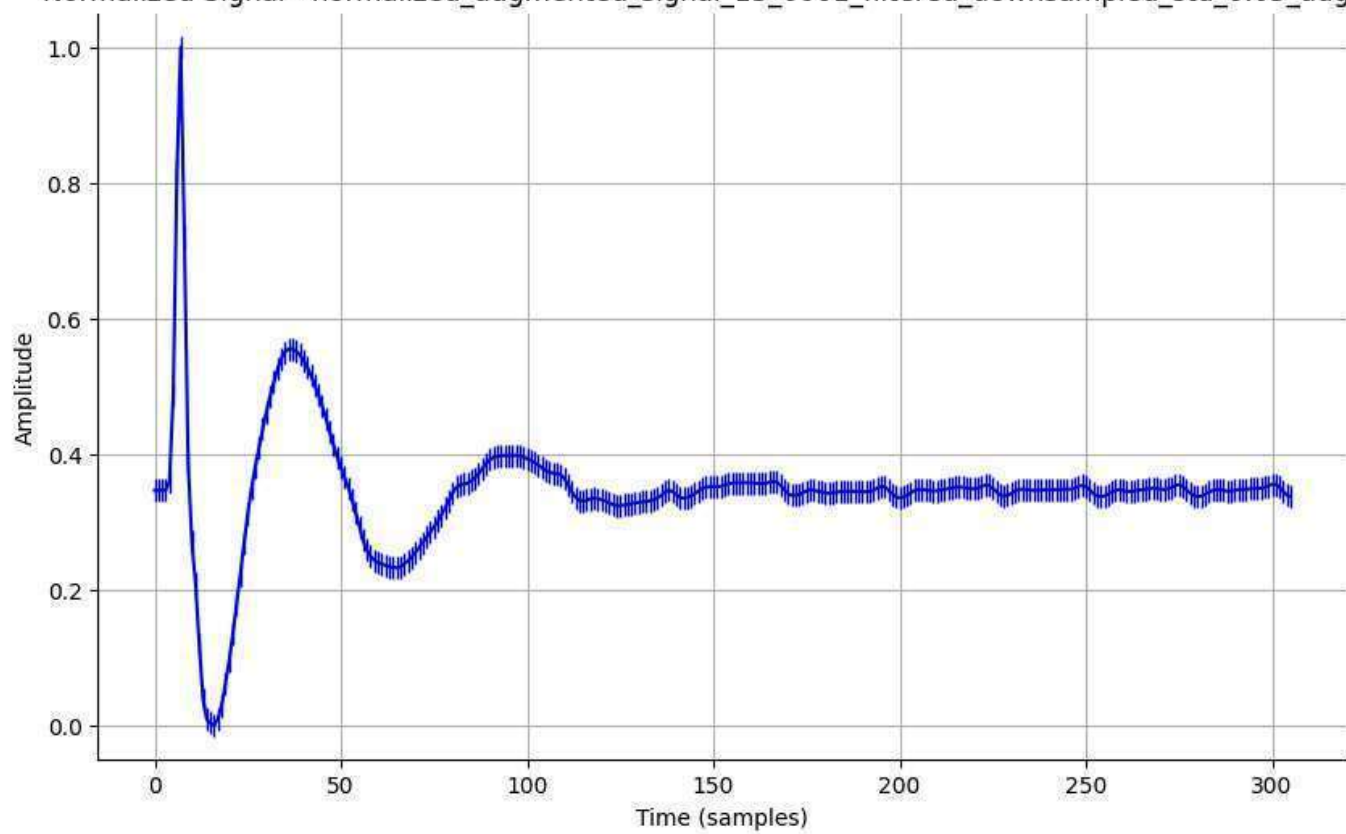
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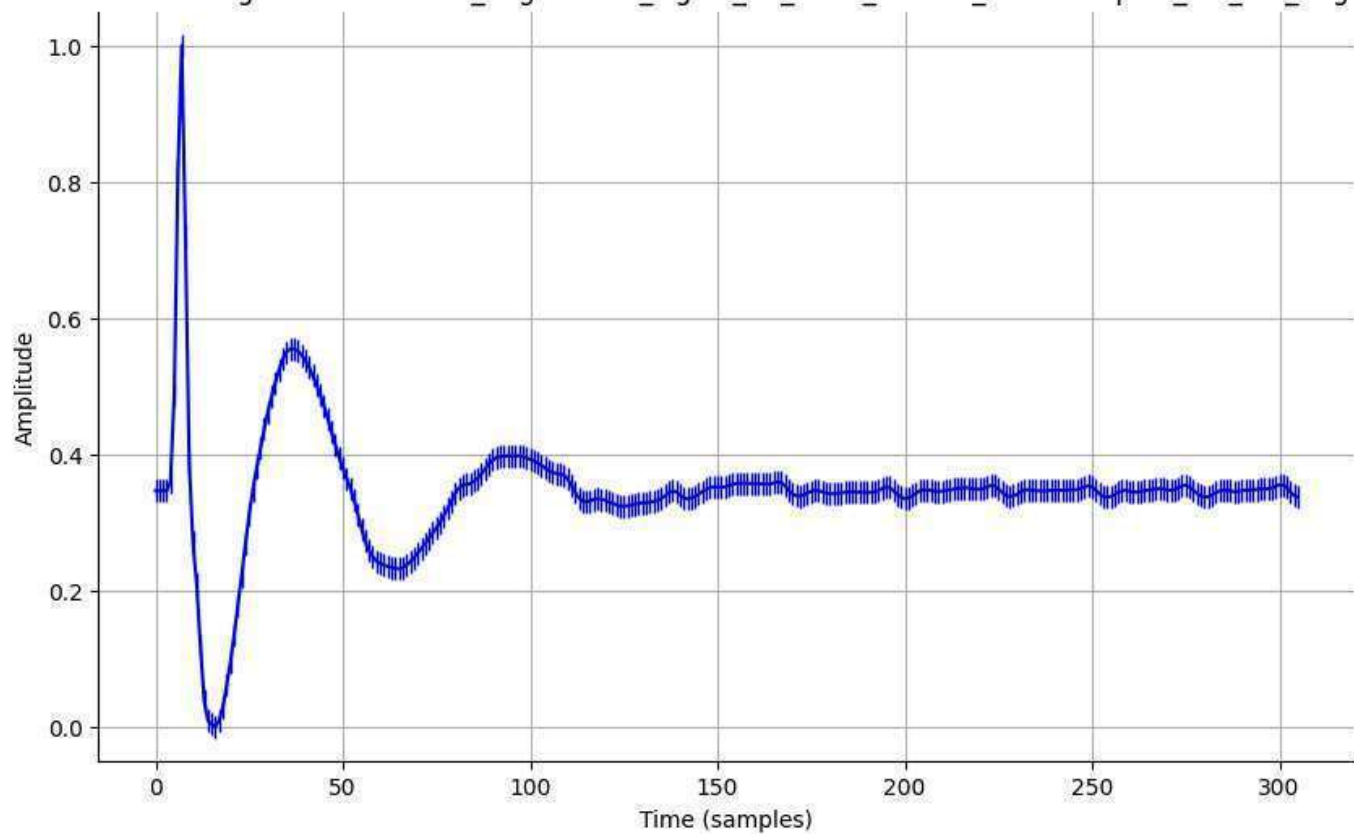
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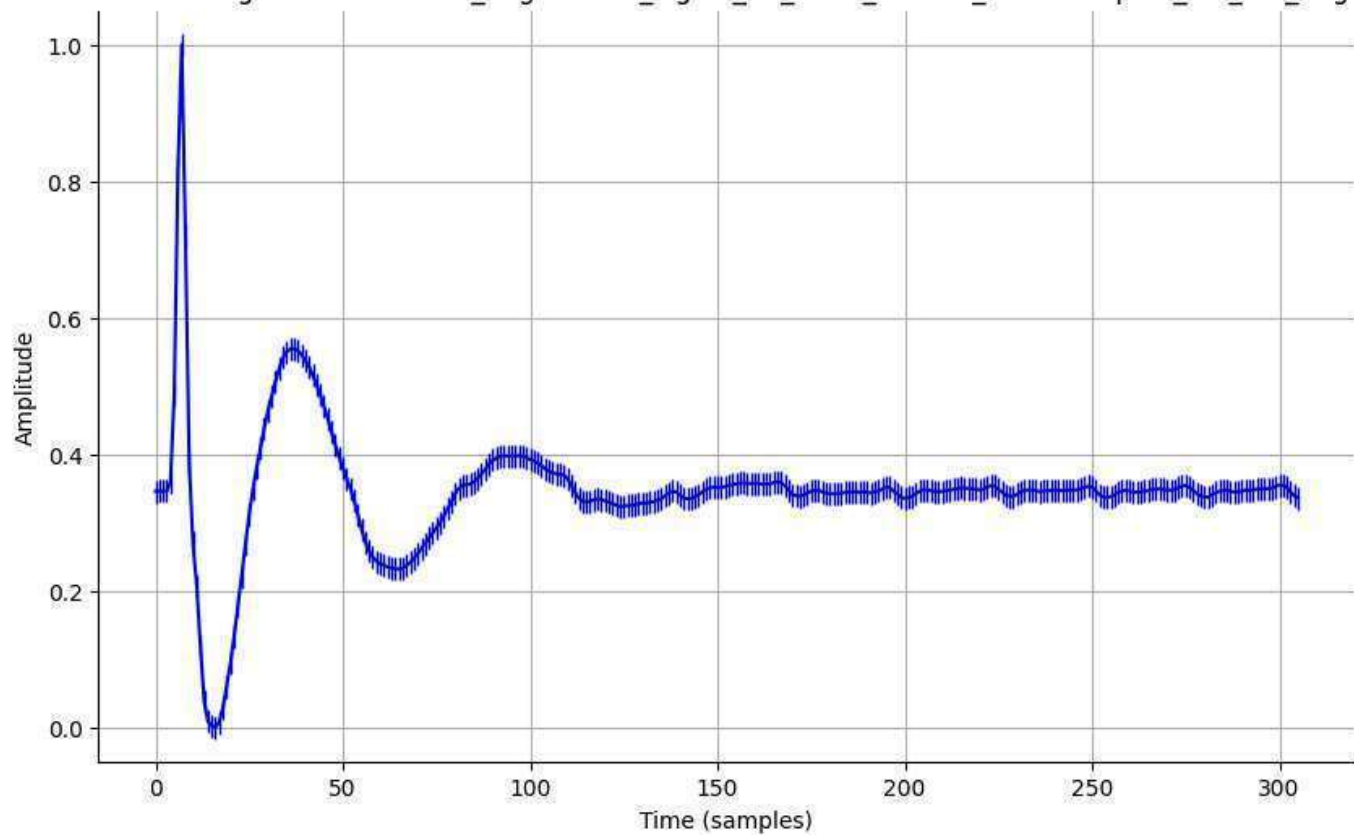
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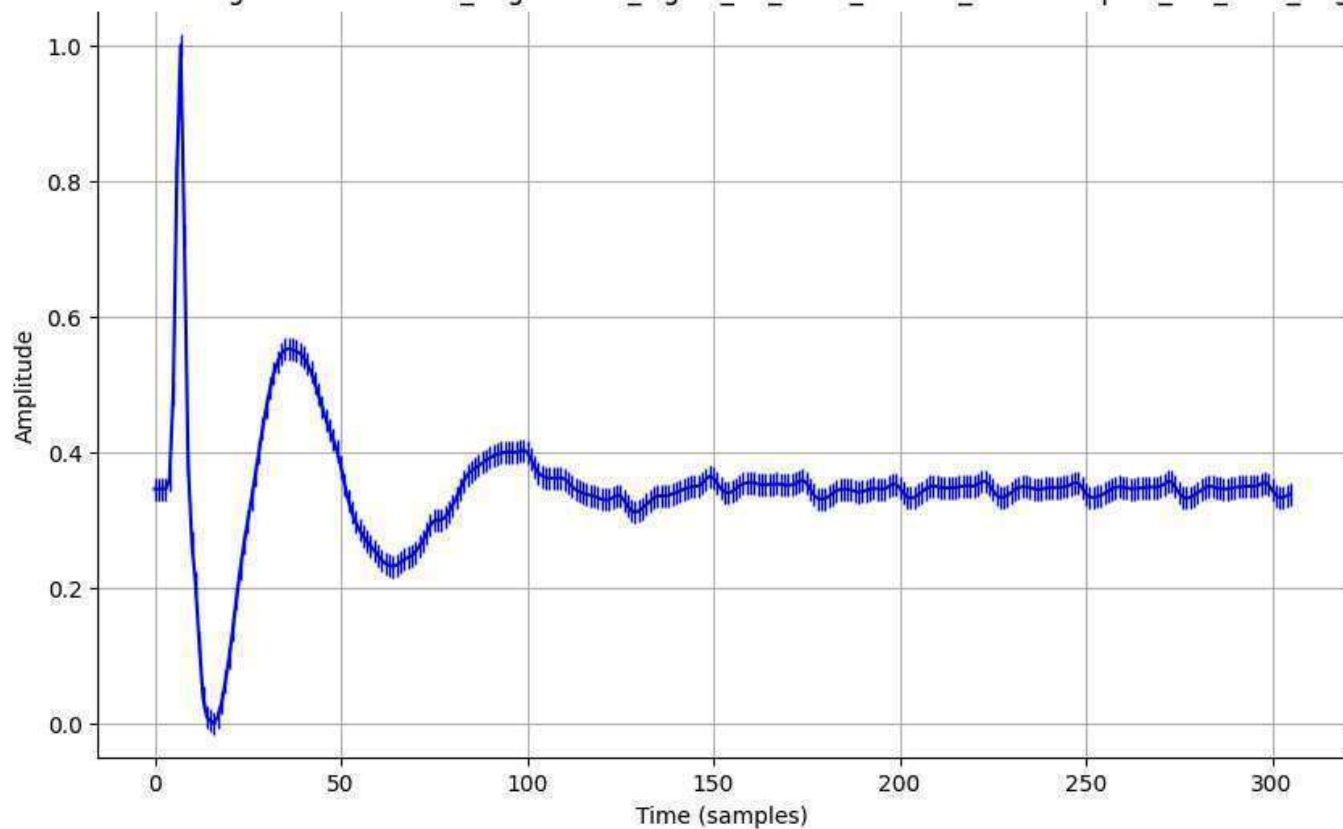
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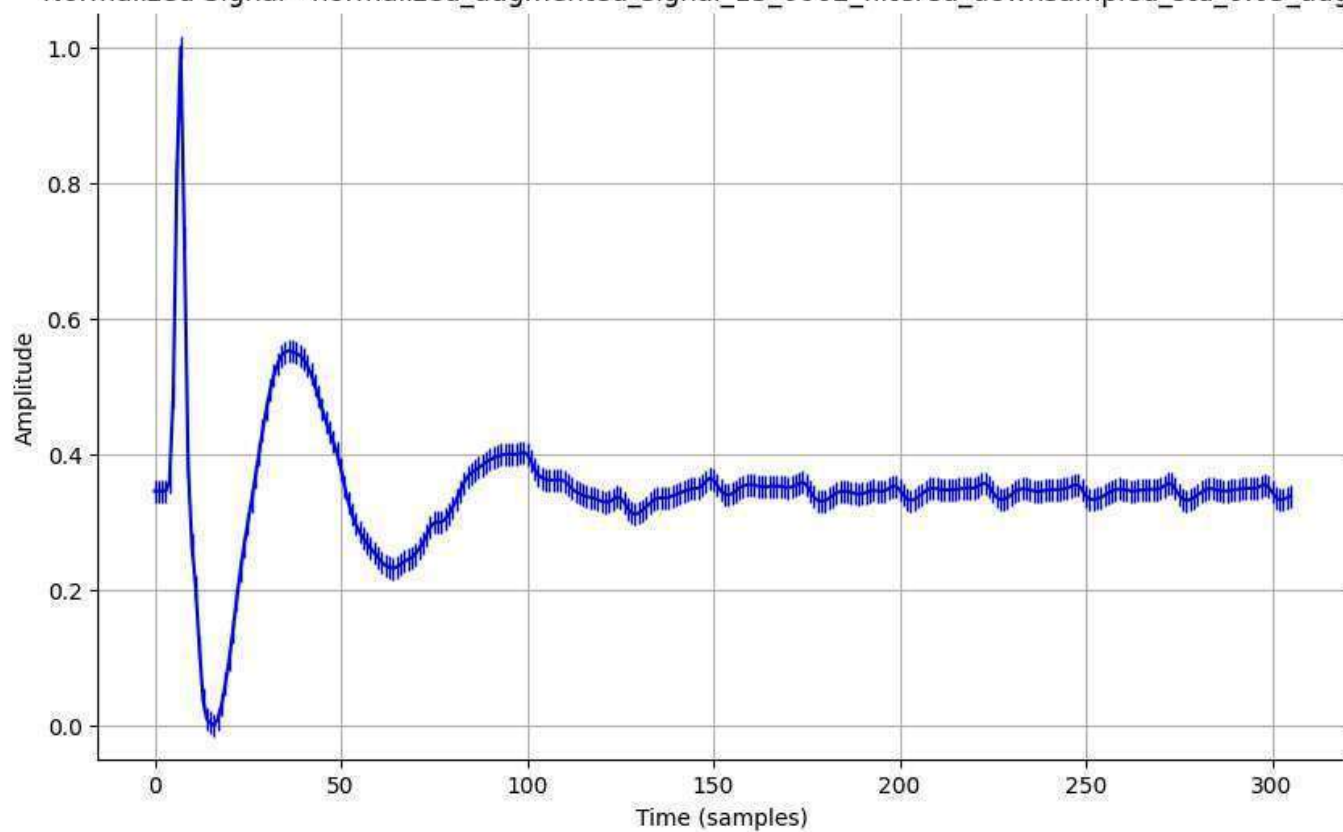
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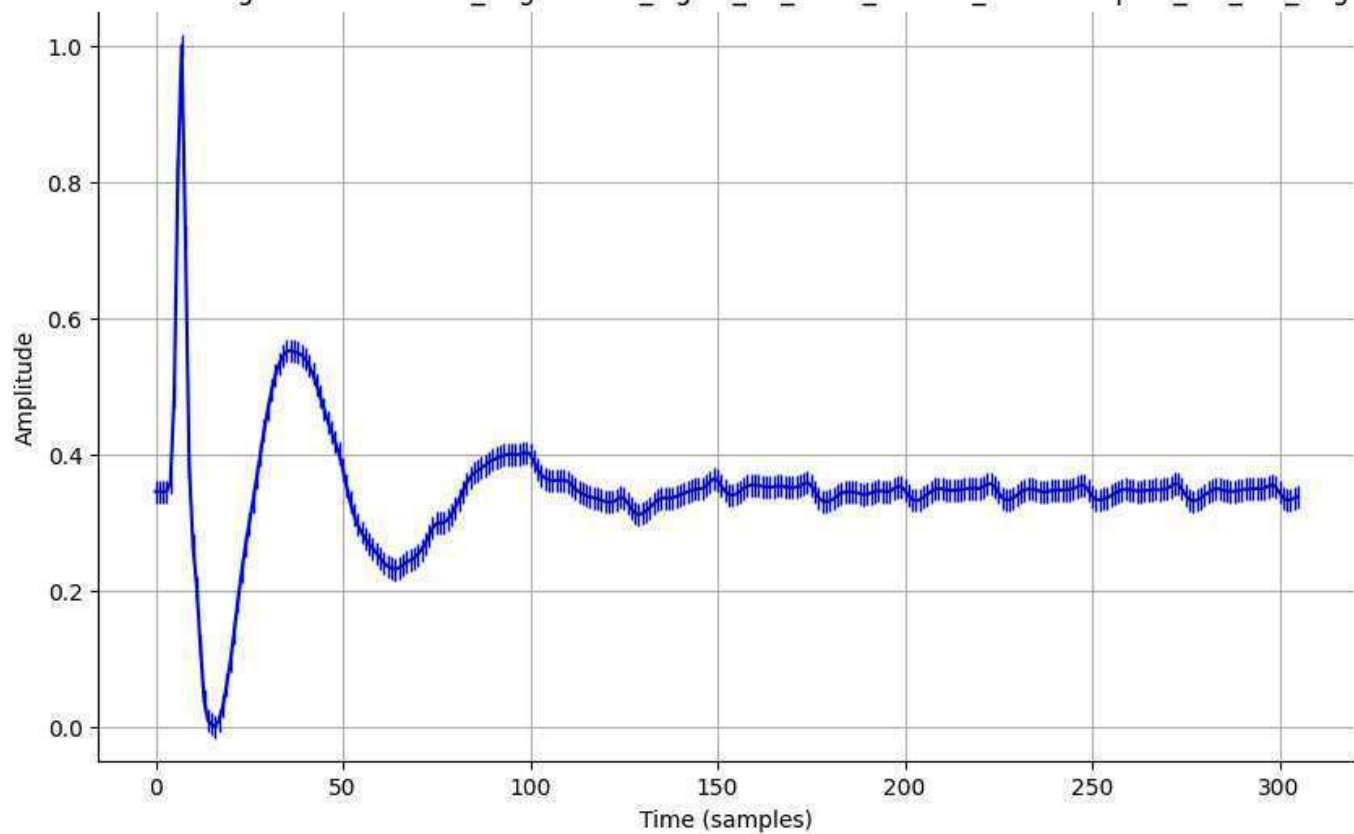
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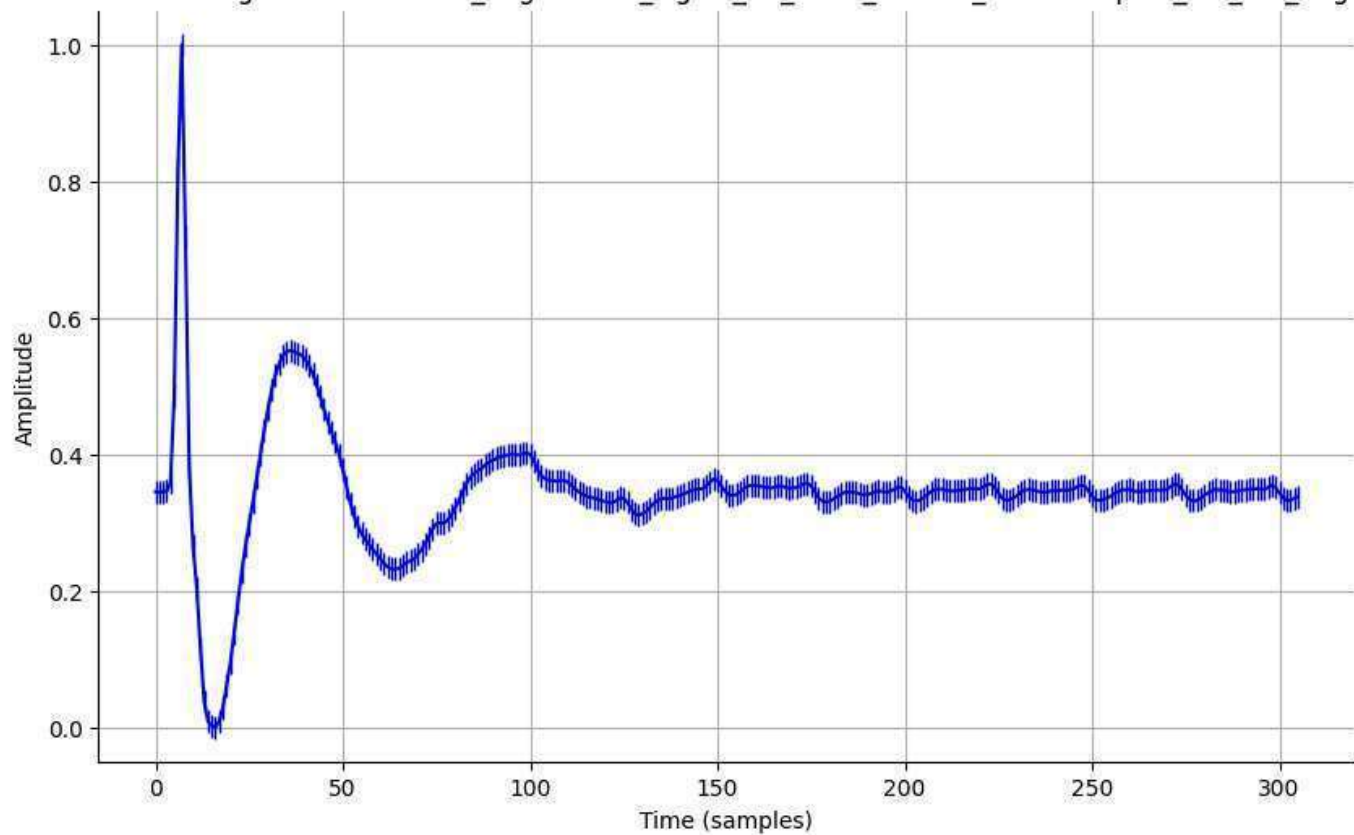
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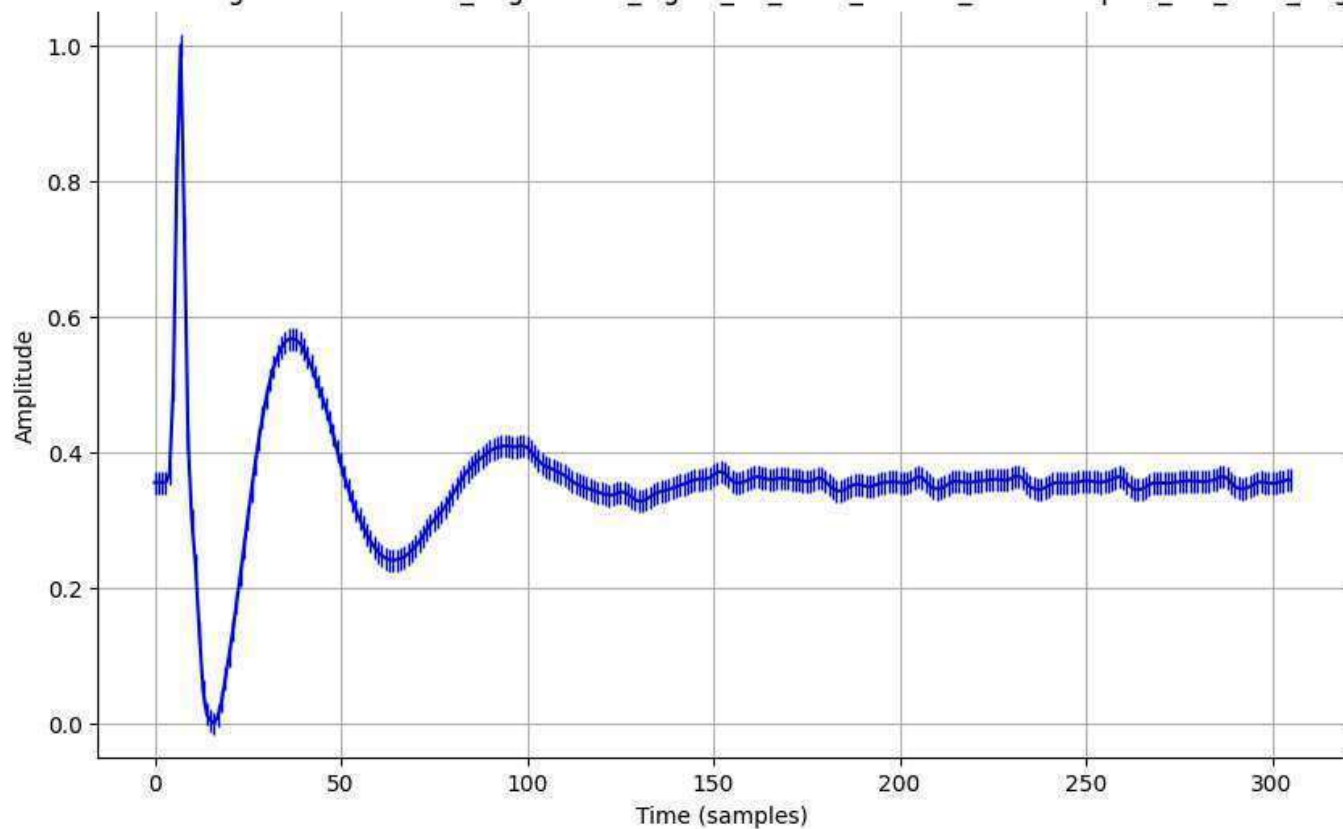
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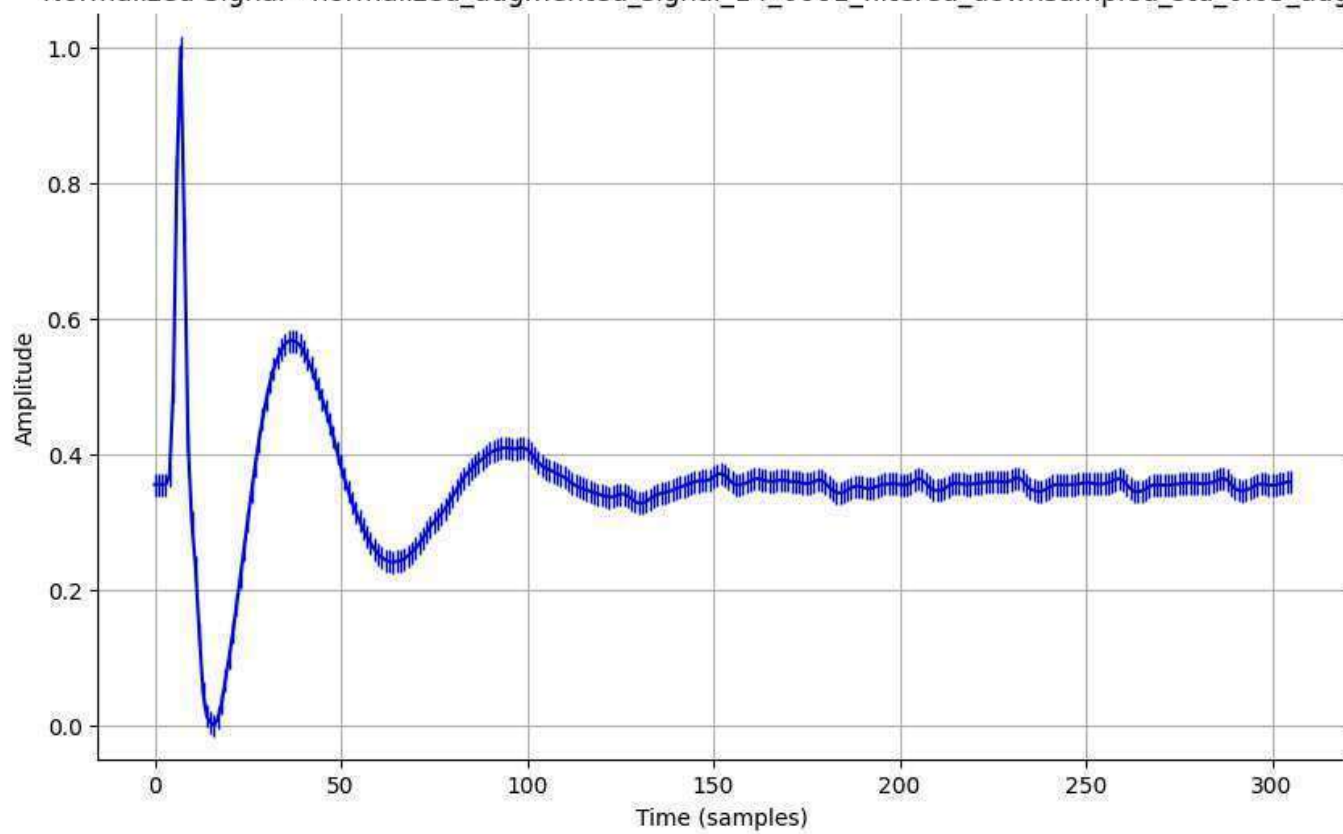
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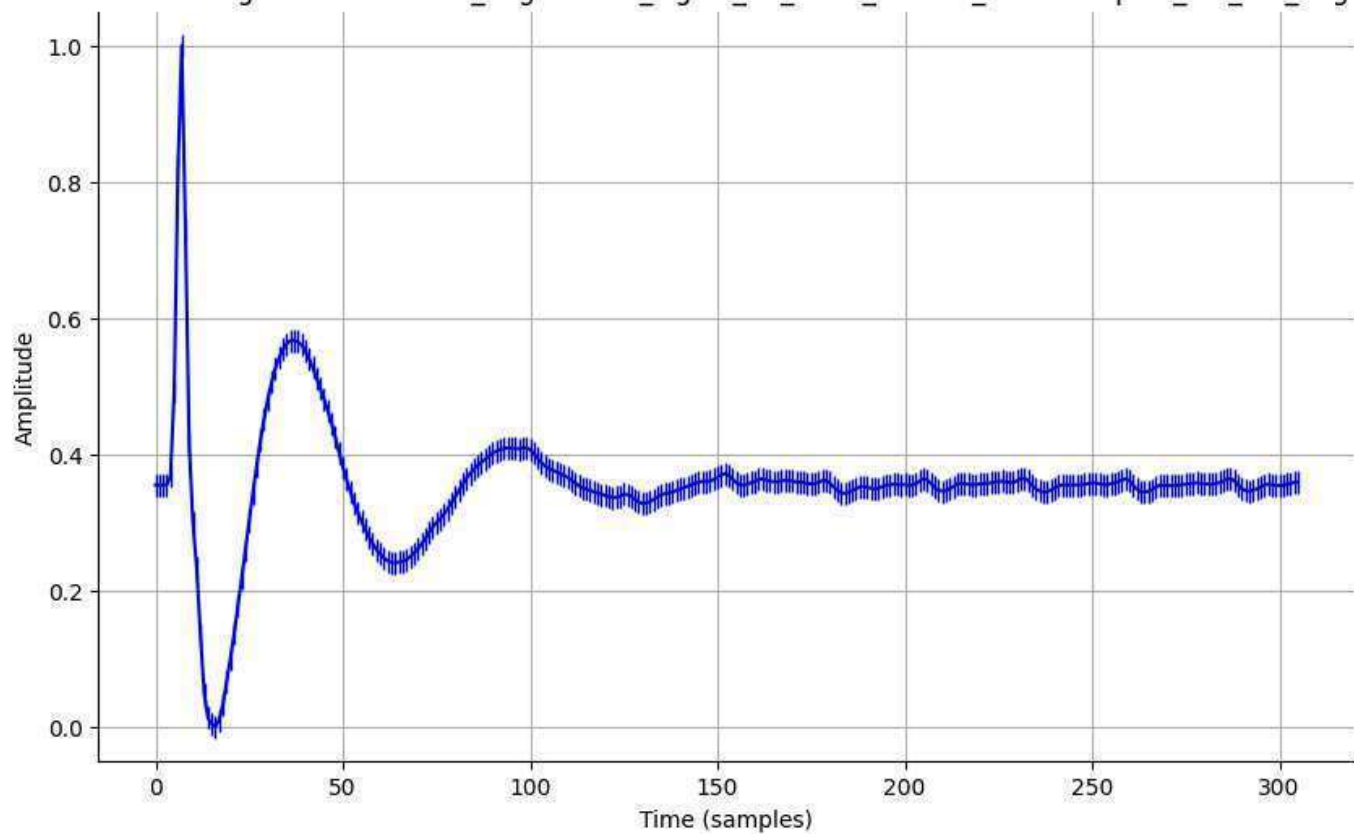
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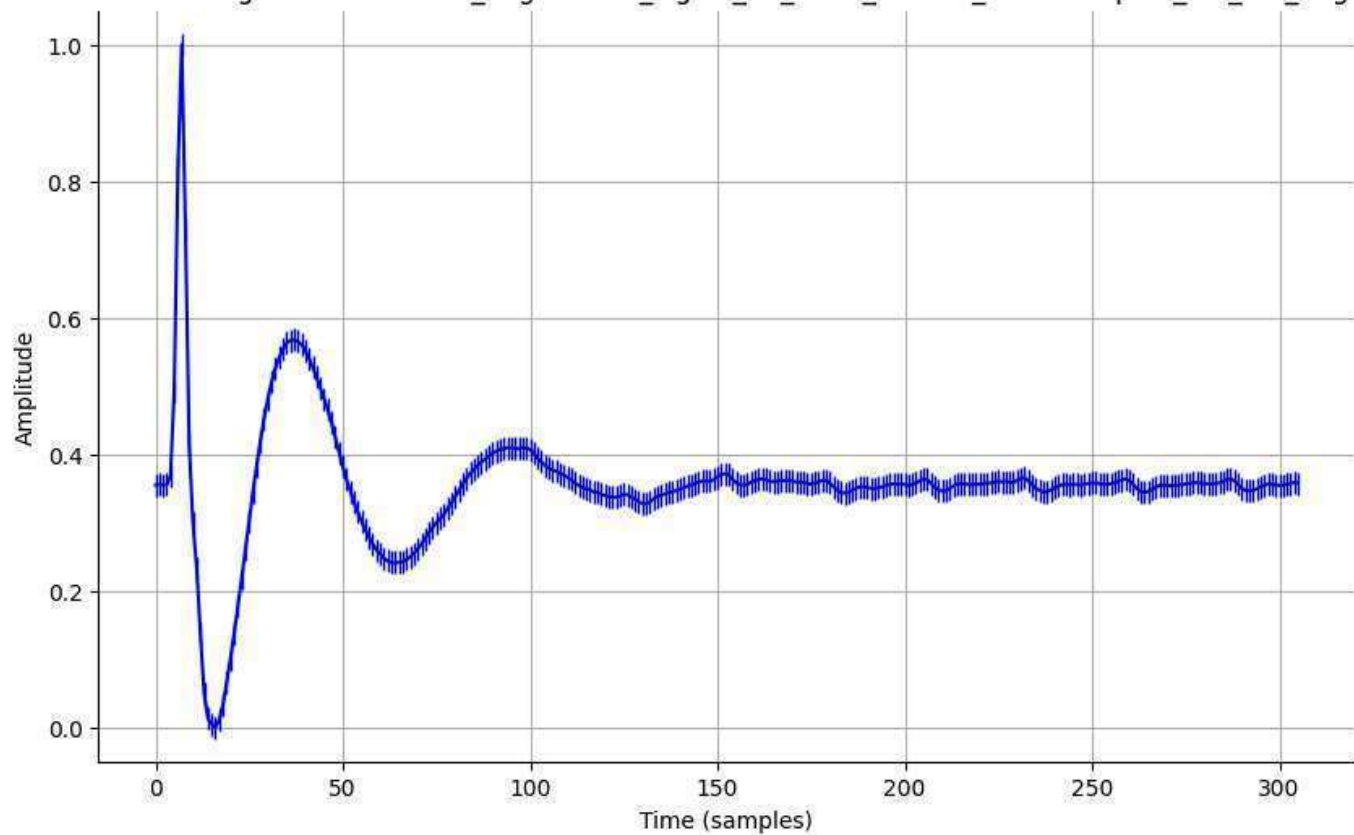
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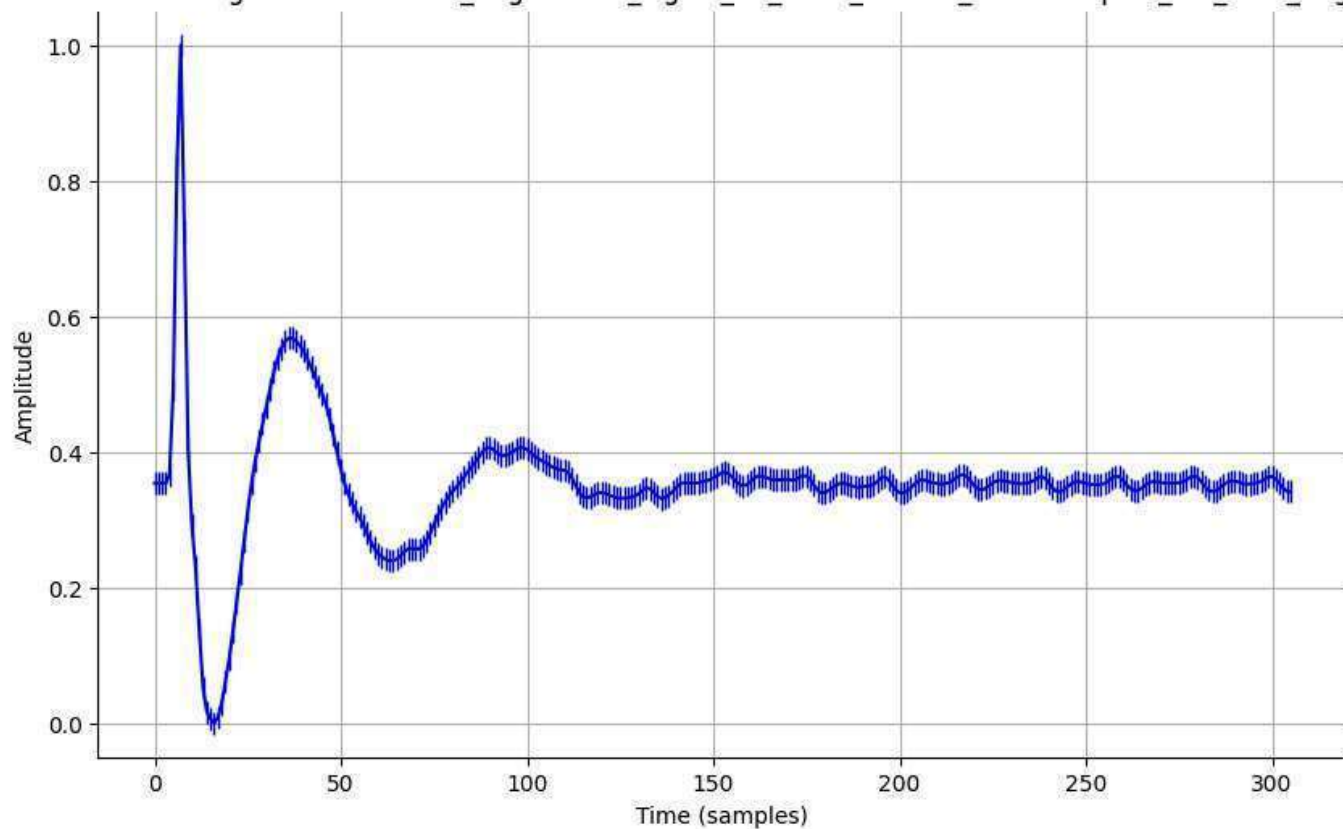
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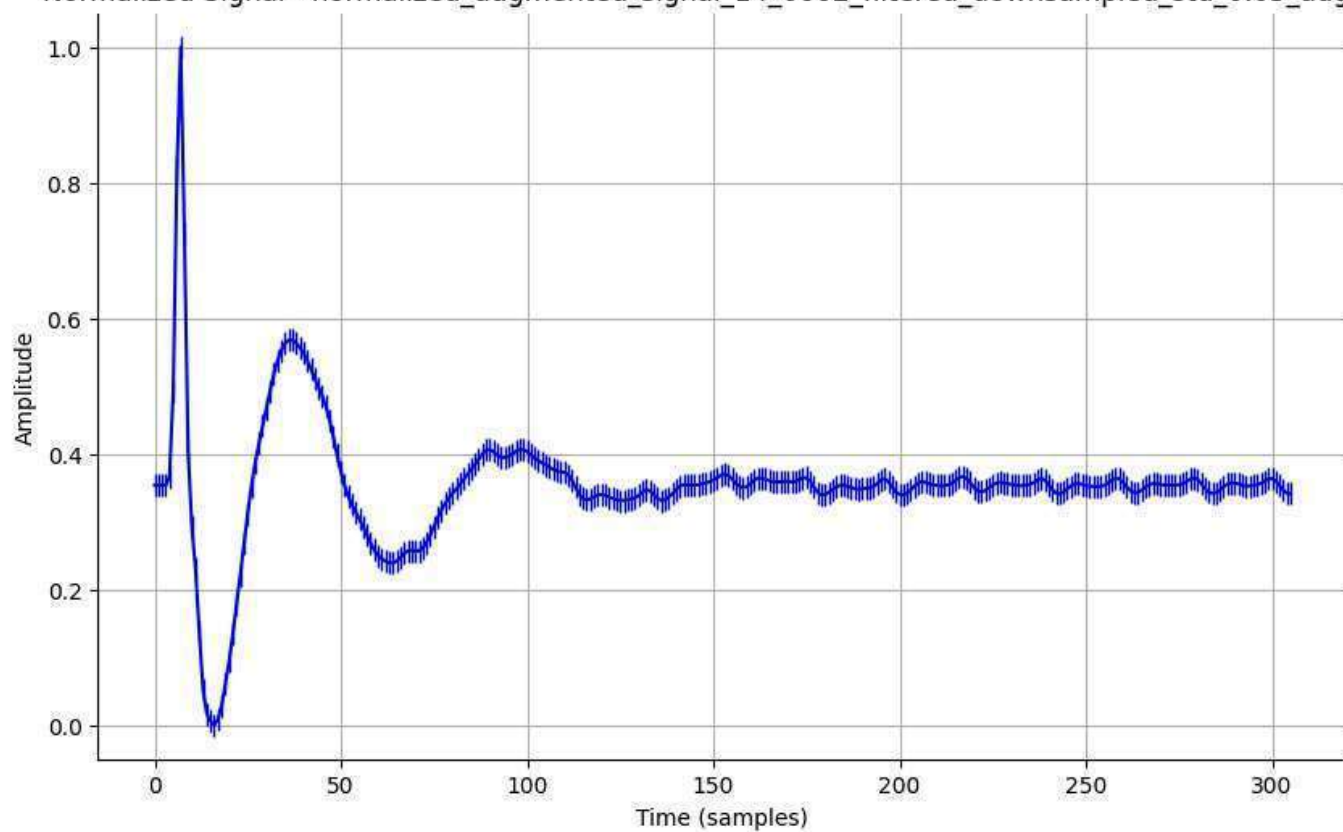
Normalized Signal - normalized_augmented_signal_14_0001_filtered_downsampled_std_0.2_aug.csv



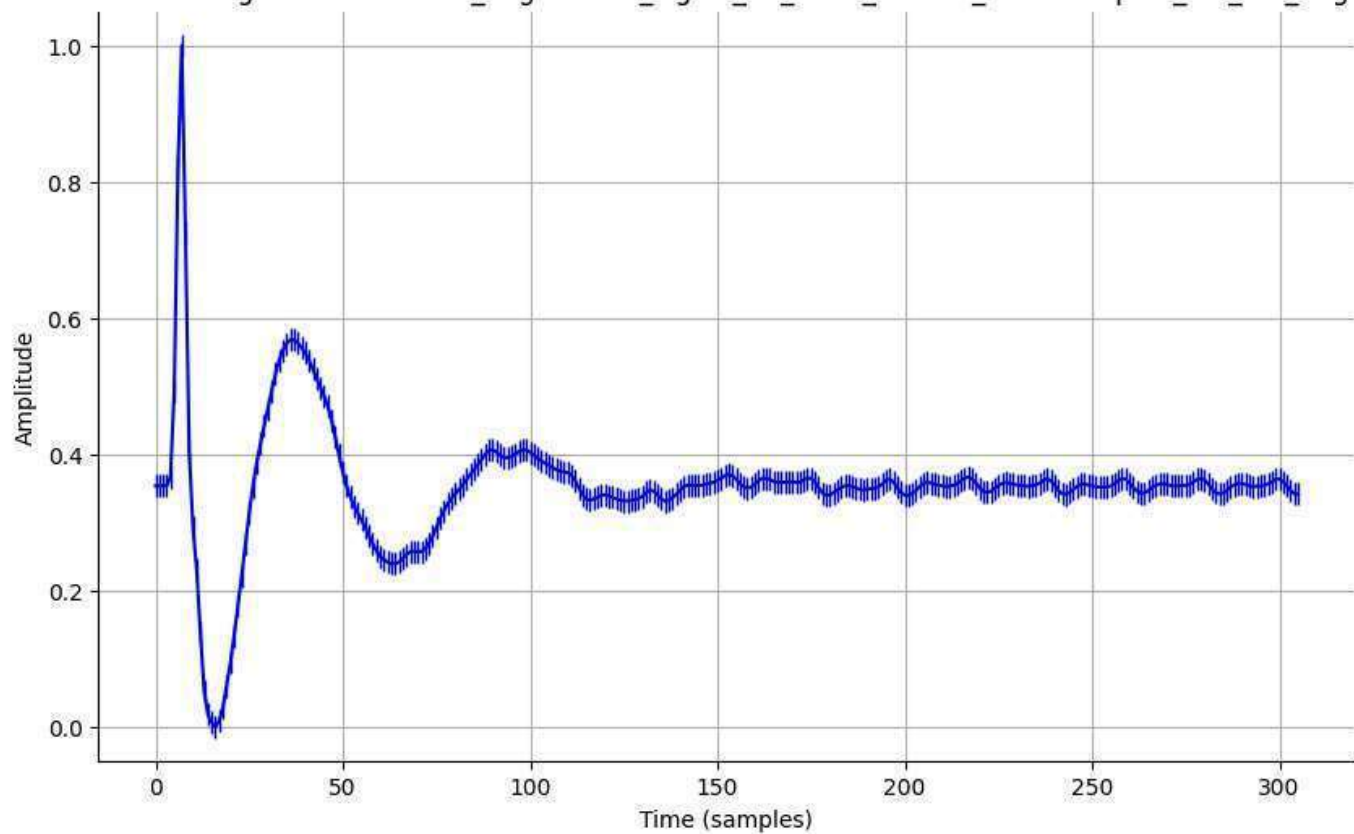
Normalized Signal - normalized_augmented_signal_14_0002_filtered_downsampled_std_0.01_aug.csv



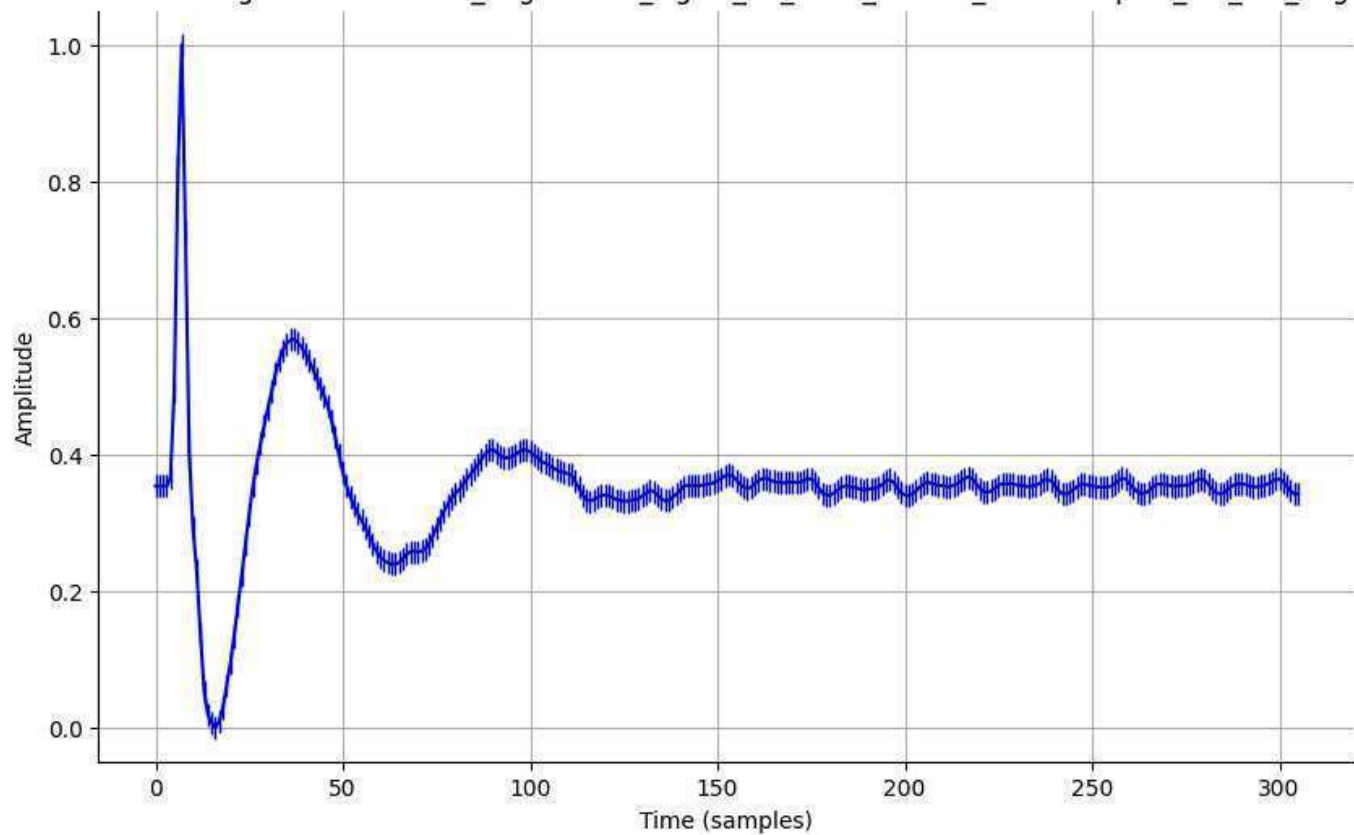
Normalized Signal - normalized_augmented_signal_14_0002_filtered_downsampled_std_0.05_aug.csv



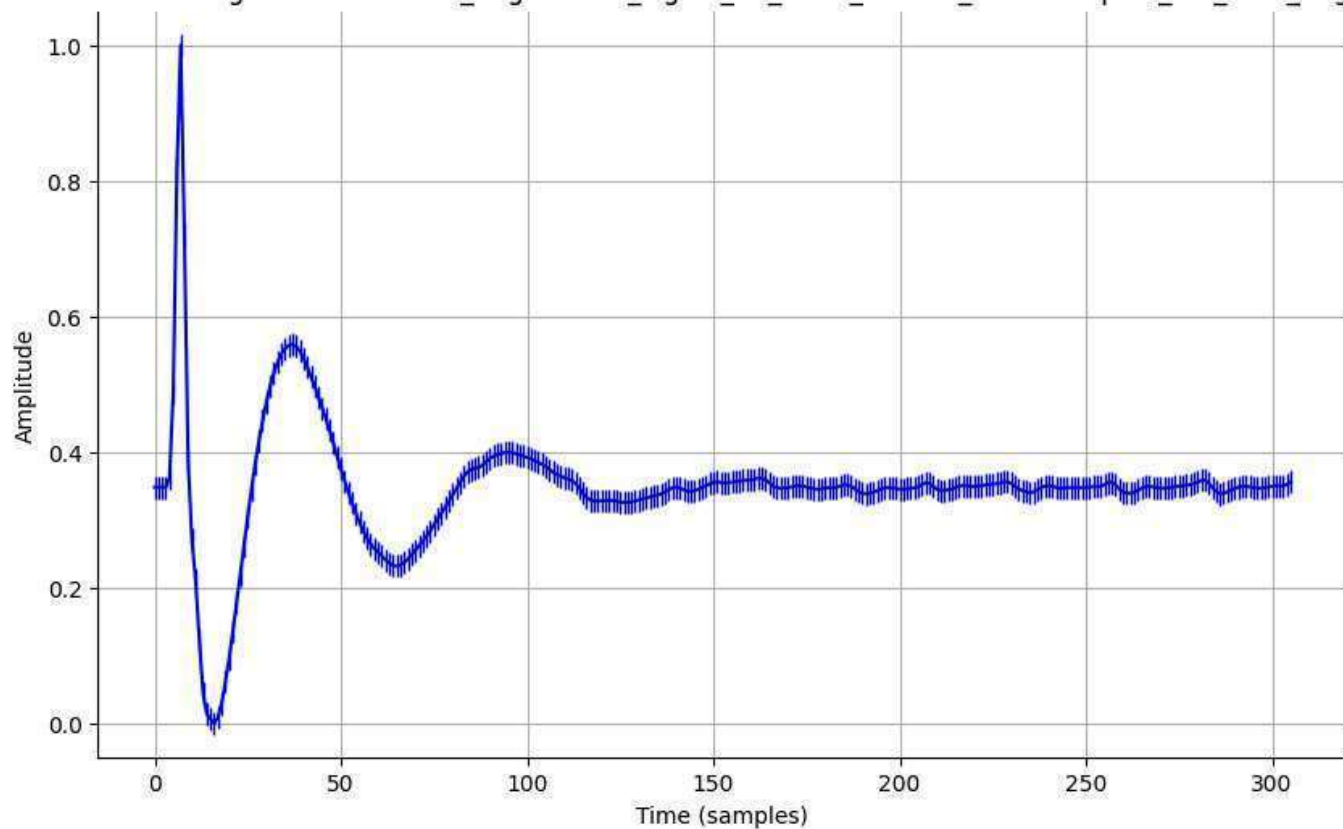
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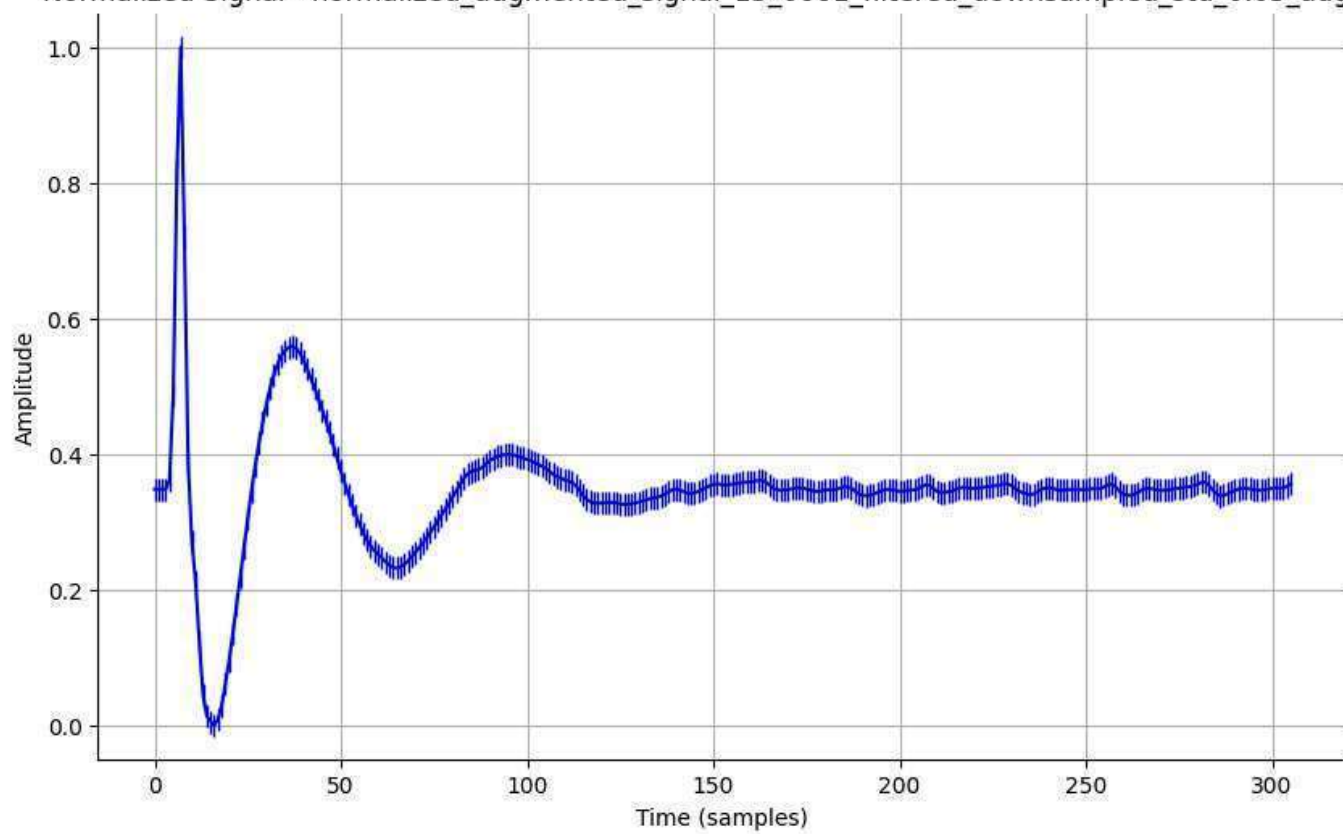
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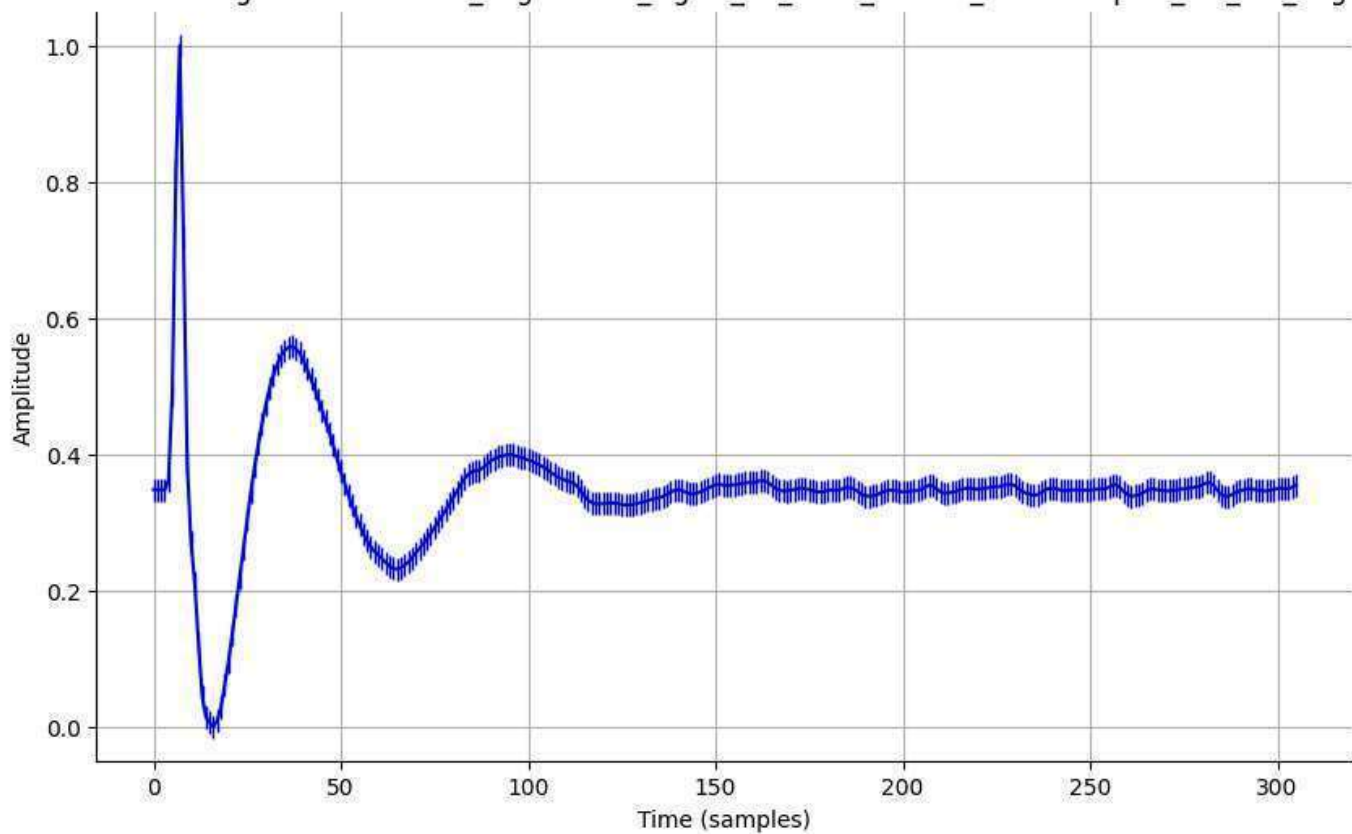
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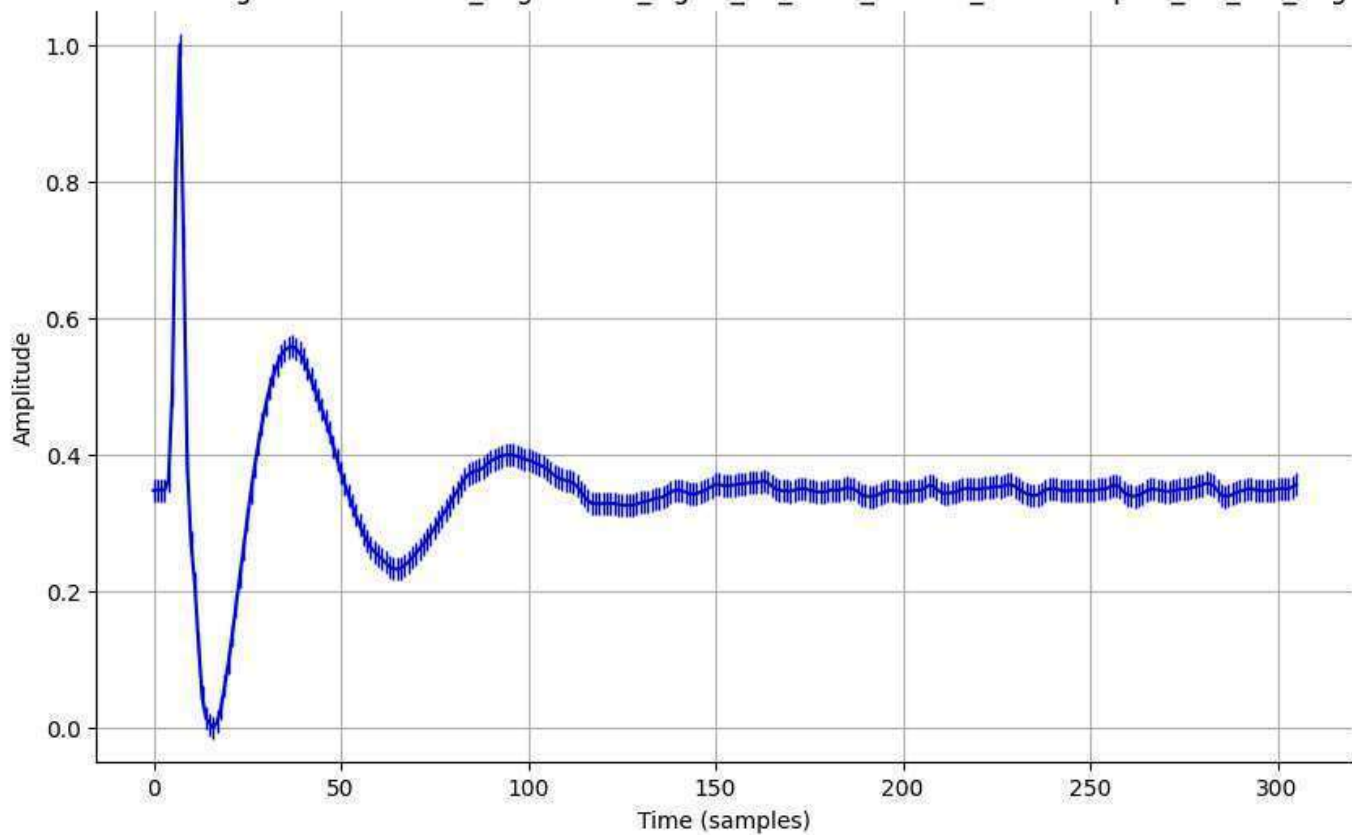
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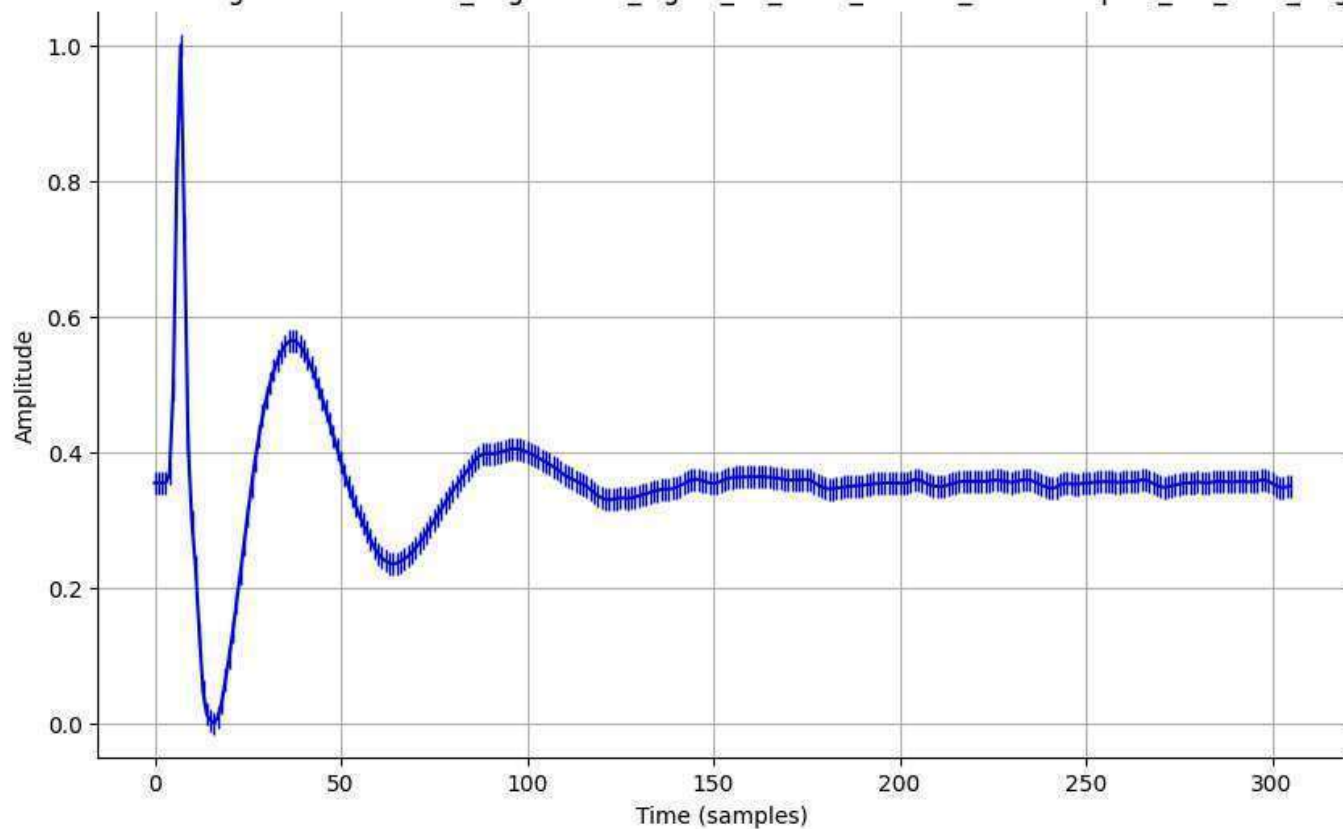
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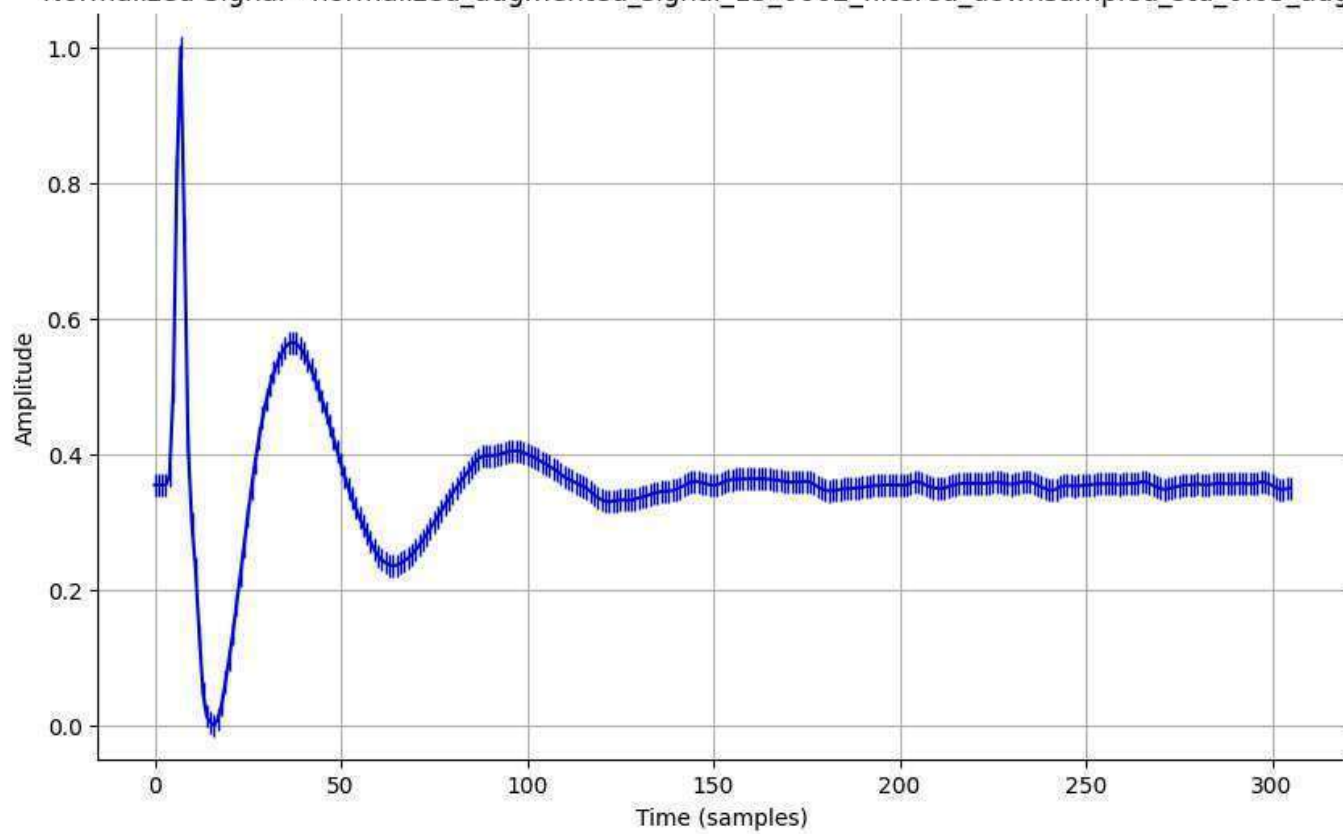
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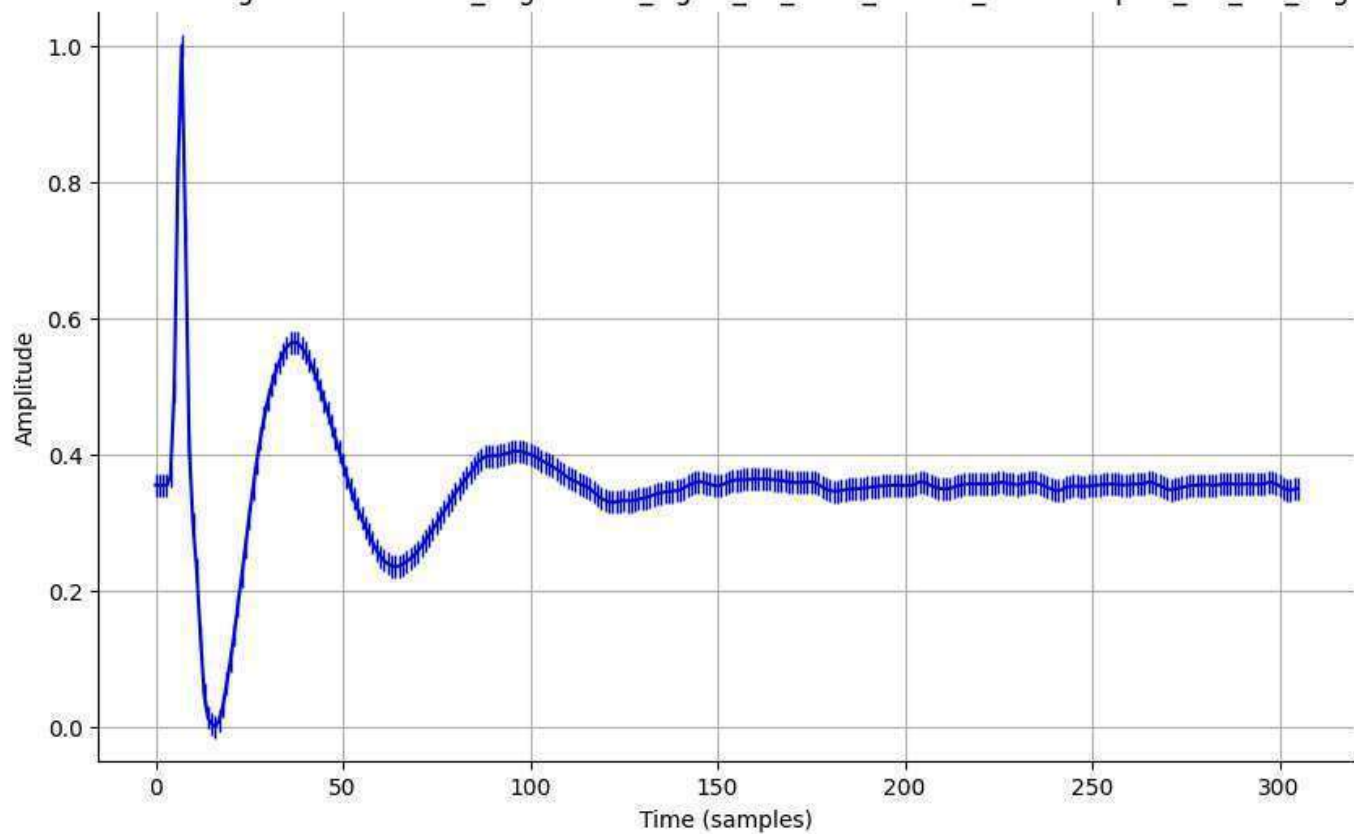
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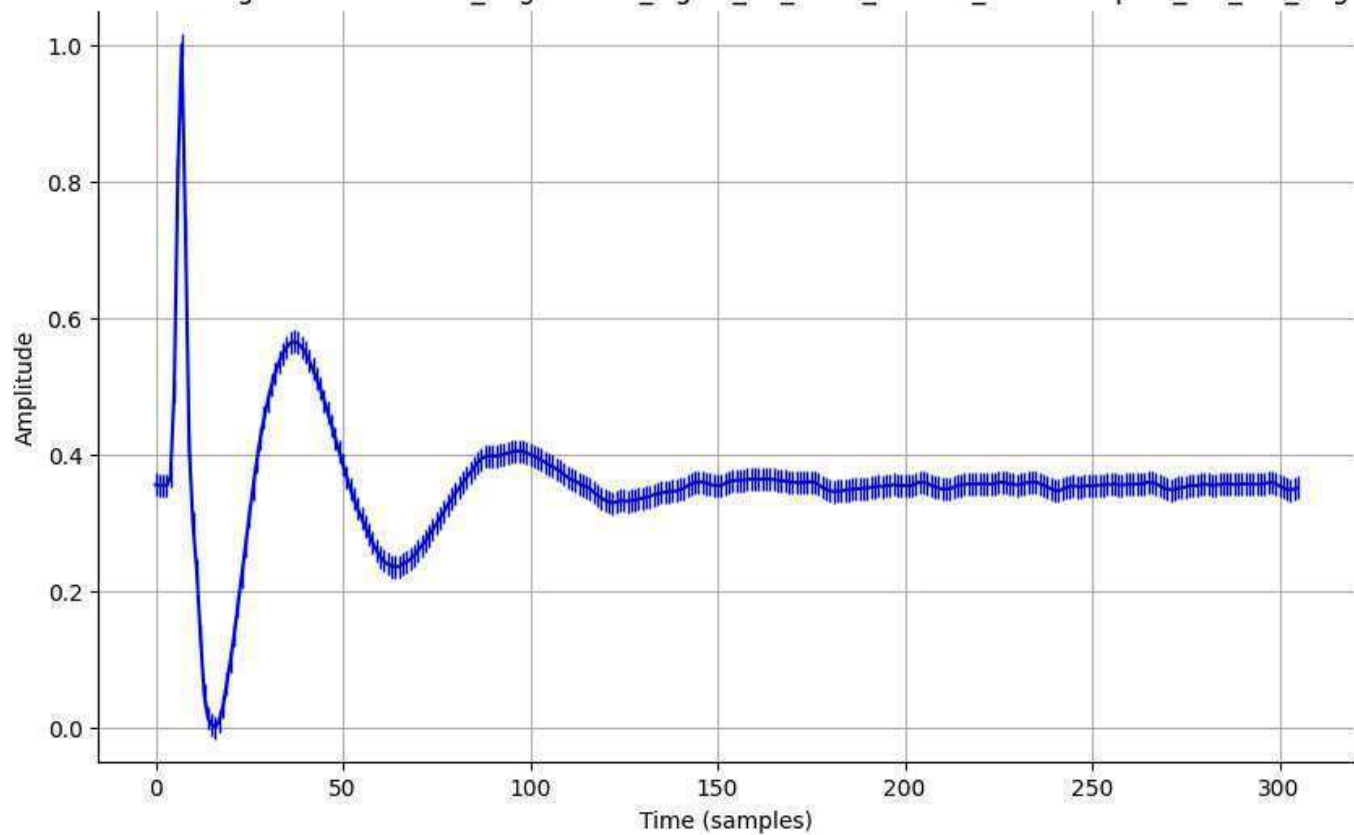
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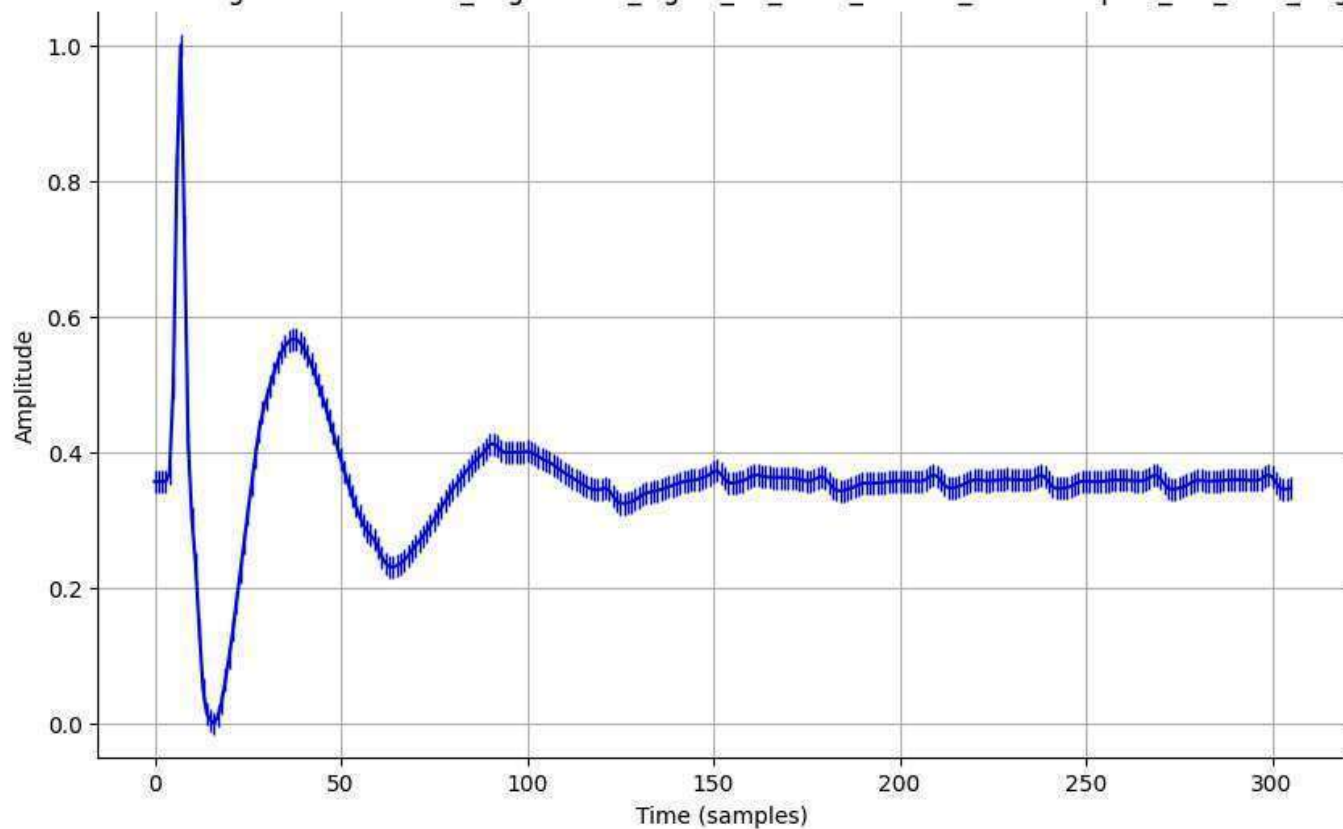
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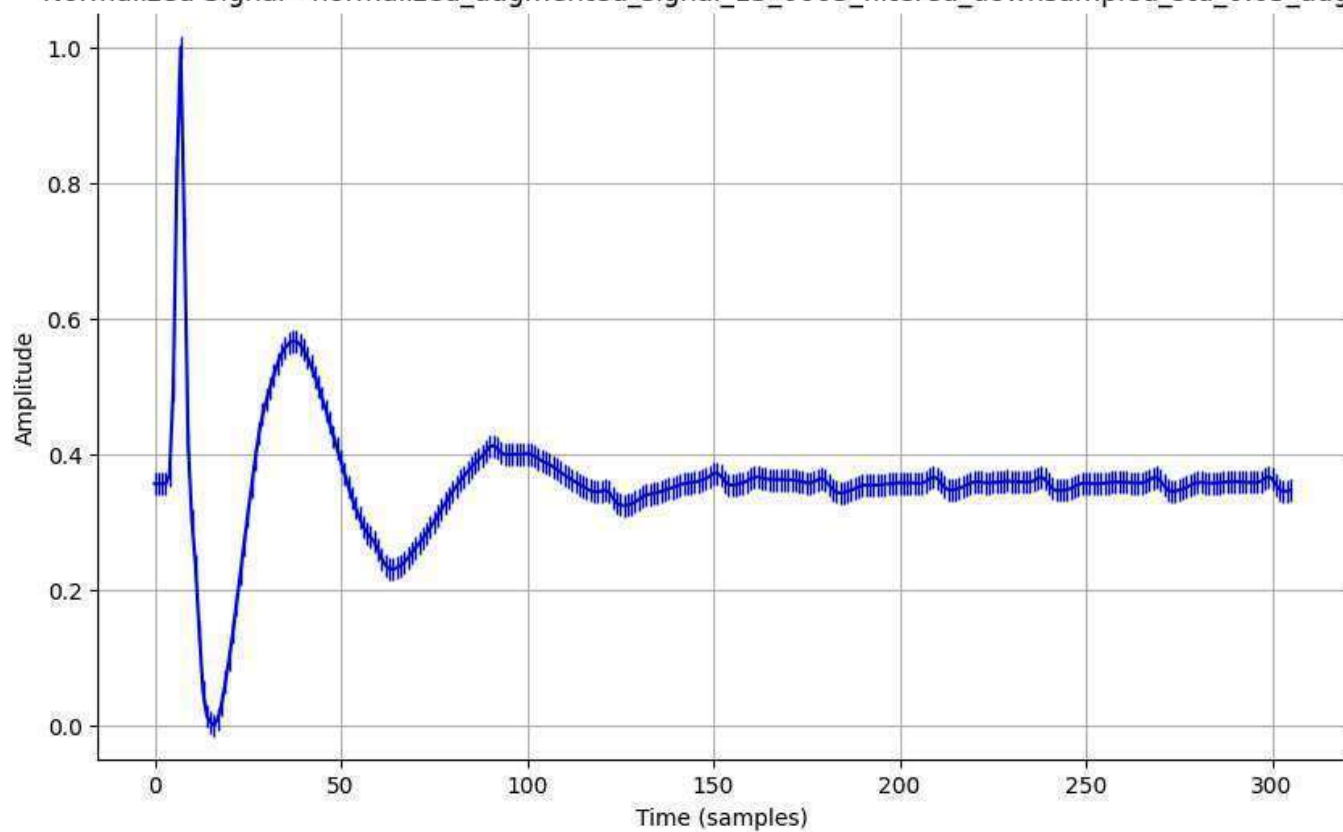
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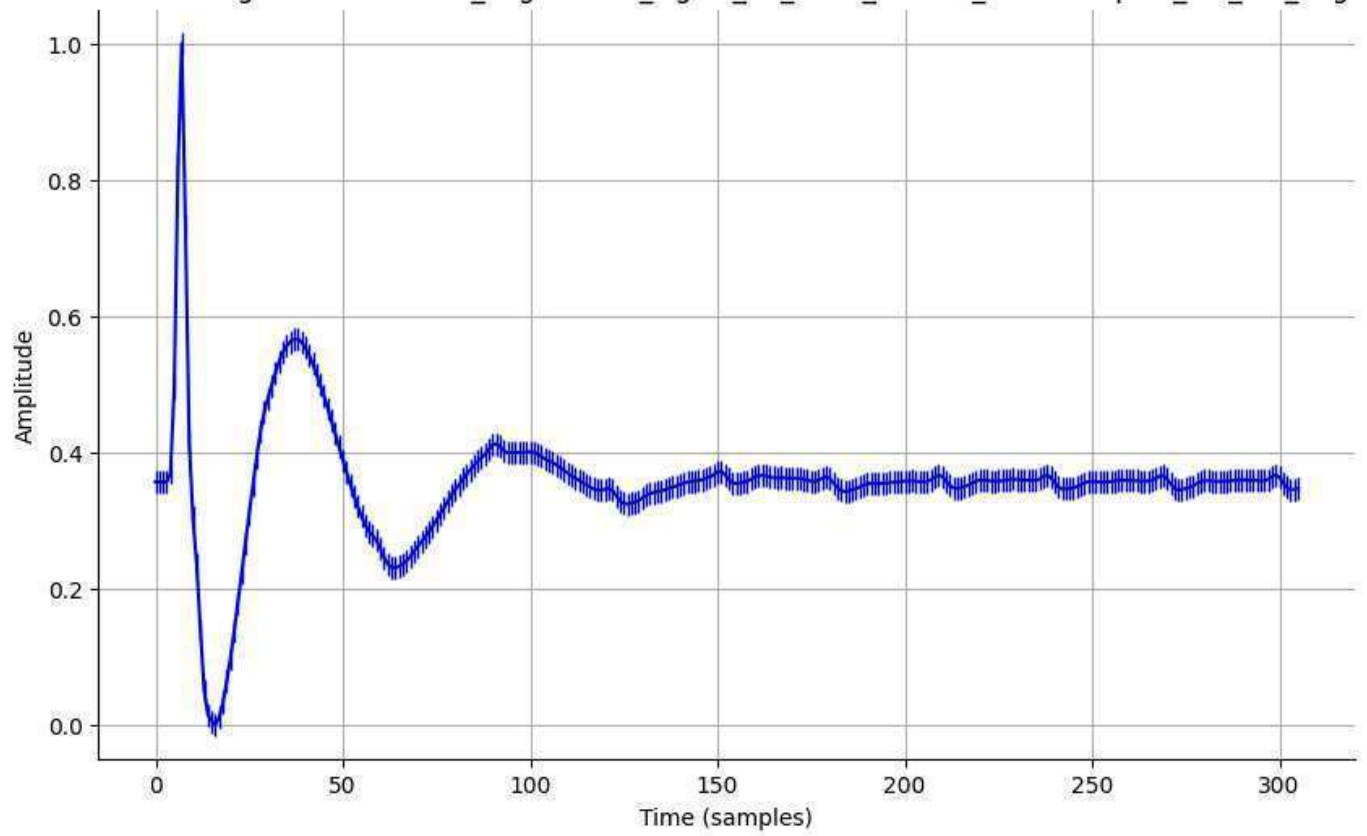
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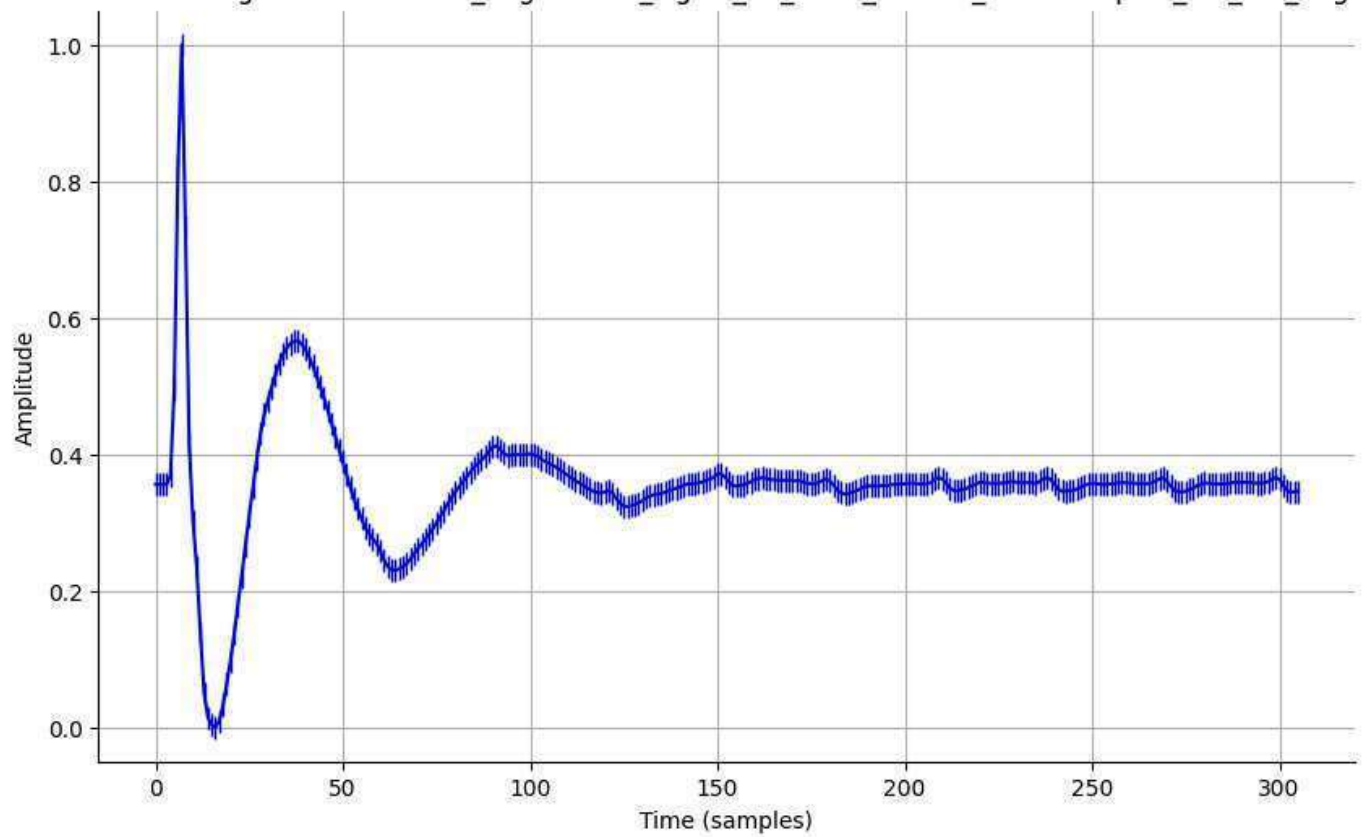
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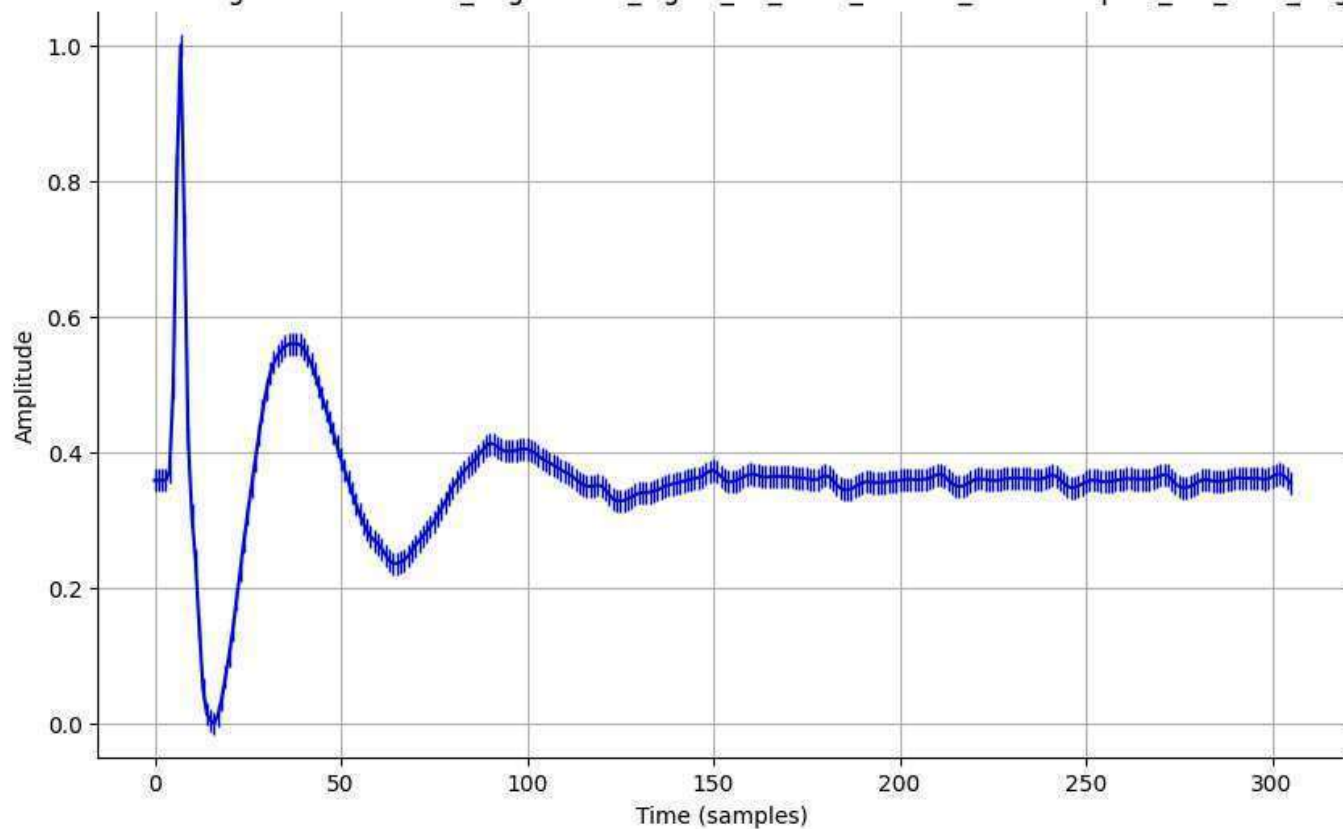
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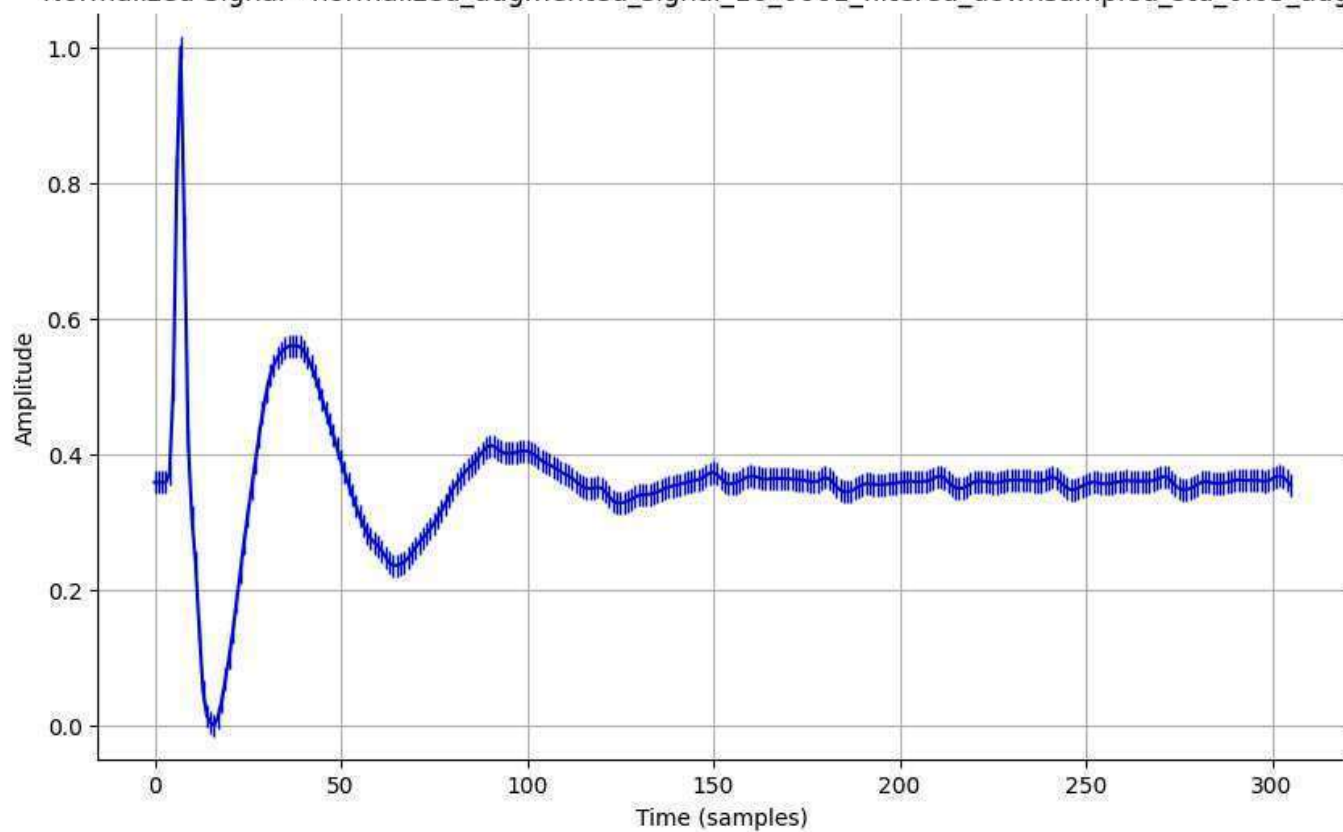
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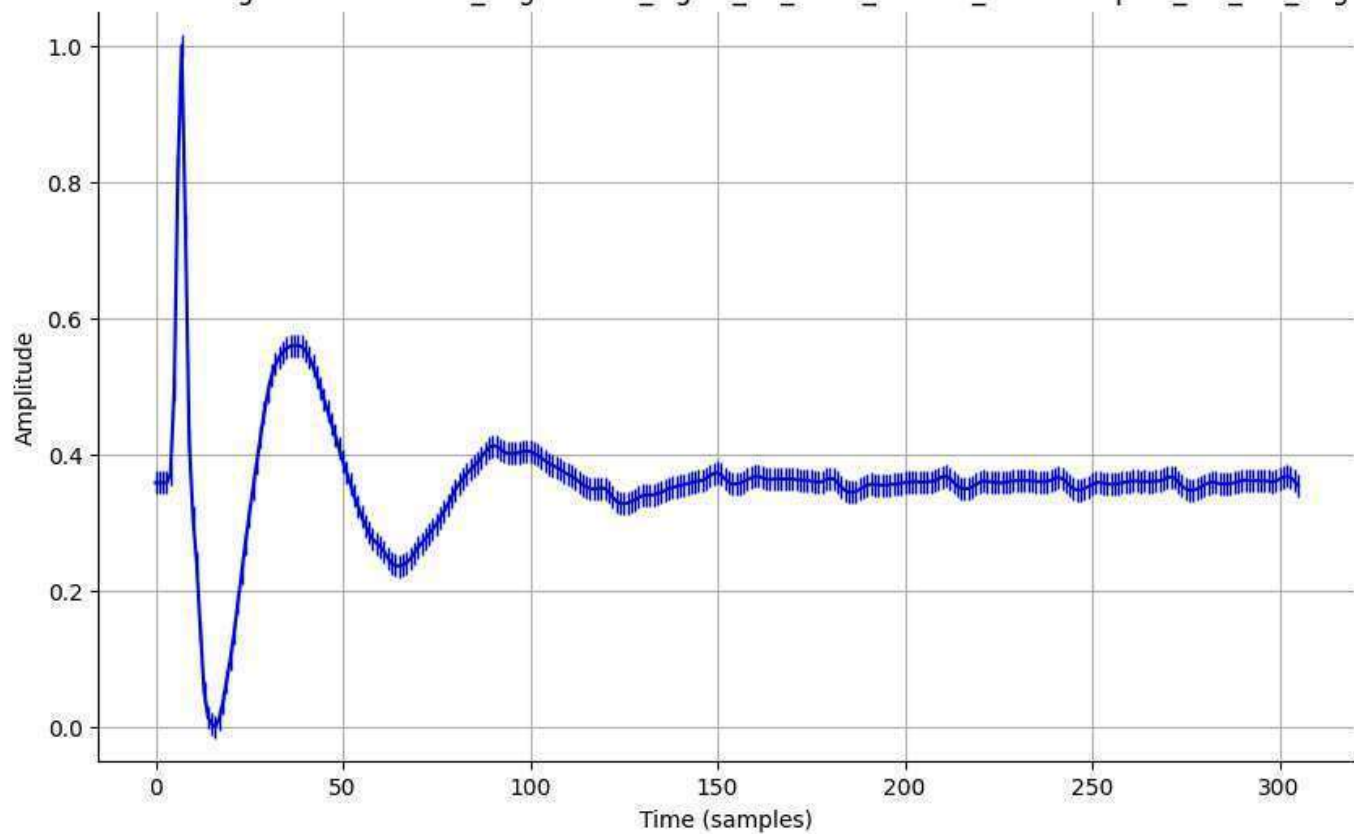
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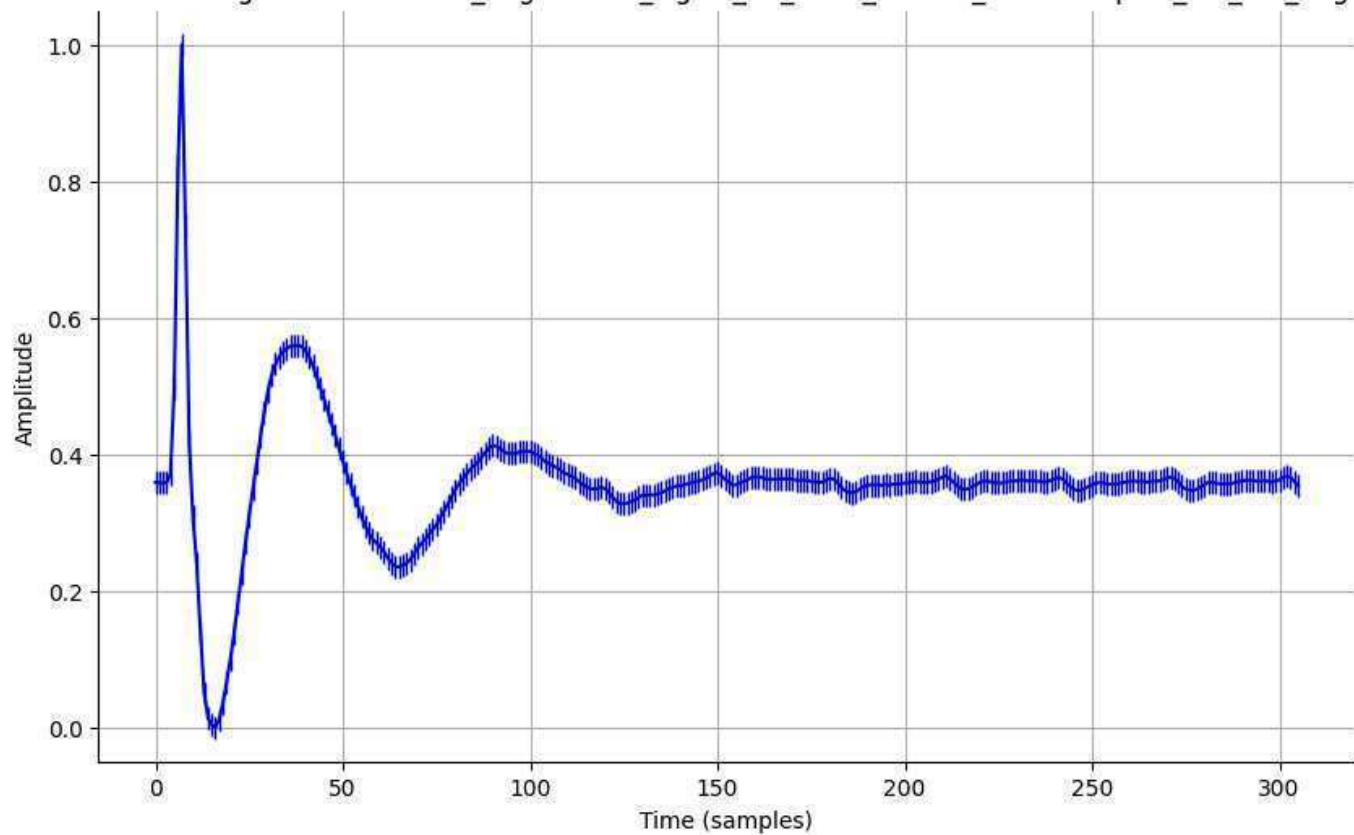
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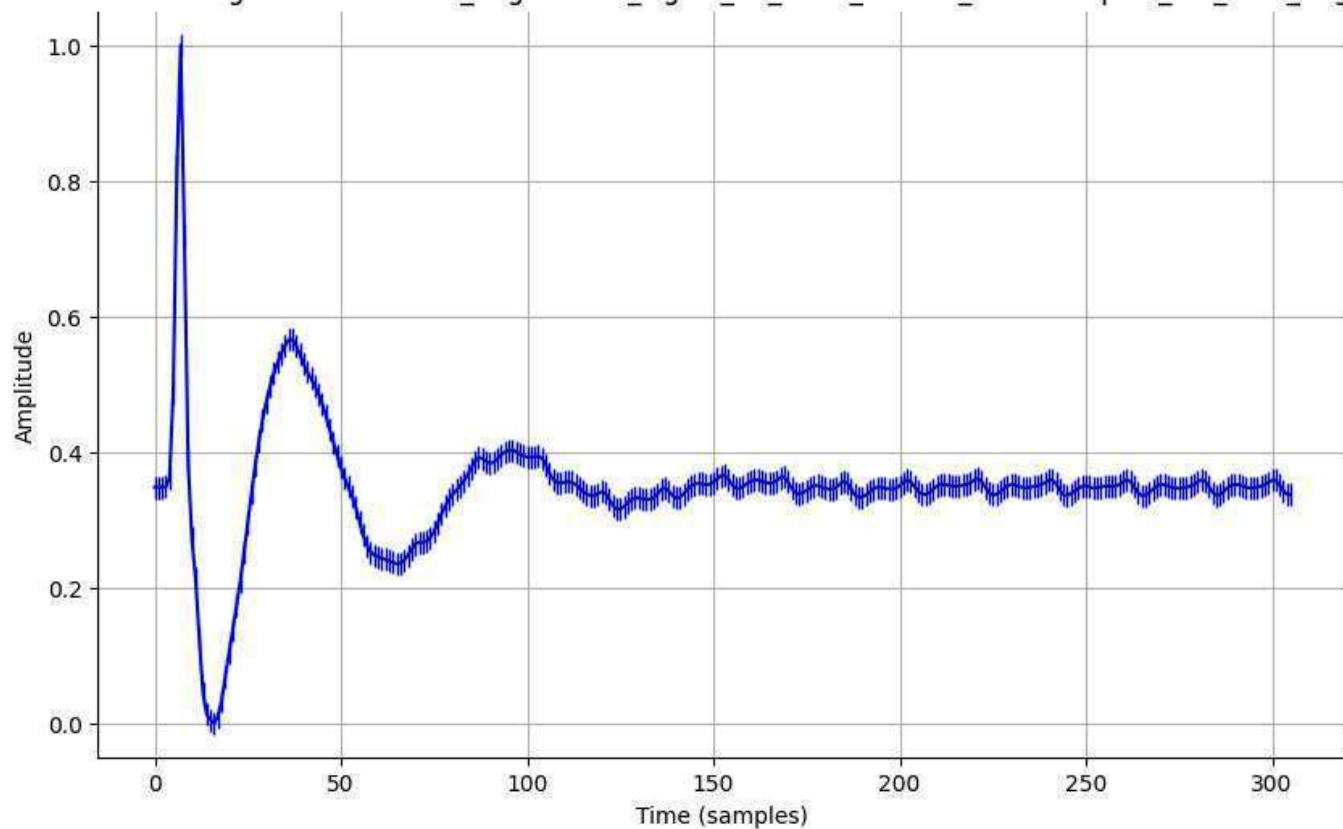
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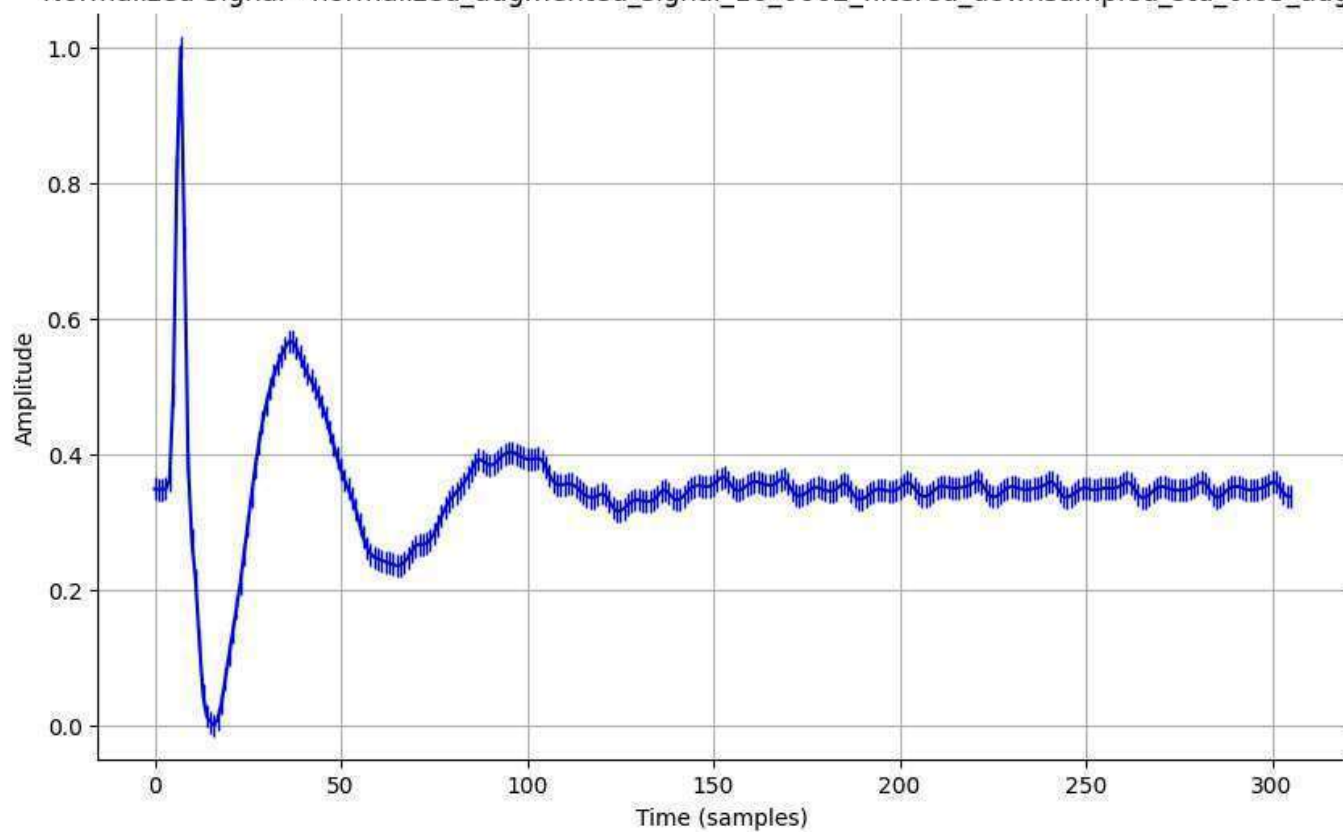
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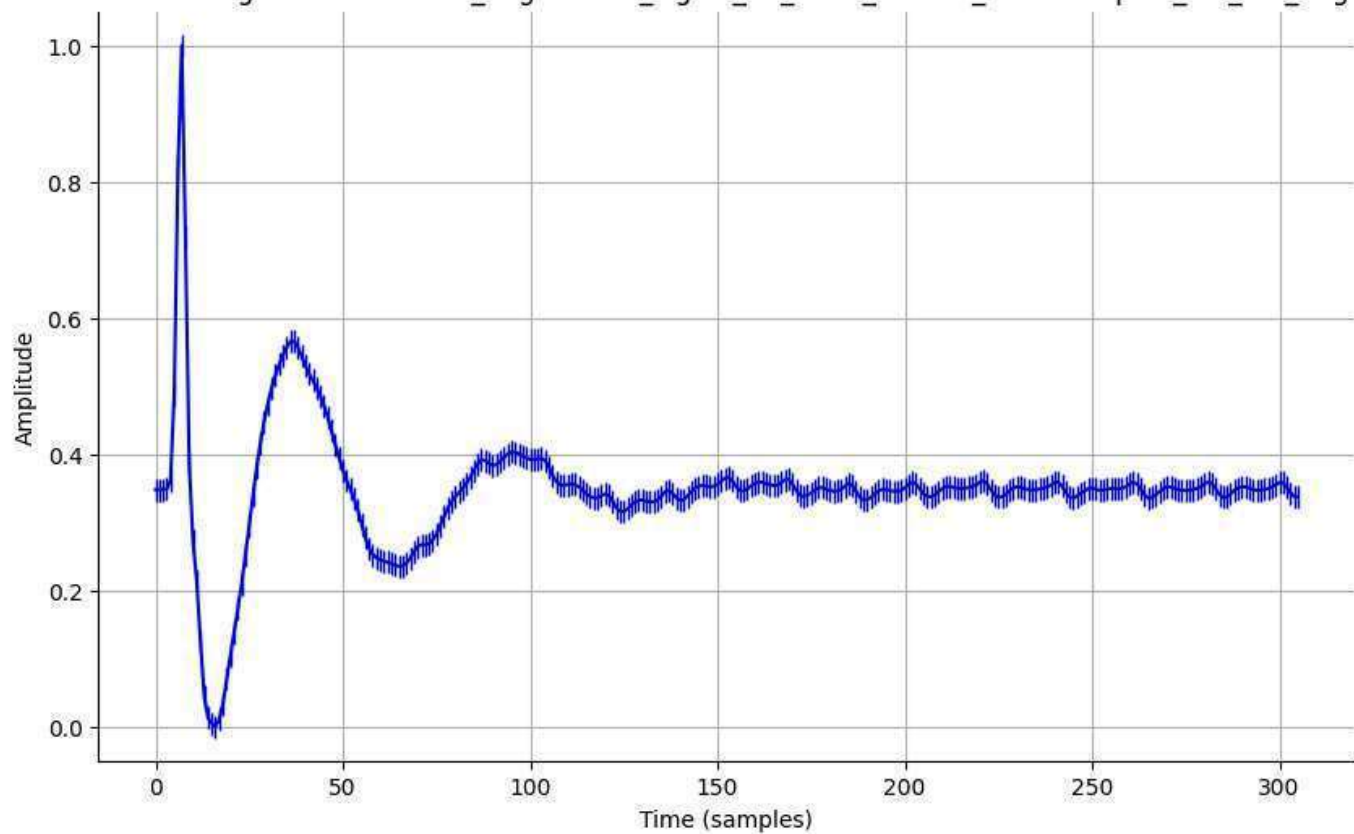
Normalized Signal - normalized_augmented_signal_16_0002_filtered_downsampled_std_0.01_aug.csv



Normalized Signal - normalized_augmented_signal_16_0002_filtered_downsampled_std_0.05_aug.csv



Normalized Signal - normalized_augmented_signal_16_0002_filtered_downsampled_std_0.1_aug.csv



Normalized Signal - normalized_augmented_signal_16_0002_filtered_downsampled_std_0.2_aug.csv

